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Crazy 2

Papers 1–4 combined · Sections 4–9

All four answer books merged section by section; overlapping answers are printed once with every fact kept, tagged P1–P4 by where they were asked.

Section	Title	Topics
4	Paediatric Urology 1	21
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Section 4 · Paediatric Urology 1

Long answer questions (LAQ)

LAQ 1 Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING

P1 · P3 · P4 **merged from 6**

Asked as: P1 LAQ "Primary (congenital) VUR: VUJ embryology, pathophysiology, natural history and management" (Describe embryological development of VU Junction and outline management of primary VUR?, Summer 2020 Paper I Q2; B/l congenital VUR - Pathophysiology and natural history, Practice Set 1 Paper I Q3f SAQ) · P3 LAQ "Management of bilateral VUR: medical treatment, surgical techniques, selection and long-term outcome" (Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR, Summer 2015 Paper III Q2; Discuss mgt. of b/l VUR, Practice Set 2 Paper III Q2; Discuss treatment of b/l VUR. Describe surgical techniques and criteria for their selection, Practice Set 2 Paper III Q3d SAQ; Discuss the management of B/l VUR and discuss its long term outcome, Practice Set 6 Paper III Q1) · P4 LAQ "VUR: etiopathogenesis, recent trends in management and STING technique" (Describe etiopathogenesis of VUR and recent trends in management, Practice Set 2 Paper IV Q2; STING technique for VUR, Summer 2020 Paper IV Q3c and Practice Set 5 Paper IV Q3e SAQ) · P3 LAQ "Development and clinical anatomy of Vesico-Ureteric junction and management of grade 3 VUR in adult?" · P3 SAQ "Endoscopic correction of VUR (STING procedure): indications and complications" (Endoscopic correction of VUR : indications and complications, Summer 2019 Paper III Q3e; Sting procedure, Winter 2024 Paper III Q3f) · P3 SAQ "Paquin and Gil-Vernet techniques of ureteric reimplantation"

"B/l congenital VUR" is read as bilateral primary (congenital) VUR (vesicoureteral reflux), often detected antenatally. STING is taken as subureteric Teflon injection, the endoscopic treatment of VUR.

Definition and epidemiology

- **VUR**: retrograde flow of urine from the bladder into the upper tract.
- **Primary**: congenitally deficient UVJ (ureterovesical junction) flap valve with a short intravesical tunnel and a normal bladder. **Secondary**: high bladder pressure (PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction)).
- Found in about **30% of children with UTI (urinary tract infection)**, about 30% of siblings and about 17% of children imaged for hydronephrosis. The younger the child, the likelier VUR.
- Infant boys predominate (higher voiding pressures, more UTI if uncircumcised); overall girls are affected twice as often. Less common in Black children.
- **Bilateral VUR is a high-risk feature:** it resolves less and scars more.
- **Infection, not sterile reflux, injures the kidney.** Infected urine plus intrarenal reflux causes scars, hypertension and CKD (chronic kidney disease); some high-grade "scars" are congenital dysplasia. So management targets infection risk, not the reflux itself.

Embryology of the VUJ (vesicoureteric junction)

1. **Ureteric bud (weeks 4–6)** arises from the caudal mesonephric (Wolffian) duct: ureter and duct form a "Y", and the common nephric (excretory) duct is the stem.
2. **Reciprocal induction:** the bud enters the metanephric blastema, branches into the collecting system and induces nephrons.
3. **Weeks 5–6:** the urorectal septum divides the cloaca into urogenital sinus (UGS) and anorectal canal.
4. **Absorption:** the common nephric duct is taken into the UGS (current view: removed by **apoptosis**), so ureter and duct open separately.
5. **Rotation (Mackie and Stephens):** the ureteric orifice moves **cranially and laterally** to the bladder base; the duct stays caudal as the ejaculatory duct, ending in the prostatic urethra (Gartner duct in females).
6. **Trigone:** classically ureteral; lineage tracing shows it is mostly bladder muscle. The **Chwalla membrane** over the orifice ruptures; its persistence explains ureterocele.
7. **Maturation:** ureteric muscle joins the detrusor as **Waldeyer's sheath**; the tunnel lengthens with growth, so VUR resolves with age.

Bud origin	Orifice	Result
Normal	Trigone corner	Tunnel length to diameter about **5:1** (Paquin); refluxing junctions about 1.4:1
Too low (early budding, early contact with UGS)	High, **lateral**	Short tunnel, so **primary VUR**; abnormal bud–metanephros contact gives **dysplasia**
Too high (late budding, late contact)	Caudal (distal), **medial**	**Ectopic ureter** (bladder neck, urethra, Wolffian structures), often obstructed
Two buds	Duplex	**Weigert–Meyer rule:** upper pole caudal-medial (ectopia, ureterocele); lower pole lateral (reflux)

- **Concept:** bud position fixes both orifice and kidney, so this single embryological fault (lateral orifice, short tunnel, dysplasia) explains why high-grade VUR and dysplasia coexist.

Clinical anatomy of the VUJ

- **Course:** juxtavesical ureter in **Waldeyer's sheath**, then an oblique intramural segment through the detrusor hiatus, then the submucosal tunnel, then the orifice (caudal and medial to the entry).
- **Intramural length 1.2–2.5 cm in adults** (0.5–0.8 cm in neonates). The **narrowest part** of the ureter (about 3–4 mm): the commonest stone impaction site.
- **Trigone:** superficial trigone (ureteric longitudinal muscle, running to the verumontanum); deep trigone (Waldeyer's sheath plus detrusor, to the bladder neck). Its base is the interureteric ridge. Waldeyer's sheath anchors the intramural ureter to the deep trigone.
- **Orifice (Lyon):** position A (medial trigone) to C (lateral); shape cone (normal), stadium, horseshoe, **golf-hole**. The more lateral and open, the more reflux.
- **Relations:** the vas (male) or uterine artery (female) crosses the ureter. **Pelvic plexus nerves lie dorsomedial to the VUJ**; wide extravesical dissection causes voiding dysfunction.
- **Clinical:** Hutch diverticulum at the hiatus; "fish-hook" ureter with BPH (benign prostatic hyperplasia).

Normal antireflux mechanism

- Balance of three factors: **ureteric peristalsis**, **UVJ anatomy** and **bladder dynamics** (low bladder pressure).
- **Passive flap valve:** the submucosal (intramural) ureter is compressed against firm detrusor backing as the bladder fills. This needs adequate tunnel length and fixation.
- **Paquin:** tunnel length to diameter about **5:1** normally versus 1.4:1 in refluxing units. Orifice shape also matters.
- **Active:** ureteric peristalsis opens the tunnel for each bolus, then it closes.
- **Bladder pressure:** high storage or voiding pressure overwhelms a normal valve.

Etiopathogenesis

1. Embryology (Mackie and Stephens): early budding gives a high, lateral orifice with a short tunnel (VUR) and dysplasia; late budding gives a distal, medial, ectopic and often obstructed ureter (see above).

2. Primary versus secondary

Primary	Secondary
Congenitally deficient UVJ: short tunnel, weak detrusor backing, lateral orifice	Normal UVJ overwhelmed by pressure: PUV, neurogenic bladder (VUR in up to 30% of spina bifida), dysfunctional voiding; or UVJ anomaly (ureterocele, duplication, paraureteral diverticulum)

- They overlap: "primary" VUR often has a treatable secondary component.

3. Bladder and bowel dysfunction (BBD): high voiding pressures and incomplete emptying distort the UVJ. Detrusor overactivity is the commonest urodynamic finding. Infant boys have high voiding pressures that settle with maturation.

4. Genetics: sibling prevalence about 30%, highest under 3 years; identical twins almost fully concordant; polygenic.

5. Infection: cystitis lowers the reflux threshold; *E. coli* endotoxin causes ureteric atony.

Pathogenesis of renal damage

	Reflux-associated dysplasia	Acquired scar
Cause	Abnormal bud (congenital)	Infected urine plus VUR
Typical patient	Boy, grade IV–V, antenatal hydronephrosis	Girl, recurrent febrile UTI, BBD
Preventable	No	Yes

- **Intrarenal reflux** through flat compound papillae at the poles needs only 2 mmHg in neonates, so scars are polar.
- **Sterile reflux rarely scars** unless pressures are high (Ransley and Risdon). Most scars follow early infections ("big bang", now debated).
- Sequelae: hypertension in **34–38%** of scarred children; CKD; poor growth; pre-eclampsia.
- **Also know:** massive bilateral VUR gives the megacystis–megaureter association.

International grading (International Reflux Study, 1981; VCUG (voiding cystourethrogram))

Grade	Findings
I	Non-dilated ureter only

Grade	Findings
II	Ureter, pelvis and calyces reached; no dilatation, sharp fornices
III	Mild–moderate dilatation of ureter, pelvis and calyces; minimal forniceal blunting
IV	Moderate dilatation and tortuosity; fornices obliterated but **papillary impressions kept**
V	Gross dilatation and tortuosity; **papillary impressions lost**; intrarenal reflux

- **Also know:** grading varies between observers; UDR (ureteral diameter ratio) predicts resolution better.

Evaluation (children)

- History of febrile UTIs and BBD; urine culture; BP (blood pressure), growth, spine, urine protein, creatinine.
- **RBUS (renal and bladder ultrasound)**: a normal scan does not exclude VUR.
- **VCUG** (gold standard): grade and urethra (excludes PUV). **Nuclear cystogram** (lower dose) for follow-up; **ceVUS (contrast-enhanced voiding urosonography)** avoids radiation.
- **DMSA (dimercaptosuccinic acid) scan**: scars and split function; a defect persisting 6 months is a scar.
- **UDR** predicts resolution and breakthrough UTI better than grade.

Natural history

- Resolution comes from **elongation of the intravesical ureter** with growth.
- Resolution falls with grade and age: grade II 63–85%, grade III about 50%, **grades IV–V and bilateral grade III rarely** (International Reflux Study). Younger age resolves better; infants with grade IV–V may still resolve fairly well.
- **Bilaterality and older age delay resolution:** persistence is highest in bilateral grade IV (AUA (American Urological Association) 1997: about 10% resolve in 5 years), and in bilateral grade III at 6–10 years.
- **Consequences of bilateral damage:** adult hypertension in 34–38% of scarred children; CKD, with reflux nephropathy a leading paediatric transplant diagnosis; poor growth; pre-eclampsia (worse with bilateral scars). Correction cannot reverse dysplasia.

Risk stratification (Hidas)

Low risk	High risk
Male, >5 years	Female, <1 year (infant)
Unilateral, grade I–II	**Bilateral**, grade III–V
No BBD, no scars	BBD, scars, multiple febrile UTIs

- Two-year breakthrough UTI on CAP (continuous antibiotic prophylaxis): 8.6% low, 26% intermediate, 62.5% high risk.

Recent trends in management

1. **Treat infection risk, not reflux:** risk stratification by sex, age, grade, laterality, BBD, scars and presentation (Hidas model).
2. **Fewer VCUGs:** ultrasound first (AAP (American Academy of Pediatrics) 2011, NICE (National Institute for Health and Care Excellence)); or "top-down" DMSA first. Contrast-enhanced voiding urosonography avoids radiation.

3. **Objective measures:** UDR predicts resolution and breakthrough UTI better than grade; machine-learning models.
4. **Selective CAP:** RIVUR halved recurrent UTI but did not reduce scarring; NNT (number needed to treat) 5 in high-risk and 18 in low-risk children. The Swedish Reflux Trial showed benefit in girls with dilating VUR.
5. **BBD first:** timed voiding, constipation treatment, anticholinergics. Topical steroid for phimosis in boys.
6. **Endoscopic injection:** Dx/HA (dextranomer/hyaluronic acid); HIT and double HIT (hydrodistension implantation technique) give about 90% resolution. Vantris works better but can obstruct late.
7. **RALUR-EV (robot-assisted extravesical reimplantation):** less pain and shorter stay; results approaching the open benchmark of **95.9%**.
8. **Less follow-up imaging:** VCUG may be omitted after surgery for low-grade VUR if ultrasound is normal.

Medical management

- **Aim:** prevent febrile UTI and scars while VUR resolves.
- **Treat BBD first:** timed voiding, hydration, constipation treatment, anticholinergics if needed. BBD raises breakthrough UTI and lowers resolution and endoscopic success.
- **CAP:** one low dose at night, a quarter of the treatment dose.

Agent	Age	Dose	Note
Trimethoprim or amoxicillin	<2 months (young infants)	—	Avoid sulfonamide (neonatal jaundice)
TMP-SMX (trimethoprim-sulfamethoxazole)	>2 months	2 mg/kg TMP	Avoid under 2 months (kernicterus); used in RIVUR
Nitrofurantoin	>2 months	1–2 mg/kg	Avoid in G6PD (glucose-6-phosphate dehydrogenase) deficiency; rare lung, liver toxicity

- **RIVUR (Randomized Intervention for Children with Vesicoureteral Reflux, 2014)**, 607 children: recurrent UTI fell from **23.6% to 12.9%** on TMP-SMX; **scarring unchanged**; most benefit with BBD and in high risk.
- **Swedish Reflux Trial** (grade III–IV, age 1–2 years): febrile UTI in girls 19% on CAP, 23% after endoscopy, 57% on surveillance; **no difference in boys**.
- **IRSC (International Reflux Study in Children) and Birmingham trials:** surgery equalled medical care for (new) scarring. **PRIVENT:** 6% absolute UTI reduction.
- **Circumcision** or topical steroid for phimosis: about 4 boys with VUR circumcised prevent one UTI (NNT about 4).
- After stopping CAP, 8–10% have a febrile UTI.
- Low-risk children: observation without CAP.
- **Guidelines:** AUA 2010/2017 recommends CAP for VUR with BBD and treating BBD first; EAU/ESPU (European Association of Urology/European Society for Paediatric Urology) give high-risk children CAP and correct early after breakthrough infection.

Indications for correction

- Breakthrough febrile UTI on CAP despite good bladder and bowel habits.

- New scars or falling function.
- Persistent high-grade (IV–V) bilateral VUR (girls traditionally corrected before puberty; benefit unproven).
- Non-adherence or intolerance to CAP; parental choice.
- **Controversy:** asymptomatic bilateral high-grade VUR in an older boy with normal kidneys can be observed.

Case: 8-year-old boy with bilateral grade IV VUR

- Exclude PUV, neurogenic bladder and BBD; DMSA, creatinine, BP.
- This rarely resolves at 8 years, but older boys have low UTI risk, so treating an **asymptomatic** boy is **controversial**.
- **No UTI, no scars:** observation or CAP with BBD therapy. **Breakthrough UTI, scars or falling function:** correct surgically.
- **Choice here: bilateral Cohen.** `Exclude PUV/neurogenic bladder, treat BBD -> febrile UTI, scars or renal loss? Yes -> bilateral Cohen (or Dx/HA) No -> observe or CAP; correct if UTI/scar`

Endoscopic correction: STING procedure

Definition and principle

- **STING (subureteric Teflon injection):** endoscopic injection of a **bulking agent** beneath the ureteric orifice. The mound supports and lengthens the intramural tunnel, restoring the **flap valve**.
- First done with Teflon by Matouschek (1981); popularised by **O'Donnell and Puri, 1986**. **Dx/HA (Deflux)** is now standard.

Bulking agents

Agent	Particle / property	Problem
Teflon (PTFE (polytetrafluoroethylene) paste)	90% <40 µm, permanent	**Migration**; never FDA (US Food and Drug Administration) approved
Dx/HA (Deflux)	Microspheres **80–250 µm**; gel resorbed	Volume loss about 23%; late recurrence; FDA approved 2001; standard agent
Polyacrylate (Vantris)	300 µm, permanent	Works better but late ureteric obstruction

- **Also know:** Macroplastique.

Indications

- Persistent primary VUR (grades II–IV) with **breakthrough febrile UTI** on prophylaxis, **new scars**, poor adherence or parental preference.
- Older girls; residual VUR after reimplantation; duplex or secondary VUR (lower success).
- **Relative contraindications:** grade V, **UVJ obstruction**, untreated BBD, large diverticulum.

Technique

1. Antibiotic at induction; offset-lens cystoscope; rigid 3.7–5 Fr needle; bladder partly filled.
2. **STING:** needle enters **2–3 mm below the orifice at 6 o'clock**, bevel up, advanced **4–5 mm** submucosally.
3. Inject slowly; a mound should appear within **0.1–0.2 mL**, else reposition.

4. Aim for a **"volcano" mound** with a slit-like orifice. Do not cauterise bleeding (the implant extrudes). Day case.

Technique	Needle entry	Result
STING	2–3 mm below the orifice, at 6 o'clock	Baseline
HIT (hydrodistension implantation technique)	Inside the hydrodistended orifice	Better than STING, all grades
Double HIT	Two intraureteric sites	About 90% resolution

Results

- Low-grade VUR about 90% after one injection.
- **Elder 2006 meta-analysis** (one injection): grade I–II about 79%, III 72%, **IV 63%**, V 51%; repeat injections raise success.
- **Swedish Reflux Trial:** 20% recurrence at 2 years after initial success.
- Less effective in adults.

Complications

- **Failure or recurrence** (commonest): repeat injection once or twice, then reimplant (reimplantation is not harder after Dx/HA).
- **New contralateral VUR** after unilateral injection.
- **Ureteric obstruction:** early or **late-onset**, so long-term ultrasound.
- Febrile UTI despite a negative VCUG.
- **Calcified mound** (over half) mimics a ureteric stone. Teflon migration.

Follow-up

- Antibiotics 3 months; ultrasound at 2–6 weeks; ultrasound with or without VCUG at 3 months.

Ureteric reimplantation: principles and techniques

- **Principles:** exclude secondary VUR; gentle handling; tension-free; straight course; tunnel **5:1** (Paquin's rule).
- Open success **95.9%** overall (Elder 1997; AUA meta-analysis benchmark).

Technique	Method / tunnel	Strength / weakness
Cohen (intravesical)	Cross-trigonal	Commonest; success over 98%; one cystotomy for both sides; later ureteroscopy or retrograde access hard
Politano–Leadbetter (intravesical)	New superomedial hiatus above the old one	Long tunnel; blind hiatus risks bowel injury
Glenn–Anderson (intravesical)	Orifice advanced towards the bladder neck	Least kinking; short tunnel
Lich–Gregoir (extravesical)	Detrusorotomy, ureter laid in	No haematuria or spasm; no cystotomy; **retention up to 10% after bilateral repair** (pelvic plexus injury); stay close to the ureter

Technique	Method / tunnel	Strength / weakness
RALUR / RALUR-EV (robot-assisted laparoscopic ureteral reimplantation, extravesical)	Y-shaped detrusorotomy	Less pain, shorter stay; results approaching open
Paquin	Combined extra- and intravesical, suprahiatal	Long tunnel; for dilated ureters
Gil-Vernet	Intravesical trigonoplasty, no tunnel	Low-grade bilateral VUR

Paquin technique

- **Combined extravesical–intravesical** reimplant described by Paquin (1959), with the 5:1 rule.
- Steps: 1. Ureter mobilised **extravesically** and divided at the bladder. 2. Bladder opened; a **new hiatus** created higher on the posterior wall (**suprahiatal**). 3. Long **submucosal tunnel** from the new hiatus down to the original orifice site, length about 5 × diameter. 4. Ureter drawn through, spatulated, sutured to mucosa and anchored to detrusor. 5. Old hiatus closed.
- **Advantages:** long tunnel under direct vision; good for **dilated ureters and megaureter** (can combine with tapering).
- **Disadvantages:** kinking or obstruction at the new hiatus.

Gil-Vernet trigonoplasty

- **Intravesical, with no tunnelling and no ureteric mobilisation.**
- Steps: 1. Bladder opened. 2. **Transverse mucosal incision** across the trigone between the two orifices. 3. Intramural ureters with their muscle sheath **advanced medially** by one non-absorbable or absorbable stitch across the midline. 4. Mucosa closed.
- **Effect:** lengthens and medialises the intravesical ureters; corrects both sides at once.
- **Indications:** low to moderate grade (I–III), usually bilateral VUR, with non-dilated ureters.
- **Advantages:** quick, simple, low obstruction risk; easy later retrograde access.
- **Disadvantages:** unsuitable for high grade, dilated ureters or paraureteral diverticulum.

Feature	Paquin	Gil-Vernet
Approach	Extra- + intravesical	Intravesical
Ureter divided	Yes	No
Tunnel	New long submucosal	None (advancement)
Best for	High grade, dilated	Low grade, bilateral

Criteria for selecting the technique

Situation	Preferred option
Grades II–IV, parents want least invasive	Dx/HA injection (lower success at higher grade)
Bilateral VUR, small or trabeculated bladder	**Cohen** (both sides through one cystotomy)
Grade V, dilated ureter	Intravesical reimplant with or without tapering (Politano–Leadbetter or Cohen)

Situation	Preferred option
Unilateral, older child	Lich–Gregoir or RALUR-EV
Bilateral extravesical	Nerve-sparing dissection; warn about retention
Duplex system	Common sheath reimplant
Failed injection	Reimplant (not harder after Dx/HA)
Neurogenic or PUV bladder (secondary VUR)	Treat the bladder first (anticholinergics, CIC (clean intermittent catheterisation))

Complications and follow-up after reimplantation

- Ultrasound at 6–12 weeks.
- Complications: transient obstruction; persistent reflux (usually resolves by 1 year); contralateral VUR (about 1 in 10, usually resolves); leak; retention after bilateral extravesical repair.
- Obstruction early (oedema): stent or nephrostomy; late (ischaemia, kink): redo.
- Lifelong BP and renal checks if scarred; scarred women risk pre-eclampsia.

Long-term outcome

Outcome	Facts
Renal scarring	About 15% get a febrile UTI within 2 years; about 15% of those scar. Risk highest under 1 year. Much "scarring" in boys with high grade is **congenital dysplasia**, not preventable by any treatment
Hypertension	Reflux nephropathy may be the commonest cause of childhood hypertension after essential hypertension (renin–angiotensin). **34–38%** of scarred children are hypertensive as adults. Correcting VUR does not lower BP; nephrectomy of a small scarred kidney helps selected cases
CKD and ESKD (end-stage kidney disease)	Now rare: RRT (renal replacement therapy) from reflux nephropathy fell from 1.9 to 0.6 per million (USRDS (United States Renal Data System), 1996–2014). Bilateral scars cause hyperfiltration, proteinuria, concentrating defect, tubular acidosis
Pregnancy	VUR in late pregnancy: pyelonephritis about one-third versus under 5%. Scars: about **3× hypertension, 8× pre-eclampsia**; bilateral worse. Impaired function: fetal loss. Reimplanted women still get UTIs
Growth; offspring	Catch-up growth after infection control; about 43% of screened offspring have VUR

- **Concept:** in adult life **the scar, not the reflux, carries the risk**, so correction does not remove it. Scarred patients need lifelong BP, proteinuria and creatinine checks.

Grade 3 VUR in an adult

- **Grade 3:** mild–moderate dilatation of ureter, pelvis and calyces, minimal forniceal blunting.
- **Adults rarely lose function from reflux alone**, unless high bladder pressure, stasis and UTI coexist. The **scar**, not the reflux, carries the risk. Spontaneous resolution is unlikely.
- **Causes:** persistent primary VUR; **secondary**: BOO (bladder outlet obstruction), neurogenic bladder, resection near the orifice, ureterocele incision, paraureteral diverticulum, transplant ureter, TB (tuberculosis).

- **Presentation:** recurrent febrile UTI, pyelonephritis (especially in pregnancy), loin pain on voiding, hypertension, proteinuria, CKD.
- **Evaluation:** VCUG (grade, bladder, diverticulum, outlet); ultrasound and **DMSA** for scars and split function; MAG3 (mercaptoacetyltriglycine) renogram if obstruction is suspected; **UDS (urodynamics)**: compliance, voiding pressure, residual; cystoscopy; blood pressure, creatinine, urine protein.
- **Treat secondary causes first** (relieve BOO; CIC and antimuscarinic for neurogenic bladder), then reassess.

Situation (primary grade 3)	Management
Incidental; no UTI, no scars	Observe: timed voiding, bowel care, prompt UTI treatment
Recurrent febrile UTI, new scars, falling function	**Correct the reflux**
Woman planning pregnancy with scars or prior pyelonephritis	Consider correction before pregnancy
Poorly functioning scarred kidney with infection or hypertension	Nephrectomy

- Low-dose prophylaxis selectively; adult evidence is limited.
- **Correction:** endoscopic Dx/HA injection (day case, repeatable, but less effective in adults); or **ureteroneocystostomy** (open, laparoscopic or robotic). Lich–Gregoir/RALUR-EV suits adults (no cystotomy; stay close to the ureter to protect nerves); Politano–Leadbetter (long tunnel); Cohen (over 98%; later retrograde access hard).
- **Psoas hitch** if length is needed; tapering seldom needed in grade 3.
- Follow-up: ultrasound at 6–12 weeks; lifelong blood pressure and renal checks if scarred.

Overall management algorithm

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Treat BBD -> low risk: observe | high risk (bilateral): CAP Breakthrough UTI / new scar / persistent IV-V -> Dx/HA injection or reimplant (Cohen) Scars -> lifelong BP and proteinuria checks
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Diagrams to draw:

- "Y" of ureteric bud and mesonephric duct with orifice migration (early-budding Y diagram).
- Long oblique tunnel versus short lateral refluxing tunnel; oblique intramural ureter in Waldeyer's sheath with submucosal tunnel and detrusor backing.
- Grades I–V.
- Cohen cross-trigonal reimplant; Gil-Vernet transverse trigonal incision and medial advancement stitch.
- Needle sites: STING below the orifice, HIT and double HIT inside the tunnel, with volcano mound.

Key points to remember:

- Low (early) bud → high lateral orifice → short tunnel → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 normal versus 1.4:1 refluxing (Paquin); adult intramural ureter 1.2–2.5 cm.
- Dysplasia (boys, congenital) versus scar (girls, infection); sterile reflux rarely scars.
- Grade V loses papillary impressions; grade IV–V and bilateral grade III rarely resolve; bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Treat BBD and give CAP first; correct for breakthrough febrile UTI, new scars or persistent grade IV–V.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- STING: O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%; Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction; calcified mound mimics a stone.
- Open reimplantation 95.9%; Cohen cross-trigonal (over 98%) suits bilateral VUR through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- Paquin: tunnel 5:1; combined, suprahiatal reimplant. Gil-Vernet: trigonoplasty, medial advancement, no tunnel; for low-grade bilateral VUR.
- 34–38% of scarred children become hypertensive; correction does not lower BP; scars bring 8× pre-eclampsia risk.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first; correct for recurrent pyelonephritis, new scars or pregnancy plans.

LAQ 2 Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods

P1 · P2 · P3 **merged from 4**

Asked as: P1 LAQ "Testicular development, gonadal descent, abnormalities of descent and bilateral undescended testis" (Write a note on the embryological descent of the gonads, explaining the various phases of descent. Outline the evaluation and management of bilateral undescended testis in a 3 year old child, Winter 2024 Paper I Q2; Describe developmental embryology of testes and its physiological descent. Mention various abnormalities associated with testicular descent, Practice Set 14 Paper I Q1) · P3 SAQ "Maldescended (undescended) testes and treatment algorithm" (Treatment algorithm of undescended testis, Summer 2018 Paper III Q3a; Maldescended testes, Practice Set 5 Paper III Q3b) · P2 SAQ "Undescended testis" · P1 SAQ "Methods of orchidopexy"

Maldescent is taken as any testis not in a stable scrotal position: undescended, ectopic, retractile and acquired.

Development of the testis

- **Indifferent gonad:** the gonadal (germinal) ridge forms at **4 weeks** on the medial urogenital ridge. It has four components: coelomic epithelium, underlying mesenchyme, adjacent mesonephros and **PGCs (primordial germ cells)**.
- PGCs arise in the **yolk sac wall in week 3** and migrate along the dorsal mesentery of the hindgut to the ridge (about 6 weeks).
- **SRY (sex-determining region Y)** on the short arm of Y, partly activated by SF1 (steroidogenic factor 1), switches on **SOX9** in pre-Sertoli cells. Sertoli cells then appear (about 40 days). Without SRY no Sertoli cells form and the gonad becomes an ovary.
- Sertoli cells enclose germ cells as **testicular cords (about 6 weeks)**, which canalise into seminiferous tubules.
- **Rete testis** forms with the cords and joins **5–12 mesonephric tubules**, which become efferent ductules. Full continuity (tubules, rete, efferent ductules, epididymis) is established by the **third month**.
- The testis turns round and a thick **tunica albuginea** forms. Testes become spherical after 7–8 weeks; ovaries stay oblong.

Hormone	Cell	Action	Male remnants
AMH (anti-Müllerian hormone)	Sertoli	Müllerian regression, weeks 8–10	Appendix testis, prostatic utricle
Testosterone	Leydig (from 8 weeks; driven by placental hCG (human chorionic gonadotrophin), later fetal LH (luteinising hormone))	Wolffian duct to epididymis, vas, seminal vesicle (weeks 8–12)	Appendix epididymis, paradidymis
DHT (dihydrotestosterone)	Formed locally by 5α-reductase type 2	External genitalia, prostate, scrotum (fused by 10–12 weeks)	–

Hormone	Cell	Action	Male remnants
INSL3 (insulin-like peptide 3)	Leydig	Gubernacular swelling (via RXFP2 receptor)	–

Gonadal descent: starting point

- At **7–8 weeks** the testis lies near the kidney (upper lumbar), held by the **cranial suspensory ligament (CSL)** above and the **gubernaculum** below. The gubernaculum appears at 7 weeks, attaches to the **epididymis** and runs to the future inguinal canal.
- **Female:** the CSL persists; the gubernaculum becomes the ovarian and round ligaments; the ovary descends only into the pelvis.

Phases of testicular descent (classic two-phase model)

	Transabdominal phase	Inguinoscrotal phase
Timing	Weeks 10–15	From the 7th month; inguinal passage rarely before 22 weeks; most testes scrotal after 27 weeks
Events	**CSL regresses**; **gubernaculum swells** (outgrowth); its cord shortens, holding the testis at the internal ring while the kidney and abdomen grow away (the kidney moves cranially)	Gubernaculum migrates **3–5 cm** to the scrotum; **processus vaginalis** forms within it; the bulb then regresses; the processus closes
Control	CSL regression by **androgen**; swelling by **INSL3** from Leydig cells, via its receptor RXFP2 (occurs even in androgen resistance)	**Androgen**, mainly indirectly via the **genitofemoral nerve (GFN)** releasing **CGRP (calcitonin gene-related peptide)**; intra-abdominal pressure
Failure gives	Intra-abdominal testis	Canalicular or ectopic testis

- **Modern view:** the "transabdominal descent" is largely relative, not true migration. Barteczko and Jacob describe **five phases** (5 weeks to 28 weeks). The gubernaculum **swells to make room** (outgrowth), then **shrinks to let the testis pass** (regression). Testosterone (peak 14–16 weeks) and INSL3 (15–17 weeks) act together on gubernacular and cremaster growth (myogenesis).
- Spontaneous postnatal descent occurs mainly in the first 3–6 months.
- **Also know:** AMH defects (persistent Müllerian duct syndrome) cause cryptorchidism; Insl3-knockout mice are cryptorchid; the scrotum forms by 10–12 weeks under DHT.

Failed step	Result
CSL retained	Intra-abdominal testis
Abnormal or ectopic gubernacular attachment	Canalicular testis; ectopic (superficial inguinal pouch, perineal, femoral, prepenile)
Processus not closed	Hernia, hydrocele (patent in 87% of unilateral cases)
Fibrous remnant of processus / processus not fully involuted	Acquired (ascending) testis; acquired cryptorchidism

Abnormalities of descent: definitions and classification

Type	Description
Undescended (cryptorchid)	Arrested on the normal path: abdominal, "peeping" at the ring, canalicular, high scrotal/prescrotal
Ectopic	Off the normal path: **superficial inguinal pouch** (commonest), perineal, femoral, prepubic, penile, contralateral scrotum (transverse, crossed)
Retractile	Scrotal testis pulled up by the cremaster; stays down after sustained traction; review yearly (risk of ascent); some become acquired
Acquired (ascending)	Documented scrotal earlier, ascends later; diagnosed at 8–11 years; fibrous processus remnant
Recurrent, secondary, iatrogenic	Re-ascent after spontaneous descent; after hernia repair or orchidopexy
Vanishing vs agenesis	Lost after formation, in utero or perinatally (torsion; monorchia, anorchia) vs never formed

- Clinically: **palpable** (canalicular, high scrotal, ectopic) or **non-palpable** (intra-abdominal, "peeping", vanished/absent).

Epidemiology and aetiology

- 1–4% of term boys (P2: 2–4% at birth, 1–2% at 3 months); up to 45% of preterm boys. Most descend by 6 months.
- 75–80% palpable; 60–70% unilateral.
- Descent needs Leydig hormones: **INSL3** (transabdominal phase) and **androgen** (inguinoscrotal phase), with the genitofemoral nerve/CGRP guiding the gubernaculum.
- Risk factors: prematurity, low birth weight, syndromes (prune belly, cerebral palsy).

Associated anomalies and sequelae

- **Testicular: TTE (transverse testicular ectopia)**, polyorchidism, splenogonadal fusion; epididymal non-fusion.
- **PMDS (persistent Müllerian duct syndrome):** mutation of AMH or its receptor (AMHR2); bilateral cryptorchidism or TTE.
- **Genital and DSD (differences of sex development):** hypospadias (12–24%); bilateral non-palpable testes need karyotype and 17-OHP (17-hydroxyprogesterone) to exclude **CAH (congenital adrenal hyperplasia)**.
- **Syndromes:** prune belly (all), Klinefelter, androgen insensitivity, hypogonadotrophic hypogonadism, cerebral palsy, gastroschisis, omphalocele, myelomeningocele, PUV (posterior urethral valves).
- **Sequelae:** subfertility (worse if bilateral), **TGCT (testicular germ cell tumour) 2–5-fold**, torsion, hernia, trauma.

Evaluation

- **History:** prematurity, testes once scrotal (acquired), hernia, family history, salt-wasting illness.
- **Examination:** warm room; supine and frog-leg, then cross-legged or squatting. Sweep ("milk") from the anterior superior iliac spine down the canal to the scrotum to find the lowest position; check the superficial inguinal pouch.

- Sustained traction fatigues the cremaster: a **retractile** testis stays in the scrotum after gentle traction.
- Note penis size, meatus (hypospadias), scrotal development, syndromic features. Contralateral hypertrophy suggests a vanished testis.
- **No ultrasound or MRI (magnetic resonance imaging)**: inaccurate (ultrasound sensitivity about 45% for non-palpable testes); they never change the operation or remove the need for surgery, and delay surgery.

Bilateral undescended testis at 3 years

- This is **late presentation**; surgery is advised from 6 months, preferably by 12 months and by 18 months at the latest.
- **Bilateral palpable:** no investigations needed; plan orchidopexy, both sides usually in one sitting.
- **Bilateral non-palpable (or with hypospadias, small phallus or micropenis): treat as DSD until proven otherwise**, with urgent work-up:
- **Karyotype**; electrolytes and **17-OHP** to exclude **CAH** in a virilised 46,XX child.
- **Hormones:** LH, FSH (follicle-stimulating hormone), testosterone, **AMH and inhibin B**. Measurable AMH or inhibin B confirms testicular tissue; high gonadotrophins with absent AMH suggest **anorchia**. The **hCG stimulation test** (testosterone rise) is now less used.
- Pelvic ultrasound for Müllerian structures.
- **Both testes very high or of doubtful viability:** stage the sides, the second at 6 months, to avoid bilateral atrophy.
- **Biopsy** only if DSD or dysgenesis is suspected. **Bilateral anorchia:** testosterone at puberty, prostheses later.
- Adjuvant LHRH (buserelin) in bilateral cases is experimental.

Management principles (AUA (American Urological Association) 2014; EAU (European Association of Urology) similar)

- **Observe to 6 months** (corrected age) for spontaneous descent; refer if still undescended.
- **Orchidopexy from 6 months, preferably by 12 months, and by 18 months at the latest.** Reason: germ-cell and Leydig-cell loss starts in infancy; early surgery restores testicular growth and lowers cancer risk. **Kollin RCT (randomised controlled trial):** surgery at 9 months gave better testicular growth than at 3 years.
- **No hormonal therapy** (hCG or LHRH (luteinising hormone-releasing hormone)) to induce descent: descent in about 20% only; hCG may increase germ-cell apoptosis.
- **Goals of surgery:** tension-free **dependent scrotal** testis with intact blood supply; ligate the patent processus.

Treatment algorithm

```
Boy with empty scrotum ├─ Bilateral non-palpable ± abnormal phallus → DSD work-up (karyotype, hormones, electrolytes) └─ Otherwise → re-examine at 6 months corrected age Still undescended → refer → examination under anaesthesia (EUA) ├─ Retractile → yearly review (risk of ascent) ├─ Palpable (undescended or ectopic) → inguinal or scrotal (Bianchi) orchidopexy └─ Non-palpable → diagnostic laparoscopy ├─ Testis seen → laparoscopic orchidopexy │ vessels short (testis not within 2–4 cm of ring) → Fowler–Stephens │ (clip testicular
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vessels; testis lives on vasal collaterals; │ one-stage, or two-stage with second stage at 6 months) or Shehata traction ├─ Vessels + vas enter ring → inguinal or scrotal exploration (excise nubbin) └─ Blind-ending vessels → stop (vanished testis)

- **Scrotal nubbin with contralateral hypertrophy** may start with scrotal exploration.
- **Postpubertal intra-abdominal testis with a normal contralateral testis: orchidectomy** (poor function, cancer risk); offer a prosthesis.

Methods of orchidopexy: palpable testis

- **Inguinal orchidopexy** (steps): 1. Groin incision. 2. Open external oblique (spare the ilioinguinal nerve). 3. Free the cord; divide the gubernaculum. 4. Separate and **ligate the sac (patent processus) at the internal ring**. 5. Retroperitoneal mobilisation; divide lateral bands. 6. **Prentiss manoeuvre** if needed (medial transposition under the inferior epigastric vessels) for extra length. 7. **Subdartos pouch.** Transalbugineal sutures are unnecessary.
- **Trans-scrotal (Bianchi) orchidopexy** for low testes: high scrotal incision; sac ligated through it; inguinal conversion 3.5–4.4%; atrophy 0.3–0.6%; less pain.

Methods of orchidopexy: non-palpable testis

Examination under anaesthesia, then laparoscopy.

Method	Principle	Success
Primary laparoscopic	Mobilise vessels, preserve vasal collaterals, new hiatus medial to epigastric vessels	>90% (97%); primary orchidopexy, when possible, is most successful
Open transabdominal (Jones and Bagley)	Preperitoneal high mobilisation	95%
Fowler–Stephens (one- or two-stage)	Divide testicular vessels high; testis survives on **vasal and gubernacular collaterals**; second stage after 6 months	60–97% (one-stage 63%, two-stage 68%)
Shehata traction	Fix testis near the opposite anterior superior iliac spine; vessels lengthen; orchidopexy at 12 weeks	84–85%; atrophy 10%
Microvascular autotransplantation	Testicular vessels to inferior epigastric vessels	96%
Orchidectomy	Postpubertal, dysgenetic or atrophic testis	—

- Fowler–Stephens is for testes **not within 2–4 cm of the internal ring**.
- **Also know:** staged (two-stage) inguinal orchidopexy; Dessanti PTFE (polytetrafluoroethylene) sleeve traction; gubernaculum-sparing technique; redo approaches (Redman lateral, Cartwright fascial patch).

Complications of orchidopexy

- **Atrophy** (highest after Fowler–Stephens), retraction or re-ascent, vas or vessel injury, haematoma, infection; torsion very rare after subdartos fixation.

Follow-up and prognosis

- Check size and position at 6 months. Offer a prosthesis for an atrophic testis.

- **Fertility:** paternity about 80% (four-fifths) after unilateral and one-third after bilateral; worse when bilateral (oligospermia 42%, azoospermia 31% in one large review).
- **Cancer:** 2–5-fold TGCT risk (P1: about 3-fold), higher with late surgery; orchidopexy after 10–11 years carries relative risk 5.8 (Walsh).
- Counsel on fertility and cancer risk; teach **self-examination** after puberty.

Diagrams to draw:

- Sagittal fetus showing CSL, gubernaculum, processus vaginalis and both phases of descent.
- Line of descent with ectopic sites.
- Laparoscopic findings at the internal ring: testis, vessels entering the ring, blind-ending vessels.
- Fowler–Stephens collateral supply: testicular artery clipped, deferential and cremasteric arteries intact.

> **Key points to remember:**
>
> - SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
> - Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
> - Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP; most testes scrotal after 27 weeks.
> - Ectopic = off the path; commonest site superficial inguinal pouch.
> - Operate from 6 months, preferably by 12 months, and by 18 months at the latest.
> - No imaging, no hormones.
> - Non-palpable: EUA then diagnostic laparoscopy; laparoscopy decides.
> - Bilateral non-palpable: DSD work-up first (karyotype, 17-OHP, AMH) to exclude CAH/DSD.
> - Subdartos pouch; Prentiss manoeuvre gives extra length.
> - Fowler–Stephens relies on vasal collaterals.

LAQ 3 Retrocaval ureter and IVC anomalous development

P1 · P3 **merged from 2**

Asked as: P1 LAQ "IVC anomalous development and retrocaval ureter" · P3 SAQ "Retrocaval ureter"

Introduction

- The **IVC (inferior vena cava)** forms between the 6th and 8th weeks from three pairs of fetal veins by anastomosis, regression and persistence. The final IVC is **right-sided**, so long left veins must cross the aorta to reach it.
- Most anomalies are silent CT (computed tomography) findings. They **bleed at surgery, mimic lymph nodes, trap the ureter and alter IVC filter placement**.

Embryology of the IVC

Three paired venous systems

- Fetal veins (posterior cardinal, supracardinal, subcardinal) form a venous collar around each ureter and ascending kidney: posterior cardinal and supracardinal veins dorsally, subcardinal veins ventrally.

Vein	Position	Adult derivatives
Posterior cardinal (first)	Dorsolateral	Iliac confluence; mostly regresses
Subcardinal	Ventromedial	Right: **suprarenal IVC**; intersubcardinal anastomosis: **left renal vein**; left: adrenal and gonadal veins
Supracardinal (last)	Dorsomedial	Right: **infrarenal IVC**; thoracic parts: **azygos and hemiazygos**

Segments of the definitive IVC (top to bottom)

1. **Hepatic**: right vitelline vein.
2. **Suprarenal**: right subcardinal vein (via the subcardinal–hepatic anastomosis).
3. **Renal**: subcardinal–supracardinal anastomosis.
4. **Infrarenal**: right supracardinal vein.
5. **Iliac confluence**: posterior cardinal veins.

Renal collar

- A venous ring encircles the aorta at renal level. The **ventral (pre-aortic) limb** persists as the left renal vein; the **dorsal (retro-aortic) limb** regresses.
- Normally the right **supracardinal** vein, which lies **dorsal** to the ureter, forms the infrarenal IVC, so the normal ureter lies in front of the IVC.

`Left-sided channels regress -> right-sided persist -> single right IVC + pre-aortic left renal vein`

Diagram to draw: three stages: paired cardinal veins, anastomoses with renal collar, definitive IVC.

Anomalies of IVC development

Anomaly	Embryological error	Key features
Retrocaval ureter ("preureteral vena cava")	Infrarenal IVC forms from a vein **ventral** to the ureter	Right only; 1 in 1500 autopsies; male 3:1; presents 3rd–4th decade. **Type I** low loop, fishhook, obstructs; **type II** high loop, sickle, mild
Circumaortic left renal vein	Dorsal limb persists with ventral	Commonest (2–16%); posterior limb runs lower into lumbar IVC
Retroaortic left renal vein	Ventral limb regresses, dorsal persists	Less common; may be compressed ("posterior nutcracker")
Left IVC	Left supracardinal persists, right regresses	Crosses to the right at renal level, in front of the aorta
Duplicated IVC	Both supracardinals persist	Left IVC joins left renal vein; double right cava can trap the ureter
Azygos continuation	Subcardinal–hepatic anastomosis fails	Large azygos; hepatic veins drain directly to right atrium; polysplenia, heterotaxy
Absent infrarenal IVC	Supracardinal and posterior cardinal failure	Deep vein thrombosis in the young

Retrocaval ureter: definition

- The **right ureter** passes **behind the IVC**, then crosses in front of it from medial to lateral to resume a normal course.
- Better termed "**preureteral vena cava**": the anomaly is venous; the ureter is trapped.
- In P3 the question was also asked as "Retrocecal ureter"; this is taken as a typo for retrocaval (circumcaval) ureter.

Retrocaval ureter: embryology

- If a vein **ventral** to the ureter persists as the infrarenal IVC, the ureter is trapped behind it.
- Campbell names a persistent **lumbar subcardinal** vein (the lumbar part of the ventral subcardinal vein; Chapter 52) or a **persistent posterior cardinal** vein (Chapter 93, and most texts); both mean that a ventral vein forms the IVC.
- Variant: double right IVC, with the ureter between the limbs.

Retrocaval ureter: incidence and presentation

- About **1 in 1500** autopsies; **male 3:1**; right side only.
- Presents in the **third–fourth decade** with right flank pain, UTI (urinary tract infection), stones or incidental hydronephrosis.
- Associations: horseshoe kidney, left renal anomalies; bilateral cases occur in situs inversus.

Retrocaval ureter: classification (Bateson and Atkinson)

	Type I (low loop), commoner	Type II (high loop)
Crossing	Lower, at L3–L4	Higher, at or near renal pelvis level
Shape	**J or "fishhook" / reverse-J**, S-curve	Smooth horizontal **sickle** curve
Obstruction	Yes, moderate–severe, at the iliopsoas edge	Mild or none (pelvis also crosses)

Retrocaval ureter: investigations

- **CT urography (computed tomography urography): investigation of choice**, showing ureter and cava together; MRI (magnetic resonance imaging) is the alternative.
- IVU (intravenous urography): fishhook; ureter not seen beyond the J hook.
- Retrograde pyelography: **S-curve** or fishhook at L3–L4.
- **Diuretic renogram:** grades the degree of obstruction and gives split function.
- Pitfall: mistaken for **PUJ (pelviureteric junction) obstruction**, so pyeloplasty fails.

Retrocaval ureter: management

- **Non-obstructed (usually type II): observe.**
- **Obstructed: dismembered ureteroureterostomy** (or ureteropyelostomy): divide the ureter, transpose it **anterior to the IVC**, and do a spatulated, tension-free anastomosis over a **double-J stent**. Excise or bypass the **aperistaltic retrocaval segment**; respect and preserve the ureteric blood supply.
- **Laparoscopic or robotic** repair (transperitoneal or retroperitoneal) is now preferred: equal success and faster recovery.
- Nephrectomy if the kidney is non-functioning.
- A pyeloplasty that misses the diagnosis fails.
- **Also know:** retroiliac ureter (L5–S1, with vasal anomalies); mimics of extrinsic ureteric crossing: spermatic / gonadal or lumbar vein, psoas (iliopsoas) tendon.

Diagram to draw: right ureter hooking behind the IVC with a fishhook pelvis.

Clinical implications of IVC anomalies

1. Diagnosis

- Venous-phase CT or MRI. A left or duplicated IVC mimics **para-aortic lymphadenopathy**; a large azygos mimics a mediastinal mass, risking needless biopsy.
- Retrocaval ureter: S-shaped or fishhook ureter at L3–L4 on retrograde pyelography; CT urography confirms; diuretic renography grades obstruction.

2. RPLND (retroperitoneal lymph node dissection)

- Preoperative CT must look for circumaortic or retroaortic veins and retrocaval ureter. A left or duplicated IVC lies in the left template; a retroaortic limb tears when lumbar vessels are divided.

3. Radical nephrectomy

- The hidden posterior limb of a circumaortic vein bleeds when the artery is approached from behind: ligate both limbs. Caval thrombus in a duplicated IVC needs rerouting of the left IVC.

4. Living donor nephrectomy

- The left kidney is preferred for its long vein. A circumaortic or retroaortic vein shortens it and adds bleeding risk; CT angiography is mandatory, and the right kidney may be chosen.

5. Retrocaval ureter repair

- As above: observe non-obstructed type II; for obstruction, divide, transpose in front of the IVC and re-anastomose over a stent, excising the aperistaltic segment.

6. Interventional and vascular

- **IVC filter:** with duplication, emboli bypass a single filter, so filter both cavae or place the filter suprarenally.
- Aortic surgery risks retroaortic vein injury. A retroaortic vein can cause haematuria or left varicocele.
- **Also know:** variable left adrenal vein entry (relevant to adrenal vein sampling); right gonadal vein drains into the right renal vein in up to 10%.

> **Key points to remember:**
> - Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
> - Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
> - Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
> - Type I fishhook obstructs; type II sickle often does not.
> - CT urography; anterior transposition and ureteroureterostomy.
> - Duplicated IVC: filter both cavae; do not mistake it for nodes.

LAQ 4 Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula

P1 · P4 **merged from 2**

Asked as: P4 LAQ "Hypospadias: present concepts in management, and hypospadias fistula" · P1 SAQ "Hypospadias repairs – single-stage versus staged techniques"

Introduction

- Read "present concepts" as the present (current) concepts in the management of hypospadias.
- Hypospadias is an arrest of the androgen-dependent "closing zipper" of urethral fold fusion (weeks 8–16). Triad: ventral ectopic meatus, ventral curvature, dorsal hooded prepuce.
- 1 in 150–300 male births; distal 70–85%, proximal 10–25%.
- **Goals:** a straight erection, a urethra to the tip (standing voiding, semen delivery) and acceptable appearance, with the fewest operations and a result lasting into adulthood.
- The choice of repair depends on **curvature after degloving**, plate quality and glans size, not on meatal position alone.

Aetiology and associations

- **Aetiology:** causes converge on deficient androgen action (placental insufficiency, assisted reproduction, genes, endocrine disruptors).
- Associations: hernia or hydrocele 9–16%, cryptorchidism about 7% (10% in proximal), enlarged utricle.

Evaluation

- **DSD (disorder of sex development) work-up** if hypospadias plus undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).
- No routine VCUG (voiding cystourethrogram) or renal ultrasound.

Classification: severity is judged in theatre, not by the meatus

- A "distal" meatus may hide absent spongiosum and severe curvature, so classify **after degloving**.

GMS (glans–meatus–shaft) score	Glans	Meatus	Shaft curvature
1	Good size, deep groove	Glanular	None
2	Adequate, grooved	Coronal	<30°
3	Small; narrow or flat plate	Mid or distal shaft	30–60°
4	Very small; flat plate	Proximal shaft, penoscrotal	>60°

- Total 3–12; a higher score predicts more complications.
- **POST (plate objective scoring tool):** a low ratio of neomeatus length to glans fusion length means a poor plate, which favours an inlay graft over plain TIP (tubularised incised

plate).

Timing and preparation

- Repair at **6–18 months** (AAP (American Academy of Pediatrics): 6–12 months), before genital awareness at about 18 months. One short anaesthetic appears low-risk.
- A straight penis that voids standing may gain little from surgery: counsel honestly.
- **Preoperative androgen** if glans width <14–15 mm: testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Evidence mixed; two RCTs (randomised controlled trials) showed fewer complications.

Curvature decides the operation

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Deglove + excise dysplastic dartos + artificial erection ├─ straight → urethroplasty (single-stage) ├─ residual <30° → dorsal midline plication (12 o'clock nerve-free zone), single-stage └─ >30° → divide plate → still >30° → ventral corporotomy ± graft → staged repair
```

- Recurrence 28% after plication versus 9% after grafting. Under-correction causes repeated surgery, so correct fully first time.

Choice of technique

Situation	Current choice
Glanular meatus, no curvature	Advancement: MAGPI (meatal advancement and glanuloplasty), MIV (M inverted V) glansplasty; selectively
Distal or midshaft, good plate	**TIP (Snodgrass)**: >90% of distal, about 80% of midshaft repairs; Thiersch–Duplay; GAP (glans approximation procedure) for a deep groove
Narrow or shallow plate	TIP with **dorsal inlay graft**; onlay island flap; Mathieu (SLAM modification)
Proximal, curvature >30°	**Two-stage**: STAG (staged tubularised autograft) or Bracka graft; Byars flaps; distal-first (Ulaanbaatar)
Proximal, curvature corrected, good tissue	One-stage onlay or TPIF (transverse preputial island flap); proximal TIP abandoned

Single-stage techniques

Group	Examples	Indication
Advancement	**MAGPI** (Duckett 1981); MIV	Glanular meatus, no curvature
Tubularisation	**TIP** (Snodgrass 1994); Thiersch–Duplay; GAP	Distal–midshaft; TIP in >90% of distal
Flap	Mathieu (flip-flap); **OIF** (onlay island flap)	Thin plate
Proximal one-stage	TPIF (Duckett tube); proximal OIF	Curvature corrected; good tissue
Graft	Dorsal buccal inlay	Narrow plate

- Pros: one anaesthetic.
- Proximal TIP is abandoned: devascularised plate, stricture, recurrent curvature.

Staged techniques

- Now the approach of choice for most proximal hypospadias.
- **Rationale (why stage):** a urethroplasty laid over a fresh corporal graft fails more often; stage 1 builds a supple, vascular bed. Stage 2 after **≥6 months**.

Type	Stage 1	Stage 2 (≥6 months)
Bracka / STAG (staged tubularised autograft)	Correct curvature, divide plate, quilt inner preputial or buccal graft	Tubularise, glansplasty, tunica vaginalis cover
Byars flaps	Split dorsal prepuce, transpose ventrally	Tubularise
Distal-first (Ulaanbaatar)	Curvature and distal urethra	Join proximal urethra

- Repeat the artificial erection before stage 2.

Single-stage versus staged: comparison of outcomes

Measure	Single-stage	Staged
Distal success	83–95%	–
Proximal complications	About 60% (long TIP, mixed)	STAG about 23% (lowest); Byars 43–81%
Recurrent curvature	Higher after plication	0–10%
Penile length, perception	Lower	Better
Extra procedures	Fewer planned	25–75% need more

Operative principles: protect blood supply

- Magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), rapidly absorbed fine sutures, no hair-bearing skin, haemostasis.
- Postoperative: stent 3–21 days by complexity, compressive dressing, oxybutynin; antibiotic prophylaxis debated.

Complications and outcomes (standardised)

- Definitions: glans dehiscence (fusion <2 mm); meatal stenosis (symptoms plus <8 Fr before puberty, <12 Fr after); recurrent curvature >30°.
- TIP meta-analysis (Pfistermuller 2015): complications 10.6%, fistula 5.7%. Proximal complications 23–68%; STAG about 23% (lowest); Byars 43–81%.
- Fistula is the commonest complication. A long, rigid, spongiosum-free neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum.
- **Also know:** BXO (balanitis xerotica obliterans): excise and stage with buccal graft; redo repairs use buccal (lip, inside glans) mucosa, Cecil scrotal parasitisation, tissue expansion.

Hypospadias fistula: definition and incidence

- A UCF (urethrocutaneous fistula) is a urethral leak anywhere below the meatus: the **commonest complication** of hypospadias repair, just under 10% at short-term follow-up.
- After TIP: 5.7% (**Pfistermuller 2015**); after proximal repair 3–45%. Usually at the original meatus or coronal margin.

Hypospadias fistula: aetiology

- **Distal obstruction** (meatal stenosis, stricture): raises voiding pressure and turbulence; the **key correctable cause**.
- **Ischaemia**, oedema, infection, haematoma, tension.
- **Technical:** overlapping suture lines, poorly inverted epithelium, slowly absorbed sutures, **no barrier layer**.
- **Patient:** proximal meatus, glans <14 mm, redo surgery. A long, rigid neourethra raises resistance (Poiseuille's law).

Hypospadias fistula: prevention

- **Vascularised barrier** (dartos; **tunica vaginalis** in proximal or redo), offset suture lines, stent in complex repairs.

Hypospadias fistula: evaluation

- Split stream / multiple streams, usually within 2 years.
- Under anaesthesia: **bougie calibration** for distal narrowing, **urethroscopy** if irregular, and a **stress test** (compress proximally and distally, inject saline or methylene blue) for hidden tracts.

Hypospadias fistula: management

- Rarely closes spontaneously. **Wait 6–12 months**; **correct distal obstruction at the same sitting**.

Fistula	Repair
Small, shaft	Excise the epithelialised tract, inverting subepithelial closure with fine absorbable suture, multiple offset layers, **dartos barrier**
Large or thin tissue	Trap-door or island skin flap plus second layer
Coronal margin (glans dehiscence)	Implies distal narrowing or poor tissue: **redo urethroplasty and glansplasty**
With diverticulum	Excise, or reuse the redundant wall as a flap

- Recurrence about **20%**; reduce with interposed vascularised layers (dartos, tunica vaginalis, Cecil modification when tissue is scarce).
- Catheter: none for simple repairs, 7–14 days for complex.
- **Also know:** preputial fistula does not involve the urethra; congenital urethral fistula is closed the same way; follow through puberty.

```
UCF → wait 6–12 months → calibrate, urethroscopy, stress test ├─ distal obstruction → correct at same sitting ├─ small → excise + layered closure + barrier flap └─ large/coronal → flap repair or redo urethroplasty
```

Follow-up and patient-reported outcomes

- About three-quarters of complications appear by 3 years; stricture and curvature appear late. Follow proximal cases **through puberty**.
- Tools: HOSE (Hypospadias Objective Scoring Evaluation, surgeon), PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation, photo panels). Decision aids reduce parental regret.

Recent advances

- Registries, AI (artificial intelligence) video review, genomic risk profiling, a quality-of-life measure, engineered tissue.

Diagrams to draw: curvature algorithm; STAG stage 1 graft and stage 2 tubularisation; fistula repair showing neourethra, dartos barrier and skin with offset suture lines.

Key points to remember:

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages: curvature >30° after degloving means stage.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal; STAG has the lowest proximal complications.
- Stage 2 after ≥6 months.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.
- Fistula: commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

LAQ 5 Wilms tumour, including bilateral Wilms tumour

P2 · P3 **merged from 2**

Asked as: P2 LAQ "Wilms tumour: classification, staging and management" · P3 SAQ "Bilateral Wilms tumour: treatment strategies"

Introduction

- Read as "Wilms tumour" (nephroblastoma); "Wilm's bladder" in one P2 wording is taken as a typo.
- WT (Wilms tumour) is an embryonal tumour of metanephric blastema. It forms 95% of kidney cancers under 15 years; median age 3.5 years; over 80% present before 5 years.

Genetics, syndromes and precursors

- **Genetics:** mostly sporadic; familial 1–2%. Genes: **WT1 (11p13)**, IGF2/11p15 (commonest change, about 70%), WTX, CTNNB1; **TP53 in anaplasia (in most anaplastic tumours)**.

Syndrome	WT risk
WAGR (Wilms, aniridia, genitourinary anomalies, intellectual disability / retardation; WT1 deletion)	50%
Denys–Drash (46,XY DSD [disorder of sex development], mesangial sclerosis; WT1 point mutation)	50%
BWS (Beckwith–Wiedemann syndrome), hemihypertrophy (11p15)	5–10%

- **Screening:** ultrasound every 3–4 months until 7 years when risk is >1%.
- **Nephrogenic rests** (precursors) occur in over one-third of WT kidneys and over 90% of bilateral cases.

Classification: histology

Group	Features	Significance
Favourable histology	**Triphasic:** blastema, epithelium (tubules), stroma; pseudocapsule	Survival >90%
Anaplasia (unfavourable)	Nuclei ≥3× neighbours (larger), hyperchromasia, abnormal mitoses; **focal** or **diffuse** (diffuse worse than focal)	Chemoresistant; poor even at stage I

- **SIOP (International Society of Paediatric Oncology) post-chemotherapy risk groups:**

Risk	Histology
Low	Completely necrotic
Intermediate	All others (regressive, mixed, epithelial, stromal, focal anaplasia)
High	**Blastemal-predominant**; diffuse anaplasia

- Residual blastema predicts relapse; stromal and epithelial tumours shrink little but are cured by excision.

Staging

COG (Children's Oncology Group): surgical-pathological, before chemotherapy

Stage	Description
I	Confined to kidney, completely resected, capsule intact, no renal sinus vessel invasion
II	Beyond kidney (renal sinus vessels, perirenal fat, vein thrombus) but completely resected
III	Residual abdominal disease: **positive nodes, any spill, prior biopsy**, positive margin, peritoneal implants
IV	Haematogenous metastases (lung commonest, liver, bone, brain)
V	Bilateral at diagnosis (each side staged separately)

- **SIOP staging** uses the same five stages but is applied **after preoperative chemotherapy**. Necrotic, chemotherapy-induced change in the sinus or perirenal fat does not upstage.
- **Biology:** combined **1p and 16q LOH (loss of heterozygosity)** predicts relapse; 1q gain is the next stratifier.

Clinical features

- Abdominal mass; haematuria about 20%; hypertension about 25% (renin).
- Look for aniridia, hemihypertrophy and varicocele. Varicocele or hepatomegaly suggests IVC (inferior vena cava) thrombus.
- Acquired von Willebrand disease in up to 8%.

Investigations

- Ultrasound with Doppler (IVC), then contrast CT (computed tomography) or MRI (magnetic resonance imaging) abdomen, then **chest CT** (lung is the commonest metastatic site; do it before anaesthesia).
- Blood count, renal and liver function, coagulation; syndrome review.

Management: two philosophies

	COG (North America)	SIOP (Europe)
Sequence	**Upfront transperitoneal radical nephrectomy**, then chemotherapy	**Preoperative VCR + AMD** (4 weeks; add DOX for 6 weeks if metastatic), no biopsy, then nephrectomy
Advantage	Untreated histology and biology for risk assignment	Tumour shrinks; spill almost zero
Risk basis	Stage + histology + LOH	Post-chemotherapy histology + stage

(VCR = vincristine; AMD = actinomycin D (dactinomycin); DOX = doxorubicin; RT = radiotherapy; NSS = nephron-sparing surgery.)

Surgical principles (radical nephrectomy)

- Transperitoneal approach; palpate renal vein and IVC before ligation; spare a tumour-free adrenal.
- **Avoid spill** (it makes the tumour stage III).
- **Sample at least 5 lymph nodes** (hilar, paracaval, para-aortic, interaortocaval); this is essential, formal dissection is not. Omitted sampling is the commonest protocol violation.
- IVC thrombus at or above the hepatic veins: preoperative chemotherapy to avoid bypass.

Postoperative therapy (COG, favourable histology)

Group	Treatment
Very low risk (<2 years, <550 g, stage I)	Nephrectomy alone (AREN0532)
Stage I–II	VCR + AMD
Stage III	VCR + AMD + DOX + flank RT
Stage IV	Three drugs + whole-lung RT; lung RT omitted if lung nodules clear by week 6 (AREN0533)
Anaplasia or 1p/16q LOH	Intensified regimens (regimen M adds cyclophosphamide + etoposide)

Bilateral WT (stage V): principles

- **Synchronous bilateral WT** is **5–7%**: **COG stage V**. Each kidney is staged separately.
- Linked to WAGR, Denys–Drash (about 50% risk) and BWS; **nephrogenic rests in over 90%**.
- Aim: **cure while preserving renal tissue (nephrons)**. Renal failure is about 15% at 15 years; ESRD (end-stage renal disease) mainly follows bilateral nephrectomy.
- **No upfront biopsy and no upfront nephrectomy.**

Bilateral WT: COG approach (AREN0534)

```
VCR + AMD + DOX × 6 weeks (no biopsy) → imaging at week 6 ├─ resectable / NSS feasible → bilateral NSS └─ not → 6 more weeks chemotherapy → surgery by week 12 poor response → bilateral OPEN biopsy → further therapy by each kidney's stage and histology ± flank RT
```

- **Why stop at 12 weeks:** in NWTS-4 (National Wilms Tumor Study 4), prolonged chemotherapy in poor responders delayed surgery and some tumours became anaplastic. A non-shrinking tumour is often differentiated and is cured by excision.
- **Why an open, bilateral biopsy:** anaplasia is discordant between kidneys in about 4 of 5.

Bilateral WT: surgery

- **Bilateral NSS** (partial nephrectomy or enucleation); frozen section; lymph node sampling; avoid spill.
- Compressed normal parenchyma is often underestimated, so try NSS in every case. Expert centres achieve bilateral NSS in over 90%.
- Enucleation: positive margins 23–36%, good outcome with flank radiotherapy.
- Unresectable both sides: bilateral nephrectomy and dialysis are rare and only for refractory tumour; transplant after **1–2 years** tumour-free.

Bilateral WT: adjuvant therapy and SIOP approach

- Adjuvant therapy is guided by the worst histology and margins: anaplasia or positive margin means intensified chemotherapy and flank radiotherapy.
- **SIOP:** same principle (no biopsy, NSS by week 12) but **two drugs, VCR + AMD**, if non-metastatic.

Unilateral nephron-sparing surgery

- For syndromic children and solitary kidney; SIOP UMBRELLA criteria are volume <300 mL, a substantial remnant and negative nodes.

Prognosis and outcomes

- Favourable histology: survival >90%.
- **AREN0534 (bilateral):** surgery by week 12 in 84%; bilateral NSS 39%; **4-year EFS (event-free survival) 82.1%, OS (overall survival) 94.9%**, versus 56% and 80.8% on NWTS-5.
- Salvage survival after relapse: 70–80% after VCR/AMD only; about 10% for anaplastic or blastemal relapse.

Late effects and follow-up

- **Late effects:** cardiotoxicity (DOX), second cancers (RT), infertility, renal failure (bilateral).
- **Follow-up (bilateral):** lifelong blood pressure, proteinuria, creatinine (renal function) and imaging, as late relapse occurs.

Diagrams to draw: COG staging diagram showing capsule, renal sinus, nodes and spill; bilateral tumours with planned partial nephrectomies; AREN0534 timeline (weeks 0, 6, 12).

> **Key points to remember:**
>
> - WT1 at 11p13 (WAGR, Denys–Drash); BWS at 11p15.
> - Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
> - Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
> - COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
> - Bilateral: stage V, 5–7%; no upfront nephrectomy or biopsy; three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12, NSS; SIOP uses two drugs.
> - AREN0534: 4-year OS about 95%; bilateral NSS to avoid ESRD.

LAQ 6 Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair

P2 · P3 **merged from 2**

Asked as: P3 LAQ "Embryology of the penis and management of proximal hypospadias (including Koyanagi repair)" · P2 SAQ "Perineo-scrotal hypospadias"

Embryology of the penis

Indifferent stage (up to 8–9 weeks)

- Cloacal folds form the **genital tubercle**; after the cloaca divides (7 weeks), **urethral folds** flank the urogenital membrane, with **labioscrotal swellings** laterally. Both sexes look identical until about 9 weeks.

Hormonal control

- SRY (sex-determining region Y) drives the testis, whose **Leydig cells (about 8 weeks)** secrete testosterone.
- **Type 2 5α-reductase** in the tubercle converts it to **DHT (dihydrotestosterone)**, which acts on androgen receptors. **DHT at 8–12 weeks** is the key mediator; the critical window is weeks 8–16. Without androgen action the default is female.
- First sign: longer anogenital distance, then tubercle elongation (sonic hedgehog-driven).

Primordium (germ layer)	Male	Female
Genital tubercle (mesoderm: corpora; ectoderm: skin, prepuce)	Glans, corpora	Clitoris
Urethral folds	Fuse to form penile urethra, median raphe	Labia minora
Labioscrotal swellings	Scrotum	Labia majora
Urethral plate (endoderm)	Whole urethra, including glanular	Vestibule

Urethral formation: the "double zipper"

1. **Opening zipper** (both sexes): the urethral plate canalises proximal to distal into a urethral groove.
2. **Closing zipper** (androgen-dependent, male only): the folds fuse in the ventral midline, forming the tubular urethra and raphe. - The **glanular urethra** forms by direct canalisation of the endodermal plate; Glenister's ectodermal ingrowth theory (1954) is outdated. - Timetable: coronal sulcus 11–12 weeks; shaft fused by 16 weeks; complete by about 18 weeks; fetal ventral curvature gone by 20 weeks. - **Prepuce** fuses dorsally first; ventral fusion awaits urethral closure, hence the **dorsal hood** in hypospadias. - Dorsal nerves run at 11 and 1 o'clock, leaving a **12 o'clock nerve-free zone** (safe plication site). - **Hypospadias = arrest of the closing zipper**: the meatus stops where closure stopped, and unfused dysplastic ventral tissue tethers the shaft into curvature.

Diagram to draw: double-zipper urethral formation.

Definition of proximal (perineoscrotal) hypospadias

- Meatus on the proximal shaft, at the penoscrotal junction, scrotum or perineum (10–25% of hypospadias).

- Perineoscrotal hypospadias is a severe form, with severe ventral curvature, a short penis, bifid scrotum and often **penoscrotal transposition**.
- Cause: failed androgen-dependent urethral fold fusion (8–16 weeks).

Evaluation

- Severity is judged **after degloving**: meatus, plate and curvature (GMS (glans–meatus–shaft) score).
- Examine stretched penile length, glans width, penoscrotal transposition, bifid scrotum and **both testes**.
- **DSD (disorder of sex development) work-up** if hypospadias plus an undescended testis (especially non-palpable), perineal meatus or ambiguity: karyotype, ultrasound for Müllerian remnants, testosterone, LH (luteinising hormone), FSH (follicle-stimulating hormone), AMH (anti-Müllerian hormone).
- DSD risk is 64% in proximal versus 7% in distal cases. A 46,XY karyotype can still hide 5α-reductase deficiency or androgen insensitivity.
- Prostatic utricle is commoner; cystoscopy in severe cases. No routine VCUG (voiding cystourethrogram) or renal ultrasound.

Preparation

- **Timing:** 6–18 months (AAP (American Academy of Pediatrics): 6–12 months).
- **Preoperative androgen** if glans width <14–15 mm: e.g. testosterone 2 mg/kg IM (intramuscular) at 5 and 2 weeks before surgery (about 4 mm glans gain). Side effects are transient; evidence is mixed, but two RCTs (randomised controlled trials) showed fewer complications.

Curvature correction (decides one- versus two-stage)

```
Deglove + excise dysplastic dartos + artificial erection ├─ straight → urethroplasty ├─ residual <30° → dorsal midline plication (12 o'clock) └─ >30° → divide urethral plate └─ still >30° → ventral corporotomy ± graft → staged repair
```

	Dorsal plication (Nesbit, Baskin)	Ventral lengthening
Method	Shortens the convex dorsal side	1–3 transverse corporotomies, or one full-thickness 9-to-3 o'clock corporotomy with dermis, tunica vaginalis or SIS (small intestinal submucosa) graft
Use	Mild (≤15–30°)	Moderate–severe
Recurrence	28%	9%

Diagram to draw: curvature algorithm.

Urethroplasty options

Approach	Technique	Comment
One-stage **proximal TIP (tubularised incised plate)**	Plate kept, incised and tubularised	Recurrent curvature, stricture: largely abandoned for severe cases
One-stage **onlay island flap**	Inner preputial flap on preserved plate	Diverticulum, fistula; higher complications

Approach	Technique	Comment
One-stage **TPIF (transverse preputial island flap, Duckett tube)**	Tube after plate division	High complications
One-stage **Koyanagi**	Meatus-based parameatal foreskin flap (see below)	No proximal circular anastomosis
Two-stage Bracka / STAG (staged tubularised autograft)	Stage 1: correct curvature, divide plate, quilt inner-prepuce or **buccal graft**. Stage 2 (≥6 months): tubularise, glansplasty, barrier layer	**Preferred for most proximal cases**; STAG lowest complications (about 23%)
Two-stage Byars	Dorsal preputial flaps moved ventrally, tubularised later	Supple flap bed; complications 43–81%

- **Why stage?** A urethroplasty laid over a fresh corporal graft fails more often; staging gives a supple, vascular bed.
- Principles: magnification, tension-free closure, offset suture lines, **vascularised barrier layer** (dartos, tunica vaginalis), fine absorbable sutures, no hair-bearing skin. Stent, compressive dressing, oxybutynin; **scrotoplasty** for transposition.

Diagram to draw: two-stage graft repair: stage 1 graft bed, stage 2 tubularisation.

Koyanagi repair: definition and principle

- **One-stage repair of proximal hypospadias** with a **meatus-based parameatal foreskin flap** (Koyanagi). Skin strips round the meatus continue into the inner prepuce on both sides; they are swung ventrally and tubularised.
- **Principle:** the neourethra stays attached to the native meatus, so there is **no circular proximal anastomosis**, and the urethral plate can be divided to correct severe chordee.

Koyanagi repair: indications and contraindications

- **Indications:** proximal hypospadias (penoscrotal, scrotal, perineal) with **severe curvature** needing plate transection, and an **ample, unoperated prepuce**.
- **Contraindications:** prior circumcision or failed repair; small glans (give preoperative testosterone); curvature needing a ventral corporal graft (stage instead).

Koyanagi repair: procedure

`Stay suture; artificial erection → incise round the meatus, along both sides of the plate to the corona and round the inner prepuce (meatus-based "wings") → deglove; divide the plate; excise dysplastic dartos → residual curvature <30° → dorsal midline plication → split the dorsal prepuce; bring both wings ventrally on dartos pedicles → join inner edges, then outer edges, over a catheter → tube → glanuloplasty, meatus at the tip → dartos barrier layer; skin cover ± scrotoplasty → stent 7–10 days; compressive dressing`

- Modified and "neo-modified" versions add flap fixation and barrier layers to reduce fistula and diverticulum.

Diagram to draw (Koyanagi): meatus-based foreskin wings split dorsally and joined ventrally into a tube.

Koyanagi repair: complications

Complication	Cause
Fistula (commonest)	Long suture line; repair ≥6 months later
Meatal stenosis, stricture	Distal flap ischaemia
Diverticulum	No spongiosum; flap not fixed to the shaft
Glans dehiscence	Tension
Residual or recurrent curvature	Plication alone in severe cases
Flap necrosis, infection, poor cosmesis	Pedicle injury

- Because proximal repairs overall have **23–68%** complications, many now prefer **two-stage repair** (STAG, Bracka).

Complications and outcomes of proximal repair

- Complications **23–68%**; 25–75% need further surgery after planned staging.
- **Fistula** is the commonest (3–45%; repair ≥6 months later; exclude distal obstruction).
- Others: glans dehiscence, meatal stenosis (<8 Fr before puberty), stricture, **diverticulum** (no spongiosum), **recurrent curvature >30°**, BXO (balanitis xerotica obliterans), hair-bearing urethra.
- A long, rigid neourethra raises resistance (Poiseuille's law), causing fistula and diverticulum.

Follow-up

- Follow through puberty; patient-reported outcomes (PPS (Penile Perception Score), HOPE (Hypospadias Objective Penile Evaluation) score).

> **Key points to remember:**
> - DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
> - Hypospadias (including perineoscrotal) plus undescended testis: DSD work-up.
> - <30° plication; >30° divide plate ± ventral graft and stage.
> - STAG/Bracka staged repair preferred; fistula commonest complication.
> - Koyanagi: one stage; meatus-based parameatal foreskin flap; plate divided; no proximal circular anastomosis.
> - Koyanagi: fistula commonest; diverticulum, meatal stenosis.

LAQ 7 Urological evaluation and management of pediatric kidney transplant patient.

P1

Type: LAQ (25 marks)
Source: 2022-Summer Paper 1, Q2

Introduction

- Children needing KRT (kidney replacement therapy) often have CAKUT (congenital anomalies of the kidney and urinary tract). The urologist must ensure that the bladder that destroyed the native kidneys does not destroy the graft.
- **Five goals:** unobstructed drainage; a capacious, low-pressure reservoir with social continence; volitional emptying; no infection; the fewest operations.

Causes of ESKD (end-stage kidney disease) in children

- Commonest age at transplant 13–17 years.
- **Younger:** dysplasia, hypoplasia, aplasia and obstructive uropathy, mainly PUV (posterior urethral valves). **Older:** FSGS (focal segmental glomerulosclerosis) and neuropathic bladder.

Pretransplant evaluation

- **Every child:** urological history, renal ultrasound, uroflow with PVR (post-void residual) if toilet-trained.
- **VCUG (voiding cystourethrogram)** only with neuropathic bladder, PUV, VUR (vesicoureteral reflux), previous lower-tract surgery, recurrent febrile UTI (urinary tract infection) or unexplained hydronephrosis.
- **Urodynamics** for neuropathic bladder, PUV or large ureterocele, refractory voiding dysfunction, high-grade hydronephrosis or recurrent UTI. They measure capacity, compliance, detrusor activity, emptying and sphincter function, and are repeated.

`History positive → ultrasound + VCUG → urodynamics Poor capacity/compliance → anticholinergic, CIC, cycling → still poor → augment Poor emptying → CIC ± catheterisable stoma Adequate → clear for transplant History negative → ultrasound + PVR → normal → clear`

- An anuric child with no urological history needs no further tests.

Pretransplant management

1. Bladder optimisation (drugs → CIC → cycling → augmentation)

- Commonest problem: low capacity, detrusor overactivity, poor compliance. High pressure harms the kidney even without reflux.
- **Target:** age-expected capacity at detrusor pressure **<40 cm H_2O**.
- Antimuscarinics (oxybutynin titrated; solifenacin), mirabegron, intradetrusor botulinum toxin; α-blockers aid emptying. Better storage may worsen emptying, so add **CIC (clean intermittent catheterisation)**. Prove volitional emptying by repeated low PVRs.
- A family's adherence to CIC predicts adherence to immunosuppression.

2. Valve bladder versus neuropathic bladder

- PUV bladder moves from overactivity to **myogenic failure**: postpone augmentation. Neuropathic bladder may deteriorate under high post-transplant urine output: reconstruct

earlier.

3. **Defunctionalised bladder** (anuria or diversion)

- **Bladder cycling:** catheter instillation of increasing saline volumes, with drugs, until age-expected capacity; confirm by urodynamics. **Never use the graft to test the bladder.**

4. **Augmentation**

- Indication: cannot reach **>75% expected capacity at <40 cm H_2O** with CIC every 3–4 hours despite maximal therapy. It does not clearly raise transplant complications.

Segment	Point
Dilated ureter (ureterocystoplasty)	Explore first: urothelium, no mucus or acidosis
Ileum, sigmoid	Most used; mucus, stones, infection, malignancy; lifelong follow-up
Stomach	Haematuria–dysuria syndrome in anuric patients; rarely used

- Continent stoma (Mitrofanoff) if the urethra is sensate or difficult; site it medial to the planned incision. Allow **6–12 weeks** healing before transplant.

5. **Infection, nephrectomy and dialysis**

- Treat stasis (double voiding, CIC); distinguish colonisation from UTI; consider circumcision in boys with valves.
- **Native nephrectomy** (default is to keep): uncontrolled hypertension, nephrotic syndrome, massive polyuria, recurrent pyelonephritis, large stones, malignancy risk, very large kidneys, massive VUR. Retroperitoneal route keeps PD (peritoneal dialysis).
- Keep a normal native ureter for later ureteroureterostomy. Intraperitoneal surgery on PD needs temporary haemodialysis.

At transplantation

- Retroperitoneal Gibson incision; children **<15 kg** may need aorta–caval anastomosis, sometimes transperitoneal.
- **Ureter:** extravesical non-refluxing ureteroneocystostomy (3–4 cm detrusorotomy) preferred; Barry tunnel; ureteroureterostomy to a normal native ureter.
- **Stent:** selective; double-J removed at 4–6 weeks, or early removal at 5–7 days.

Post-transplant complications

Complication	Rate	Management
Ureteric obstruction	6%	Usually distal ischaemic stenosis; >half within 100 days. Stent or nephrostomy; balloon dilatation; open repair (native ureter, bladder flap)
Urine leak	3%	Early; drain creatinine; catheter, nephrostomy, stent, then reimplantation
Graft VUR	2% (significant)	Febrile UTI → reimplant; asymptomatic → observe; injection works in one-third
Stones	2%	Treat if >1 cm and growing, obstructing or infected

- **Rising creatinine with hydronephrosis:** catheter drainage 1–2 weeks (bladder cause?) → biopsy if rejection is suspected → stent or nephrostomy. Obstruction and rejection can coexist.

- Also: lymphocele, **PTLD (post-transplant lymphoproliferative disorder)**, graft thrombosis (a leading cause of early graft loss), late infection reflecting bladder dysfunction.

Follow-up

- Lifelong ultrasound, PVR and bladder surveillance; recurrent UTIs lower graft GFR (glomerular filtration rate).

Diagram to draw: extravesical ureteroneocystostomy with detrusor flaps closed over the ureter.

> **Key points to remember:**
> - Bladder target: age-expected capacity at <40 cm H_2O; augment if <75% despite CIC.
> - PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.
> - Never test a bladder with a graft; cycle the defunctionalised bladder.
> - Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

LAQ 8 Embryology and management of ectopic ureter.

P3

Type: LAQ (25 marks)
Source: 2018-Summer Paper 3, Q1

Definition

- An **ectopic ureter** is any ureter, single or duplex, that does **not open on the trigone**. True ectopia means an orifice **at or distal to the bladder neck**.
- Commoner in girls. In girls most ectopic ureters drain the **upper pole of a duplex kidney**; in boys most are **single-system**.

Embryology

- **Week 4–6:** the **ureteric bud** arises from the **mesonephric (wolffian) duct** and induces the metanephric blastema to form the kidney.
- The duct segment distal to the bud is absorbed into the **urogenital sinus**; the ureteric orifice then moves cranially and laterally, the wolffian orifice caudally to the verumontanum.
- **Mackie–Stephens bud theory:**

Bud position on wolffian duct	Absorption	Orifice	Result
Too caudal	Early	High and lateral, short tunnel	VUR (vesicoureteral reflux)
Too cranial	Late or never	Low and medial (bladder neck, urethra) or stays on wolffian duct	**Ectopic ureter**, often obstructed

- A bud from the wrong place meets poor blastema, so **renal dysplasia increases with the degree of ectopia**.
- **Duplex system:** the upper-pole bud is more cranial, absorbed later, and ends **lower and more medial** (**Weigert–Meyer law**): upper pole obstructs; lower pole refluxes.

Sites of ectopia (the key sex difference)

	Female	Male
Sites (commonest first)	Urethra, vestibule, vagina; rarely cervix, uterus, rectum (via **Gartner duct**)	Posterior urethra, seminal vesicle; also vas, ejaculatory duct
Relation to sphincter	Can open **below** the external sphincter	Always **above** the external sphincter
Continence	**Continuous dribbling with normal voiding**	**Never incontinent**
Typical presentation	Constant dampness, perineal pus, UTI (urinary tract infection)	**Epididymo-orchitis**, UTI, flank pain, hydronephrosis

- **Zinner syndrome:** ectopic ureter into the seminal vesicle with ipsilateral renal agenesis or dysplasia and a seminal vesicle cyst.

Clinical features

- **Prenatal hydronephrosis**: most are detected this way.

- **Girl:** wet day and night despite normal voids; she is never dry for 30–60 minutes. Often misdiagnosed as voiding dysfunction.
- **Infection:** low-grade fever with spikes; urine culture may be negative because the infected moiety does not drain into the bladder.
- **Boy:** recurrent epididymitis in a young boy demands an upper-tract ultrasound.
- **Also know:** pain then perineal discharge; bilateral single-system ectopia with small bladder and wide bladder neck.

Evaluation

Test	What it shows
Ultrasound (first)	Dilated upper pole and ureter behind the bladder; parenchyma. Bladder view separates it from ureterocele
VCUG (voiding cystourethrogram)	Lower-pole or contralateral VUR ("drooping lily"), reflux into the ectopic ureter
DMSA (dimercaptosuccinic acid)	Gold standard for moiety function ("some" versus "none"); MAG3 (mercaptoacetyltriglycine) if drainage needed
MR (magnetic resonance) urography	Non-dilated upper pole, small dysplastic kidney, complex anatomy
Examination / cystovaginoscopy	Perineal orifice between urethra and vagina, cannulated for contrast

- **Also know:** clear urine leaking despite methylene blue in the bladder suggests an ectopic ureter; CT for a small unseen kidney.

Management

Goals: preserve renal function; remove infection, obstruction and reflux; achieve continence; with least morbidity. Start antibiotic prophylaxis in the neonate with significant dilation.

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Neonate with sepsis or massive dilation → end cutaneous ureterostomy (or nephrostomy); DMSA later DUPLEX system Upper pole non-functioning → upper-pole heminephroureterectomy Upper pole worth keeping → ureteroureterostomy (UU) / pyeloureterostomy + lower-pole VUR → common sheath reimplant (or lower-pole reimplant + distal UU) SINGLE system Functioning → ureteral reimplant (± tapering) Non-functioning → nephroureterectomy (laparoscopic) Bilateral → bladder neck reconstruction ± augmentation
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- **Upper-pole heminephroureterectomy** (open, laparoscopic or robotic): upper-pole ureter as a handle, passed behind the hilum; ligate polar vessels; excise the pole; close over fat. Dissect on the ureter wall to protect the **lower-pole blood supply**. The stump is rarely a problem.
- **Ureteroureterostomy (UU):** end-to-side upper into lower pole ureter, distal (inguinal incision) or proximal (preferred with big size mismatch). **Stent the recipient lower-pole ureter** cystoscopically for identification. Now used even for non-functioning poles; feared hypertension has not been shown.
- **Common sheath reimplantation:** do not separate the two ureters (they share blood supply).

Complications

- Loss of lower-pole function (most serious); urinoma (up to 20% after minimally invasive partial nephrectomy, mostly harmless); stump infection; new reflux; persistent incontinence.

Recent advances

- Robotic UU and heminephrectomy; single-site surgery; non-dismembered side-to-side ureterovesicostomy in infants.

Diagram to draw: duplex kidney showing Weigert–Meyer orifices, and female versus male ectopic sites relative to the external sphincter.

> **Key points to remember:**
> - Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
> - Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
> - DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
> - Protect the lower-pole blood supply; stent the recipient ureter for UU.

Short answer questions (SAQ)

SAQ 1 TIP (Snodgrass) repair for hypospadias

P1 · P3 **merged from 2**

Asked as: P3 SAQ "Snodgrass (TIP) repair for hypospadias, including extended indications" · P1 SAQ "TIP repair in hypospadias"

Introduction

- "Shodgrass" in one P3 wording is taken as the Snodgrass TIP repair, and "distal hypos." as distal hypospadias.
- **TIP (tubularised incised plate) urethroplasty** was described by Snodgrass in 1994, built on Rich's hinging incision (1989).
- Designed for distal hypospadias, it is now used for **over 90% of distal and about 80% of midshaft repairs**.
- Distal hypospadias forms 70–85% of cases.

Principle

- A **deep midline relaxing incision** of the urethral plate widens a narrow plate so that it can be tubularised **without tension**.
- The incised raw area **heals by re-epithelialisation** without a graft, giving a **vertical slit-like meatus**.

Indications, prerequisites and assessment

- Distal and midshaft hypospadias, and redo cases with a healthy plate, when curvature is mild (**<30°** after degloving) and corrected by degloving or dorsal plication **without dividing the plate**.
- Supple, well-vascularised plate of adequate width and depth, and an adequate glans.
- **Glans width <14 mm** raises complications (glans width ≥14 mm is needed): give preoperative testosterone to enlarge it.

- **POST (plate objective scoring tool):** a low ratio (AB/BC ratio) of neomeatus length to glans fusion length means a poor plate, which favours a dorsal inlay graft.
- Timing: 6–12 months of age.
- Not for severe curvature needing plate division; proximal TIP is largely abandoned.

Extended indications

Indication	Rationale	Caveat
Midshaft hypospadias	Same principle	Good results
Proximal hypospadias with mild curvature, good plate	Single stage; no flaps	More fistula, stricture, recurrent curvature; Campbell-Walsh authors now prefer staging
Redo or failed repairs with preserved plate	Plate often survives	Worse than primary; inlay graft if scarred
MIP (megameatus intact prepuce)	Wide, deep plate	GAP (glans approximation procedure) alternative
Narrow or shallow plate: TIP + **dorsal inlay graft** ("Snodgraft")	Preputial or buccal graft fills the incision, so it cannot contract	Less meatal stenosis
Adults, circumcised boys	No preputial tissue needed	Success similar

- **Also know:** in redo cases with poor blood supply a dorsal buccal inlay beats onlay or tube; incise-and-graft is also used in adult urethral stricture (Asopa dorsal inlay) and female urethroplasty.

Contraindications

- Curvature >30° needing plate division; absent or scarred plate; penoscrotal transposition. Use staged repair instead (Bracka, STAG [staged tubularised autograft], Byars).

Operative steps

1. Glans stay (traction) suture; tourniquet.
2. U-incision around the plate and meatus; circumscribe and deglove; artificial erection.
3. Separate (mobilise) the glans wings deeply from the plate.
4. **Midline incision of the plate** from inside the meatus to the end of the plate, down to the corpora.
5. Tubularise over a **6–8 Fr stent** with subepithelial 6-0/7-0 absorbable sutures, from distal to proximal, placing the first stitch about 3 mm (2–3 mm) short of the plate end to leave an oval meatus.
6. **Dartos flap** (dorsal preputial) over the neourethra as a barrier layer.
7. Two-layer glansplasty; mucosal collar; skin closure with Byars flaps, or preputioplasty.
8. Stent 5–7 days; compressive dressing; oxybutynin for bladder spasm.

Complications and outcomes

- Pfistermuller meta-analysis (2015, 4,675 boys): complications 10.6%, reoperation 4.5%; **fistula 5.7%**, **meatal stenosis 3.6%**, stricture 1.3%.
- **Meatal stenosis is commoner than after Mathieu**; glans dehiscence also occurs.
- Distal repair success 83–95%.

- **Proximal and redo cases do significantly worse**; a dartos barrier layer improves results and reduces fistula.

Modifications

- **Dorsal inlay graft (Snodgraft)**: inner preputial skin graft in the incised bed for a shallow plate.
- Lateral plate incisions to 12 mm width.
- **Also know:** MAGPI (meatal advancement and glanuloplasty incorporated), Mathieu, Thiersch–Duplay, onlay flap.

Diagram to draw: cross-sections showing plate incision, inlay graft, tubularisation and dartos cover.

> **Key points to remember:**
> - Snodgrass 1994; incise and tubularise; heals by epithelialisation.
> - Slit-like meatus; dartos barrier reduces fistula.
> - Glans <14 mm and poor plate raise complications.
> - Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
> - Extended to midshaft, selected proximal, redo, MIP and adults.
> - Key condition: curvature <30° without plate division.
> - Poor plate: dorsal inlay graft ("Snodgraft").
> - Proximal and redo TIP fail more often.

SAQ 2 Glandular hypospadias

P1

Type: SAQ (10 marks)
Source: Practice Set 11 Paper 1, Q3b

"Glandular" is read as glanular hypospadias: meatus on the glans.

Definition

- The mildest **distal (anterior)** hypospadias: the meatus opens on the ventral glans, proximal to the normal tip.
- Hypospadias occurs in 1 in 150–300 male births; distal forms are 70–85%.
- Usual features: **dorsal hooded foreskin**, ventral glans tilt, little or no curvature. **MIP (megameatus intact prepuce)** variant (up to 5%): wide meatus with normal foreskin.

Embryology

- Urethral folds close from proximal to distal during weeks 8–16 under **DHT (dihydrotestosterone)**.
- Glanular urethra: ectodermal ingrowth (classic) or endodermal plate ("two-zipper", current view). Arrest at the last step gives a glanular meatus.

Evaluation

- Examine meatus, plate, glans size, curvature, torsion; retract the foreskin before any **newborn circumcision**.
- **GMS (glans–meatus–shaft) score** (each 1–4, total 3–12) predicts complications.

- No workup needed for isolated distal hypospadias; DSD (difference of sexual development) workup if a testis is undescended.

Indications and timing

- Deflected or spraying stream, meatal stenosis, curvature, cosmesis. A straight penis that voids standing may need no surgery; counsel parents.
- **Age 6–18 months** (AAP, American Academy of Pediatrics, 6–12 months).

Operations

Procedure	Suitable for	Key point
Urethromeatoplasty	Stenotic meatus, blind pit	Heineke–Mikulicz: vertical incision, horizontal closure
MAGPI (meatal advancement and glanuloplasty, Duckett 1981)	Glanular meatus, mobile skin	Advance meatus, two-layer glanuloplasty; retracts if used more proximally
MIV (M inverted V) glansplasty	Glanular or coronal	Avoids MAGPI retraction
GAP (glans approximation)	Wide deep groove, fish-mouth meatus (MIP)	Tubularise and close glans wings
TIP (tubularised incised plate, Snodgrass 1994)	Narrow plate	Most used; slit-like meatus
Mathieu flip-flap; Thiersch–Duplay	Selected	Stenosis, fistula

- Preputioplasty or circumcision; dartos barrier layer.

Complications

- Meatal stenosis or retraction, fistula, **glans dehiscence**, residual curvature, poor cosmesis. Distal repairs succeed in about 80%.

Diagram to draw: MAGPI steps (meatal advancement, glans wings closed beneath).

> **Key points to remember:**
> - Glanular = most distal hypospadias; dorsal hood, minimal curvature.
> - MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
> - Repair at 6–18 months; may be left alone if function is normal.

SAQ 3 Vanishing testes

P1

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 1, Q3b

Definition

- A testis that **developed normally and was then lost**, usually from **torsion or a vascular accident in fetal life**, after genital masculinisation but before scrotal fixation.
- Unilateral loss gives **monorchia**; bilateral loss gives **anorchia** (testicular regression syndrome), which is rare.
- Excised "nubbins" contain haemosiderin.

Incidence

- **15–40% of nonpalpable testes** at surgery.

Clinical features

- **Contralateral compensatory hypertrophy:** volume >2 mL (sensitivity 72%, specificity 100%) or length about 19–20 mm suggests a solitary testis.
- No palpable intrascrotal "appendage" tissue; sometimes a scrotal nubbin.
- These aid counselling but **do not replace exploration**.
- **Anorchia:** micropenis in nearly half.

Evaluation

- **Imaging** (ultrasound, MRI [magnetic resonance imaging]) is not recommended: it does not change the operation.
- **Bilateral nonpalpable:** exclude DSD (difference of sex development) and CAH (congenital adrenal hyperplasia) first.
- Anorchia: high FSH (follicle-stimulating hormone) and LH (luteinising hormone); low AMH (anti-Müllerian hormone) and inhibin B; no rise after hCG (human chorionic gonadotrophin). Most reliable during mini-puberty.

Management: diagnostic laparoscopy (after examination under anaesthesia)

Laparoscopic finding	Action
Blind-ending vessels and vas above a closed ring	Diagnosis confirmed; stop
Vessels and vas entering the internal ring	Inguinal or scrotal exploration; excise the nubbin
Testis seen	Orchidopexy (single stage or Fowler–Stephens)
Neither vas nor vessels	Search up to the kidney (agenesis very rare)

- **Excise the remnant:** viable germ cells or tubules in 5–15%.
- **Scrotal-first exploration** is reasonable with a palpable nubbin and contralateral hypertrophy (length ≥1.8 cm); laparoscopy is then needed in about a quarter.
- **Contralateral fixation:** debated; not required by the AUA (American Urological Association) algorithm.
- **Prosthesis:** offer at least 6 months after scrotal surgery or after puberty, inguinal approach.
- **Anorchia:** testosterone from puberty. **Also know:** agenesis shows ipsilateral Müllerian persistence.

Diagram to draw: laparoscopic view of blind-ending vessels at a closed ring.

> **Key points to remember:**
> - Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
> - Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
> - Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
> - Anorchia: high FSH/LH, low AMH and inhibin B.

SAQ 4 Pelvicalyceal anomalies other than PUJ obstruction

P1

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 1, Q3d

Introduction

- Congenital pelvicalyceal variants occur in about 4% of people.
- **Key question:** is there obstruction? Dilatation without obstruction needs no treatment; stasis causes stones and infection.

Anomalies

Anomaly	Features	Management
Calyceal diverticulum	Transitional-lined intrarenal cavity joined to a calyx (Type I, commoner, small) or pelvis (Type II, larger, symptomatic) by a narrow neck; 4.5 per 1000; congenital (persistent 3rd–4th generation ureteric branches) or acquired. **Stones in almost 40%**; milk of calcium	Asymptomatic: observe. Pain, infection, stones: ureteroscopic neck incision, **percutaneous ablation with fulguration**, laparoscopic marsupialisation; SWL (shock wave lithotripsy) ineffective
Hydrocalycosis	Dilated major calyx connected to the pelvis; infundibular obstruction by vessel, stenosis, scarring or achalasia	Symptomatic only: infundibulotomy or dilatation
Megacalycosis	**Non-obstructive**; malformed papillae; **12–20 calyces**; pelvis normal; male 6:1; normal diuretic renogram	**None**; may coexist with segmental megaureter
Infundibular / infundibulopelvic stenosis	Narrow infundibula ± small pelvis; usually bilateral, with VUR (vesicoureteric reflux), dysplasia; over a third reach renal failure (hyperfiltration)	Monitor creatinine and eGFR (estimated glomerular filtration rate); ACE (angiotensin-converting enzyme) inhibitors; surgery only if obstruction harms function: endoscopic, calicocalicostomy, ureterocalicostomy
Extrarenal pelvis	Large pelvis outside the sinus; mimics dilatation	None
Bifid pelvis	About 10%; normal variant	None
Unipapillary kidney	Single papilla and calyx; rare; usually hypoplastic kidney	None

- **Also know:** acquired infundibular stenosis (tuberculosis, post-PCNL (percutaneous nephrolithotomy)); Fraley syndrome (crossing artery on upper infundibulum).

Evaluation

- **Ultrasound:** cyst-like space; milk of calcium layers and shifts with position.
- **CT (computed tomography) urography or MRU (magnetic resonance urography): delayed contrast pooling** in a diverticulum.
- **Diuretic renogram (MAG3 (mercaptoacetyltriglycine)):** separates obstruction (hydrocalycosis) from non-obstruction (megacalycosis, extrarenal pelvis).
- **Retrograde pyelography:** maps the neck.

Differential diagnosis

- Obstructive hydronephrosis, papillary necrosis, pyelonephritic clubbing, tuberculosis, renal cyst.
- **Stones:** retrograde first (find, dilate or laser-incise the ostium); percutaneous for lower-pole diverticula.

Diagram to draw: kidney showing diverticulum with narrow neck, hydrocalyx and megacalyces.

> **Key points to remember:**
> - Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
> - Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
> - Infundibulopelvic stenosis: bilateral, renal failure risk.

SAQ 5 Retractile testis?

P2

Type: SAQ (10 marks)
Source: 2020-Summer Paper 2, Q3b

Definition

- A **descended testis** that moves up out of the scrotum because of an active **cremasteric reflex**. It can be brought to the bottom of the scrotum by hand and **stays there**, at least for a while, after release.
- It is not undescended, but may be a mild form of the same disease and can later become an acquired undescended testis.

Epidemiology

- Common: suprascrotal in up to one-third of 4-year-olds; 11–15% of boys up to 11 years. Nearly all are scrotal after 12 years.

Examination (diagnosis is clinical)

- Warm room and hands, relaxed child, soap or gel; supine frog-leg (thigh abduction inhibits the cremasteric reflex), cross-legged or squatting.
- One hand sweeps from the internal ring down the canal; the other pulls the testis to the scrotal bottom. **Sustained gentle traction tires the cremaster.**
- Note size, mobility, contralateral testis and hernia.

Testis	Finding	Action
Retractile	Reaches scrotal bottom and stays	Observe yearly
Gliding (unstable high scrotal)	Pulled down but springs back	Treat as undescended
Undescended (palpable)	Cannot reach mid-scrotum	Orchidopexy by 18 months
Ascending (acquired)	Previously scrotal, now not	Orchidopexy

Natural history

- Ascent: 2.6% in a prospective birth cohort; 7–32% in specialist series.
- Operated retractile testes show small volume, patent processus vaginalis and epididymal anomalies, suggesting a continuum with cryptorchidism.

Management

- No ultrasound (adds nothing); no hormones (hCG [human chorionic gonadotrophin] neither separates nor treats).
- Reassure parents; **examine yearly** (AUA [American Urological Association] 2014) until puberty, because some testes ascend.
- **Orchidopexy** if it ascends, cannot be brought down, rests mostly high or becomes smaller. If unsure, examine under anaesthesia.

Diagram to draw: inguinoscrotal sketch: testis drawn by traction to the scrotal bottom.

> **Key points to remember:**
> - Retractile = comes down and stays down.
> - Gliding testis springs back: treat as undescended.
> - Yearly examination until puberty; operate if it ascends.

SAQ 6 Retroileal Ureter

P2

Type: SAQ (10 marks)
Source: 2025-Winter Paper 2, Q3b

*"Retroileal" is taken as **retroiliac ureter**, the nearest real entity; retrocaval ureter is compared below (full answer: 2017-Winter Paper 3, Q3d).*

Definition

- The ureter passes **behind the common iliac artery** and is compressed there.
- Better called "**preureteral iliac artery**": the anomaly is presumed vascular; the ureter is trapped.

Embryology

- Normally the **ventral root of the umbilical artery** is replaced by a more **dorsal** branch, which lies behind the ureter.
- If the **ventral root persists** and the dorsal root fails, the ureter is trapped behind the artery.

Features

- **Very rare**; either side; **bilateral** cases reported.
- Obstruction at **L5–S1**.
- **Associated anomalies are common**, especially **vasal and mesonephric duct ectopia**: ureter and vas draining together into the posterior urethra; vasa ending in the ureters; ectopic vaginal ureter.

Presentation and diagnosis

- Flank pain, UTI (urinary tract infection) or hydroureteronephrosis down to the iliac crossing.
- **CT urography (computed tomography urography) with arterial phase** or MRI (magnetic resonance imaging) shows ureter and artery together; **diuretic renogram** for function and drainage.
- Caution: a vessel indenting an already dilated ureter may not be the cause.

Management

- Non-obstructed: observe.
- Obstructed: **divide the ureter, re-route it in front of the artery** and do a spatulated ureteroureterostomy (or reimplantation) over a double-J stent; open, laparoscopic or robotic.

Comparison with retrocaval ureter

	Retroiliac	**Retrocaval (circumcaval)**
Vessel	Common iliac artery	IVC (inferior vena cava)
Embryology	Persistent ventral umbilical artery root	Persistent subcardinal vein
Side	Either, may be bilateral	Right
Level	L5–S1	L3–L4, "fishhook"
Associations	Vasal/mesonephric ectopia	Horseshoe kidney, left renal anomalies
- **Also know:** distal ureter obstruction by uterine, umbilical, obturator or hypogastric vessels.		

Diagram to draw: ureter behind the common iliac artery at L5–S1, dilated above.

> **Key points to remember:**
> - Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
> - Obstruction at L5–S1; either side or bilateral.
> - Look for vasal and ureteric ectopia.
> - Retrocaval: right, L3–L4, fishhook, 1 in 1500.

SAQ 7 Medullary sponge kidney

P2

Type: SAQ (10 marks)
Source: Practice Set 9 Paper 2, Q3e

Definition

- **MSK (medullary sponge kidney; precalyceal canalicular ectasia):** dilated (ectatic) distal **collecting ducts** with small cysts and diverticula **confined to the medullary pyramids**. Nephrons are initially normal.
- Usually **non-heritable** (sporadic); usually bilateral.

Pathogenesis of stones and nephrocalcinosis

- **Stasis and infection** in the ectatic ducts favour crystal deposition.
- **Tubular defects:**
- **Hypercalciuria** in one-third to one-half (mostly absorptive; sometimes renal leak).
- **Incomplete distal RTA (renal tubular acidosis)** in 30–40%, shown on acid loading. This causes **hypocitraturia** and alkaline urine.
- Impaired urinary concentration.

- Mutations in H^+-ATPase genes (ATP6V1B1, ATP6V0A4) support an MSK–distal RTA link.
- Result: calcium stones within dilated ducts and medullary **nephrocalcinosis**.

Clinical features

- Often silent for life; presents **after 20 years**.
- **Renal colic 50–60%**, UTI (urinary tract infection) 20–33%, gross haematuria 10–18%.
- Found in **2.6–21% of stone formers**, more often women.

Diagnosis

- **Urographic triad:** 1. Enlarged kidneys, with or without papillary calcification. 2. Elongated papillary tubules or cavities filling with contrast: **"bristles on a brush"** (paintbrush); when calcified, a **"bouquet of flowers"**. 3. Papillary blush and persistent medullary opacification.
- CT urography now replaces IVU (intravenous urography); plain CT shows medullary nephrocalcinosis.
- **Metabolic work-up:** 24-hour urine (calcium, citrate, pH), serum bicarbonate and potassium.
- **Differentials:** papillary necrosis, other medullary nephrocalcinosis (distal RTA, hyperparathyroidism), calyceal diverticulum.

Management

Agent	Target	Note
Potassium citrate	Hypocitraturia, hypercalciuria	**First choice**; reduces calciuria and recurrence
Thiazide	Hypercalciuria	Useful even without hypercalciuria
Inorganic phosphate	Hypercalciuria	If thiazide unsuitable; avoid with urease infection
Fluids, low-sodium diet	Volume, calciuria	All patients
- Treat infection; stones by SWL (shock-wave lithotripsy) or ureteroscopy; stones inside ducts may be unreachable.		
- **Prognosis:** usually benign; renal failure is rare.		

Diagram to draw: pyramid with ectatic ducts filled with contrast ("brush") and small calculi.

> **Key points to remember:**
> - Ectatic collecting ducts; sporadic; "bristles on a brush".
> - Hypercalciuria, incomplete distal RTA, hypocitraturia.
> - Potassium citrate is first-line prevention.

SAQ 8 Megameatus intact prepuce

P2

Type: SAQ (10 marks)
Source: Practice Set 15 Paper 2, Q3f

Definition

- **MIP (megameatus intact prepuce)** is a hypospadias variant (up to 5%) with a **wide, "fish-mouth" glanular or coronal meatus**, a **completely formed prepuce** and **no significant curvature**. Described by Duckett and Keating (1989).

Embryology

- Normally the prepuce closes ventrally only after the urethral folds fuse (about 20 weeks), so 95% of hypospadias have a hooded, incomplete foreskin.
- In MIP the foreskin is complete, showing that **preputial and urethral development can dissociate**.

Clinical features

- Hidden under a normal foreskin; often found on **retraction before circumcision** or during circumcision.
- Wide, deep glans groove with a large, patulous meatus; well-developed spongiosum.
- Symptoms: spraying or deflected stream; cosmetic concern.
- **Advice:** retract the foreskin before every newborn circumcision. If found, the circumcision may be completed; prior circumcision does not harm later repair.

Evaluation

- Examine meatal site and size, glans and plate, curvature (artificial erection at surgery). GMS (glans–meatus–shaft) score.
- No routine upper-tract imaging for isolated distal hypospadias.

Management

- **Indication:** cosmesis and a straight stream; repair is optional. Usual timing 6–18 months.
- **Pyramid procedure (Duckett and Keating):** 1. Glans traction suture; retract prepuce. 2. Incise around the megameatus in a **pyramid shape**, apex on the ventral shaft. 3. Dissect the wide distal urethra free of the glans wings. 4. **Excise a ventral wedge** of redundant urethra to taper it. 5. Tubularise over a catheter; cover with dartos. 6. Approximate glans wings in two layers (glansplasty); preserve or reconstruct the prepuce.
- **Alternatives:** GAP (glans approximation procedure; pyramid = GAP with intact prepuce), TIP (tubularised incised plate) urethroplasty, Thiersch–Duplay tubularisation, Mathieu.

Complications

- Fistula, meatal stenosis or regression, glans dehiscence; generally low after distal repair.

Diagram to draw: fish-mouth glanular meatus with intact prepuce, and the pyramid incision.

> **Key points to remember:**
> - Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
> - Often found at circumcision; retract foreskin first.
> - Pyramid repair: dissect, taper and tubularise; glansplasty.

SAQ 9 Techniques of chordee correction

P2

Type: SAQ (10 marks)
Source: Practice Set 17 Paper 2, Q3b (related: Practice Set 10 Paper 1, Q3d)

Definition and causes

- Chordee: **ventral penile curvature**, with or without hypospadias. It, not meatal position, decides one-stage versus staged repair.
- Causes: tethered ventral skin, fibrous dysplastic dartos and Buck's fascia, short urethral plate, **corporal disproportion**.
- Devine–Horton (1973) graded the dysplastic layers: I spongiosum, II Buck's and dartos, III dartos only, IV corporal disproportion.

Assessment

- Parents' photographs; **artificial erection** in theatre: Gittes–McLaughlin (tourniquet at the base plus saline), corporal compression, or prostaglandin. A tourniquet can hide basal curvature.
- Grade: mild <30°, moderate 30–45°, severe ≥45° (goniometer; eye underestimates).

Stepwise correction

```
Deglove to penoscrotal junction + excise dysplastic dartos → erection test ├─ Straight → urethroplasty ├─ Residual <30° → dorsal plication └─ >30° → divide urethral plate → still >30° → ventral lengthening (stage)
```

	Dorsal shortening (plication)	**Ventral lengthening (corporotomy ± graft)**
Methods	**Nesbit:** excise dorsal tunical ellipses, close transversely. **Baskin:** single midline plication in the **12 o'clock nerve-free zone**. Heineke–Mikulicz dorsal incision closed transversely; 16-dot plication	1–3 transverse **full-thickness corporotomies** (STAG, staged tubularised autograft); single **9-to-3 o'clock corporotomy** with graft: **dermis, tunica vaginalis**, SIS (small intestinal submucosa); dermal graft 20% larger than defect
Use	≤15–30°	Moderate–severe after plate division
Recurrence	28% (one series)	9%
Risk	Shortening; dorsal nerve injury off midline	Longer, staged; theoretical erectile dysfunction

- **Plate division** (transection) releases a short plate; not dividing it is an independent risk factor for recurrence. Repair is then staged.
- **Also know:** "fairy cuts" (superficial tunical incisions); lateral or dorsal curves need multipoint plication; correct **torsion** by dartos flap rotation.
- **Rule:** correct fully the first time and repeat the erection test before urethroplasty. Curvature may recur at puberty.

Diagram to draw: penile cross-section with dorsal nerves at 11 and 1 o'clock and midline plication at 12.

> **Key points to remember:**
> - Artificial erection after degloving decides the technique.
> - <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
> - Plication recurrence 28% vs grafting 9%.

SAQ 10 Congenital ureteral anomalies

P3

Type: SAQ (10 marks)
Source: 2017-Summer Paper 3, Q3f

Embryology basis

- The **ureteric bud** arises from the **mesonephric (wolffian) duct** at about week 5 and induces the metanephric blastema.
- Anomalies arise from **abnormal budding**, **failed distal canalisation**, **abnormal muscle maturation** or **persistent embryonic vessels**.

Classification

Category	Anomaly	Key points
Number	**Duplication**: incomplete (bifid, Y-ureter) or complete	About 1 in 125; unilateral 6× bilateral. Lower pole: VUR (vesicoureteric reflux), PUJ (pelviureteric junction) obstruction. Upper pole: ectopia, ureterocele
Termination (orifice)	**Ectopic ureter**	Opens at or below the bladder neck. **Females**: bladder neck to vagina, uterus; **continuous dribbling with normal voiding**. **Males**: always above external sphincter (posterior urethra, seminal vesicle, vas); **never incontinent**; epididymitis. Zinner syndrome
Structure	**Ureterocele**	Cystic dilation of distal submucosal ureter (Chwalle membrane). Intravesical vs ectopic (Glassberg); cecoureterocele (Stephens)
	PUJ obstruction	Intrinsic aperistaltic segment (infants) or crossing lower-pole vessel (older children)
	Megaureter (≥7 mm)	Obstructed, refluxing, non-refluxing non-obstructed, refluxing-obstructed; primary obstructive = adynamic distal segment
	Primary VUR	Short submucosal tunnel, lateral orifice
Position	**Preureteral (retrocaval) vena cava**	Persistent subcardinal vein traps the right ureter; fishhook (S) shape at L3–L4; type I obstructs, type II does not
	Retroiliac ureter; herniation	Rare

Weigert–Meyer law

- In complete duplication the **upper-pole ureter opens lower and more medial** (ectopic, obstructs, ureterocele); the **lower-pole ureter opens higher and lateral** (refluxes).
- Reason: the upper-pole bud arises more cephalad on the wolffian duct, is incorporated later and ends more caudally.
- **Also know:** blind-ending ureter, inverted-Y ureter, triplication, congenital stricture, ureteral valves, fibroepithelial polyp.

Evaluation

- Ultrasound, VCUG (voiding cystourethrogram), **DMSA (dimercaptosuccinic acid)** scan, MR (magnetic resonance) urography.

Management principles

- **Ureterocele:** endoscopic incision, upper-pole heminephrectomy or ureteroureterostomy.
- **Ectopic ureter:** heminephroureterectomy if non-functioning; otherwise ureteroureterostomy or reimplantation.
- **PUJ obstruction:** pyeloplasty. **Megaureter:** observe most; taper and reimplant (5:1) if function falls.
- **Retrocaval ureter:** divide, bring anterior to the cava, excise the aperistaltic segment, reanastomose.

Diagram to draw: complete duplication showing Weigert–Meyer orifice positions.

> **Key points to remember:**
> - Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
> - Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
> - Retrocaval ureter is a venous anomaly; always right-sided.

SAQ 11 Autosomal recessive polycystic kidney disease.

P3

Type: SAQ (10 marks)
Source: 2018-Summer Paper 3, Q3c

Definition and genetics

- **ARPKD (autosomal recessive polycystic kidney disease)**: bilateral, symmetrical renal enlargement from **dilated collecting ducts**, **always with congenital hepatic fibrosis**.
- Gene **PKHD1 on chromosome 6p**; protein **fibrocystin (polyductin)** on the **primary cilium** of tubular and bile-duct cells, so it is a **ciliopathy**.
- Incidence about 1 in 10,000–50,000 live births. Siblings have a 25% risk.
- "Infantile" is a misnomer: a **spectrum**; earlier onset means more severe disease.

Pathology

- Radially arranged dilated collecting ducts (**microcysts**).
- **Liver:** periportal fibrosis, biliary ectasia; **Caroli syndrome** if intrahepatic ducts dilate.

Clinical features

Age	Features
Fetus	Large echogenic kidneys, **oligohydramnios** → Potter facies, limb deformity, **pulmonary hypoplasia**
Neonate	Huge, smooth, hard flank masses; respiratory distress; rising creatinine; **30–50% die early**
Child	Hypertension, CKD (chronic kidney disease), growth failure, anaemia, osteodystrophy
Older child	**Portal hypertension**: varices, splenomegaly, cholangitis; liver function stays normal

- No association with renal tumours.

Investigations

- **Ultrasound:** bilateral very large, **homogeneously hyperechoic** kidneys (brighter than liver) from countless microcyst interfaces; discrete cysts appear with age.
- **Parents' ultrasound** normal (excludes ADPKD); **PKHD1 genetic testing**; liver ultrasound.
- **Differential:** neonatal ADPKD (autosomal dominant PKD), bilateral hydronephrosis, MCDK (multicystic dysplastic kidney), glomerulocystic disease, bilateral tumours, renal vein thrombosis.

Management (no cure)

- **Neonatal:** ventilation, fluid and electrolyte care, antihypertensives (ACE (angiotensin-converting enzyme) inhibitors).
- **CKD care:** nutrition, growth hormone, anaemia and bone treatment; dialysis and **kidney transplantation**.
- **Nephrectomy** (breathing, feeding, blood pressure, room for peritoneal dialysis): cautiously; **very early bilateral nephrectomy** (first 3 months) had the worst neurological outcomes (ARegPKD registry).
- **Liver:** variceal sclerotherapy, portosystemic shunt; **combined liver–kidney transplant** for severe liver disease.
- **Genetic counselling** and prenatal diagnosis.

ARPKD versus ADPKD

	ARPKD	ADPKD
Gene	PKHD1 (chromosome 6)	PKD1 (16), PKD2 (4)
Age	Perinatal	30–50 years
Cysts from	Collecting duct	Whole nephron
Ultrasound	Symmetric, homogeneous, echogenic	Large discrete cysts, asymmetric
Liver	Fibrosis (always)	Cysts
Other	None	Berry aneurysm, mitral prolapse

Diagram to draw: cut kidney with radial dilated collecting ducts.

> **Key points to remember:**
> - PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
> - Huge homogeneous echogenic kidneys with oligohydramnios.
> - Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

SAQ 12 Ureterocele

P3

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 3, Q3c (ectopic ureter also answered in 2018-Summer Paper 3, Q1)

Definition

- Cystic dilatation of the distal ureter, in the bladder or spanning bladder neck and urethra; **never wholly urethral**, never joined to Wolffian structures.

- Commoner in girls (about 4:1); about 80% drain the **upper pole of a duplex kidney** (Weigert–Meyer rule: upper pole orifice caudal-medial); about 10% bilateral.
- **Cause:** persistent **Chwalla membrane**; upper pole often dysplastic.

Classification

System	Basis	Types
Glassberg (most useful)	Position	**Intravesical** vs **ectopic (extravesical)**: part at bladder neck or urethra
Stephens	Orifice morphology	Stenotic, sphincteric, sphincterostenotic, **cecoureterocele** (cavity extends under urethra), blind, non-obstructed
Churchill	Function	Upper pole only / whole kidney / contralateral at risk

Clinical features

- Antenatal hydronephrosis (most); febrile UTI (urinary tract infection) or urosepsis; adult single-system type with stone.
- **Prolapse:** smooth interlabial mass in girls; BOO (bladder outlet obstruction).
- **No incontinence when untreated** (unlike female ectopic ureter).

Evaluation

- **Ultrasound:** thin-walled cyst in bladder, dilated upper pole and ureter.
- **VCUG (voiding cystourethrogram):** lower pole reflux ("drooping lily"), eversion (weak floor), prolapse.
- **DMSA (dimercaptosuccinic acid):** upper pole function; cystoscopy before incision.

Management

- **Goals:** preserve function; remove infection, obstruction, reflux; keep continence.
- **CAP (continuous antibiotic prophylaxis)** from birth if hydronephrotic.
- **Observation:** small, non-obstructing, low-grade reflux, non-functioning upper pole.

Option	Indication / result
TUI (transurethral incision)	Sepsis, neonates, intravesical type. Low transverse full-thickness incision; relieves obstruction in 78–97%; new reflux 0–50%. Intravesical without reflux: 70–80% cured; ectopic: secondary surgery up to 70%
Upper pole heminephrectomy	Non-functioning upper pole, no reflux; secondary surgery 40–50%
Ureteroureterostomy / pyeloureterostomy	Functioning upper pole
Excision + common sheath reimplant ± bladder neck repair	Persistent reflux, ectopic or cecoureterocele; persisting reflux 5–10%
Total reconstruction	Older child, non-functioning upper pole, lower pole reflux

- **Also know:** "watering-can" laser punctures (less reflux); pseudoureterocele; ureterocele disproportion.

Diagram to draw: duplex kidney with upper pole ureterocele and lower pole reflux.

> **Key points to remember:**
>
> - Upper pole of duplex; never wholly urethral; Chwalla membrane.
> - Intravesical vs ectopic predicts TUI success.
> - TUI relieves obstruction in 78–97%; new reflux 0–50%.

SAQ 13 Primary obst. megaureter.

P3

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 3, Q3f

Assumed meaning: primary obstructive megaureter (POM).

Definition

- **Megaureter:** ureter diameter **≥7 mm** (normal up to 16 years is 5.0–6.5 mm). It is a descriptive term.
- **Classification:** primary (intrinsic) or secondary (neurogenic bladder, outlet obstruction); and **obstructed, refluxing, non-refluxing non-obstructed, or refluxing and obstructed**.
- **POM:** an **adynamic juxtavesical segment** at the UVJ (ureterovesical junction) blocks peristalsis and causes functional obstruction above.

Pathogenesis

- Ureteric muscle matures cranially to caudally; the juxtavesical segment matures last, even after birth. This explains spontaneous resolution.
- Excess collagen, thick muscle sleeve, abnormal interstitial cells of Cajal.

Clinical features

- 10–23% of antenatal upper-tract dilatation. Boys, **left side**; **bilateral 25%**; contralateral dysplasia or obstruction 10–15%.
- Mostly asymptomatic; or UTI (urinary tract infection), pain, haematuria, stones.

Evaluation

- **Ultrasound:** hydroureteronephrosis.
- **VCUG (voiding cystourethrogram):** excludes reflux and PUV (posterior urethral valves).
- **MAG3 (mercaptoacetyltriglycine) renogram:** split function; washout T½ unreliable (false positive up to 44%).
- MRU (magnetic resonance urography) for anatomy.

Management

Conservative (most)

- CAP (continuous antibiotic prophylaxis) in infants; serial ultrasound and renograms; most resolve.
- Diameter <10 mm: virtually never needs surgery; ≥10 mm: a substantial minority do.
- Follow to early adulthood.

Indications for surgery

- Recurrent febrile UTI; progressive dilatation; **DRF (differential renal function) <40%**; **fall in DRF ≥5%**.

Infant (small bladder, huge ureter): temporise

- Cutaneous ureterostomy; refluxing reimplant; nephrostomy; **endoscopic balloon dilatation with double-J** for 2–6 months (70–80% success; complications up to a third).

Definitive repair (child)

- Excise stenotic segment → straighten → **taper** → **non-refluxing reimplant at 5:1** (Cohen or Politano–Leadbetter; extravesical for huge ureters). Stent afterwards.

Tailoring	Method	Use
Starr plication	Imbricate lateral wall	Moderate (<1.75 cm)
Kalicinski folding	Exclude and fold lumen	Moderate
Hendren excision	Excise lateral wall	Massive; risk of ischaemia

- Success over 90%.
- **Complications:** obstruction (stent first), VUR (vesicoureteral reflux; observe or inject), persistent dilatation.

Diagram to draw: dilated ureter above narrow adynamic segment; tapered reimplant.

> **Key points to remember:**
> - Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
> - Surgery: UTIs, progression, DRF <40% or fall ≥5%.
> - Excise, taper, reimplant 5:1; success >90%.

Rapid Revision Sheet · Section 4

Primary VUR: VUJ embryology/anatomy, pathophysiology, management (incl. bilateral, adult grade 3), reimplantation techniques and STING (LAQ)

- Low (early) bud → high lateral orifice → short tunnel → VUR plus dysplasia; high (late) bud → ectopia.
- One embryological fault: lateral orifice, short tunnel, dysplasia.
- Tunnel length to diameter 5:1 normal versus 1.4:1 refluxing (Paquin); adult intramural ureter 1.2–2.5 cm.
- Dysplasia (boys, congenital) versus scar (girls, infection); sterile reflux rarely scars.
- Grade V loses papillary impressions; grade IV–V and bilateral grade III rarely resolve; bilateral grade IV persists most; bilateral VUR is high risk.
- RIVUR: CAP halves recurrent UTI, not scarring.
- Treat BBD and give CAP first; correct for breakthrough febrile UTI, new scars or persistent grade IV–V.
- Trend: risk-stratified care, fewer VCUGs, BBD first, endoscopic or robotic correction.
- STING: O'Donnell and Puri 1986; needle 2–3 mm below orifice at 6 o'clock.
- Dx/HA 80–250 µm; HIT better than STING; double HIT about 90%; Elder 2006: grade IV 63% after one injection.
- Watch for late obstruction; calcified mound mimics a stone.
- Open reimplantation 95.9%; Cohen cross-trigonal (over 98%) suits bilateral VUR through one cystotomy.
- Bilateral extravesical (Lich–Gregoir) repair risks retention up to 10%.
- Paquin: tunnel 5:1; combined, suprahiatal reimplant. Gil-Vernet: trigonoplasty, medial advancement, no tunnel; for low-grade bilateral VUR.
- 34–38% of scarred children become hypertensive; correction does not lower BP; scars bring 8× pre-eclampsia risk.
- In adults the scar, not the reflux, carries the risk; exclude secondary causes first; correct for recurrent pyelonephritis, new scars or pregnancy plans.

Testicular descent and undescended testis: evaluation, treatment algorithm and orchidopexy methods (LAQ)

- SRY → SOX9 → Sertoli cells (AMH); Leydig cells → testosterone and INSL3.
- Transabdominal: CSL regression plus INSL3-driven gubernacular swelling.
- Inguinoscrotal: androgen via genitofemoral nerve (GFN) and CGRP; most testes scrotal after 27 weeks.
- Ectopic = off the path; commonest site superficial inguinal pouch.
- Operate from 6 months, preferably by 12 months, and by 18 months at the latest.
- No imaging, no hormones.
- Non-palpable: EUA then diagnostic laparoscopy; laparoscopy decides.
- Bilateral non-palpable: DSD work-up first (karyotype, 17-OHP, AMH) to exclude CAH/DSD.
- Subdartos pouch; Prentiss manoeuvre gives extra length.
- Fowler–Stephens relies on vasal collaterals.

Retrocaval ureter and IVC anomalous development (LAQ)

- Infrarenal IVC = right supracardinal; suprarenal = right subcardinal; hepatic = vitelline.
- Circumaortic left renal vein is commonest (2–16%): ligate both limbs.
- Retrocaval ureter: venous anomaly (persistent subcardinal vein); right side only; 1 in 1500; male 3:1; transpose anterior to IVC.
- Type I fishhook obstructs; type II sickle often does not.
- CT urography; repair by anterior transposition and ureteroureterostomy.
- Duplicated IVC: filter both cavae; do not mistake it for nodes.

Hypospadias: present concepts, single-stage vs staged repairs, and hypospadias fistula (LAQ)

- Classify after degloving (GMS); curvature, not meatus, decides one versus two stages.
- Curvature >30° after degloving: stage.
- TIP for distal; staged graft repair (STAG/Bracka) for severe proximal; STAG has the lowest proximal complications.
- Stage 2 after ≥6 months.
- Hypospadias plus undescended testis: DSD work-up.
- Follow through puberty; use patient-reported outcomes.
- Fistula: commonest complication, just under 10%; TIP 5.7%.
- Exclude distal obstruction; wait 6–12 months.
- Layered closure with barrier flap; recurrence about 20%.

Wilms tumour, including bilateral Wilms tumour (LAQ)

- WT1 at 11p13 (WAGR, Denys–Drash); BWS at 11p15.
- Anaplasia (≥3× nuclei, hyperchromasia, abnormal mitoses) is unfavourable; TP53.
- Any spill, biopsy or positive node = stage III; sample ≥5 nodes.
- COG operates first; SIOP gives chemotherapy (VCR + AMD) first.
- Bilateral: stage V, 5–7%; no upfront nephrectomy or biopsy.
- Bilateral: COG three drugs for 6 weeks (reassess at 6 weeks), surgery by week 12, NSS; SIOP two drugs.
- AREN0534: 4-year OS about 95%.
- Bilateral NSS to avoid ESRD.

Proximal (perineoscrotal) hypospadias: penile embryology and management including Koyanagi repair (LAQ)

- DHT at 8–12 weeks; opening then closing zipper; 12 o'clock nerve-free zone.
- Hypospadias (including perineoscrotal) plus undescended testis: DSD work-up.
- <30° plication; >30° divide plate ± ventral graft and stage.
- STAG/Bracka staged repair preferred; fistula commonest complication.
- Koyanagi: one stage; meatus-based parameatal foreskin flap.
- Koyanagi: plate divided; no proximal circular anastomosis.
- Koyanagi: fistula commonest; diverticulum, meatal stenosis.

Urological evaluation and management of pediatric kidney transplant patient. (LAQ)

- Bladder target: age-expected capacity at <40 cm H_2O; augment if <75% despite CIC.
- PUV bladder heads to myogenic failure; neuropathic bladder needs earlier reconstruction.

- Never test a bladder with a graft; cycle the defunctionalised bladder.
- Stenosis 6%, leak 3%, stones 2%, significant reflux 2%.

Embryology and management of ectopic ureter. (LAQ)

- Ectopia = orifice at or below the bladder neck; bud too cranial on the wolffian duct.
- Girls: continuous dribbling with normal voiding; boys: never incontinent, epididymitis.
- DMSA decides keep (UU/reimplant) versus remove (heminephroureterectomy).
- Protect the lower-pole blood supply; stent the recipient ureter for UU.

TIP (Snodgrass) repair for hypospadias (SAQ)

- Snodgrass 1994; incise and tubularise; heals by epithelialisation.
- Slit-like meatus; dartos barrier reduces fistula.
- Glans <14 mm and poor plate raise complications.
- Fistula 5.7%, meatal stenosis 3.6% (Pfistermuller 2015).
- Extended to midshaft, selected proximal, redo, MIP and adults.
- Key condition: curvature <30° without plate division.
- Poor plate: dorsal inlay graft ("Snodgraft").
- Proximal and redo TIP fail more often.

Glandular hypospadias (SAQ)

- Glanular = most distal hypospadias; dorsal hood, minimal curvature.
- MAGPI only for a truly glanular meatus; GAP for MIP; TIP is the workhorse.
- Repair at 6–18 months; may be left alone if function is normal.

Vanishing testes (SAQ)

- Prenatal torsion after masculinisation; 15–40% of nonpalpable testes.
- Contralateral hypertrophy (>2 mL) suggests it, but explore anyway.
- Blind-ending vessels confirm; excise remnant (germ cells 5–15%).
- Anorchia: high FSH/LH, low AMH and inhibin B.

Pelvicalyceal anomalies other than PUJ obstruction (SAQ)

- Diverticulum: Type I to calyx, Type II to pelvis; stones in 40%.
- Megacalycosis: non-obstructive, 12–20 calyces, no treatment.
- Infundibulopelvic stenosis: bilateral, renal failure risk.

Retractile testis? (SAQ)

- Retractile = comes down and stays down.
- Gliding testis springs back: treat as undescended.
- Yearly examination until puberty; operate if it ascends.

Retroileal Ureter (SAQ)

- Retroiliac = preureteral iliac artery; persistent ventral umbilical artery root.
- Obstruction at L5–S1; either side or bilateral.
- Look for vasal and ureteric ectopia.
- Retrocaval: right, L3–L4, fishhook, 1 in 1500.

Medullary sponge kidney (SAQ)

- Ectatic collecting ducts; sporadic; "bristles on a brush".

- Hypercalciuria, incomplete distal RTA, hypocitraturia.
- Potassium citrate is first-line prevention.

Megameatus intact prepuce (SAQ)

- Wide glanular meatus, intact prepuce, no curvature; up to 5% of hypospadias.
- Often found at circumcision; retract foreskin first.
- Pyramid repair: dissect, taper and tubularise; glansplasty.

Techniques of chordee correction (SAQ)

- Artificial erection after degloving decides the technique.
- <30°: dorsal plication; >30°: divide plate ± ventral graft and stage.
- Plication recurrence 28% vs grafting 9%.

Congenital ureteral anomalies (SAQ)

- Upper pole obstructs (ectopia, ureterocele); lower pole refluxes.
- Female ectopic ureter: continuous dribbling with normal voiding; male never incontinent.
- Retrocaval ureter is a venous anomaly; always right-sided.

Autosomal recessive polycystic kidney disease. (SAQ)

- PKHD1, chromosome 6, fibrocystin; always congenital hepatic fibrosis.
- Huge homogeneous echogenic kidneys with oligohydramnios.
- Neonates die of pulmonary hypoplasia; survivors develop portal hypertension.

Ureterocele (SAQ)

- Upper pole of duplex; never wholly urethral; Chwalla membrane.
- Intravesical vs ectopic predicts TUI success.
- TUI relieves obstruction in 78–97%; new reflux 0–50%.

Primary obst. megaureter. (SAQ)

- Megaureter ≥7 mm; POM = adynamic juxtavesical segment.
- Surgery: UTIs, progression, DRF <40% or fall ≥5%.
- Excise, taper, reimplant 5:1; success >90%.

Section 5 · Paediatric Urology 2

Long answer questions (LAQ)

LAQ 1 Bladder embryology and exstrophy-epispadias complex

P1 · P2 · P3 **merged from 3**

Asked as: P3 LAQ "Bladder and ureter development, exstrophy–epispadias complex and its management" · P1 LAQ "Describe embryology of bladder. What are the proposed theories for extrophy epispadias complex? Describe skeletal abnormalities." · P2 SAQ "Embryogenesis of urinary bladder"

Introduction

- Assumed meanings: "ectopia vesicae" means classic bladder exstrophy (CBE); "single stage repair of exstrophy of UB" means single-stage (complete primary) repair of exstrophy of the urinary bladder, mainly CPRE (complete primary repair of exstrophy, Mitchell).
- **EEC (exstrophy–epispadias complex)** is one spectrum: glanular epispadias, then **CBE (classic bladder exstrophy)** (over 50%), then cloacal exstrophy.
- CBE incidence about 1 in 46,000; male:female 2.3:1. Recurrence in offspring of an affected parent: 1 in 70.

Development of the ureter

- **Weeks 4–6:** the **ureteric bud** arises from the distal **Wolffian (mesonephric) duct**. GDNF (glial cell line-derived neurotrophic factor) from the metanephric mesenchyme induces a single bud (extra buds cause duplication).
- **Reciprocal induction:** the bud invades the metanephric blastema. The bud stalk forms the ureter; its branches form the pelvis, calyces and collecting ducts; the blastema forms the nephrons.
- **Wall:** urothelium, lamina propria, smooth muscle and serosa, the same plan as the bladder.
- **Chwalla membrane** at the future orifice normally ruptures; failure is the classic explanation of ureterocele.

Development of the bladder: division of the cloaca

- The **cloacal membrane** (bilaminar ectoderm–endoderm) occupies the infraumbilical wall. **Mesenchyme grows in** between its layers and forms the lower abdominal muscles and pelvic bones.
- **Cloacal division (weeks 5–6; P2: weeks 4–6):** the **urorectal septum** (Tourneux and Rathke folds) grows caudally and divides the endodermal cloaca into a dorsal **anorectal canal** (rectum, posterior) and a ventral **UGS (urogenital sinus)** (bladder, anterior). It meets the cloacal membrane, which perforates to form the urogenital and anal openings.
- Paired genital tubercles fuse **cephalad** to the membrane.
- **Wolffian (mesonephric) duct entry** is the landmark that divides the UGS:
- **Cranial vesicourethral canal** (continuous with the allantois) forms the **bladder and pelvic urethra**.
- **Caudal (definitive) UGS** forms the phallic urethra in males, and the vestibule and distal vagina in females.

Development of the bladder: wall layers

- **Urothelium:** UGS endoderm.
- **Lamina propria, detrusor, adventitia / serosa:** surrounding splanchnic mesenchyme, by epithelial–mesenchymal signalling (epithelium and mesenchyme induce each other).

Trigone and ureterovesical junction (weeks 5–6)

- The **common excretory duct** (Wolffian duct distal to the bud) dilates and is absorbed into the UGS; **apoptosis** separates the ureter from the Wolffian duct.
- The arms rotate (**Mackie–Stephens**): the ureteric orifice moves **cranially and laterally**; the Wolffian duct ends caudally as the ejaculatory duct at the verumontanum.
- **Trigone muscle:** classic view, ureteric extension; current view (mouse lineage tracing), mostly bladder smooth muscle.

Bud origin	Orifice	Result
Too low (early contact)	High, lateral, short tunnel	VUR (vesicoureteral reflux)
Too high (late contact)	Caudal, medial	Ectopic ureter, often obstructed
Either	Abnormal bud–blastema contact	Dysplasia (VUR and dysplasia coexist)

- **Weigert–Meyer:** in duplex systems the upper-pole orifice is caudal and medial.

Allantois, urachus and fetal bladder function

- **Months 4–5:** the bladder descends into the pelvis; the allantoic duct elongates and obliterates as the **urachus** (median umbilical ligament).
- Urine production starts at about **week 10**. The fetal bladder is seen in about half at 10 weeks and almost all by 13 weeks; it empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

Bladder anomalies explained by embryology

Anomaly	Basis
Exstrophy	Cloacal membrane not reinforced by infraumbilical mesoderm
Urachal (patent urachus, sinus, cyst, diverticulum)	Failed obliteration at different levels
Agenesis, duplication	UGS or trigone failure; septation
Congenital (Hutch) diverticulum	Detrusor deficiency at the ureteric hiatus
VUR, ectopic ureter	Abnormal bud position

Developmental basis of EEC: core defect

- The **cloacal membrane is not reinforced by mesodermal ingrowth**. This mesoderm normally forms the lower abdominal muscles and pelvic bones, so the thin membrane ruptures (**Marshall–Muecke** overdevelopment theory).
- **Timing and extent of rupture decide the variant:** before septum descent gives cloacal exstrophy; after gives CBE; limited rupture gives epispadias.

Developmental basis of EEC: theories

Theory	Proposed defect	Result
Cloacal membrane overdevelopment (Marshall–Muecke)	Abnormally large membrane in week 4 blocks medial mesenchymal migration	Rupture before septum descent: cloacal exstrophy; after: bladder exstrophy; limited: epispadias
Lateral body-wall fold failure	Folds fail to meet in midline	Abdomen and pelvis: cloacal; pelvis alone: classic
Caudal genital hillocks	Genital tubercles fuse **below** rather than above the membrane	Dorsal urethral defect (epispadias)
Abnormal body-stalk insertion	Caudal insertion prevents mesenchyme interposition	Superficial unstable membrane disintegrates
Bony pelvis first	Pelvic ring primordium fails to rotate	Pelvic structures cannot meet, so the bladder herniates

- All theories converge on **absent infraumbilical midline mesoderm**.

Aetiology of EEC

- Multifactorial (inconsistent twin concordance). **22q11.2 duplication** is the commonest variant; **ISL1** susceptibility gene; p63 under-expressed in exstrophic urothelium.
- **IVF (in vitro fertilisation)** raises risk about 7.5-fold and early high-dose progesterone about tenfold; maternal smoking is linked to cloacal exstrophy; folate to the mildest form.

Resulting anatomy in classic exstrophy

- Exposed (open) bladder plate; epispadias; low umbilicus; anterior anus; wide levator hiatus; inguinal hernias.
- **Male genitalia:** anterior corpora about 50% shorter; short, upturned penis with dorsal chordee.
- **Female:** bifid clitoris, short vagina, anterior stenotic introitus; later prolapse.
- **Bladder:** immature, not neuropathic (raised type III collagen falls after closure). Ureters enter with no obliquity, so **VUR (vesicoureteric reflux) occurs in 100%** after closure.

Skeletal abnormalities: classic bladder exstrophy

3D (three-dimensional) CT (computed tomography) studies

- **Rotational:** posterior pelvis externally rotated (mean 12° each side); anterior segment externally rotated (mean 18° each side); sacroiliac joint rotated; acetabular and femoral retroversion.
- **Dimensional: pubic diastasis** (mean 4.8 cm); **anterior pubic segment shortened by 30%**; intertriradiate distance increased by about 31%; posterior segment length normal; larger sacrum.
- Bone histology is normal, so growth potential is normal.
- **Clinical effects:** short pendular penis (anterior segment short and splayed); wide hips, waddling gait and out-toeing in early childhood. These cause little disability and improve with growth.
- **Pelvic floor:** levator ani displaced posteriorly and flattened; anterior anus; later pelvic organ prolapse.

- **Spine:** minor variants (spina bifida occulta, lumbarisation, sacralisation) fairly common; scoliosis and dysraphism uncommon.

Skeletal abnormalities: cloacal exstrophy and isolated epispadias

- **Cloacal exstrophy:** widest, **asymmetric** diastasis (about twice CBE), sacroiliac malformation, hip dislocation.
- **Spinal anomalies in 85–100%** (lumbar myelodysplasia commonest; lipomeningocele); MRI (magnetic resonance imaging) for all.
- Limb defects: **clubfoot** (commonest), limb shortening, tibial and fibular deformity.
- **Isolated epispadias:** male symphysis widened (less than CBE); female symphysis usually closed or narrow.

Management: prenatal and at birth

- **Prenatal:** prenatal diagnosis in only about 25%. Ultrasound shows no bladder filling, low umbilicus, wide pubic rami, small genitalia and an enlarging lower abdominal mass. Counsel; deliver at an exstrophy centre.
- **At birth:** tie the cord with silk (no clamp); cover the plate with plastic wrap; saline irrigation at nappy change. Renal ultrasound; multidisciplinary team.
- **Assess:** template size and polyps (**≥3 × 3 cm** suits newborn closure), diastasis and pelvic malleability under anaesthesia, urethral plate and penile length, associated anomalies.
- Small template, penoscrotal duplication, hypoplastic bladder: **delayed closure at 6–9 months** (delay closure rather than do a bad early closure).

Pelvic osteotomy and immobilisation

- **Purpose:** closing the pelvic ring places the bladder neck deep in the pelvis, reduces tension and helps continence.
- **Indications:** closure **after 72 hours**, diastasis **>4 cm**, stiff (poorly malleable) pelvis, failed closure, or cloacal exstrophy (always).
- **Combined bilateral anterior innominate + vertical iliac osteotomy** (most used) with external fixator and Buck traction for 4–6 weeks.
- Under 72 hours with malleable pubes: closure without osteotomy, with modified Bryant traction for 4 weeks.
- Spica casting fails more often (failure with external fixation plus traction 7.5% versus spica 47%).

Modern staged repair of exstrophy (MSRE)

- Aim of closure: convert exstrophy into incontinent epispadias, with enough outlet resistance for bladder growth but not enough to harm the kidneys.

Stage	Age	Procedure	Purpose
1	Newborn	Bladder, posterior urethra and abdominal wall closure; ± osteotomy	Converts to complete epispadias; outlet resistance stimulates bladder growth
2	6–12 months	Epispadias repair (**modified Cantwell-Ransley**), after testosterone	Raises capacity; straightens penis

Stage	Age	Procedure	Purpose
3	5–7 years (another source: 4–7 years)	BNR (bladder neck reconstruction): modified **Young-Dees-Leadbetter (YDL)** + bilateral **Cohen** reimplantation	Continence; needs capacity **≥100 mL** and a motivated child

- **Closure keys: radical division of the urogenital diaphragm fibres** so the vesicourethral unit sits deep in the pelvis (missing this is the commonest error); ureteric stents; suprapubic tube 4 weeks; pubic nylon mattress suture.
- **Alternatives:** CPRE (Mitchell) with penile disassembly; Kelly radical soft-tissue mobilisation; Erlangen total repair (single stage); Mainz ureterosigmoidostomy. Most still need an outlet procedure.

Single-stage repair (CPRE and others)

- Single-stage repairs aim for fewer operations and continence without bladder neck repair.

Repair	Features
CPRE (Mitchell)	Newborn closure + penile disassembly + bladder neck tailoring
Erlangen ("total")	All in one at about 8 weeks, no osteotomy
Combined closure + epispadias repair	Delayed or failed closures, boys >6 months

- **Selection:** template ≥3 × 3 cm, elastic, few polyps; experienced team. Not for penoscrotal duplication, hypoplastic bladder or bilateral hydronephrosis.
- **CPRE technique:** 1. Combined general and epidural anaesthesia; **osteotomy if >72 hours** (3 days) old, diastasis >4 cm or stiff pelvis. 2. Incise around the plate; free the bladder; **divide the urogenital diaphragm fibres** to the levator hiatus so the unit sits deep. 3. **Penile disassembly:** right and left hemicorporoglanular bodies plus a urethral wedge on spongiosum (now often left attached to the glans). 4. Urethra placed ventrally between the corpora; corpora joined; meatus usually **hypospadiac**. 5. Bladder and neck closed and tapered; ± bilateral ureteric reimplantation (BUR-CPRE). 6. Pubic No. 2 nylon mattress suture; suprapubic catheter 4 weeks, ureteric stents. 7. Immobilise: external fixation plus traction (failure 7.5% versus spica 47%).
- **Complications:** same frequency as MSRE but **more serious**: dehiscence, prolapse, vesicocutaneous fistula, **glans or corporal loss** (ischaemia), early reimplantation for VUR or hydronephrosis; retention in girls.

Follow-up, failures and salvage

- **Follow-up:** prophylaxis; ultrasound; yearly cystogram and cystoscopy under anaesthesia.
- Dehiscence or prolapse: wait 4–6 months, then reclose with osteotomy.
- Failed BNR (no 3-hour dry interval by 2 years): bulking, redo YDL, or **augmentation + bladder neck closure + Mitrofanoff** (continent catheterisable stoma).

Cloacal exstrophy (OEIS) management

- OEIS: omphalocele, exstrophy, imperforate anus, spinal defects.
- Stabilise; neurosurgery (spinal defects 85–100%); separate bowel from hemibladders; colostomy; osteotomy always; gender congruent with karyotype; continence mostly by augmentation with CIC (clean intermittent catheterisation).

Outcomes and long-term issues

- Daytime dryness after MSRE about **70–74%** (another source: 70–75%), versus about 56% after CPRE with YDL; most still need bladder neck surgery after CPRE.
- **Bladder capacity at neck repair is the main predictor**; each failed closure lowers it. **Best continence predictors: template size at birth and a successful first closure.**
- **Also know:** variants (superior vesical fissure, duplicate exstrophy, pseudoexstrophy); adenocarcinoma risk (cytology, cystoscopy; colonoscopy after ureterosigmoidostomy); reduced male fertility and uterine prolapse in adults; female epispadias; psychosocial support.

Diagrams to draw: cloaca divided by the urorectal septum (vesicourethral canal, UGS, allantois) with failed mesodermal ingrowth of the cloacal membrane; Wolffian duct with ureteric bud and its absorption into the UGS; exstrophy pelvis with diastasis; MSRE timeline; penile disassembly into two hemicorpora and urethral wedge.

> **Key points to remember:**
>
> - Bud from Wolffian duct; position decides VUR or ectopia.
> - Urorectal septum weeks 5–6 (P2: 4–6); mesonephric duct entry is the landmark; vesicourethral canal forms the bladder; urachus from allantois.
> - Trigone: common excretory duct absorption; mostly bladder muscle (current view).
> - Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.
> - Marshall–Muecke: overdeveloped cloacal membrane, failed mesodermal reinforcement; rupture timing decides variant.
> - CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis about 4.8 cm; VUR in 100%.
> - Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.
> - MSRE: closure at birth, epispadias repair 6–12 months, YDL / BNR at 5–7 years at ≥100 mL.
> - Osteotomy if closure after 72 h or diastasis >4 cm.
> - CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
> - Capacity ≥100 mL, template size and a successful first closure predict continence.

LAQ 2 Prune belly syndrome

P1 · P3 **merged from 2**

Asked as: P1 LAQ "Prune Belly Syndrome. Antenatal diagnosis, pathophysiology and postnatal management." · P3 SAQ "Prune belly syndrome"

Definition

- **PBS (prune belly syndrome)**, also called Eagle–Barrett or triad syndrome: **deficient abdominal wall muscles**, **bilateral intra-abdominal testes** and a **dilated, poorly contracting urinary tract**.
- 1 in 29,000–44,000 live births; **95% male**.
- **Survival depends on renal dysplasia and lung hypoplasia, not on the abdominal wall.**

Antenatal diagnosis

- **Ultrasound** may suggest PBS from 11–14 weeks, but the classic signs are not consistent even at 30 weeks.
- **Findings:** huge bladder filling the abdomen; bilateral hydroureteronephrosis; urachal diverticulum; dilated posterior urethra; echogenic kidneys with cortical cysts (dysplasia); irregular abdominal circumference; fetal ascites; oligohydramnios when severe.
- **Differential diagnosis:** PUV (posterior urethral valves), urethral atresia, megacystis–megaureter.
- Fetal dilatation does **not** predict postnatal renal function.
- **Prenatal management:** intervention has no proven renal benefit. Vesicoamniotic shunting is justified only for **urethral atresia with progressive oligohydramnios**, or to decompress the abdomen and avoid dystocia. Termination is hard to justify except with early, severe oligohydramnios. Plan delivery in a centre with neonatal intensive care.

Diagram to draw: fetal ultrasound sketch: huge bladder, dilated posterior urethra, bilateral hydroureteronephrosis.

Pathogenesis (theories, none proven)

Theory	Mechanism
Early (fetal) urethral obstruction	Back-pressure dilates the tract and stretches the wall; but most have no obstruction after birth
Lateral plate mesoderm defect	This mesoderm forms ureters, bladder, prostate, urethra and gubernaculum, so one defect explains the triad
Intrinsic urinary tract (ureteric) defect	Primary ureteric dilatation and fetal ascites
Yolk sac defect	Primary defect of the yolk sac

- **Genetics:** mostly sporadic, normal karyotype; smooth-muscle genes **MYOCD (myocardin)** and **FLNA (filamin A)** in multiplex families.

Pathophysiology and features by organ

- **Kidneys:** dysplasia in about 50% (Potter type II from mesenchymal defect; type IV cystic from obstruction).

- **Ureters:** long and tortuous, worse distally. Collagen replaces smooth muscle, so walls do not coapt and peristalsis fails: **a failed pump, not a blocked pipe**. The result is stasis and infection. **VUR (vesicoureteric reflux) in about 75%.** The proximal ureter has better muscle.
- **Bladder:** large, smooth, collagen-rich; urachal pseudodiverticulum (urachus patent in 25–30%); splayed trigone. Normal compliance but poor contractility, so voiding is "unbalanced", with residual urine.
- **Prostate and urethra:** prostatic hypoplasia dilates the posterior urethra; distal obstruction in about 20%; urethral atresia; **megalourethra** (scaphoid: spongiosum deficient; fusiform: all erectile bodies, more severe).
- **Testes:** intra-abdominal; azoospermia usual; paternity only by sperm retrieval and ICSI (intracytoplasmic sperm injection).
- **Extra-genitourinary (about 75%):** pulmonary hypoplasia; cardiac about 10%; gastrointestinal at least 30% (malrotation); orthopaedic up to 65% (talipes, hip dysplasia).

Classification (Woodard)

Category	Features	Outlook	Management
I	Oligohydramnios, **pulmonary hypoplasia**, dysplasia, urethral atresia	Perinatal death	Supportive; bladder drainage only
II	Full triad, moderate or unilateral dysplasia / moderate uropathy, **no pulmonary hypoplasia**	Risk of CKD (chronic kidney disease)	Individualised (controversial, debated group)
III	Mild or incomplete; **stable function**	Good; majority	Surveillance

- **Also know:** RUBACE score (Renal, Ureter, Bladder, Abdominal wall, Cryptorchidism, Extra-genitourinary) grades morbidity / severity in survivors; "pseudo-prune" presents late with renal failure.

Postnatal management: neonatal

Team: neonatology, nephrology, urology, cardiology.

1. Heart and lungs first: **chest radiograph** for pneumothorax and hypoplasia.
2. Creatinine, BUN (blood urea nitrogen), electrolytes; follow the **trend**, because the first value reflects the mother.
3. Renal and bladder ultrasound once stable.
4. Antibiotic prophylaxis; **circumcision**; strict asepsis.
5. **VCUG (voiding cystourethrogram)** only with renal insufficiency or suspected outlet obstruction.
6. **DMSA (dimercaptosuccinic acid) at 4–6 weeks** for split function; MAG3 (mercaptoacetyltriglycine) for drainage.
7. Outlet obstruction: catheter or suprapubic tube.

Diagram to draw: VCUG showing large bladder and dilated posterior urethra.

Postnatal management: urinary tract

- **Conservative** for category III: surveillance, treat infection.
- **Surgery for:** recurrent upper tract infection, sepsis, progressive dilatation or rising creatinine / renal decline, high-grade VUR, true obstruction.

- **Temporary diversion:** wide **cutaneous vesicostomy (Blocksom)**. Avoid high ureterostomy, which wastes the good proximal ureter.
- **Reconstruction after 3–6 months** (lung maturity): excise the distal ureter, taper and reimplant the healthier proximal ureter; reduction cystoplasty gives little lasting benefit (capacity recurs). **CIC (clean intermittent catheterisation)** or Mitrofanoff for poor emptying.

Orchidopexy

- **Bilateral transabdominal at about 6 months**, often with other surgery; later, **staged Fowler–Stephens** (better than single-stage) or laparoscopic.

Abdominoplasty

- Aims: cosmesis; better cough, voiding / emptying and defecation.

Technique	Key feature
Randolph	Transverse; excises the worst lower wall
Ehrlich	Vertical; umbilicus kept; vest-over-pants overlap
Monfort	Vertical; umbilicus kept; central fascial plate with lateral flaps overlapped; transperitoneal access, good exposure for combined surgery

- **Also know:** Furness (extraperitoneal plication); Dénes; corset.

Prognosis and follow-up

- About 20% die perinatally; **up to 30% develop chronic renal failure**.
- **Nadir creatinine <0.7 mg/dL at 1 year** predicts stable function.
- Lifelong timed voiding, cultures and imaging; transplant when needed.

> **Key points to remember:**
> - Triad; 95% male; dysplasia and lungs decide survival.
> - Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
> - Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
> - Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

LAQ 3 Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency

P1 · P3 **merged from 2**

Asked as: P1 LAQ "Describe neurophysiology of lower urinary tract and various types of voiding dysfunction in children" · P3 SAQ "DD of frequency and urgency in 3 yr male child and treatment"

Introduction

- The LUT (lower urinary tract) has two opposite functions: **storage** at low pressure with a closed outlet, and **emptying** by a coordinated detrusor contraction with a relaxed outlet. Bladder and sphincter act reciprocally.

Neurophysiology: peripheral innervation

	Parasympathetic	Sympathetic	Somatic
Origin	**S2–S4** (sacral parasympathetic nucleus)	Thoracolumbar (T10–L2)	**Onuf nucleus** (S2–S4 ventral horn)
Nerve	Pelvic	Hypogastric	Pudendal
Bladder	Contracts (ACh [acetylcholine] on **M3**; ATP)	Relaxes (noradrenaline on **β3**)	–
Outlet	Relaxes urethra (**NO**, nitric oxide)	Contracts bladder neck (**α1**)	Contracts EUS (external urethral sphincter; nicotinic)
Role	Voiding	Storage	Voluntary closure

- All three nerves carry afferents. **Aδ fibres** (tension receptors in muscle) signal normal fullness. **Unmyelinated C fibres** are "silent" normally; after inflammation, SCI (spinal cord injury) or obstruction they are recruited and drive overactivity (the basis of capsaicin and resiniferatoxin).
- Detrusor has more M2 receptors, but **M3 mediates contraction**.

Neurophysiology: reflex and central control

- **Storage (spinal reflexes):** low afferent activity triggers sympathetic outflow (detrusor inhibition, bladder-neck closure) and the **guarding reflex** (pudendal; sphincter activity rises with filling).
- **Voiding (spinobulbospinal reflex):** high afferent activity reaches the **PAG (periaqueductal grey)** and then the **PMC (pontine micturition centre, Barrington nucleus)**; the sphincter relaxes first, then the detrusor contracts. The PMC is an on–off switch.
- **Forebrain decides:** the insula senses fullness; the **medial prefrontal cortex** tonically holds the switch "off" until voiding is socially appropriate.
- Lesion patterns: above the pons, overactivity with a coordinated sphincter; between pons and sacral cord, overactivity with **DSD (detrusor–sphincter dyssynergia)**; sacral or peripheral, underactive bladder.
- **Central transmitters:** glutamate excitatory; GABA (γ-aminobutyric acid) and glycine inhibitory; serotonin inhibits voiding; dopamine tonically inhibits the PMC (lost in Parkinson disease).

`Filling → Aδ afferents → sacral cord → PAG low traffic: spinal storage reflexes (sympathetic + guarding) high traffic + cortex "permits": PMC → EUS relaxes → detrusor contracts`

Diagram to draw: bladder with pelvic, hypogastric and pudendal nerves, plus brainstem loop (PAG–PMC).

Maturation in children

- Bladder visible on fetal ultrasound by 14 weeks. Infants void reflexly but **wake before voiding** (cortical arousal gates voiding).
- Infant voiding is often interrupted (transient dyscoordination), with detrusor pressure often **>100 cmH2O**; frequent small voids largely disappear by **2 years**.
- Capacity rises: **EBC (expected bladder capacity) = (age + 2) × 30 mL** (infants: weight in kg × 7 mL).
- Toilet training: bowel control, then daytime bladder, then night-time bladder. Persisting dyscoordination after toddlerhood is dysfunction, not immaturity.

Definitions: frequency and urgency

- **Frequency:** ≥8 voids a day (ICCS, International Children's Continence Society). **Urgency:** sudden, compelling need to void.
- At 3 years toilet training may still be ongoing, so decide first whether the pattern is abnormal.

Types of voiding dysfunction: non-neurogenic (functional) LUT dysfunction (ICCS)

Type	Key feature
OAB (overactive bladder)	Urgency ± frequency (≥8 voids a day), urgency incontinence; tower flow
Voiding postponement	Habitual holding; holding manoeuvres; infrequent voids (≤3 a day)
Dysfunctional voiding	Habitual sphincter/pelvic-floor contraction during voiding in a neurologically normal child; **staccato** flow
Underactive bladder	Needs abdominal straining to void; interrupted flow
Giggle incontinence	Complete emptying with laughter
Vaginal reflux	Small leak just after voiding in girls
Enuresis	Wetting in sleep, ≥5 years; monosymptomatic or non-monosymptomatic

- **BBD (bladder and bowel dysfunction):** LUT dysfunction plus constipation; rectal loading worsens symptoms and UTI (urinary tract infection).
- **Hinman (non-neurogenic neurogenic bladder):** the severe end of dysfunctional voiding, with trabeculated bladder, VUR (vesicoureteric reflux) and renal damage.

Types of voiding dysfunction: neurogenic LUT dysfunction

- Causes: **myelomeningocele** (commonest), occult dysraphism / **tethered cord**, sacral agenesis, anorectal malformation, cerebral palsy, SCI, transverse myelitis, pelvic tumour surgery (sacrococcygeal teratoma).

- Pattern = detrusor (overactive or underactive) × outlet (competent, incompetent or dyssynergic). Bony level does not predict bladder function.
- **Hostile bladder: DLPP (detrusor leak point pressure) ≥40 cmH2O** (McGuire 1981), poor compliance, or NDO (neurogenic detrusor overactivity) with DSD.

Types of voiding dysfunction: anatomical

- **PUV (posterior urethral valves)** "valve bladder": overactivity, then poor compliance, then myogenic failure; ectopic ureter.

Differential diagnosis of frequency and urgency (3-year-old boy)

Group	Causes and clues
Infection	**UTI**: fever, dysuria; balanoposthitis; viral haemorrhagic cystitis
Functional	**BBD**/OAB; voiding postponement; **extraordinary daytime urinary frequency** (sudden very frequent small voids, no nocturia or wetting, often after stress; self-limiting)
Bowel	**Constipation**: rectal mass presses on bladder; commonest associated cause
Obstruction	**PUV**, mild late form; **meatal stenosis** (circumcised boys; thin deflected stream); severe phimosis
Neurogenic	Occult spinal dysraphism, tethered cord (sacral dimple, hair tuft, gait or foot abnormality)
Bladder irritants	Stone, foreign body, hypercalciuria, pinworms, rarely **rhabdomyosarcoma**
Polyuria	Diabetes mellitus, diabetes insipidus, chronic kidney disease
Behavioural	Anxiety, family stress, developmental disorders

Evaluation

- **History:** onset, nocturia, wetting, stream, fever, **bowel habit (Bristol stool chart)**, thirst, stressors.
- **Examination:** abdomen, meatus and foreskin, **lumbosacral spine**, lower-limb neurology.
- **Urinalysis and culture**; glucose; specific gravity >1.005 makes diabetes insipidus unlikely.
- **Bladder and bowel diary.**
- **Ultrasound** of kidneys and bladder: wall thickness, residual, dilation, stones.
- **Uroflowmetry with PVR (post-void residual) and patch EMG (electromyography)** if the child cooperates: bell (normal), tower (OAB), staccato (dysfunctional voiding), interrupted (underactive/DSD), plateau (obstruction).
- **Selective:** VCUG (voiding cystourethrogram) for suspected PUV or febrile UTI with abnormal ultrasound; spinal MRI (magnetic resonance imaging) for neurological signs; urine calcium/creatinine ratio.
- Invasive (video)urodynamics only for neurogenic, anatomical or refractory cases.

Diagram to draw: flowchart: urinalysis, then ultrasound with residual, then targeted tests.

Management

- **Treat constipation first:** disimpaction, then maintenance **polyethylene glycol**, diet and toilet routine.

- **Standard urotherapy** for 4 weeks: timed voiding every 2–3 hours, relaxed posture, adequate fluids, avoid caffeine.
- **UTI:** antibiotics per culture.
- **OAB persisting after urotherapy: oxybutynin 0.3–0.6 mg/kg/day** in 2–3 doses (maximum 15 mg/day; watch for constipation); mirabegron; botulinum toxin; parasacral TENS (transcutaneous electrical nerve stimulation).
- **Dysfunctional voiding: biofeedback**; α-blocker; sphincter botulinum toxin if refractory.
- **Extraordinary daytime frequency:** reassurance; resolves in weeks to months.
- **Enuresis:** alarm or desmopressin.
- **Meatal stenosis:** meatotomy. **PUV:** endoscopic valve ablation. **Tethered cord:** neurosurgical release.
- **Neurogenic: CIC (clean intermittent catheterisation; Lapides 1971) + antimuscarinic**, then intradetrusor botulinum toxin, then augmentation.

> **Key points to remember:**
> - S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
> - PMC = switch; medial prefrontal cortex holds it off.
> - EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
> - DLPP ≥40 cmH2O = hostile.
> - Always ask about bowels; treat constipation first.
> - Frequency/urgency: rule out UTI, PUV, meatal stenosis and occult dysraphism.
> - Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

LAQ 4 Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies

P1 · P3 **merged from 2**

Asked as: P3 LAQ "Antenatal hydronephrosis (UTD) and fetal uropathy (LUTO)" · P1 SAQ "Anomalies detected on Prenatal scan between 16th and 20th weeks of gestation"

Introduction and terminology

- Assumed meaning: HN (hydronephrosis) and ANH (antenatal hydronephrosis) are now called **UTD (urinary tract dilation)**.
- Major congenital anomalies occur in about **2–3% of births**. Urinary tract anomalies are among the commonest detected prenatally (commonly quoted as 20–30% of all prenatal anomalies).
- **UTD is the second commonest prenatal anomaly after cardiac defects, in 1–3% of pregnancies.**
- Mostly UTD on the 20-week scan, usually mild and transient: mild 57–88%, moderate 10–30%, severe 1.5–13% of UTD. The serious group is **LUTO (lower urinary tract obstruction)**: PUV (posterior urethral valves), urethral atresia, PBS (prune-belly syndrome).

The prenatal anomaly scan (16–20 weeks)

- The **anomaly scan is done at 18–20 weeks** (ACOG (American College of Obstetricians and Gynecologists)); by 20 weeks most urological anomalies are visible.
- Screening US (ultrasound) at 18–20 weeks detects dilation well (sensitivity about 90%) but predicts the diagnosis poorly.
- **Why this window:** fetal urine production begins at about 14 weeks, and after about 16 weeks amniotic fluid is mainly fetal urine, so **oligohydramnios reflects renal failure or outlet obstruction** and causes **pulmonary hypoplasia**. The bladder should be seen filling and emptying.
- Survey kidneys, collecting system, ureters, bladder, genitalia, amniotic fluid, abdominal wall and spine. AFI (amniotic fluid index): normal 8–18 cm; oligohydramnios <5–6 cm.

Anomalies detected and their signs

Anomaly	Signs
PUV (posterior urethral valves)	Male; thick bladder; **keyhole sign**; bilateral hydroureteronephrosis; oligohydramnios
Bilateral renal agenesis	**Anhydramnios after 16 weeks**, no bladder, no renal arteries: lethal Potter sequence
MCDK (multicystic dysplastic kidney)	Non-communicating cysts, no central pelvis
ARPKD (autosomal recessive polycystic kidney disease)	Large, homogeneously **bright** kidneys; oligohydramnios
Bladder exstrophy	**Bladder never seen**, low cord insertion, lower abdominal mass, wide pubis
Prune belly	Thin-walled large bladder, empty scrotum
Myelomeningocele	Spinal defect, lemon and banana signs

- **Also know:** duplex kidney with ureterocele (intravesical cyst); cloacal exstrophy (adds omphalocele, spinal defects); hydrocolpos; ambiguous genitalia; renal ectopia.

Diagram to draw: fetal keyhole sign.

Causes of antenatal UTD (SFU [Society for Fetal Urology] consensus)

Cause	Incidence
Transient/physiological dilation	50–70%
UPJ (ureteropelvic junction; PUJ, pelviureteric junction) obstruction pattern	10–30%
VUR (vesicoureteral reflux)	10–40%
UVJ (ureterovesical junction) obstruction/megaureter	5–15%
MCDK	2–5%
Duplex system, ureterocele, ectopic ureter	5–7%
LUTO (PUV, prune-belly)	0.1–0.5%

- The chance of postnatal pathology rises with grade: about 12% (mild) to 88% (severe). UPJ obstruction and PUV track grade; **VUR is not predicted by grade**.

Grading systems

APD (anteroposterior renal pelvic diameter, APRPD): transverse image at the hilum. APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.

Grade	Second trimester	Third trimester
Mild	4 to <7 mm	4 to <9 mm
Moderate	7–10 mm	9–15 mm
Severe	>10 mm	>15 mm

SFU (postnatal only): 1 pelvis barely split; 2 pelvis and major calyces; 3 minor calyces uniformly dilated; 4 plus parenchymal thinning.

UTD consensus 2014 (risk stratification): six parameters: APD, central/peripheral calyces, parenchymal thickness, parenchymal appearance, ureter, bladder (plus amniotic fluid prenatally).

Prenatal parameter	A1 low risk	A2–3 increased risk (any one)
APD 16–27 weeks	4 to <7 mm	≥7 mm
APD ≥28 weeks	7 to <10 mm	≥10 mm
Calyces	Central or none	Peripheral
Parenchyma, ureter, bladder	All normal	Abnormal
Amniotic fluid	Normal	Unexplained oligohydramnios

- **A1:** one repeat US after 32 weeks. **A2–3:** US every 4 weeks to delivery. Both: postnatal US after 48 hours.

Prenatal algorithm (unilateral UTD, male fetus)

`Anomaly scan: unilateral UTD → Detailed scan: contralateral kidney, ureter, bladder wall and emptying, posterior urethra (keyhole), amniotic fluid, other anomalies ├─ Bilateral / thick bladder / keyhole / oligohydramnios ─► LUTO`

pathway │ (fetal medicine; karyotype; Ruano staging; possible shunt) ├─ A1 ─► repeat US after 32 weeks └─ A2–3 ─► US every 4 weeks to delivery → Normal delivery at local centre; no fetal intervention for unilateral UTD → Postnatal US after 48 hours (earlier if solitary kidney or suspected LUTO)

- Repeat scans because the lesion may progress and a second anomaly is more likely. **Fetal MRI** (magnetic resonance imaging) only for complex anatomy.
- Counsel the parents: most cases resolve; prognosis with a normal opposite kidney is excellent.

Antenatal management of fetal uropathy (suspected LUTO): evaluation

- Male fetus, thick-walled "keyhole" bladder, bilateral hydroureteronephrosis, oligohydramnios: LUTO until proved otherwise.
- Anomaly scan, **fetal echocardiogram, karyotype** (one-fifth of megacystis is aneuploid); fetal MRI if unclear.
- **Serial fetal urine** (3 samples, 48–72 hours apart). Favourable: Na <100 mEq/L, Cl <90 mEq/L, osmolality <200 mOsm/L, β2-microglobulin <6 mg/L; not validated.
- **Ruano staging:** I mild, normal fluid: observe. **II severe, function preserved** (oligohydramnios, favourable urine): intervene. III cysts, dysplasia, unfavourable urine: intervention for the lungs only.

Antenatal management of fetal uropathy: options

Option	Notes
Observation, serial scans	Most cases
VAS (vesicoamniotic shunt)	Dislodgement, blockage, preterm labour
Fetal cystoscopy + laser valve ablation	Diagnoses and treats; fistula up to 10%
Serial amnioinfusion	Lungs in anhydramnios; under trial
Early delivery / termination	Mature lungs, falling fluid / poor prognosis

- Shunt or fetal cystoscopy is done at 18–24 weeks in severe LUTO.
- **PLUTO (2013):** shunt versus conservative (31 fetuses): better perinatal survival, **no renal benefit**.

Antenatal management of fetal uropathy: counselling and delivery

- Multidisciplinary team: fetal medicine, paediatric urology, nephrology, neonatology.
- Intervention improves survival, not renal function; complications **over 40%**.
- Deliver where the bladder can be decompressed at once; prophylaxis, then VCUG (voiding cystourethrogram).

Postnatal algorithm

- **First US after 48 hours**, because neonatal dehydration under-reads dilation. Postnatal normal: APD <10 mm, no calyceal or ureteral dilation.

	P1 (SFU 1–2)	P2 (SFU 3)	P3 (SFU 4)
Findings	APD 10–15 mm, central calyces	APD ≥15 mm, peripheral calyces or dilated ureter	Abnormal parenchyma or bladder
Repeat US	3–6 months	Individualised	1 month

	P1 (SFU 1–2)	P2 (SFU 3)	P3 (SFU 4)
Antibiotic prophylaxis	No	Discretion; **yes if ureter >7 mm**	Yes
VCUG	No	Discretion	Yes
MAG3 (mercaptoacetyltriglycine) renogram	No	Discretion	If persistent

- **Male-specific points:** VCUG confirms a normal posterior urethra. **Circumcision** protects (NNT [number needed to treat] 10). UTI (urinary tract infection) risk: female sex, uncircumcised boys, high grade, dilated ureter.
- Prophylaxis: amoxicillin or a first-generation cephalosporin from birth; nitrofurantoin and trimethoprim only after 8 weeks.
- MAG3: DRF (differential renal function) <40%, T½ >20 minutes, or delayed retention suggest obstruction. MRU (magnetic resonance urography) for anatomy; DMSA (dimercaptosuccinic acid) for MCDK or duplex function.
- **Surgery indications (UPJ):** rising APD, DRF <40% with poor drainage, falling DRF, symptoms (UTI, stone, pain). Megaureter surgery: DRF <40%, febrile UTI; defer reimplant beyond 1 year.
- Outcome: SFU 1 resolves in nearly all by 2.5 years; SFU 2 about 95% by 4 years.

Bilateral dilation and PUV: how they differ

- Bilateral hydroureteronephrosis, thick bladder, keyhole sign and oligohydramnios mean LUTO until proved otherwise.
- At birth: 5 Fr feeding tube, prophylaxis, creatinine, early VCUG; prenatal intervention improves survival but not renal function (PLUTO).

Diagram to draw: UTD A1/A2–3 prenatal and P1–P3 postnatal flowchart; keyhole bladder with bilateral hydroureteronephrosis and shunt.

> **Key points to remember:**
> - UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
> - Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
> - ANH = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
> - UTD A2–3 = APRPD ≥7 mm (16–27 weeks) or any abnormal feature.
> - Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
> - First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.
> - Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
> - Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
> - ARPKD large bright kidneys; MCDK non-communicating cysts.
> - Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
> - PLUTO: survival up, renal function unchanged.

LAQ 5 Embryological development of urinary bladder and management of Hutch Diverticulum?

P3

Type: LAQ (25 marks)
Source: 2021-Summer Paper 3, Q1 (bladder diverticulum also answered in 2016-Summer Paper 2, Q3c)

A. Embryology of the bladder

1. Division of the cloaca (weeks 4–6)

- The **urorectal septum** (Tourneux and Rathke folds) divides the endodermal cloaca into a dorsal **anorectal canal** and a ventral **UGS (urogenital sinus)**. The cloacal membrane perforates to form the urogenital and anal openings.
- The **Wolffian (mesonephric) duct entry** is the landmark:
- **Cranial vesicourethral canal** (continuous with the allantois) → bladder and pelvic urethra.
- **Caudal (definitive) UGS** → phallic urethra in males; distal vagina and vestibule in females. The whole female urethra comes from the pelvic part.

2. Wall layers

- **Urothelium:** endoderm of the UGS.
- **Lamina propria, detrusor and adventitia:** surrounding splanchnic mesenchyme. Epithelium and mesenchyme induce each other (epithelial–mesenchymal signalling).

3. Trigone (weeks 5–6)

- The **common excretory ducts** (mesonephric duct distal to the ureteric bud) dilate and are absorbed into the UGS. **Apoptosis** separates the ureters from the mesonephric ducts.
- Ureteric orifices migrate **cranially and laterally**; the Wolffian ducts end caudally as ejaculatory ducts at the verumontanum.
- Classic view: trigone muscle is ureteric (Wolffian) tissue. Current view (mouse lineage tracing): **mostly bladder smooth muscle**, minor ureteric contribution.
- **Bud position (Weigert–Meyer):** bud too low → lateral orifice, short tunnel → VUR (vesicoureteral reflux); too high → caudal, medial, ectopic orifice.

4. Allantois and urachus

- **Months 4–5:** the bladder descends into the pelvis. The allantoic duct elongates and obliterates into the **urachus** (median umbilical ligament).

5. Function

- Urine production starts at **week 10**. Fetal bladder is seen from 10 weeks, almost always by 13 weeks, and empties every 15–20 minutes. After **16 weeks** amniotic fluid is mainly fetal urine.

B. Anomalies explained by embryology

Anomaly	Embryological basis
Agenesis	Cranial UGS atrophy, or trigone not formed; survival only with ectopic ureters
Exstrophy	Cloacal membrane not reinforced by mesoderm

Anomaly	Embryological basis
Duplication	Complete or incomplete; sagittal or coronal
Urachal (patent urachus, sinus, cyst, vesicourachal diverticulum)	Urachus not obliterated at different levels
Congenital diverticulum	Focal detrusor deficiency, often at the ureteric hiatus

C. Hutch diverticulum: definition

- **Paraureteral diverticulum**: mucosa herniates through the **ureteric hiatus**, between the intravesical ureter and the roof of the hiatus, superolateral to the orifice, sparing the trigone. **Hutch** divided them into primary and secondary.

	Primary (congenital)	Secondary (acquired)
Cause	Congenitally deficient detrusor at the hiatus	High pressure: PUV (posterior urethral valves), BOO (bladder outlet obstruction), neurogenic bladder (described in paraplegics), dysfunctional voiding, prior surgery
Bladder	Smooth; single, often intermittent	Trabeculated; one of many

D. Pathophysiology

- No muscle backing → **VUR**; no muscle in the sac → cannot empty → stasis, UTI (urinary tract infection), stones; tumour in adults.
- An enlarging sac damages **Waldeyer's sheath** and pulls the ureter out → UVJ (ureterovesical junction) dysfunction, reflux, later **obstruction** and renal damage.
- Large sacs compress the bladder neck → retention (vicious circle).

E. Evaluation

- **US (ultrasound):** sac, residual urine, hydronephrosis.
- **VCUG (voiding cystourethrogram), gold standard:** oblique and post-void films show size, VUR and whether the ureter enters the sac. Repeat if suspicion persists (sac is dynamic).
- **DMSA (dimercaptosuccinic acid):** scars; **MAG3 (mercaptoacetyltriglycine) renogram** if obstruction.
- **Urodynamics** if secondary; cystoscopy (orifice, neck); in adults enter the sac, cytology.

F. Management

1. Treat the cause first (secondary): valve ablation, BOO surgery, CIC (clean intermittent catheterisation) and antimuscarinics. Remodelling may make excision unnecessary.

2. Primary diverticulum:

Situation	Management
Small, asymptomatic, no VUR	Observe
VUR, ureter beside the sac	As primary VUR (resolves like VUR of the same grade); many excise if it persists. Dx/HA (dextranomer/hyaluronic acid) injection possible
Ureter enters the sac	VUR will not resolve → surgery

Situation	Management
Breakthrough febrile UTI, new scars, large sac with residual, stones, obstruction, retention	**Diverticulectomy + ipsilateral reimplantation**

3. Operative steps (open, laparoscopic or robotic): `Cystoscopy ± ureteric catheter → cystotomy → evert sac or dissect it from its pseudocapsule (0.5 cm mucosal cuff round the neck) → close detrusor defect in two layers (firm backing) → reimplant ureter: Cohen cross-trigonal or Politano–Leadbetter (tunnel 5:1) → catheter ± suprapubic tube, drain`

- Extravesical or combined approaches suit selected cases. In adults, endoscopic outlet surgery is done first.

G. Complications and follow-up

- Ureteric injury, leak, persistent VUR, obstruction, recurrence (syndromic patients).
- US at 6–12 weeks; VCUG for febrile UTI; urodynamics if voiding is abnormal.

Also know: posterolateral multiple diverticula (Ehlers–Danlos, Williams, Menkes); hypoplasia; megacystis.

Diagram to draw: cloacal division by the urorectal septum; Hutch sac beside the ureteric orifice.

> **Key points to remember:**
> - Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
> - Hutch: herniation through the ureteric hiatus → VUR.
> - Ureter entering the sac: VUR will not resolve.
> - Secondary: treat the outlet first; else diverticulectomy + reimplantation.

LAQ 6 Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias)

P3

Asked as:

- Discuss development of urethra. Classify congenital anomalies and discuss management of epispadias — *Practice Set 6 Paper III Q2 (LAQ)*
- Isolated penopubic Epispadias. — *Summer 2022 Paper III Q3f (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)
(isolated penopubic epispadias also answered in 2022-Summer Paper 3, Q3f)

Development of the urethra

1. Division of the cloaca (weeks 5–6)

- The urorectal septum divides the cloaca into the dorsal anorectal canal and the ventral **UGS (urogenital sinus)**.
- Wolffian duct entry is the landmark. Above it, the **vesicourethral canal** forms the bladder and pelvic urethra. Below it, the **definitive UGS** forms the phallic urethra in males, and the urethra and vestibule in females.
- UGS also forms the prostate (buds at 10 weeks) and Cowper and Littre glands; the **utricle** is Müllerian.
- The epithelium of the **entire male urethra is endodermal** (UGS origin).

2. Penile urethra: the "double zipper" (androgen-dependent)

- Indifferent genital tubercle until 8–9 weeks. Testosterone (Leydig cells, 8 weeks) is converted to DHT (dihydrotestosterone) by **type 2 5α-reductase**.
- The **urethral plate** (endoderm) forms in the tubercle.
- **Opening zipper:** the plate canalises proximal to distal into a urethral groove bounded by the urethral folds.
- **Closing zipper:** the folds fuse in the ventral midline, forming the tubular urethra and median raphe.
- **Glanular urethra:** formed by direct canalisation of the plate inside the glans (current endodermal view; Glenister's ectodermal ingrowth theory is outdated).
- **Timetable:** coronal sulcus 11–12 weeks; fusion complete 16 weeks; urethra complete 18 weeks; fetal curvature gone 20 weeks.
- The prepuce forms with the glanular urethra, so hypospadias has a hooded foreskin.
- **Female:** the vestibular plate canalises but the folds do not fuse, giving an open vestibule and labia minora.

Classification of congenital urethral anomalies

Mechanism	Anomaly
Failed ventral fold fusion	**Hypospadias** (1 in 250 males; commonest at shaft–glans junction); congenital urethral fistula
Dorsal defect	**Epispadias** (exstrophy–epispadias complex)

Mechanism	Anomaly
Obstruction	**PUV (posterior urethral valves)**, Young type 1 (95%) and type 3; **anterior urethral valves** with diverticulum; urethral atresia or hypoplasia; meatal stenosis
Duplication	**Urethral duplication**: sagittal; ventral channel is functional; Effmann type I blind, II complete, III with caudal duplication
Dilatation	**Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); prune-belly association
Cystic or remnant	Cowper's syringocele; enlarged utricle (proximal hypospadias); parameatal cyst

Epispadias (including isolated penopubic epispadias): definition, embryology, classification and clinical features

- Part of the **exstrophy–epispadias complex**: a **dorsal urethral defect** (defect of the dorsal urethral wall): the roof is replaced by a broad (dorsal) mucosal strip (urethral plate). Cause (embryology): absent mesodermal reinforcement of the cloacal membrane; the genital tubercles fuse **caudal** to it, giving a dorsal urethral defect.
- **Male (1 in 117,000; incidence of isolated male epispadias about 1 in 117,000), classification by meatus:** glanular, penile, **penopubic** (subsymphyseal) (about 70% of male cases are complete (penopubic), with incontinence). **Female (1 in 484,000), Davis:** vesicular (patulous), subsymphyseal, retrosymphyseal (incontinent).
- **Features (clinical):** penopubic meatus; outlet may admit a finger → **total incontinence**. **Short, broad, upturned penis with dorsal chordee**, like exstrophy; pubic diastasis (less than exstrophy); small bladder from no outlet resistance; **VUR (vesicoureteral/vesicoureteric reflux) 30–40% (male), 30–75% (female)**. Female (Davis retrosymphyseal type): bifid clitoris, depressed mons, incontinence.

Evaluation

- (Renal) ultrasound; VCUG (voiding cystourethrogram); cystoscopy and capacity under anaesthesia.
- **Key decision:** epispadias or exstrophy pathway. Treat as exstrophy (osteotomy and closure) if pubic diastasis is over 3 cm, the intrasymphyseal bar is poor, sphincter is absent or capacity is tiny (tiny bladder).

Management of epispadias

- **Aims:** continence, safe upper tracts, a straight functional penis of good length.
- **Preoperative testosterone** 2 mg/kg intramuscularly (intramuscular), 5 and 2 weeks before urethroplasty.

Stage 1: penile reconstruction and urethroplasty (6–12 months)

- Release chordee: divide suspensory ligaments and scar; free corpora from the inferior pubic rami.
- **Modified Cantwell–Ransley:** outline an 18-mm plate to the tip; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; free the plate from the corpora (distal 1 cm left on the glans); tubularise over 8 Fr; transpose it (the tubularised plate is moved) **ventral to the corpora**, which are rotated medially ± cavernocavernostomy; close glans wings; Z-plasty at the base; stent 10–12 days.
- **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias.

- Outlet resistance rises, so capacity grows.

Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)

- **YDL (Young–Dees–Leadbetter)** bladder neck repair + Marshall–Marchetti–Krantz suspension + **ureteric reimplantation**. Osteotomy usually unnecessary.
- **Good capacity:** single-stage repair possible.

Female epispadias: at 6–12 months excise the urethral roof, close over 10 Fr, unite clitoral halves and labia minora, monsplasty; YDL at 4–5 years.

Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.
- Early fistula up to (about) one-fifth (at the penile base); long-term fistula and stricture about 5% (stricture rare); erections and paternity (usually) preserved.
- Better than exstrophy (bladder never exposed or scarred). Failures (failure): bulking injection (injectables), augmentation, continent stoma.

Diagram to draw: double zipper (canalisation then fusion); penopubic epispadias (dorsal penopubic meatus and plate).

> **Key points to remember:**
> - Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
> - About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
> - Diastasis over 3 cm (>3 cm) → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
> - Continence after YDL: males 47–82%, females 70–87%.

Short answer questions (SAQ)

SAQ 1 Posterior urethral valves: embryology, management and valve bladder

P1 · P2 · P3 **merged from 3**

Asked as: P3 SAQ "Embryology and Management of P.U. valve patient." · P1 SAQ "Embryology of Posterior Urethral Valves." · P2 SAQ "Valve bladder: management"

Introduction

- PUV (posterior urethral valves): the commonest cause of bladder outlet obstruction in boys (1.4–2.2 per 10,000 births).
- Morgagni described them (1769); Hugh Hampton Young named and classified them (1919).

Normal development

- **Weeks 5–6:** the **urorectal septum** divides the cloaca into the anorectal canal (rectum) and the **urogenital sinus**.
- The **Wolffian (mesonephric) ducts** join the cloaca just before division. Their orifices normally migrate to the **verumontanum**, leaving small folds (plicae colliculi).

- The **urogenital membrane** normally disappears completely.
- Fetal urine starts at 10–14 weeks; after 16 weeks amniotic fluid is mainly fetal urine. So obstruction in this period damages bladder, kidneys and lungs.

Embryology of the valves: Young types and their origin

Type	Anatomy	Embryology
I (about 95%)	Leaflets from the verumontanum fuse anteriorly just proximal to the external sphincter	Abnormal anterior / anterolateral insertion of the Wolffian duct ends into the cloacal wall: hypertrophied inferior urethral crest
II	Folds from verumontanum to bladder neck	Not real (Stephens 1955): secondary bladder-neck hypertrophy
III (5–10%)	Annular (ring) diaphragm with a central hole	**Persistent urogenital membrane**

Other theories and genetics

- Hypertrophied mucosal folds (oldest); cloacal remnants; Wolffian–Müllerian maldevelopment.
- **COPUM (congenital obstructing posterior urethral membrane, Dewan):** one oblique membrane, torn by the first catheter into apparent type I leaflets. Plausible but unproven.
- **Cobb's collar (Moormann's ring):** a congenital ring-like narrowing of the proximal bulbar urethra, just distal to the external sphincter; some authors group it with type III valves.
- **Genetics:** mostly sporadic; chromosome 11 duplications; copy number variants (worse renal outcome); TBX5, PTK7.

Diagram to draw: posterior urethra with type I leaflets versus a type III membrane, with keyhole VCUG.

Consequences of obstruction

- Thick bladder, dilated posterior urethra (**keyhole sign**), hydroureteronephrosis, **renal dysplasia**, oligohydramnios, pulmonary hypoplasia.
- Pop-offs (VUR (vesicoureteral reflux), urinoma, ascites) relieve pressure.
- Dysplasia is fixed in utero; ablation cannot reverse it.

Antenatal management

- Bilateral hydroureteronephrosis, thick bladder, **keyhole sign**, oligohydramnios.
- **Vesicoamniotic shunt** helps survival (lungs), not kidneys (PLUTO trial, 2013).

Neonatal management

1. **Drain the bladder:** 5–7 Fr feeding tube (coudé if it curls); minimal balloon inflation.
2. **Stabilise:** ventilation; correct **post-obstructive diuresis**, hyponatraemia, acidosis, hyperkalaemia; antibiotics.
3. **Confirm:** VCUG (voiding cystourethrogram), renal ultrasound, MAG3 (mercaptoacetyltriglycine) at 2–3 months.
4. **Primary endoscopic valve ablation** (preferred): laser, Bugbee or hook at 5 and 7 o'clock ± 12; VCUG within 1 month.
5. **Vesicostomy** if the urethra is too small or drainage fails. **Upper-tract diversion** rarely: no proven renal benefit.

6. **Circumcision:** cuts febrile UTI (urinary tract infection) sharply. - **VUR** (50–80%) is secondary; treat the bladder, not the ureter.

Prognosis

Nadir creatinine (first year)	ESRD (end-stage renal disease) risk
< 0.4 mg/dL	Nil
0.4–0.69	2%
0.7–0.99	27%
> 1.0	Up to 100%

- Lifetime ESRD risk 20–50%.

Valve bladder: definition

- **Valve bladder syndrome** (Mitchell, 1982): upper tracts and renal function worsen after successful PUV ablation, without residual obstruction, because of bladder dysfunction.

Valve bladder: pathophysiology (vicious cycle)

``` Detrusor hypertrophy, collagen, low compliance → poor sensation, rising residual

- polyuria (nephrogenic diabetes insipidus) → bladder never empty → pressure transmitted to dilated upper tracts → they refill the bladder → renal damage ```

- Three processes: **polyuria, poor sensation, residual urine**; together they keep the upper tracts dilated.
- Age pattern: hyperreflexia in infancy, better compliance in childhood, then a large atonic bladder (myogenic failure) after puberty.

**Diagram to draw:** vicious-cycle diagram of the valve bladder.

### Valve bladder: clinical features and evaluation

- Late continence (about 5.5 years), incontinence, UTI, worsening hydronephrosis, rising creatinine. **New incontinence after being dry is a red flag.**
- Uroflowmetry and PVR (post-void residual) at each visit; ultrasound; creatinine; voiding diary.
- **Videourodynamics** when the picture changes and before transplantation.

### Valve bladder: management (escalating ladder)

1. **Behavioural:** adequate fluids, timed and double voiding, biofeedback.
2. **Drugs:** anticholinergic (oxybutynin 0.1 mg/kg twice daily) for a small high-pressure bladder; α-blocker for high residual. Stop oxybutynin if residual rises.
3. **CIC (clean intermittent catheterisation)** if residual rises despite drugs.
4. **Overnight bladder drainage** (catheter for 7–10 hours) for worsening hydronephrosis, UTI or renal function; it gives bladder and kidneys a daily rest.
5. **Mitrofanoff appendicovesicostomy** if urethral CIC is difficult (sensate urethra, high bladder neck).
6. **Augmentation** (rare) for a small, high-pressure bladder despite the above: **ureterocystoplasty** with the dilated ureter is preferred, avoiding bowel (mucus, acidosis, stones). - Reflux is treated through the bladder; reimplantation is rarely needed.
```

Long-term care and transplantation

- **CKD (chronic kidney disease)** care.
- ESRD in 20–50%. Before **transplantation**, optimise the bladder (videourodynamics, CIC or overnight drainage); graft survival is then similar.

> **Key points to remember:**
> - Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
> - Urorectal septum divides the cloaca at weeks 5–6.
> - COPUM: one membrane torn into leaflets by catheterisation.
> - Dysplasia is fixed in utero; ablation cannot reverse it.
> - Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
> - Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
> - Valve bladder: polyuria, poor sensation, residual urine; overnight drainage and CIC break the cycle.
> - Ureteric augmentation rather than bowel; optimise before transplant.

SAQ 2 Ultrasonography in neonates

P1 · P3 **merged from 2**

Asked as: P1 SAQ "USG in neonates" · P3 SAQ "HSG in neonates"

Introduction

- Assumed meaning: ultrasonography (USG) of the neonatal urinary tract, including antenatal hydronephrosis follow-up. "HSG" (hysterosalpingography) in the P3 wording is not done in neonates, so it is taken as USG.
- First-line imaging, used for antenatal **UTD (urinary tract dilation)** (1–3% of pregnancies), febrile UTI (urinary tract infection), mass and poor stream.

Why ultrasound first

- No radiation or sedation; repeatable. Shows **anatomy, not function**.

Timing and technique

- **First scan after 48 hours**: earlier, physiological (relative) dehydration under-reads dilation.
- **Scan early if outlet obstruction (PUV, posterior urethral valves) is suspected or the kidney is solitary**; catheterise in suspected obstruction.
- Higher-frequency probes than in adults; kidneys supine and prone; length against age nomograms; bladder full and after voiding; Doppler.
- Measure **APD (anteroposterior pelvic diameter)** on a transverse mid-kidney view; assess calyces, parenchyma, ureters and bladder wall (thick if over 3 mm when full).

Normal neonatal kidney (pitfalls)

- Cortex **iso- or hyperechoic** to liver.
- **Prominent hypoechoic pyramids** are normal and mimic dilated calyces or cysts (not hydronephrosis).
- Normal postnatal **APD <10 mm**, with no calyceal or ureteric dilation.

Grading

- **SFU (Society for Fetal Urology) grades:** 0 none; 1 pelvis only; 2 major calyces; 3 major and minor calyces; 4 grade 3 plus parenchymal thinning.
- **UTD 2014, prenatal:** A1 (APD 4 to <7 mm before 28 weeks, 7 to <10 mm after) versus A2–3 (larger, or any other abnormality).

Antenatal hydronephrosis follow-up (UTD 2014, postnatal)

UTD risk	Findings	Action
P1 (SFU 1–2)	APD 10 to <15 mm, central calyces	Repeat US at 3–6 months; no VCUG (voiding cystourethrogram) or prophylaxis
P2 (SFU 3)	APD ≥15 mm, peripheral calyces or dilated ureter	Repeat US; VCUG and prophylaxis at discretion / optional (give prophylaxis if ureter >7 mm)
P3 (SFU 4)	Abnormal parenchyma or bladder	Repeat US at 1 month; VCUG, prophylaxis, MAG3 (mercaptoacetyltriglycine) scan

Diagram to draw: SFU grades 0–4 as kidney outlines.

Specific neonatal findings

Condition	Ultrasound finding
PUJO (pelviureteric junction obstruction)	Dilated pelvis, normal ureter; pyramids under 3 mm predict pyeloplasty
VUR (vesicoureteral reflux)	Normal first scan in up to 25%
PUV	Thick bladder, dilated posterior urethra (**keyhole sign**), bilateral hydroureteronephrosis; VCUG before discharge
Duplex anomalies	Ureterocele (thin-walled intravesical cyst), ectopic ureter, megaureter
MCDK (multicystic dysplastic kidney)	Non-communicating cysts of varying size, no central pelvis, scant parenchyma
ARPKD (autosomal recessive polycystic kidney disease)	Bilateral large, bright (echogenic) kidneys
Renal vein thrombosis	Large echogenic kidney, no venous flow; haematuria, mass, thrombocytopenia, hypertension
Adrenal haemorrhage	Avascular, evolving suprarenal mass (neuroblastoma is vascular)
Other masses	Mesoblastic nephroma
Spine	Occult dysraphism before arch ossification (up to 6 months)
DSD (differences of sex development)	Müllerian structures, gonads

- **Also know:** hydrometrocolpos, scrotal masses, twinkling artefact (stones).

Limitations

- Normal US does not exclude VUR. Function needs MAG3 renography (after renal maturation) or DMSA (dimercaptosuccinic acid).

Recent advances

- ceVUS (contrast-enhanced voiding urosonography).

> **Key points to remember:**
> - First postnatal scan after 48 h, except suspected PUV; normal APD <10 mm.
> - Echogenic cortex and prominent pyramids are normal in neonates.
> - UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
> - Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

SAQ 3 Nocturnal enuresis

P1 · P3 **merged from 2**

Asked as: P1 SAQ "Nocturnal enuresis" · P3 SAQ "Enuresis"

Definition (ICCS [International Children's Continence Society])

- Intermittent urinary incontinence **during sleep** in a child **aged 5 years or more**.
- **Primary:** never dry. **Secondary:** wetting after at least **6 months** dry (look for stressors, diabetes, UTI [urinary tract infection], neurological disease).
- **MNE (monosymptomatic):** no daytime LUTS (lower urinary tract symptoms). **NMNE (non-monosymptomatic):** with daytime LUTS.

Epidemiology

- About **5–10% at 7 years**; falls with age, with spontaneous resolution about 15% a year; about 2% of adults; familial.

Pathophysiology: three-way mismatch

1. **Nocturnal polyuria** (poor or reduced night-time ADH [antidiuretic hormone] rise).
2. **Small (reduced) nocturnal bladder capacity** or DO (detrusor overactivity).
3. **Failure to arouse** from sleep. - **Why sleep apnoea matters:** obstructed breathing makes arousal difficult and releases atrial natriuretic peptide, which raises urine output. - Comorbidities: constipation, sleep-disordered breathing, ADHD (attention deficit hyperactivity disorder).

Diagram to draw: triad: nocturnal polyuria, small bladder capacity, arousal failure.

Evaluation

- **History:** day symptoms, bowels, snoring, fluids, family history.
- **Examination:** spine, genitalia, abdomen for faecal loading; urinalysis.
- **Bladder diary:** daytime voided volumes against EBC (expected bladder capacity) = (age + 2) × 30 mL in the chapter (ICCS uses [age + 1] × 30 mL); night urine output by weighing nappies.
- Uroflowmetry, PVR (post-void residual) and ultrasound only for NMNE or before adding anticholinergics. No invasive tests in MNE.

Management: general

- **Treat daytime LUTS and constipation first** (in NMNE).
- Education removes blame; standard urotherapy. Waking, fluid restriction alone and punishment do not work.

Management: first line

	Alarm	Desmopressin
Target	Arousal (conditioning)	Night urine output
Use	Nightly for months; parent wakes child; continue to 14 dry nights	0.2–0.4 mg tablet or 120–240 µg melt, 1 hour before bed
Response	50–70%; durable	Rule of thirds; immediate but relapses off the drug
Caution	Needs motivated family	Restrict / limit evening fluids (hyponatraemia)

- Overlearning (extra evening fluid once dry) reduces alarm relapse.

Management: failure and resistant cases

- **Failure:** switch or combine.
- **Second line:** add an anticholinergic (oxybutynin) after checking PVR and bowels.
- **Resistant:** imipramine (cardiotoxic; ECG [electrocardiogram]), off-label β3-agonists; treat sleep apnoea (adenotonsillectomy).

Adult enuresis

- New enuresis in an adult may signal high-pressure chronic retention: catheterise and treat the obstruction.

> **Key points to remember:**
> - ≥5 years; secondary = after 6 months dry.
> - Alarm (durable) and desmopressin (rule of thirds) are first line.
> - Treat daytime symptoms and constipation first.

SAQ 4 Paediatric neurogenic bladder (myelodysplasia): assessment, management and bladder myoplasty

P1 · P3 **merged from 2**

Asked as: P3 SAQ "Neurogenic bladder: prenatal/neonatal assessment and bladder myoplasty" · P1 SAQ "Neurogenic bladder in myelodysplasia"

Introduction

- Myelodysplasia (NTD [neural tube defect], mainly **myelomeningocele**, an open NTD) is the commonest cause of neurogenic LUT (lower urinary tract) dysfunction in children; other causes are occult dysraphism, sacral agenesis and anorectal malformation.
- Lesions: lumbosacral 47%, lumbar 26%, sacral 20%.
- **Bony or motor level does not predict bladder function**, and the lesion can progress (re-tethering), so every newborn needs urodynamic testing.
- About 10% of deaths in SB (spina bifida) are urological. Folic acid 400 µg daily before conception prevents 50–70% of NTDs.

Patterns of dysfunction

- Detrusor: overactive (NDO, neurogenic detrusor overactivity) or underactive. Outlet: competent, incompetent or dyssynergic.

- Newborns: most have detrusor contractions; over a third are acontractile. Needle EMG (electromyography): intact sacral arc about 40%, partial denervation 25%, complete 35%.

Prenatal assessment

- **Maternal serum AFP (alpha-fetoprotein)** is raised in open defects; amniotic AFP and acetylcholinesterase confirm.
- **Ultrasound:** spinal defect; "lemon" and "banana" signs of Chiari II; ventriculomegaly; hydronephrosis, thick bladder. Fetal MRI (magnetic resonance imaging) for level.
- Multidisciplinary counselling at a fetal centre.
- **MOMS (Management of Myelomeningocele Study) 2011:** closure **before 26 weeks** reduced shunt need (about 40% versus 82%) and improved motor and cognitive scores, but caused prematurity (34 versus 37 weeks) and uterine thinning. **No proven bladder benefit**: CIC (clean intermittent catheterisation) and reconstruction rates unchanged.

Neonatal assessment (baseline; EAU [European Association of Urology] 2022)

Test	Timing / purpose
Examination	Spine, legs, anal tone, perineal sensation, bladder
Renal and bladder ultrasound	Before discharge, then 6-monthly for 2 years (serially); hydronephrosis, wall thickness
Residual urine	Start CIC if not emptying
Creatinine	Within 3 months (cystatin C later; better with low muscle mass)
VCUG (voiding cystourethrogram) or video-UDS	Reflux; dilating VUR (vesicoureteric reflux) in about 7%
UDS (urodynamics) with EMG	Within 3 months, once stable after closure
DMSA (dimercaptosuccinic acid)	Baseline scarring

Urodynamic risk (McGuire 1981; UMPIRE [Urologic and Renal Protocol for the Newborn and Young Child with Spina Bifida])

Risk	DLPP (detrusor leak point pressure) or end-filling pressure
High (hostile)	40 cm H_2O or more
Intermediate	25–39 cm H_2O, NDO or poor compliance
Low (safe)	Under 25 cm H_2O

- Poor compliance and DSD (detrusor–sphincter dyssynergia) also predict upper-tract damage.
- Repeat studies: the lesion can change (re-tethering).

Diagram to draw: neonatal UDS trace with DLPP over 40 cm H_2O.

Management: kidneys first, then continence

- **Early CIC (Lapides 1971) plus oxybutynin** for high-risk (hostile) bladders: hydronephrosis 0–15% versus 18–80% expectant; ESRD (end-stage renal disease) falls from about 30% to under 2% (P1: 0–1.6%).
- Other drugs: solifenacin, mirabegron; **botulinum toxin A** about 10 U/kg (maximum 360 U), continence 65–87%.

- Surgery: vesicostomy (bridge); **augmentation cystoplasty**; Mitrofanoff channel; bladder-neck sling, reconstruction or artificial sphincter.
- VUR: CIC and antimuscarinic first; avoid Credé; reimplant during augmentation if grade ≥III.
- **Also know:** latex allergy; bowel programme; sexual dysfunction (about 75% of men have ED [erectile dysfunction]; sildenafil helps); precocious puberty.

Bladder myoplasty: definition

- The P3 wording "bladder myeloplsty" is assumed to mean "bladder myoplasty" (a muscle procedure on the bladder wall).
- **Detrusor myoplasty:** wrapping a **vascularised, innervated skeletal muscle flap** round the bladder so that voluntary contraction of that muscle empties an **acontractile bladder**.

Bladder myoplasty: indications

- **Acontractile or underactive detrusor** with catheter dependence: lower motor neuron lesions (spina bifida, cauda equina injury), myogenic failure.
- Needs good bladder capacity and compliance, and a competent but not obstructing outlet.

Latissimus dorsi detrusor myoplasty (LDDM; Stenzl)

- **Free latissimus dorsi flap:** artery and vein joined to the **inferior epigastric vessels**; motor nerve joined to an **intercostal nerve**.
- Muscle wrapped **spirally over more than three-quarters of the bladder** and anchored to pelvic floor fascia and ligaments.
- The patient voids by contracting the transferred muscle (abdominal effort).
- **Results:** most voided with low residual; bladder contractility index rose to the strong range.
- **Complications** in about one-third: thromboembolism, abscess, wound infection; none long term.
- **Rectus abdominis detrusor myoplasty:** simpler, less morbid; small spinal cord injury series with spontaneous voiding.
- **Status: investigational.** Standard care stays **clean intermittent catheterisation**.

Diagram to draw: latissimus flap spiralled round the bladder with vascular and nerve anastomoses.

Other "myoplasty": detrusor myotomy/myectomy (autoaugmentation)

- Purpose is the opposite: **increase capacity and compliance** in a small, poorly compliant or overactive bladder.
- **Technique (Cartwright and Snow, 1989):** strip and excise the detrusor over the dome, leave the mucosa intact, and anchor the detrusor edges (psoas hitch) so the mucosa bulges as a **wide-mouthed diverticulum**.
- **Pros:** urothelial lining, no mucus, no metabolic change, no bowel.
- **Cons:** seldom adds significant capacity; compliance gain may not last; **not recommended for adult detrusor overactivity**.
- **Also know:** seromuscular enterocystoplasty covers the bulge with demucosalised sigmoid.

Alternatives

- **Also know:** sacral neuromodulation, sacral anterior root stimulation, reduction cystoplasty (abandoned), muscle-derived stem cells (research).

Key points to remember:

- Level does not predict bladder: UDS within 3 months.
- DLPP 40 cm H_2O or more = hostile (McGuire 1981); early CIC + antimuscarinic protects kidneys.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

SAQ 5 Anterior urethral valve

P2

Type: SAQ (10 marks)
Source: Practice Set 17 Paper 2, Q3c

Definition and epidemiology

- AUV (anterior urethral valve): a congenital obstructing fold or cusp in the anterior urethra (bulbar, penoscrotal or penile). It is the **commonest congenital obstructive lesion of the anterior urethra** but **25–30 times less common than PUV (posterior urethral valves)**.
- Often linked to an **anterior urethral diverticulum**: the distal lip of the diverticulum fills and acts as a valve, or a semilunar fold hangs from the roof.

Embryology (unclear)

- Hypoplastic corpus spongiosum over the segment suggests a place on the **hypospadias spectrum**, or faulty union of urethral mucosa and fossa navicularis epithelium; a ruptured dilated Cowper (bulbourethral) gland duct is also proposed.

Clinical features (vary with severity and age)

- Antenatal hydronephrosis, LUTO (lower urinary tract obstruction) in severe cases.
- **Ventral penile bulge** during voiding, with urine expressed by compressing the shaft; post-void dribbling, weak stream, incontinence.
- Palpable bladder, UTI (urinary tract infection), renal insufficiency.

Evaluation

- **VCUG (voiding cystourethrogram):** dilated anterior urethra proximal to the valve, diverticulum, trabeculated bladder, bladder diverticula, often VUR (vesicoureteric reflux).
- Ultrasound, creatinine; urethroscopy confirms.
- **Differential:** Cowper syringocele, megalourethra, acquired diverticulum, stricture.

Management

- **Premature or small infant:** vesicostomy until the urethra takes a cystoscope.
- **Most cases: endoscopic valve ablation** (Bugbee electrode or laser), usually definitive.
- **Large diverticulum:** open excision and urethroplasty over a catheter.
- Then manage VUR, infection and bladder dysfunction.

AUV versus PUV

Feature	AUV	PUV
Frequency	25–30 times rarer	Commonest infant outlet obstruction
Site	Bulbar, penoscrotal, penile	From the verumontanum
Sign	Penile bulge	Weak stream, palpable bladder
Bladder dysfunction	Up to 80%	Near universal
Renal outcome	78% normal function	ESRD (end-stage renal disease) 20–50%

Prognosis

- Depends on pre-treatment creatinine and GFR (glomerular filtration rate). Azotaemia, VUR and UTI together raise the risk of poor renal outcome 25-fold.

Diagram to draw: VCUG sketch: dilated bulbar urethra with diverticulum ending at a valve lip.

> **Key points to remember:**
> - 25–30 times rarer than PUV; diverticulum common.
> - Penile bulge on voiding; diagnosis by VCUG.
> - Endoscopic ablation usually definitive; 78% keep normal renal function.

SAQ 6 CSIC.

P3

Type: SAQ (10 marks)
Source: 2018-Summer Paper 3, Q3e

*Taken to mean **CISC – clean intermittent self-catheterisation**.*

Definition and principle

- Regular, **clean (not sterile)** passage of a catheter by the patient (or carer) to empty the bladder, then removal.
- **Lapides (1971–72)**: infection comes from **stasis and overdistension** (poor bladder-wall blood flow), not from bacteria alone. Regular complete emptying protects the bladder wall and upper tracts, even though bacteriuria develops.
- Gold standard for chronic emptying failure.

Indications

- **Neurogenic bladder:** spina bifida, spinal cord injury, multiple sclerosis, cauda equina, diabetic cystopathy.
- **Underactive or acontractile detrusor**; chronic retention; post-operative retention.
- **Detrusor–sphincter dyssynergia**: CISC bypasses the obstructed sphincter.
- After **augmentation or continent diversion**; **valve bladder**; high residual after botulinum toxin.
- **Contraindications:** poor hand function or cognition without a carer, urethral false passage or stricture, unwillingness.

Technique

- Wash hands → clean meatus with water → lubricated catheter (or pre-lubricated **hydrophilic** catheter) → pass until urine flows → drain fully → withdraw slowly.
- Sizes: adults about 12–14 Fr; children by age (8–10 Fr).
- **Frequency:** 4–6 times a day, keeping each volume **below about 400–500 mL** (child: below expected bladder capacity).
- Combine with **antimuscarinics** for overactivity; bladder diary.

Catheters

Type	Note
Plain PVC (polyvinyl chloride), reusable after washing	Cheap; common in India
Hydrophilic-coated single-use	Less urethral trauma; longer time to first UTI (urinary tract infection)
Compact or closed-system	Convenience, outdoor use

Children

- Start **early (infancy)** in spina bifida with hostile bladder; families adapt best; child involved from 3–5 years.
- **Mitrofanoff appendicovesicostomy** (1980) if urethral catheterisation is difficult (wheelchair, sensate urethra, girls).

Complications

- **Asymptomatic bacteriuria** (common): **do not treat or give prophylaxis**; treat only symptomatic UTI.
- Urethral trauma, bleeding, **false passage**, stricture, epididymitis, bladder stones (hair, mucus), rarely lost catheter.

Outcome

- Less pyelonephritis, epididymo-orchitis and urosepsis than indwelling catheters; preserves the upper tracts; with drugs, continence in many.

Diagram to draw: neurogenic bladder management ladder: CISC + antimuscarinic → botulinum toxin → augmentation + channel.

> **Key points to remember:**
> - Lapides: overdistension, not bacteria, causes infection; clean is enough.
> - 4–6 times daily, under about 500 mL each.
> - Do not treat asymptomatic bacteriuria.
> - Mitrofanoff when urethral access is hard.

SAQ 7 Discuss diagnostics and operative pediatric laparoscopy in urology

P3

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 3, Q3a

Evolution

- Diagnostic (non-palpable testis, 1976) → nephrectomy (1994) → pyeloplasty (1995) → robotic (2002).

Diagnostic laparoscopy

- **Non-palpable testis** (procedure of choice, after examination under anaesthesia): note vessels, vas, testis and internal ring.

Finding	Meaning
Blind-ending vessels and vas, closed ring	Vanishing testis: stop
Vessels and vas entering ring	Explore groin; excise remnant
Intra-abdominal testis	Orchidopexy (primary or Fowler-Stephens)
- **DSD (differences of sex development):** gonadal inspection and biopsy, Müllerian structures.	

Operative laparoscopy

- **Testis:** primary orchidopexy (>90% success); Fowler-Stephens one- or two-stage; gonadectomy in DSD.
- **Kidney:** nephrectomy, heminephrectomy, **pyeloplasty** (commonest MIS (minimally invasive surgery) procedure).
- **Ureter and bladder:** extravesical or vesicoscopic (cross-trigonal) reimplantation, Mitrofanoff, augmentation, bladder neck reconstruction, utricle and urogenital sinus.
- **Also know:** RPLND (retroperitoneal lymph node dissection) for paratesticular rhabdomyosarcoma.

Technical principles

- **Access:** open Hasson or Veress at umbilicus (Palmer's point if scarred); equal safety.
- 3–5 mm ports; robots use 8 mm. Transperitoneal (most space), retroperitoneal (spares peritoneum, useful in peritoneal dialysis), vesicoscopic.
- **Pressures:** CO_2 absorption, reduced venous return, raised airway pressure; reversible oliguria (no fluid bolus).

Age	0–1 month	2–12 months	1–2 years	2–8 years
Max IAP (intra-abdominal pressure)	≤6 mmHg	≤8	≤10	≤12
- **Positioning:** modified flank for kidney; supine steep Trendelenburg for pelvis. Infants: minimise inhalational anaesthetic, rectal tube to reduce bowel distension.				
- **Only absolute contraindication:** inability to tolerate pneumoperitoneum.				

Complications

- Right common iliac vessel injury (behind umbilicus; commonest major), inferior epigastric (commonest minor), bowel injury, subcutaneous emphysema, **port-site hernia 0.5–3%** (close fascia).

Pros and cons

- Shorter stay, better cosmesis, deep pelvic access; pain benefit unproven. Robot: cost, learning curve; infant working space only about 1 L.

Diagram to draw: laparoscopic view of internal ring with blind-ending vessels.

> **Key points to remember:**
> - Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
> - Pressure ceiling 6–12 mmHg by age.
> - Right common iliac at Veress entry; close port fascia.

SAQ 8 Vesicostomy.

P4

Type: SAQ (10 marks)
Source: 2016-Summer Paper 4, Q3f

Definition

- **Cutaneous vesicostomy**: the bladder dome is brought to the lower abdominal skin as an incontinent stoma draining into the nappy. It "vents" the bladder, protecting the upper tracts; usually a **temporary** bridge to reconstruction.

Indications

- **PUV (posterior urethral valves)**: infant too small for a resectoscope; upper tract deterioration despite drainage or ablation.
- **Prune-belly syndrome**: acute renal failure, urosepsis, urethral atresia. A larger stoma is needed because stenosis is common.
- **Neurogenic bladder**: high pressure despite CIC (clean intermittent catheterisation) and drugs, or CIC impossible; worsening hydronephrosis; febrile UTI (urinary tract infection) with high-grade reflux.
- **Also know:** urethral atresia; MMIHS (megacystis-microcolon-intestinal hypoperistalsis syndrome); urogenital sinus anomalies.

Blocksom technique (Duckett modification)

1. Transverse 2 cm incision midway between umbilicus and pubis. (In prune-belly, excise any urachal diverticulum.)
2. Incise the rectus sheath; separate the recti.
3. Fill the bladder; mobilise the **dome** up to the urachus, sweeping peritoneum off.
4. Bring the **dome** (not the anterior wall) to the wound. This reduces prolapse of the posterior wall.
5. Anchor detrusor to rectus fascia; stoma should take about 24 Fr.
6. Open the dome; suture mucosa to skin. No catheter needed.

Advantages

- Simple; no catheter or appliance.
- The bladder still fills and empties through the stoma at low leak pressure, so compliance is **not** lost; it remains usable for reconstruction or transplantation.

Complications

- **Stomal stenosis**, mucosal **prolapse**, peristomal dermatitis, stones, UTI; leakage harder to manage as the child grows.

Closure

- When the cause is treated (valve ablation, urethral growth) or at definitive reconstruction; check bladder function first (cystogram or urodynamics).
- Excise the stoma, free the bladder from fascia, close in two layers, catheter drainage for a few days.

Alternatives

Option	Use
Cutaneous ureterostomy or pyelostomy	Rare; persistent upper tract obstruction
Ileovesicostomy	Unable to catheterise; 10–15 cm ileum
Appendicovesicostomy (Mitrofanoff, 1980)	Continent catheterisable channel

Diagram to draw: sagittal view of bladder dome sutured to fascia and skin.

> **Key points to remember:**
> - Blocksom: dome-based stoma midway between umbilicus and pubis.
> - Bladder keeps cycling; compliance preserved.
> - Stenosis and prolapse are the main complications.

SAQ 9 Management of Defecation Disorders resulting into LUTS

P4

Type: SAQ (10 marks)
Source: 2025-Winter Paper 4, Q3f

Assumed meaning: bladder and bowel dysfunction, where constipation or defecation disorders cause LUTS (lower urinary tract symptoms).

Why the bowel causes LUTS

- **Shared space:** a loaded rectum compresses the bladder, reducing capacity and obstructing outflow.
- **Shared nerves:** bladder and bowel afferents converge on the same spinal segments; cross-organ sensitisation causes urgency and detrusor overactivity.
- **Shared muscle:** a pelvic floor that contracts during defecation (**dyssynergic defecation**) also contracts during voiding (**dysfunctional voiding**) → high PVR (post-void residual), UTI (urinary tract infection), reflux.
- Result: urgency, daytime incontinence, enuresis, retention, recurrent UTI.

Evaluation

- Stool frequency, pain, **Bristol chart**; 3-day bladder and bowel diary.
- **Rome IV (child):** at least 2, weekly for 1 month: ≤2 defecations/week; faecal incontinence weekly; retentive posturing; painful or hard stools; large rectal mass; large-diameter stools.
- **Rome IV (adult), for 3 months:** ≥2 of straining, hard stools, incomplete evacuation, blockage, manual manoeuvres (each in >25% of defecations), <3 stools/week.

- Examine abdomen, spine and perineum; exclude organic causes (Hirschsprung disease, hypothyroidism, neurogenic).
- **Also know:** rectal diameter ≥30 mm on ultrasound; no routine X-ray; manometry if refractory.

Management

Rule: treat the bowel first (EAU, European Association of Urology/ESPU, European Society for Paediatric Urology).

1. **Disimpaction:** oral PEG (polyethylene glycol) 3350, 1–1.5 g/kg/day for 3–6 days; enemas if needed.
2. **Maintenance:** PEG 0.4 g/kg/day (range 0.2–0.8), at least 2 months; lactulose alternative; stimulants (senna) for expulsion.
3. **Urotherapy:** toileting after meals (gastrocolic reflex); footstool posture; normal fluid and fibre.
4. **Biofeedback, pelvic floor physiotherapy** for dyssynergia.
5. **Refractory: transanal irrigation** (water via rectal cone; a quarter to a third stop); **ACE (antegrade continence enema, Malone)**; botulinum toxin to the sphincter; sacral or tibial neuromodulation. - **Adults:** laxatives, prucalopride or linaclotide, biofeedback; rectocele repair; irrigation in neurogenic bowel. - **Caution:** anticholinergics worsen constipation; LUTS return.

Diagram to draw: pelvic cross-section showing loaded rectum compressing bladder.

> **Key points to remember:**
> - Treat the bowel first.
> - Rome IV: ≥2 criteria for 1 month (child).
> - PEG disimpaction 1–1.5 g/kg/day for 3–6 days.

Rapid Revision Sheet · Section 5

Bladder embryology and exstrophy-epispadias complex (LAQ)

- Bud from Wolffian duct; position decides VUR or ectopia.
- Urorectal septum weeks 5–6 (P2: weeks 4–6); mesonephric duct entry is the landmark.
- Vesicourethral canal forms the bladder; urachus from allantois.
- Trigone: common excretory duct absorption; mostly bladder muscle (current view).
- Urachus forms at months 4–5; urine from week 10; amniotic fluid mainly urine after 16 weeks.
- Marshall–Muecke: overdeveloped cloacal membrane, failed mesodermal reinforcement; rupture timing decides variant.
- CBE: external rotation 12°/18°, anterior pubis 30% short, diastasis about 4.8 cm; VUR in 100%.
- Cloacal exstrophy: spinal anomalies 85–100%; osteotomy always.
- MSRE: closure at birth, epispadias repair 6–12 months, YDL / BNR at 5–7 years at ≥100 mL.
- Osteotomy if closure after 72 h or diastasis >4 cm.
- CPRE: closure + penile disassembly in newborn. Risks: glans loss, hypospadias, VUR.
- Capacity ≥100 mL predicts continence; template size and first closure success predict continence.

Prune belly syndrome (LAQ)

- Triad; 95% male; dysplasia and lungs decide survival.
- Ureter is a failed pump (collagen); VUR about 75%; keep the proximal ureter.
- Woodard I–III; vesicostomy for drainage; orchidopexy at about 6 months.
- Nadir creatinine <0.7 mg/dL is favourable; up to 30% reach renal failure.

Voiding dysfunction in children: neurophysiology and differential diagnosis of frequency/urgency (LAQ)

- S2–S4 pelvic (voiding), T10–L2 hypogastric (storage), Onuf/pudendal (EUS).
- PMC = switch; medial prefrontal cortex holds it off.
- EBC = (age + 2) × 30 mL; staccato = dysfunctional voiding.
- DLPP ≥40 cmH2O = hostile; treat bowel first.
- Always ask about bowels; treat constipation first.
- Rule out UTI, PUV, meatal stenosis and occult dysraphism.
- Extraordinary daytime frequency: no nocturia, normal tests, self-limiting.

Antenatal hydronephrosis, fetal uropathy and prenatal scan anomalies (LAQ)

- UTD: 1–3% of pregnancies; second commonest anomaly after cardiac.
- Anomaly scan 18–20 weeks; after 16 weeks amniotic fluid = fetal urine.
- ANH (antenatal hydronephrosis) = UTD; APD normal <4 mm before 28 weeks, <7 mm after, <10 mm postnatally.
- UTD A2–3 = APRPD ≥7 mm or any abnormal feature.
- Transient dilation 50–70% (commonest cause); UPJ obstruction commonest surgical cause; VUR is not predicted by grade.
- First postnatal US after 48 hours; A1: one scan after 32 weeks; A2–3 scan every 4 weeks.

- Risk-based care: P1 no prophylaxis or VCUG; prophylaxis if ureter >7 mm or P3; P3 prophylaxis, VCUG, MAG3.
- Keyhole + oligohydramnios = PUV; absent bladder = exstrophy or agenesis.
- ARPKD large bright kidneys; MCDK non-communicating cysts.
- Favourable urine: Na <100, Cl <90, osmolality <200, β2-microglobulin <6.
- PLUTO: survival up, renal function unchanged.

Embryological development of urinary bladder and management of Hutch Diverticulum? (LAQ)

- Urorectal septum weeks 4–6; vesicourethral canal → bladder; urachus at months 4–5.
- Hutch: herniation through the ureteric hiatus → VUR.
- Ureter entering the sac: VUR will not resolve.
- Secondary: treat the outlet first; else diverticulectomy + reimplantation.

Development of the urethra, congenital urethral anomalies and management of epispadias (including isolated penopubic epispadias) (LAQ)

- Whole male urethra is endodermal UGS; fold fusion complete at 16, urethra at 18 weeks.
- About 70% of male epispadias are penopubic and incontinent; VUR 30–40%.
- Diastasis over 3 cm (>3 cm) → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL capacity.
- Continence after YDL: males 47–82%, females 70–87%.

Posterior urethral valves: embryology, management and valve bladder (SAQ)

- Type I (95%): abnormal Wolffian duct insertion; type III: persistent urogenital membrane; type II not real.
- Urorectal septum divides the cloaca at weeks 5–6.
- COPUM: one membrane torn into leaflets by catheterisation.
- Dysplasia is fixed in utero; ablation cannot reverse it.
- Catheter, stabilise, then primary valve ablation; vesicostomy if not possible.
- Nadir creatinine below 0.7 mg/dL predicts good renal outcome.
- Valve bladder: polyuria, poor sensation, residual urine.
- Valve bladder: overnight drainage and CIC; overnight drainage breaks the cycle.
- Ureteric augmentation rather than bowel; optimise before transplant.

Ultrasonography in neonates (SAQ)

- First postnatal scan after 48 h, except suspected PUV; normal APD <10 mm.
- Echogenic cortex and prominent pyramids are normal in neonates.
- UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
- Keyhole sign + thick bladder = PUV; spinal US only up to 6 months.

Nocturnal enuresis (SAQ)

- ≥5 years; secondary = after 6 months dry.
- Alarm (durable) and desmopressin (rule of thirds) are first line.
- Treat daytime symptoms and constipation first.

Paediatric neurogenic bladder (myelodysplasia): assessment, management and bladder myoplasty (SAQ)

- Level does not predict bladder: UDS within 3 months.
- DLPP 40 cm H_2O or more = hostile (McGuire 1981); early CIC + antimuscarinic.
- Early CIC plus antimuscarinic protects kidneys.
- MOMS: fewer shunts, no bladder benefit.
- LDDM: free latissimus to inferior epigastric vessels and intercostal nerve; for acontractile bladder.
- Investigational; clean intermittent catheterisation remains standard.
- Autoaugmentation: detrusor excised, mucosa bulges; little lasting gain.

Anterior urethral valve (SAQ)

- 25–30 times rarer than PUV; diverticulum common.
- Penile bulge on voiding; diagnosis by VCUG.
- Endoscopic ablation usually definitive; 78% keep normal renal function. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

CSIC. (SAQ)

- Lapides: overdistension, not bacteria, causes infection; clean is enough.
- 4–6 times daily, under about 500 mL each.
- Do not treat asymptomatic bacteriuria.
- Mitrofanoff when urethral access is hard.

Discuss diagnostics and operative pediatric laparoscopy in urology (SAQ)

- Diagnostic laparoscopy is the procedure of choice for non-palpable testis.
- Pressure ceiling 6–12 mmHg by age.
- Right common iliac at Veress entry; close port fascia. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Vesicostomy. (SAQ)

- Blocksom: dome-based stoma midway between umbilicus and pubis.
- Bladder keeps cycling; compliance preserved.
- Stenosis and prolapse are the main complications.

Management of Defecation Disorders resulting into LUTS (SAQ)

- Treat the bowel first.
- Rome IV: ≥2 criteria for 1 month (child).
- PEG disimpaction 1–1.5 g/kg/day for 3–6 days. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Section 6 · Female Urology

Long answer questions (LAQ)

LAQ 1 Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair

P1 · P2 **merged from 3**

Asked as: P1 LAQ "Female pelvic floor anatomy: prolapse, SUI surgery and fertility-sparing treatment of procidentia" · P1 SAQ "Describe clinical presentation of pelic floor prolapse. diagnostic evaluation and brief discussion of various surgical options." · P2 SAQ "Vaginal prolapse repair."

Introduction and definitions

- Assumed meanings: "procedentia" is read as procidentia (complete uterine prolapse); "fertility sparing" / "fertility preserving" means uterus-preserving treatment for a woman who wants future pregnancy; "pelic floor prolapse" and "vaginal prolapse repair" mean pelvic organ prolapse (POP) in women and the surgical repair of vaginal wall and vault prolapse.
- **SUI (stress urinary incontinence):** involuntary leakage on effort, coughing or sneezing.
- **POP (pelvic organ prolapse):** descent of the anterior wall, posterior wall, uterus/cervix or vault. Anterior is commoner than posterior, which is commoner than apical. It is a disease only if symptomatic: a quality-of-life condition.
- Pelvic organs are supported by the **levator ani**, the **endopelvic fascia and its ligaments** and the **perineal membrane and perineal body**, anchored to the **bony pelvis**. Failure causes POP and SUI. Every SUI operation uses one of these anchors.

Bony pelvis and surgical anchors

- **Bony landmarks** and dangers: **sacrospinous ligament** (pudendal bundle), **Cooper's ligament** (bleeding), **sacral promontory** (left common iliac vein, right ureter).
- **Sacrospinous ligament (SSL)**, 7–8 cm, ischial spine to sacrum: the pudendal nerve and vessels pass behind the spine, the sacral plexus just posterior.
- **Sacral promontory:** left common iliac vein and right ureter each about 2.7 cm away.

Pelvic floor muscles: pelvic diaphragm (levator ani + coccygeus)

Muscle	Attachment	Function
Puborectalis	Back of pubis, loops (as a sling) behind the anorectal junction	Anorectal angle; anorectal sling
Pubococcygeus (pubovaginal part)	Posterior pubis and arcus tendineus to viscera (vagina, perineal body, anus) and anococcygeal raphe	Slings urethra and vagina; closes hiatus, lifts urethra
Iliococcygeus	ATLA (arcus tendineus levator ani), ischial spine to raphe, coccyx	Horizontal **levator plate**
Coccygeus (not levator; separate)	Ischial spine to sacrum, coccyx	Lies on sacrospinous ligament

- Levator tone (tonic contraction) closes the **urogenital hiatus**, through which all POP passes; the uterus and upper vagina rest on the levator plate.
- Levator is palpable through the lateral vaginal walls for PFMT (pelvic floor muscle training) assessment.
- **Muscle fails first, fascia second:** once injury (birth injury, avulsion) opens the hiatus, ligaments carry the load and fail; the fascia then carries the whole load and stretches.
- Levator defects in up to **20% of primiparous women**; avulsion predicts advanced prolapse.

Endopelvic fascia and ligaments

- **Cardinal ligament** (side wall to cervix; carries the uterine artery) and **uterosacral ligament** (sacrum to cervix): the **cardinal–uterosacral complex** suspends the apex.
- The ureter is 4 cm from the uterosacral ligament near the sacrum but 0.9 cm near the cervix.
- **Pubocervical (pubovesical) fascia** attaches laterally to the **ATFP (arcus tendineus fasciae pelvis)**, the "white line" from the lower pubis to just above the ischial spine.
- **Rectovaginal fascia** supports the posterior wall.
- **Pubourethral ligaments** suspend the urethra from the pubis and anchor the mid-urethra; **urethropelvic ligaments** anchor it to the ATFP.

Perineal membrane, perineal body and urethra

- The perineal membrane spans the pubic arch and is pierced by the urethra and vagina; above it lies the striated urogenital sphincter. The perineal body anchors the distal vagina.
- **Striated urogenital sphincter:** sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**.
- Urethra 4–5 cm; the continence zone is the **proximal and mid-urethra**. The **oestrogen-sensitive vascular submucosa** gives up to about 30% of closure force.

DeLancey's levels of support (1992)

Level	Region	Structures	Loss produces
I – suspension	Cervix/upper third	Cardinal–uterosacral complex	**Uterine or vault prolapse**, enterocele
II – attachment	Middle third	Paravaginal attachment to ATFP, levator fascia	Cystocele (central or paravaginal), rectocele; anterior or posterior wall prolapse
III – fusion	Distal third	Urethra, perineal membrane, perineal body	**Urethral hypermobility**, perineal descent, distal rectocele

- **Central cystocele:** midline weakness, rugae lost. **Paravaginal cystocele:** ATFP detachment (Richardson 1981), rugae preserved.
- Level I is the keystone: always correct the apex.

Diagram to draw: sagittal female pelvis with levator plate and DeLancey levels I–III, with sacrospinous and sacrocolpopexy fixation points; hammock cross-section.

Innervation

- **Pudendal nerve (S2–S4):** sphincters and perineal muscles; runs through Alcock's canal near the ischial spine; stretched in delivery.

- **Nerve to levator ani (S3–S5):** runs on the muscle's pelvic surface, separate from the pudendal; neuropathy in about a quarter of primiparous women.
- Autonomic pelvic splanchnic and hypogastric nerves: injured in radical pelvic surgery.

Theories of continence and the operations they produced

Theory	Mechanism	Surgical answer
Pressure transmission (Enhorning 1961)	Hypermobile urethra drops out of the abdominal pressure zone	Restore position: Burch, MMK (Marshall–Marchetti–Krantz)
Hammock (DeLancey 1994)	Urethra lies on anterior vaginal wall and endopelvic fascia attached to ATFP and levator; abdominal pressure compresses it against this firm backing; repositioning not essential	Restore backing
Integral theory (Petros & Ulmsten 1990)	Pubourethral ligament laxity at the **mid-urethra**	Tension-free mid-urethral sling
Intrinsic sphincter deficiency (ISD) (McGuire 1978)	Urethra fails to close	Sling, bulking, AUS (artificial urinary sphincter)

- ISD: VLPP (Valsalva leak point pressure) <60 cmH_2O or maximal urethral closure pressure <20 cmH_2O. Most SUI includes some ISD. So slings work in hypermobility, but Burch does poorly in severe ISD with a fixed urethra.

Why prolapse occurs: risk factors (Bump & Norton)

- Muscle fails first (birth trauma, levator avulsion, denervation). The hiatus widens and the ligaments carry the whole load and stretch.
- **Predisposing:** genetics, collagen disorders (Ehlers–Danlos, Marfan).
- **Inciting: vaginal delivery** (strongest; forceps worse), parity, hysterectomy.
- **Promoting:** obesity, constipation, cough.
- **Decompensating:** ageing.

Clinical presentation of prolapse

- **Vaginal bulge** (most specific), pressure, heaviness, worse by evening.
- **Splinting/digitation** to void or defecate (high-grade POP).
- Urinary: voiding difficulty (urethral kinking), OAB (overactive bladder); SUI in about 40%, which may be **occult**.
- Bowel (rectocele): obstructed defecation, stool trapping.
- Sexual dysfunction, discharge, ulceration; procidentia may cause hydronephrosis.

Examination

- Lithotomy (supine) and **standing** with Valsalva / strain; Sims or split speculum for each compartment; lateral sulci (paravaginal defect); rectovaginal examination.
- **Cough stress test**, repeated with prolapse reduced (**occult SUI**); **Q-tip** angle change >30° = hypermobility; levator strength and avulsion.

POP-Q (Pelvic Organ Prolapse Quantification, Bump 1996)

- Nine points in cm relative to the **hymen (0)**: inside (above) negative, outside positive; measured on maximal strain, recorded on a 3 × 3 grid.

Point	Definition
Aa / Ap	Anterior / posterior wall 3 cm inside hymen (−3 to +3)
Ba / Bp	Most distal point of anterior / posterior wall (−3 if no prolapse)
C	Cervix or vault cuff
D	Posterior fornix (omitted after hysterectomy)
gh, pb, tvl	Genital hiatus, perineal body, total vaginal length (at rest)

Stage	Most distal point (leading edge)
0	No prolapse
I	>1 cm above hymen (< −1 cm)
II	Within ±1 cm of hymen (−1 to +1 cm)
III	>1 cm below hymen (> +1 cm) but < (tvl − 2) cm
IV	Complete eversion (≥ tvl − 2)

- **Also know:** Baden–Walker grades 0–4; stage I–II often normal (POSST study).

Investigations

- Urinalysis, PVR (post-void residual), bladder diary.
- **Cough test or urodynamics with prolapse reduced** for occult SUI; urodynamics also if ISD or mixed symptoms.
- Renal ultrasound in procidentia (hydronephrosis).
- Dynamic MRI or translabial ultrasound if complex or recurrent.
- A pessary trial confirms that the POP causes the symptoms.

Principles of prolapse repair

- Operate for bother, not anatomy. The first operation gives the best chance.
- Identify each compartment by POP-Q. **The apex is the cornerstone:** failing to support it greatly raises recurrence; uncorrected level I loss causes recurrent prolapse after SUI repair.
- **Three categories: restorative** (native tissue), **compensatory** (graft or mesh, for weak tissue or failed repair), **obliterative**.
- Vaginal route: fewer complications, quicker recovery. Abdominal route: more durable.
- Screen for **occult SUI** with prolapse reduced and consider a concomitant sling (OPUS, CARE trials). Finish with **cystoscopy** to confirm ureteric efflux.

Surgical options by compartment

Compartment	Operations	Key points
Anterior (cystocele)	**Anterior colporrhaphy** (midline plication of pubocervical fascia); **paravaginal repair** (fascia reattached to the ATFP)	Central versus lateral defect; does not support the apex

Compartment	Operations	Key points
Apical, vaginal (uterus, vault)	**Uterosacral suspension** (sutures; ureter injury 1–11%; cystoscopy); **sacrospinous fixation** (suture 1.5–3 cm medial to the ischial spine, avoiding the pudendal bundle; gluteal pain 15%); iliococcygeus suspension; McCall culdoplasty; vaginal hysterectomy; uterus-sparing hysteropexy; Manchester	OPTIMAL trial: uterosacral = sacrospinous at 2 years
Apical, abdominal (compensatory)	**Sacrocolpopexy** (open, laparoscopic, robotic): macroporous polypropylene mesh from vaginal apex to the anterior longitudinal ligament over S1 / sacral promontory	Most durable (>90%); mesh exposure about 3–10% (P2: about 10% long term, CARE)
Posterior (rectocele)	**Midline fascial plication** without levatorplasty; site-specific repair; perineorrhaphy (levels II–III)	Levator plication causes dyspareunia; constipation may persist
Obliterative	**Le Fort** (partial, uterus in situ, lateral channels) or total colpocleisis	Elderly, not sexually active; success >90%; coital regret ≤13%

- Cochrane 2016: vaginal apical repair has more recurrence and dyspareunia than sacrocolpopexy, but suits many older women.
- **April 2019:** the US FDA (Food and Drug Administration) ordered transvaginal synthetic mesh for (anterior) POP withdrawn; slings unaffected.
- Anterior colporrhaphy is not an SUI operation.

Importance of anatomy in SUI surgery

Operation	Anatomy used	Anatomical hazards
Burch colposuspension	Space of Retzius (avascular: symphysis in front, bladder behind, endopelvic fascia below). Paravaginal fascia at mid-urethra and bladder neck sutured to **Cooper's (iliopectineal) ligament**	Bleeding from Cooper's ligament and perivesical veins; ureteric kinking (cystoscopy); needs a mobile vagina
Paravaginal repair / VOS (vagino-obturator shelf)	Lateral vaginal sulcus to **ATFP** or obturator fascia (level II)	Obturator vessels and nerve
Retropubic MUS (mid-urethral sling), TVT (tension-free vaginal tape; Ulmsten 1995)	Tape under mid-urethra (integral theory), through **endopelvic fascia** and Retzius; suprapubic exits about 2 cm from midline	Bladder perforation (4.5%), vessels, bowel
TOT (transobturator tape; Delorme 2001)	Through obturator foramen / membrane, **obturator internus and externus**; exit at adductor longus tendon notch	Groin pain (6.4%), obturator neurovascular bundle; bladder injury rare (0.6%)
SIMS (single-incision mini-sling)	Anchors in obturator internus or endopelvic fascia	Works by restricting mobility
Autologous PVS (pubovaginal sling)	Rectus fascia at the **bladder neck** (proximal urethra), arms to rectus	Voiding dysfunction; best for ISD (compression)
Bulking agents	**Submucosa** coaptation	Repeated injections

- **Why mid-urethra:** it is the zone of maximal closure pressure and pubourethral support. A loose tape lets the posterior urethral wall be compressed and kinked during stress.
- **TOMUS (2010):** retropubic 80.8% versus transobturator 77.7% objective success, not different.
- **SISTEr:** autologous sling more successful than Burch at 2 years, with more voiding dysfunction.
- **Also know:** needle suspensions abandoned (sutures pulled through).

Diagram to draw: cross-section of urethra on the vaginal hammock (DeLancey) with Burch, bladder-neck sling and mid-urethral sling sites.

Procidentia: definition and assessment

- **Procidentia:** complete uterine (or vault) prolapse with the uterus wholly outside the hymen, **POP-Q stage IV**.
- Assess all compartments, occult SUI with prolapse reduced, PVR, hydronephrosis (ureteric kinking at the hiatus), cervical length and ulceration.
- Abnormal bleeding or cervical pathology needs gynaecological evaluation first, because it may argue against keeping the uterus.

Fertility-sparing treatment of procidentia

Principle: restore **level I** support while keeping uterus and cervix. Defer definitive surgery until the family is complete where possible, because pregnancy and delivery cause recurrence.

1. Conservative (first line)

- PFMT, weight loss, treat constipation.
- **Pessary:** space-filling type (Gellhorn, cube) for procidentia; removed for conception; review about 6-monthly for discharge and erosion.

2. Uterus-preserving surgery

Operation	Technique	Fertility points
Sacrospinous hysteropexy (vaginal)	Cervix sutured into the SSL **1.5–3 cm medial to the ischial spine**	Non-inferior to vaginal hysterectomy with uterosacral suspension (SAVE U trial); later data suggest superiority. Buttock pain, pudendal injury
Uterosacral hysteropexy (vaginal or laparoscopic)	Shorten and reattach uterosacral ligaments to cervix, sutures lateral to medial	No mesh; cystoscopy for ureteric kinking
Sacrohysteropexy (open, laparoscopic, robotic)	Mesh from posterior cervix to anterior longitudinal ligament over the promontory	Most durable apical support; later delivery usually by caesarean
Manchester (Fothergill)	Cervical amputation, cardinal ligaments sutured in front of the stump, anterior colporrhaphy	Suits elongated cervix, but risks **cervical stenosis, incompetence, miscarriage, preterm labour**: avoid if pregnancy wanted

- **Also know:** Indian abdominal (cervicopexy) slings for young nulliparous prolapse (Shirodkar, Purandare, Khanna, Virkud). **Le Fort colpocleisis** keeps the uterus but not fertility (elderly only).

3. Choosing `Procidentia, wants pregnancy -> PFMT + pessary -> fails -> sacrospinous hysteropexy or laparoscopic sacrohysteropexy (avoid Manchester) -> recurrence after childbearing -> hysterectomy + vault suspension or sacrocolpopexy`

Complications and follow-up

- Bladder, ureter or rectal injury; retention; de novo SUI; dyspareunia.
- Recurrence (mostly posterior overall; anterior after sacrospinous fixation).
- Mesh exposure: 3.4–5.4% after sacrocolpopexy in meta-analyses (P1 SAQ: about 3–10%; P2: about 10% long term, CARE); ureteric kinking.
- The retained uterus needs cervical screening.

> **Key points to remember:**
>
> - Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
> - DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone: always restore apical support.
> - Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
> - Integral theory: mid-urethral sling; hammock: backing, not repositioning.
> - Burch to Cooper's ligament; TOT through obturator foramen; PVS at bladder neck.
> - Bulge is most specific; splinting means high grade.
> - Central cystocele rugae lost, paravaginal preserved; POP-Q stage II = −1 to +1 cm (±1 cm of hymen).
> - Sacrocolpopexy most durable; colpocleisis for frail elderly.
> - Cystoscopy after repair; screen for occult SUI.
> - Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
> - Manchester amputates the cervix: poor choice for future pregnancy.

LAQ 2 Vesicovaginal fistula: management, timing of repair, eponymous techniques and recurrent VVF

P1 · P2 · P3 **merged from 3**

Asked as: P3 LAQ "Vesicovaginal fistula (VVF): management options and early vs delayed repair" · P2 SAQ "VVF repair: early versus delayed timing, and surgery for recurrent VVF" · P1 SAQ "Eponymous techniques of VVF repair"

Definition and aetiology

- VVF (vesicovaginal fistula): an epithelialised bladder–vagina tract (communication) causing continuous leakage; the commonest acquired urinary fistula (>75% of genitourinary fistulae).
- **Developing world (low-resource settings):** obstetric (obstructed labour) in >90%; the fetal head impacted for days causes pressure necrosis.
- **Developed world (high-resource settings):** iatrogenic in >75%, mainly after hysterectomy, usually supratrigonal at the vaginal cuff.
- Other causes: radiation, malignancy, mesh, foreign body.
- Common mechanism: **ischaemic necrosis**; continuous urine flow keeps the tract open until it epithelialises with time.
- Time to presentation: cystotomy, days; thermal injury, 7–10 days; radiation, months to years.

Classification

Goh (distance from meatus)	Size	Tissue
1: >3.5 cm; 2: 2.5–3.5; 3: 1.5–2.5; 4: <1.5 cm	A <1.5 cm; B 1.5–3 cm; C >3 cm	i mild fibrosis; ii moderate/severe; iii radiation, ureteric, circumferential, previous repair
- **Also know:** Waaldijk (closing mechanism involvement); WHO (World Health Organization) simple vs complex.		

Evaluation (guides choice)

- Speculum; fluid creatinine.
- **Dye tests:** single (blue backfill, tampon); **double** (oral phenazopyridine plus blue): orange = UVF (ureterovaginal fistula), blue = VVF.
- **Cystoscopy and vaginoscopy:** number, size, relation to ureteric orifices and bladder neck, mesh; **biopsy the tract if any history of cancer** (the fistula may signal recurrence).
- **Upper tract imaging (CTU, computed tomography urography) is mandatory:** to exclude a coexisting ureteric (ureterovaginal) fistula (up to 12–25%).
- Urodynamics: capacity, compliance, SUI (stress urinary incontinence) in 10–15%.
- Prepare: treat UTI (urinary tract infection), optimise nutrition, vaginal oestrogen for atrophy; counsel about residual incontinence, dyspareunia and possible reoperation.

Principles of management

- Optimise nutrition, treat infection, drain continuously, relieve obstruction, exclude malignancy.
- Surgery: excise scar, wide mobilisation, tension-free, multilayer, **non-overlapping**, watertight closure; remove foreign material; consider interposition; drain afterwards. **The first repair has the best chance.**
- At repair: treat infection; ureteric stents if within 1 cm of a ureter; vaginal or abdominal route (open, laparoscopic, robotic); **interposition flap** (Martius, omentum, peritoneum) if irradiated or recurrent; catheter drainage for about 2 weeks (≥10–14 days).

Option 1: Conservative and endoscopic (conservative trial while waiting)

- **Catheter drainage** cures about 15%. Best if small (<1 cm, ideally <3 mm), fresh, started (catheterised) within **3 weeks**, and non-irradiated. Once epithelialised (≥6 weeks), closure is unlikely. EAU (European Association of Urology): if tried, at least 12 weeks.
- **Fulguration** plus 2 weeks' drainage for tiny tracts <3.5 mm.
- **Sealants** (fibrin glue, cyanoacrylate, collagen): tiny series; not recommended by EAU.

Timing: early versus delayed repair (the debate)

	Early repair	**Delayed repair**
Definition (varies)	Days, <6 weeks or <3 months	Traditionally ≥3 months
Rationale	Clean iatrogenic tissue heals well; ends wetness and distress early	Inflammation settles, necrotic tissue sloughs, margins become clear
Evidence	**Comparable closure** in most series; Waaldijk: immediate obstetric repair gave very high first-attempt closure	One small series: more recurrence when repaired <6 weeks
Drawback	Friable tissue if inflamed	Prolonged distress, no better outcome

- Some authors avoid the **4–12 week** window (active healing phase).
- **EAU consensus:** no minimum delay if there is **no active infection and no recent irradiation**.
- **Timing by cause:**
- **Clean iatrogenic / post-hysterectomy:** early repair.
- **Obstetric with necrosis (necrotic edges):** catheter trial, then repair at about 3 months.
- **Radiation:** wait **≥12 months** after radiotherapy; biopsy to exclude recurrence; interposition flap; consider diversion if the bladder is poor.
- **Recurrent (redo) fistula:** wait until inflammation settles (commonly about 3 months); radiation ≥12 months after radiotherapy; failed radiation repairs were redone after **≥6 months**.

Option 2: Surgical route

Vaginal	**Abdominal (open, laparoscopic, robotic)**
Low, distal or trigonal, accessible apex	Supratrigonal, narrow vagina, no descent

Vaginal	Abdominal (open, laparoscopic, robotic)
Ureters distant	Ureter involved, needs reimplantation
Less pain, blood loss, shorter stay	Augmentation, **Tanagho bladder flap** or diversion possible
Flap-splitting, offset flaps	Large (≥4 cm primary, **≥2 cm recurrent**), radiation
Failed **Latzko** can be cured by a second Latzko	**O'Conor** bivalving finds the scarred plane
Risk: vaginal shortening	
- First-attempt closure about **90% by either route**; choose the route the surgeon knows best.	

Eponymous and named techniques: transvaginal

- **Sims:** first successful systematic vaginal repair (silver wire).
- **Flap-splitting (Raz):**
- Ureteric stents if within 1 cm; Foley through the tract for traction.
- Inverted J or U incision with a 1–3 mm cuff; raise anterior and posterior vaginal flaps.
- Layer 1: bladder (3-0 or 4-0), including the tract edge. Layer 2: perpendicular perivesical layer. Fill test 200–300 mL.
- Optional Martius or peritoneal flap; advance the opposite vaginal flap ≥2 cm beyond. Closure ≥90%.
- **Latzko (1942), high partial colpocleisis:**
- Denude vaginal epithelium 1–2 cm around the tract; the bladder is **not opened or dissected**.
- Close the vaginal walls over the tract in layers.
- ≥90% success in non-irradiated post-hysterectomy apical VVF; favoured for radiation VVF and tracts near the ureters. Drawback: **vaginal shortening and dyspareunia**.
- **Also know:** vaginal cuff excision (all apical scar excised as an inverted cone).

Eponymous and named techniques: abdominal

- **O'Conor and Sokol (1951), suprapubic extraperitoneal:**
- Midline cystotomy at the dome, **bivalved down** to the fistula; ureteric catheters.
- Excise the tract; separate bladder from vagina 2–3 cm beyond.
- Close the vagina; **omental or peritoneal flap** between; close the bladder in layers; suprapubic catheter.
- Modifications: **transvesical** (limited cystotomy) and **extravesical** (no cystotomy; fewer LUTS [lower urinary tract symptoms]).
- **Laparoscopic (1994) and robotic (2005):** equal closure (85–100%) with less blood loss, shorter stay and less morbidity; vaginal stent, backfill, ICG (indocyanine green) perfusion check. Robotic repair of recurrences gives high closure.

Interposition flaps (radiation, recurrent, ischaemic or large VVF)

Flap	Blood supply	Route	Use
Martius (1928) (labial fibrofatty pad)	Internal pudendal (inferior), external pudendal (superior)	Vaginal	Mid–distal VVF, bladder neck; does not reach the apex
Peritoneal (vaginal)	Local; pouch of Douglas fold	Vaginal or abdominal	High apical post-hysterectomy VVF
Omentum (J-flap)	Gastroepiploic arteries	Abdominal	Radiation, recurrent, large
Gracilis, Singapore, rectus	Myocutaneous	Vaginal/perineal	Refractory, large tissue loss
- RCT (randomised controlled trial): Martius added nothing in simple VVF, with donor-site pain in nearly a third.			
- **Interposition is widely used in recurrent fistula** (accepted, though not proven superior).			
- **Also know:** rectus abdominis flap (last resort).			

Recurrent VVF

- Persists after a failed repair; redo tissue is scarred and poorly vascularised, so the first repair has the best chance.
- Failure causes: tension, ischaemia, overlapping suture lines, infection, poor drainage, **malignancy, radiation**, missed second tract.
- Before redo: biopsy, cystoscopy and vaginoscopy, CT urography and dye test to exclude a **ureterovaginal fistula**, urodynamics; treat infection, optimise nutrition.
- Route: abdominal for recurrent ≥2 cm; a failed Latzko can be redone vaginally; robotic repair gives high closure.
- Interposition flap; catheter ≥10–14 days, then cystogram.
- More than one repair predicts residual incontinence.

Postoperative care

- Catheter ≥10–14 days (urethral and/or suprapubic); antimuscarinic or β3-agonist for spasm; cystogram before removal; stents out at 4–6 weeks.

Option 3: Urinary diversion (when repair is unwise)

- Multiple failures, severe radiation damage, massive tissue loss: conduit or continent reservoir/diversion.
- Unfit: ureteric occlusion with nephrostomies.
- Resource-poor: ureterosigmoidostomy as a last resort.

Outcomes

- Residual urgency or incontinence despite closure (Goh 3–4 predicts; more than one repair predicts); retention (vertical bladder closure); dyspareunia (narrow vagina, fistula ≥3 cm).

ICS (International Continence Society): pelvic floor physiotherapy.

Diagrams to draw

- Flap-splitting repair: perpendicular bladder layers, Martius flap, advanced vaginal flap.
- Latzko denuded area versus flap-splitting layers.
- Timing timeline: early window, 4–12 week zone, 3 months (obstetric), 12 months (radiation).
- Martius flap between bladder and vaginal closures.

Key points to remember:

- Image upper tracts; exclude ureteric fistula and cancer (biopsy before redo).
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- First repair has the highest success.
- Interposition flap (Martius, peritoneum, omentum) for complex and recurrent VVF.
- Diversion after multiple failures or severe radiation.
- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

LAQ 3 Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula

P3 **merged from 2**

Asked as: P3 LAQ "Urological complications in OBS-GYN and uretero-vaginal fistula" · P3 LAQ "Urological complications of OBGY surgery and urethrovaginal fistula"

Assumed meaning: OBS-GYN / OBGY = obstetric and gynaecological practice and surgery.

Part A: Urological complications in OBS-GYN

- The bladder, distal ureter and urethra lie against the cervix, uterine vessels and anterior vaginal wall. Risk rises with distorted anatomy: fibroids, endometriosis, cancer, previous caesarean, radiation, heavy bleeding, prolapse or mesh surgery.

By setting

Setting	Complications
Obstetric	**Obstructed labour** (fetal head impacted for days): pressure necrosis, so large VVF (vesicovaginal fistula), urethral loss, ureteric involvement; over 90% of VVF in the developing world. **Caesarean**: bladder injury, ureteric injury, **vesicouterine fistula** (Youssef syndrome: cyclical haematuria, amenorrhoea, continence preserved). Post-partum or post-epidural **retention** (overdistension)
Gynaecological surgery	**Bladder injury** (commonest organ). **Ureteric injury** (over 90% distal third; ligation commonest). **Fistulae**: VVF (over 75% follow hysterectomy in the developed world), UVF (ureterovaginal fistula), urethrovaginal fistula
Radical hysterectomy	Pelvic plexus injury: retention, poor compliance; manage with CIC (clean intermittent catheterisation)
Prolapse and continence surgery	Ureteric kinking at apical suspension; mesh or sling exposure, perforation, outlet obstruction, pain
Radiotherapy	Radiation cystitis, VVF, ureteric stricture

By complication: key facts

Complication	Key facts
Bladder injury (commonest organ injured)	Caesarean 0.27–0.3%; hysterectomy about 0.3–0.75% (laparoscopic highest); retropubic tape 3–9%. Repair: debride, 2-layer closure, fill test, catheter 7–14 days; check ureters if trigonal
Ureteric injury	Gynaecology causes over half of iatrogenic injuries; over 90% distal third. Hysterectomy overall 0.2–2.4%: vaginal 0.02–0.5%, abdominal 0.03–2%, minimally invasive 0.13–6%; prolapse surgery 1.7–4.3%. **Ligation** is the commonest gynaecological mechanism
Fistulae	VVF about 0.8 per 1000 hysterectomies; UVF; vesicouterine fistula after caesarean (Youssef syndrome); urethrovaginal fistula
Voiding dysfunction	Radical hysterectomy (pelvic plexus injury) 16–80% historically: retention, poor compliance; CIC and wait. Post-partum or post-epidural overdistension
Mesh and sling	Exposure, erosion into bladder or urethra, outlet obstruction, pain, dyspareunia
Radiation	Radiation cystitis; VVF in 0.6–2% after cervical cancer radiotherapy; ureteric stricture

- **Ureteric danger sites (four):** pelvic brim (near the infundibulopelvic ligament); **under the uterine artery** in the cardinal ligament; lateral to (beside) the uterosacral ligament; anterolateral vaginal fornix near the UVJ (ureterovesical junction). Most injuries are in the **distal 6 cm**.
- **Prevention and recognition:** haemostasis and anatomy; know the danger sites; prophylactic stents only selectively, if high risk (they aid recognition but do not prevent injury); cystoscopy with dye (IV indigo carmine or oral phenazopyridine) after hysterectomy where indicated, minimally invasive hysterectomy, and prolapse or continence surgery.
- Only about one-third of ureteric injuries are recognised at open surgery. A missed injury presents with fever, ileus, flank pain, urinoma or leakage; do CTU (computed tomography urography) with delayed films, then retrograde stenting. Distal repair: ureteroneocystostomy ± psoas hitch or Boari flap (see Part B).

Part B: Ureterovaginal fistula (UVF)

Definition

- Abnormal communication between the ureter and the vagina, causing continuous leakage.

Aetiology and pathogenesis

- Usually an **unrecognised injury to the distal third of the ureter** during pelvic surgery. About **three-quarters follow an often uneventful abdominal hysterectomy**.
- **Mechanisms:** ligation, crush, laceration, thermal injury, **devascularisation** or ischaemia. The injured segment necroses and leaks into the vaginal cuff, which is the weakest point.
- Diagnosis is delayed in 18–88%; **delayed recognition raises fistula risk about fourfold**.
- Other causes: obstetric, cancer, radiation, endometriosis, foreign body.

Clinical features

- **Continuous leakage with normal voiding** (the other kidney fills the bladder), usually appearing **within 4 weeks**, often with fever, flank pain or ileus, as thermal or ischaemic injury necroses.

	VVF	UVF
Voiding	Reduced	Normal
Upper tract	Normal	Hydronephrosis common
First line	Catheter, repair	Stent

Investigations (confirm before treatment)

Test	Method and reading
Double dye test	Oral phenazopyridine 100–200 mg (orange urine); after 1 hour, blue dye instilled into the bladder; tampon; walk about. **Orange = UVF**; blue = VVF; both = both
Three-swab test	Blue dye in the bladder; three swabs stacked in the vagina. Blue upper swab = VVF; **wet, unstained swab = UVF**
CT urography, delayed phase	Hydronephrosis, extravasation; MR (magnetic resonance) urography if contrast is contraindicated
Cystoscopy + RGP (retrograde pyelogram)	Abrupt cut-off or contrast leak; stent at the same sitting

Test	Method and reading
- **Exclude a coexisting VVF** (up to 25% of VVF have a UVF) and bilateral injury.	

Management

- **Goals:** dryness and preservation of renal function. Management has moved from nephrectomy, through reimplantation, to endourology.
- **1. Endourological (first line): stent first**
- **Retrograde double-J stent as soon as possible** if a wire passes. Stenting **within 2 weeks** of detection works best, succeeding far more often than at 2–6 weeks, because the ureter heals over the stent before fibrosis.
- Wire will not pass: **PCN (percutaneous nephrostomy)** with antegrade stent, or rendezvous.
- Stent for **4–6 weeks**, then remove and re-image. Most stentable fistulae close; a stricture may follow.
- **2. Surgical repair (stent fails or impossible)**
- **Timing:** no consensus. Early repair equals delayed: early repair took less time and blood loss with equal success; a systematic review found equal success before and after 6 weeks. Delay only if tissue is still inflamed.
- **Ureteroneocystostomy** (distal third, so usually feasible):
  1. Transperitoneal or extraperitoneal approach; identify the ureter above the fistula.
  2. Divide above the diseased segment; dissect as distally as possible for length; **preserve periureteric adventitia**, especially laterally where vessels enter.
  3. Mobilise the bladder; spatulate; tension-free mucosa-to-mucosa anastomosis over a double-J stent; refluxing or tunnelled.
  4. Drain; catheter 7–14 days.
- **Adding length:**

Technique	Extra reach
Psoas hitch (sutures parallel to psoas fibres, spare genitofemoral nerve)	Up to about 5 cm
Boari flap (base ≥4 cm, length:base ≤3:1)	10–15 cm
Downward nephropexy; ileal ureter	Longer defects
- Open, **laparoscopic and robotic repair give equal results**; vaginal reimplantation reported. **Success ≥90%.**	
- The vaginal opening is left to heal; no separate vaginal repair is needed.	

Algorithm `Leak + normal voiding -> exclude VVF (dye test, cystoscopy) -> CT urography / RGP -> wire/stent passes? Yes -> double-J 4-6 wk -> re-image -> healed / persistent No -> PCN + antegrade stent -> fails -> ureteroneocystostomy + psoas hitch / Boari flap`

Complications and follow-up

- Hydronephrosis, infection, ischaemic stricture.

- Ultrasound 1–2 weeks after stent removal; some add a renal scan at 6 weeks. Stricture usually appears within a year but may appear **up to 30 months** later, so follow beyond this.

Part C: Urethrovaginal fistula

Aetiology

- Developing world: **obstructed labour**, often with a VVF; also trauma, radiation, catheter injury.
- Developed world (iatrogenic): **urethral diverticulectomy** (fistula in 1–8%; causes 20–25% of urethrovaginal fistulae), slings, bulking agents, anterior repair or mesh, indwelling catheters (pressure necrosis), pessaries, radiation, trauma; also gender-affirming and paediatric genital surgery.

Clinical features

- Depend on site relative to the sphincter. **Distal** fistula: often asymptomatic, split stream, vaginal voiding, post-void dribble. **Proximal** (bladder neck): continuous leakage or SUI (stress urinary incontinence).
- Dyspareunia, recurrent UTI (urinary tract infection), perineal irritation; partner discomfort with exposed mesh.

Evaluation

- **Speculum examination** usually suffices; **methylene blue test** if rugae hide it (distal stain on tampon). Assess prolapse, SUI, atrophy.
- **VCUG (voiding cystourethrogram), lateral view;** VUDS (videourodynamics) for associated SUI or detrusor overactivity.
- **Cystourethroscopy** (flexible is better: outflow at the lens distends the urethra): size, site, bladder neck, exposed mesh, other fistulae. **About 20% have a separate VVF.** Image the upper tract.
- Goh location (distance from external meatus): type 1 >3.5 cm, 2: 2.5–3.5, 3: 1.5–2.5, 4: <1.5 cm. Closer means worse closure and continence.
- Optimise: treat UTI, vaginal oestrogen, nutrition; biopsy if previous cancer.

Management

- Asymptomatic distal fistula: may need no repair. Timing: early if tissue is healthy; otherwise after inflammation settles.
- **Standard: transvaginal rotational-flap repair:**

Step	Details
1. Exposure	Lithotomy (prone gives better anterior exposure); cystourethroscopy, cannulate tract; suprapubic catheter (Lowsley tractor), urethral catheter
2. Incision	Hydrodissect; circumscribe tract with 1–2 mm cuff, minimal excision; inverted **J- or U-shaped** vaginal flap incision
3. Flaps	Raise vaginal flap; mobilise periurethral tissue widely to bladder neck; open retropubic space if sling planned
4. Closure	**Close urethra** over a urethral sound: **4-0 absorbable** sutures, transverse or longitudinal; **second layer perpendicular**; buccal graft or flap onlay for large defects; leak test

Step	Details
5. Interposition	**Martius flap** (labial fat pad, external pudendal pedicle) most often; omentum, gracilis, rectus fascia occasionally
6. Cover	Advance vaginal flap beyond closure (2-0)

- **Autologous pubovaginal sling** for SUI or bladder-neck involvement; **never synthetic** (erodes the ischaemic urethra). Remove any mesh in the field.
- **Near-total urethral loss:** tubularised anterior bladder flap neourethra. Irreparable cases: diversion.
- **After repair:** urethral catheter out at 2–3 days; suprapubic catheter ≥14 days; lateral VCUG before removal; nothing per vagina for 6 weeks.

Outcomes and complications

- Closure **90–95%**, but continence only **50–80%**; postoperative SUI in up to half. Sling then or staged later.
- Urethral stricture, recurrence, dyspareunia.

Diagrams to draw

- Ureter passing under the uterine artery; double dye test; ureteroneocystostomy with psoas hitch.
- Urethrovaginal repair: inverted-U vaginal flap, two perpendicular urethral layers, Martius flap.

> **Key points to remember:**
>
> - Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
> - Ureteric injury is mostly distal and missed.
> - UVF: distal ureteric injury at hysterectomy; continuous leak within 4 weeks with normal voiding.
> - Orange tampon = UVF on double dye test; exclude coexisting VVF.
> - Stent early (within 2 weeks); remove at 4–6 weeks.
> - Reimplant with psoas hitch or Boari flap if stenting fails; success ≥90%; follow 30 months.
> - Urethrovaginal fistula: diverticulectomy is the commonest Western cause; obstetric in developing countries.
> - Look for a coexisting VVF (about 20%).
> - Layered closure plus Martius flap; autologous sling only.
> - Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

LAQ 4 Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery

P3 **merged from 2**

Asked as: P3 LAQ "Female pelvic floor, continence mechanism, classification and management of SUI" · P3 LAQ "SUI: evaluation in females, role of UDS, and surgical therapy in males"

Definition and mechanism

- **SUI (stress urinary incontinence)**: involuntary leakage on effort, exertion, coughing or sneezing (ICS, International Continence Society).
- Continence needs **urethral pressure greater than bladder pressure** at rest and during stress.
- Support comes from the **levator ani, endopelvic fascia and ligaments, perineal membrane and perineal body**.
- Two mechanisms: **urethral hypermobility** (loss of the suburethral support, the DeLancey "hammock") and **ISD (intrinsic sphincter deficiency)**. Most women have some ISD.

Anatomy of the female pelvic floor

1. Pelvic diaphragm

Muscle	Attachment	Function
Puborectalis	Back of pubis, loops behind anorectal junction	Anorectal angle
Pubococcygeus (pubovaginal)	Pubis, ATFP (arcus tendineus fasciae pelvis) to viscera, raphe	Slings urethra and vagina
Iliococcygeus	Arcus tendineus levator ani, ischial spine to coccyx	Horizontal **levator plate**
Coccygeus (not levator)	Ischial spine to sacrum, coccyx	Lies on sacrospinous ligament
- Nerve to levator ani **(S3–5)** on its pelvic surface. Resting tone closes the **urogenital hiatus**; widening of the hiatus or tilting of the plate causes prolapse. Muscle fails first, then fascia carries the load.		

2. Perineal membrane and perineal body

- The perineal membrane spans the pubic arch; urethra and vagina pierce it. Above it lies the **striated urogenital sphincter**: sphincter urethrae (mid-urethra), **compressor urethrae** and **urethrovaginal sphincter**. The perineal body anchors the distal vagina.

3. Endopelvic fascia and ligaments

- **Pubocervical fascia** attaches laterally to the **ATFP** (pubis to ischial spine); **rectovaginal fascia** posteriorly; **cardinal–uterosacral complex** at the apex; **pubourethral ligaments** anchor the mid-urethra.

4. DeLancey levels (1992)

Level	Structures	Defect
I suspension	Cardinal–uterosacral	Apical prolapse, enterocele
II attachment	ATFP, levator fascia	Cystocele (central or paravaginal), rectocele
III fusion	Perineal membrane, perineal body	**Urethral hypermobility**, perineal descent

Continence mechanism in the female

1. Intrinsic urethral factors (smooth and striated sphincters, vascular submucosa)

- **Mucosal seal**: folded urothelium coapts. The **oestrogen-dependent submucosal vascular plexus** provides up to **30%** of closure force.
- **Smooth muscle** (internal sphincter, bladder neck and proximal urethra): involuntary tone. The female bladder neck is weak and often open in continent women.
- **Striated urogenital sphincter**: the rhabdosphincter, mostly **slow-twitch** fibres, gives tonic mid-urethral closure. Distal parts are the **compressor urethrae** and **urethrovaginal sphincter**.
- The **continence zone is the proximal and mid-urethra.**

2. Support (extrinsic)

- The urethra lies on the **anterior vaginal wall and endopelvic fascia**, attached laterally to the **ATFP** and levator ani.
- **Pubourethral ligaments** anchor the mid-urethra; **levator ani** (pubococcygeus) tone keeps the backing firm.

3. Neural

- **Pudendal nerve (S2–4)** drives the sphincter (**guarding reflex**). Urethral pressure rises **just before** a cough reaches it, so closure is active, not just transmitted.

4. Theories of continence and SUI

Theory	Mechanism	Surgical implication
Pressure transmission (Enhorning 1961)	Hypermobile urethra drops out of the abdominal pressure zone	Restore position (Burch)
Hammock (DeLancey 1994)	Abdominal pressure compresses the urethra against a firm suburethral layer; a lax layer fails	Restore backing
Integral theory (Petros & Ulmsten 1990)	Pubourethral ligament laxity at the mid-urethra	Tension-free MUS (mid-urethral sling)
ISD (McGuire & Lytton 1978)	Urethra itself fails to close (VLPP [Valsalva leak point pressure] <60, MUCP [maximum urethral closure pressure] <20 cmH_2O)	Sling, bulking, AUS (artificial urinary sphincter)
- Most SUI has some ISD. **Hypermobility alone never causes SUI.**		

Classification of SUI

1. Blaivas & Olsson (1988), on video-urodynamics

Type	Bladder neck at rest	On stress
0	Closed, above lower border of symphysis	Descends and opens, but no leak seen (history of SUI)
I	Closed, above symphysis	Descends **<2 cm**; leaks; little or no cystocele
IIA	Closed, above symphysis	Descends **>2 cm** with cystourethrocele; leaks
IIB	Closed but **at or below** lower border of symphysis	Proximal urethra opens; leaks with or without further descent
III	**Open at rest** without detrusor contraction	Leaks with minimal stress = **ISD**

2. Functional: hypermobility versus ISD

	Urethral hypermobility	ISD
Q-tip test	Angle change **>30°**	Often fixed urethra
VLPP / ALPP	**≥90 cmH_2O**	**<60 cmH_2O** (60–90 equivocal)
MUCP	Normal	**≤20 cmH_2O**
Cause	Childbirth, ageing, obesity	Previous surgery, radiation, neuropathy, ageing

3. Other

- **Severity (Stamey):** grade 1 leak on cough or sneeze; grade 2 on minimal effort (walking, rising); grade 3 at rest.
- **Index (uncomplicated)** versus **complicated** SUI (recurrent, prolapse, neurogenic, high PVR [post-void residual], radiation); pure versus mixed incontinence.

Evaluation of SUI in women

History

- Leakage with cough or activity means SUI; after urgency, UUI (urgency urinary incontinence); both, MUI (mixed). A positive SUI history predicts urodynamic SUI in about three-quarters.
- Severity (pads/day), bother; voiding symptoms, prolapse, obstetric history, prior surgery or radiation, neurological disease, drugs.
- **Questionnaires:** ICIQ-UI SF (International Consultation on Incontinence Questionnaire–UI Short Form); UDI-6 (Urogenital Distress Inventory: bother); IIQ-7 (Incontinence Impact Questionnaire: impact). **Bladder diary** (3 days).

Examination (comfortably full bladder)

- Neurological (S2–S4 sensation, anal tone, bulbocavernosus reflex); atrophy; prolapse by **POP-Q (Pelvic Organ Prolapse Quantification)**.
- **Cough stress test:** cough supine, then standing if negative. Leakage synchronous with cough confirms SUI. Reduce prolapse to unmask **occult SUI**.
- **Q-tip test:** swab to bladder neck, Valsalva; rotation **>30°** = hypermobility. Visual mobility examination is now routine.

Tests

Test	Method and use
Urinalysis	Required; exclude infection, haematuria
PVR	Required before surgery; ultrasound; repeat if high (no agreed cut-off)

Test	Method and use
1-hour pad test (ICS)	500 mL fluid, standard exercises; >1.4 g positive; 1–10 g mild, 11–50 g moderate, >50 g severe
24-hour pad test	Normal activities; more reproducible
Cystoscopy	Haematuria, irritative symptoms, previous mesh

Role of urodynamics (UDS) in SUI

- UDS (urodynamic studies) assess storage and voiding. The key question in SUI is: **will the result change the operation?** Pure, demonstrable SUI gains little. **UDS is not routine.**

Parameter	Method and meaning
USI (urodynamic stress incontinence)	Leak with raised abdominal pressure, no detrusor contraction
ALPP/VLPP (abdominal/Valsalva leak point pressure)	Standing, from 150 mL, Valsalva or cough; **<60 cmH_2O = ISD**, 60–90 equivocal, ≥90 = little ISD (McGuire 1993)
MUCP	Urethral pressure profile; a low MUCP (≤20 cmH_2O) also suggests ISD; does not diagnose or grade SUI
Filling cystometry	DO (detrusor overactivity), compliance, capacity; cough-induced DO mimics SUI
Pressure–flow study	Voiding dysfunction, poor contractility (retention risk after sling)
Prolapse reduction	Occult SUI
- If no leak with the catheter in place, repeat with it removed.	

Evidence

- **VALUE (ValUE) trial (2012), 630 women** with uncomplicated SUI: office evaluation alone was **non-inferior** to office evaluation plus UDS before surgery (success 77.2% vs 76.9%).
- **Dutch VUE study:** immediate midurethral sling was non-inferior to UDS-tailored treatment.
- **SISTEr:** UDS did not predict outcome.

Guidelines

- **AUA (American Urological Association)/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023:** UDS may be omitted in the index patient with demonstrable SUI.
- **UDS advised in non-index patients:** unclear diagnosis or SUI not demonstrated; mismatch between symptoms and findings; urgency-predominant MUI; neurogenic bladder; raised PVR or voiding dysfunction; prolapse stage ≥3 without demonstrable SUI; failed previous anti-incontinence or prolapse surgery.
- **EAU (European Association of Urology):** no routine UDS for uncomplicated SUI; do not use leak point or urethral pressure to grade severity.
- **Men:** before AUS or sling for post-prostatectomy incontinence, UDS confirms USI and finds DO, poor compliance or weak contractility. The MASTER trial recruited men with USI.

Conservative management (first line)

- Weight loss, fluid and caffeine adjustment.
- **Supervised PFMT (pelvic floor muscle training)** for at least 3 months (Cochrane: more cure than no treatment).

- **Pessary**; vaginal oestrogen for atrophy.
- **Duloxetine** (serotonin–noradrenaline reuptake inhibitor): increases Onuf's nucleus drive to the sphincter; approved in Europe, not the USA.

Surgical options for female SUI: principles

Operation	Principle (how it works)	Best for	Key facts
MUS: retropubic (TVT, tension-free vaginal tape, Ulmsten 1995), transobturator (TOT; Delorme 2001; inside-out de Leval), single-incision	Tension-free type I polypropylene tape under the **mid-urethra** acts as a new pubourethral ligament; the urethra kinks against it during stress	First-line index SUI	**First line** (AUA/SUFU strong recommendation). **TOMUS**: retropubic 80.8% vs transobturator 77.7% cure (equivalent); retropubic often preferred in ISD. Retropubic: more bladder perforation; transobturator: more groin pain
SIMS (single-incision mini-sling)	Short tape, vaginal only	Option	Non-inferior to TOT for subjective cure in most studies; conditional recommendation (2023)
Autologous pubovaginal sling (PVS) (rectus fascia or fascia lata)	At the **bladder neck**, minimal tension; restores the hammock, giving bladder-neck support and compression	**All types, including ISD**, recurrent SUI, neurogenic SUI, failed MUS, after mesh removal, mesh-averse	Gold standard. **SISTEr**: sling beat Burch (47% vs 38%) but more voiding dysfunction and UTI (urinary tract infection)
Burch colposuspension (open, laparoscopic or robotic)	Paravaginal fascia beside the mid-urethra and bladder neck sutured to **Cooper's ligament**; lifts the bladder neck back into the pressure zone and gives a stable backing	Hypermobility; women declining mesh; at abdominal surgery	85–90% at 1 year, about 70% at 5 years. Needs a mobile urethra; poor in ISD
Paravaginal repair, VOS (vagino-obturator shelf)	Vagina to ATFP or obturator fascia		Flatter, less obstructive
MMK (Marshall–Marchetti–Krantz)	Bladder neck to symphysis periosteum	Historical	Osteitis pubis 0.7–2.5%; abandoned
Urethral bulking agents	Submucosal injection at bladder neck or proximal urethra improves coaptation	ISD with fixed urethra; frail patients; minimal surgery	Less durable; repeat injections
AUS	Bladder-neck cuff; constant circumferential closure	Severe refractory ISD, fixed urethra	Second-line; erosion, infection, revision

Operation	Principle (how it works)	Best for	Key facts
- **Also know:** needle suspensions abandoned (sutures pull through); anterior colporrhaphy is not an SUI operation; CARE (prophylactic Burch at sacrocolpopexy); Ward–Hilton (TVT = colposuspension).			

Burch steps (algorithm): Pfannenstiel → space of Retzius → stay lateral to urethra → 2–4 sutures each side (mid-urethra, then bladder neck) → Cooper's ligament → tie loosely (two fingers behind pubis) → **cystoscopy** for ureteric efflux.

Retropubic vs transobturator MUS

	Retropubic	Transobturator
Bladder perforation	4.5%	0.6%
Voiding dysfunction (TOMUS surgery)	More (2.7% vs 0%)	Less
Groin pain	1.3%	6.4%
ISD, long-term durability	Often preferred	Possibly more repeat surgery

Complications of female SUI surgery

- **MUS complications:** perforation, retention, de novo urgency, mesh exposure (about 2%), pain.
- Bladder trocar injury: re-pass and continue (cystoscopy after every pass); **urethral injury: abandon mesh**.
- Voiding dysfunction (sling too tight or proximal): urethrolysis or sling incision; de novo urgency; UTI; haematoma.
- **Mesh complications** (IUGA/ICS [International Urogynecological Association/International Continence Society] terms): exposure, extrusion, perforation into bladder or urethra, pain; manage by excision. FDA (Food and Drug Administration) 2019 stopped transvaginal mesh for prolapse, not slings.

Male SUI: surgical therapy

- **Aetiology:** commonest after **radical prostatectomy (RP)**, which removes the internal sphincter, leaving continence to the rhabdosphincter. Also TURP (transurethral resection of the prostate), radiation, pelvic fracture.
- **Work-up:** 24-h pad weight (mild <150 g, moderate <400 g, severe above); standing cough test; **cystoscopy** (anastomotic stricture, sphincter coaptation); UDS if mixed or poor contractility.
- **Timing:** PFMT first; no surgery for 6–12 months, as continence keeps improving (other P3 answer: continence improves for up to 2 years, but do not delay surgery beyond 6 months for severe leak).

Device	Mechanism	Use and results
Transobturator sling (AdVance XP)	Repositions the bulb 3–4 cm proximally; needs residual sphincter	Mild–moderate leakage, no radiotherapy; cure about 60%
Quadratic sling (Virtue)	Compression plus elevation	Mild; limited data
Adjustable sling (ATOMS, adjustable transobturator male system)	Cushion adjusted via scrotal port	Mild–moderate; about two-thirds dry
ProACT balloons	Bladder-neck compression	More erosion
Bulking agents	Submucosal bulk	Small leaks, unfit men; rarely durable; not advised
AUS	Circumferential compression	**Gold standard**; all severities, moderate–severe, after radiation; about 80% continent

AMS 800 AUS

- **Components:** urethral **cuff** (bulbar, usually 4.0–4.5 cm); **PRB (pressure-regulating balloon) 61–70 cmH_2O** (51–60 high risk; 71–80 bladder neck) in a preperitoneal pocket; scrotal **pump**. Filled with 23 mL.
- **Technique:** perineal incision; cuff on the proximal bulbar urethra; **deactivate for 4–6 weeks**.
- **Needs** dexterity and cognition. Social continence (0–1 pad) 55–77%; satisfaction about 90%; lifespan 7–10 years.
- **Complications:** infection 1–3% (explant; reimplant after 3–6 months); **erosion** 2–15% (explant, catheter about 3 weeks); urethral **atrophy** (late recurrent leak); mechanical failure; retention.
- **Erosion risks:** radiation, prior erosion, urethroplasty; use a **transcorporal cuff** (tunica adds bulk) or lower PRB. Atrophy: smaller or tandem cuff.

Algorithm (AUA/SUFU 2019) `Severe SUI or any radiotherapy -> AUS Moderate, no radiotherapy -> AUS or male sling Mild, no radiotherapy -> sling (or PFMT, AUS) Sling failure -> AUS`

- **MASTER RCT (randomised controlled trial):** sling was **non-inferior** to AUS for continence at 12 months; AUS remains preferred for heavy leak (>250 g/day) or radiation.

Diagrams to draw

- DeLancey levels and hammock cross-section.
- Sites of Burch sutures, bladder-neck sling and mid-urethral tape; sagittal pelvis showing retropubic and transobturator tape paths.
- Cystometrogram trace with abdominal pressure spike and leak (ALPP).
- AMS 800 (cuff, pump, PRB) and male transobturator sling path.

Key points to remember:

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD.
- Q-tip >30° = hypermobility; VLPP / ALPP <60 cmH_2O = ISD; MUCP ≤20 cmH_2O.
- Burch corrects position; slings and AUS correct function.
- VALUE (ValUE) trial: UDS not needed in uncomplicated (index) SUI.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.
- AUS PRB 61–70 cmH_2O; deactivate 4–6 weeks.
- Radiation or severe male SUI: AUS first; MASTER: sling non-inferior.

LAQ 5 Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management

P3

Asked as:

- Etio-pathogenesis, diagnosis and management of Recto-urethral fistula? — *Summer 2020 Paper III Q1 (LAQ)*
- Etiopathogenesis, diagnosis and management of Recto-Prostatic fistula — *Winter 2025 Paper III Q1 (LAQ)* (also answered in 2020-Summer Paper 3, Q1, as recto-urethral fistula)

Type: LAQ (25 marks)

Assumed meaning (recto-prostatic): rectourethral fistula at the level of the prostatic urethra or the vesicourethral anastomosis.

Definition

- **RUF (rectourethral fistula):** a communication between the posterior urethra / prostatic urethra (or the vesicourethral anastomosis after prostatectomy) and the distal rectum.
- Rare but very morbid. Almost never seen in females, because the vagina lies between urethra and rectum.

Aetiology

Group	Causes
Congenital	Anorectal malformation in boys (rectobulbar or rectoprostatic fistula)
Surgical (commonest acquired)	**RP (radical prostatectomy)**; historically TURP (transurethral resection of the prostate) and open (simple) prostatectomy
Radiation and energy	**EBRT (external beam radiotherapy) plus brachytherapy** (commonest), then brachytherapy alone, then pelvic radiotherapy; cryotherapy; HIFU (high-intensity focused ultrasound): low after primary, up to about 9% after salvage HIFU
Others	Trauma (penetrating injury, pelvic fracture); Crohn disease; advanced rectal or prostate cancer

Pathogenesis

- **Surgical:** the rectum is torn during apical or posterior dissection behind Denonvilliers fascia, especially with small glands and salvage surgery. An unrecognised tear (near the apex) leaks urine (urine leaks into it), an abscess forms and the tract epithelialises.
- Rectal injury at RP: laparoscopic about 1%, open and robotic under 0.5%, salvage RP about 7%. (Another source: rectal injury is commonest with retropubic and laparoscopic RP, lowest with robotic RP, and was over 15% in historical open salvage series.)
- **Radiation or ablation:** obliterative endarteritis makes the tissue hypoxic and fibrotic. Necrosis follows months to years later. These fistulae are large and heal poorly.
- **Congenital:** abnormal cloacal development leaves the rectum ending in the posterior urethra.

Classification: Martini (after prostate cancer treatment)

Axis	Category	Definition
Stage (size on CT (computed tomography) urography or cystography)	I / II / III	≤1.5 cm / >1.5 cm / any size with sphincter damage
Grade (tissue)	0	Surgical injury, no prior non-surgical treatment
	1	Prior radiation, brachytherapy, HIFU or other energy
	2	Salvage RP or salvage ablation
- **Tissue quality (grade), not size, chiefly decides the repair.**		

Clinical features

- **Faecaluria and pneumaturia** (most typical); urine per rectum on voiding.
- Recurrent UTI (urinary tract infection), dysuria, urosepsis; coexisting SUI (stress urinary incontinence).

Diagnosis

- **DRE (digital rectal examination):** may feel the defect on the anterior rectal wall.
- **Cystoscopy and proctosigmoidoscopy:** show the tract and tissue vitality; **biopsy** to exclude recurrent cancer.
- **VCUG (voiding cystourethrogram) or RUG (retrograde urethrogram):** usually conclusive; gives size and site for planning.
- Upper-tract imaging to exclude ureteric injury; pelvic MRI (magnetic resonance imaging) for complex cases.
- **UDS (urodynamics)** with rectal packing: fistula repair alone will not cure severe SUI; an AUS (artificial urinary sphincter) may be needed later; a later AUS erodes more often.

Prevention

- At RP: a recognised tear without gross spillage → two-layer (multilayer) closure, air test, **omental or peritoneal interposition**, drain, antibiotics; colostomy if (previously) irradiated.
- Radiotherapy: a **hydrogel rectal spacer** lowers rectal dose and late rectal toxicity (randomised trial: late rectal toxicity 2.0% vs 7.0%).

Management

- **Principles:** control sepsis; divert urine ± faeces; let inflammation settle (about 12 weeks); exclude cancer; then excise the tract and close urethra and rectum separately in **tension-free, non-overlapping layers with vascularised (interposition) tissue interposed**.
- **Conservative:** no sepsis and no faecaluria → catheter (urethral or suprapubic), low-residue diet ± colostomy. A simple fistula may close in about 12 weeks; irradiated or complex ones rarely do.
- **Surgery:** no single accepted operation; choice depends on site, radiation and prior repair. **A diverting colostomy usually comes first.**

Approach	Essentials	Strengths	Weaknesses
Perineal (Young; Goodwin)	Separate and close urethra and rectum separately; **gracilis flap** between (or dartos; buccal graft)	High closure; most stomas reversed	Urinary incontinence common
Transsphincteric (York-Mason)	Prone jackknife; sphincter divided in tagged layers; through anterior rectal wall; three-layer closure	Avoids scarred plane; spares sexual function	Poor membranous exposure; no flap; faecal incontinence up to 20%
Abdominal (transvesical, extravesical, O'Conor-like) / robotic	Omental flap	Large or irradiated fistulae, extensive resection	Morbid; poor deep-pelvic view; small robotic series
Laparoscopic / robotic	Open pouch of Douglas; close rectum in two layers; omental or peritoneal flap; can combine with perineal gracilis	Deep-pelvic vision	Small series only
Salvage	Cystoprostatectomy, ileal conduit ± permanent colostomy	Unreconstructable irradiated field	Major surgery

`RUF → sepsis control, catheter ± colostomy, biopsy small, non-irradiated, no faecaluria → conservative ~12 weeks persistent, grade 0 → perineal (gracilis) or York-Mason irradiated or large → abdominal/robotic (omentum) or perineal (gracilis) unreconstructable → urinary + faecal diversion Congenital → newborn colostomy → posterior sagittal anorectoplasty`

- **PSARP (posterior sagittal anorectoplasty)** separates the rectum from the urethra and places it within (in) the sphincter complex.

Complications

- **Recurrence** (the greatest challenge), urinary or faecal incontinence, urethral stricture, ED (erectile dysfunction), wound infection, permanent stoma.

Diagram to draw: sagittal pelvis showing (with) the fistula and a gracilis flap between urethra and rectum.

> **Key points to remember:**
> - Most acquired RUFs follow prostate cancer treatment.
> - Martini: stage = size; grade = tissue (surgery, radiation, salvage).
> - VCUG or RUG plus endoscopic biopsy to exclude cancer.
> - Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

Short answer questions (SAQ)

SAQ 1 Leak point pressures and ALPP

P1 · P2 **merged from 2**

Asked as: P1 SAQ "Leak point pressures (VLPP, ALPP, DLPP) in SUI: clinical implications" · P2 SAQ "ALPP"

Definition

- **LPP (leak point pressure):** the pressure at which leakage is first seen, spontaneous or provoked. There are two independent types:

Feature	ALPP (abdominal LPP), incl. VLPP (Valsalva LPP) and CLPP (cough LPP)	DLPP (detrusor LPP)
Definition	Total Pves (vesical pressure) at leak during raised abdominal pressure, **no detrusor contraction**	Lowest Pdet (detrusor pressure) at leak, **without contraction or straining**
Measures	Sphincter strength against abdominal pressure	Bladder pressure against outlet resistance
Bad result	**Low = bad** (weak sphincter)	**High = bad**: >40 cmH_2O means upper-tract risk; hydronephrosis or reflux in 85% (McGuire 1981)
Used in	SUI (stress urinary incontinence)	Neurogenic or low-compliance bladder

- **ALPP (abdominal leak point pressure):** the **intravesical pressure** at which urine leaks because of raised abdominal pressure, **without a detrusor contraction**. It measures the urethra's ability to resist abdominal pressure: **lower ALPP = weaker sphincter**.
- Provoked by Valsalva (**VLPP**, the intravesical pressure at which urine leaks during Valsalva) or cough (**CLPP**).
- Seen only in SUI; there is **no normal value** (continent women never leak).
- Newer term: "urodynamic stress test" (ICS, International Continence Society).

Technique (VLPP/ALPP)

1. Small dual-lumen urethral catheter plus rectal catheter; zero to atmosphere.
2. Patient **standing or sitting**; fill to **150 mL** (original method; some use 200 mL).
3. Ask for a **graded Valsalva** or cough; record the **lowest total Pves at visible leak** (include baseline, not the change).
4. No leak: repeat every 50 mL to capacity; then, at capacity, **remove the urethral catheter** and read pressure from the rectal line.
5. Report volume, catheter size and provocation (Valsalva or cough).

Interpretation (McGuire 1993)

VLPP / ALPP (cmH_2O)	Inference
<60	**ISD (intrinsic sphincter deficiency)**
60–90	Equivocal; mixed
≥90 (>90)	**Urethral hypermobility**, little ISD

Clinical implications and uses of ALPP

- Separates ISD from hypermobility; a low VLPP/ALPP (ISD) traditionally favours a sling or bulking agent over colposuspension.
- Post-prostatectomy incontinence: many men leak only once the urethral catheter is removed.

Limitations

- Not standardised: falls with **larger catheters** and **higher volumes**.

- **VLPP is lower than CLPP** in the same woman (VLPP < CLPP), so state the method.
- Invalid during DO (detrusor overactivity) or poor compliance; prolapse raises it or masks the leak (reduce it).
- ISD and hypermobility coexist; VLPP/ALPP alone **does not predict surgical outcome** well.
- **VALUE trial (Nager 2012):** for uncomplicated SUI, office evaluation was non-inferior to urodynamics before surgery. AUA (American Urological Association)/SUFU (Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2023 SUI guideline: urodynamics optional in the index patient.
- **EAU (European Association of Urology):** do not use leak point pressure to grade severity.

DLPP: clinical implications

- **McGuire (1981):** DLPP **>40 cmH_2O** in myelodysplastic children leads to hydronephrosis or VUR (vesicoureteral reflux) in **85%**.
- A strong outlet keeps the patient dry but transmits pressure to the kidneys; a weak outlet "pops off" safely.
- **Hostile bladder** (≥40 cmH_2O): CIC (clean intermittent catheterisation) plus antimuscarinic; botulinum toxin, augmentation or vesicostomy if needed.
- **Caveat:** 40 is not a validated "safe" value; 20 cmH_2O is more sensitive in children.

Also know

- DO leak point pressure (ICS 2017); MUCP (maximum urethral closure pressure) ≤20 cmH_2O suggests ISD.

Diagram to draw

- Pves, Pabd and Pdet traces marking ALPP (leak on Valsalva with flat Pdet) and DLPP.

> **Key points to remember:**
> - VLPP/ALPP: Pves at leak with Valsalva or strain, no detrusor contraction; start at 150 mL.
> - ALPP low = weak sphincter; DLPP high = upper-tract danger.
> - VLPP <60 ISD; 60–90 equivocal; ≥90 (>90) hypermobility.
> - VLPP < CLPP; catheter size and volume alter it.
> - DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
> - Lower DLPP is safer; no validated safe cut-off.

SAQ 2 Mid-urethral slings: TOT and TVT

P1 · P2 **merged from 2**

Asked as: P2 SAQ "Mid-urethral slings for SUI: TOT, TVT and TOT versus TVT" · P1 SAQ "TOT in SUI"

Definition and history

- Both are polypropylene **MUS (mid-urethral slings)** for female SUI (stress urinary incontinence), based on the **integral theory** (Petros and Ulmsten, 1990): a loose mid-urethral tape replaces the pubourethral ligaments as a backboard on straining.
- **TVT (tension-free vaginal tape, retropubic):** retropubic polypropylene MUS; Ulmsten, 1995–1998.

- **TOT (transobturator tape):** a synthetic MUS of macroporous monofilament polypropylene, passed through the obturator foramina instead of the retropubic space. Delorme (2001) designed it to avoid bladder, bowel and vessel injury.

Mechanism

- **Integral theory:** pubourethral ligaments, the suburethral vaginal hammock and pubococcygeus maintain continence at the mid-urethra. Hypermobility is a sign of their damage, not the cause.
- The loose tape at the mid-urethra supports the **posterior urethral wall**, which is compressed against it on straining (dynamic kinking); it is not tight at rest.
- TOT anchors in the obturator muscles (**obturator internus and externus**).

Material

- **Type 1 (Amid) macroporous monofilament polypropylene**, pores >75 µm: allows macrophage entry and fibrous ingrowth, so it resists infection.

Indications and evaluation

- SUI or stress-predominant mixed incontinence after failed PFMT (pelvic floor muscle training); index SUI with urethral hypermobility.
- TVT preferred over TOT in **ISD (intrinsic sphincter deficiency)** or recurrent SUI.
- History, examination, **objective stress test**, urinalysis, PVR (post-void residual); urodynamics only if complicated.
- Counsel about permanent mesh, mesh risks and alternatives (colposuspension, autologous sling, bulking) (AUA/SUFU [American Urological Association/Society of Urodynamics and Female Urology]).

Technique: differences

	TVT (retropubic)	TOT (transobturator)
Path	Vagina to retropubic space to suprapubic skin, hugging the pubis; cystoscopy essential	Labiocrural fold to obturator membrane to vagina (outside-in), or reverse (inside-out)
Structures at risk	Bladder, bowel, iliac and epigastric vessels	Obturator nerves and vessels, vaginal fornix

TVT technique

1. Local, spinal or general anaesthesia; lithotomy.
2. Two suprapubic stab incisions 2 cm lateral to midline; 1.5 cm midline vaginal incision 1.5 cm from the meatus; minimal dissection, endopelvic fascia not opened.
3. **Bottom-up** 5 mm trocar hugging the back of the pubis, through the retropubic space to the skin (catheter guide deflects the bladder).
4. **Cystoscopy with a 70° lens, full bladder**, after each pass; if perforated, re-pass more laterally.
5. Tension **loosely** with a clamp between tape and urethra; remove sheath.

TOT technique (outside-in)

1. Lithotomy with hips hyperflexed; 1.5 cm midline vaginal incision 1.5 cm below the meatus; dissect laterally to the ischiopubic ramus.

2. Groin stab incision in the labiocrural fold, at the notch between inferior pubic ramus and **adductor longus tendon**.
3. Trocar perforates the obturator membrane (felt as a give), turns medially onto the finger and exits the vaginal incision.
4. Cystourethroscopy; check the vaginal fornices (sulci).
5. Tension-free: a clamp or scissors pass easily between tape and urethra. - **Inside-out** (de Leval; vagina to thigh): fewer vaginal perforations, more voiding dysfunction.

Outcomes

- TVT: Ulmsten (1998): 91% cured at 1 year. **Ward–Hilton:** equal to colposuspension at 5 years. Long-term cure varies 48–97% by definition.
- **TOMUS (Richter, 2010):** 597 women; 12-month objective cure **80.8% retropubic vs 77.7% transobturator** (not significantly different, equivalent); voiding dysfunction needing surgery 2.7% vs 0%.
- TOMUS at 5 years: 51.3% vs 43.4%; equivalence not shown; satisfaction similar.
- **Cochrane review (Ford, 2017):** cure similar up to and beyond 5 years; repeat surgery possibly commoner after TOT.
- TOT is poorer with **ISD** or a **fixed urethra**: retropubic sling preferred.

Complications: comparison (Cochrane, 2017)

Complication	TVT (retropubic)	TOT
Bladder perforation	**4.5%**	**0.6%**
Groin pain	1.3%	**6.4%**
Voiding dysfunction, bleeding, vascular injury	More	Less
Mesh exposure	About 2%	About 2%
Long-term repeat surgery	Less	Possibly more

TVT complications

Complication	Notes
Bladder perforation	4.5% (TOT 0.6%); re-pass and continue
Urethral injury	Abandon mesh; staged repair
Bleeding/haematoma	2–3%; rarely bowel or vascular injury
Voiding dysfunction	Surgery needed 2.7%; incise tape in midline within 3 months
Mesh exposure	About 2%
Others	De novo urgency, suprapubic pain (groin pain less than TOT)

TOT complications

- Groin or thigh pain (usually temporary; NSAIDs [non-steroidal anti-inflammatory drugs]), vaginal mesh exposure, voiding dysfunction (incise sling within 4 weeks), de novo urgency, UTI (urinary tract infection), rarely thigh abscess.
- Urethral injury during either sling: **do not place mesh** (AUA guideline).

Choosing between TVT and TOT

- **TVT preferred: ISD** (ALPP [abdominal leak point pressure] <60 cmH_2O or MUCP [maximum urethral closure pressure] <20 cmH_2O), fixed urethra, recurrent SUI. Failure is 12 times likelier after TOT with low leak point pressure (O'Connor, 2006).
- **TOT preferred:** previous retropubic surgery, obesity, avoiding bladder injury.
- AUA: either may be offered to the index patient.
- **Also know:** SIMS (single-incision mini-slings; non-inferior to TOT); mesh warnings (FDA [Food and Drug Administration] 2010; UK pause 2018): counsel about alternatives (Burch colposuspension, autologous fascial sling, bulking agents).

Diagram to draw

- Coronal pelvis: U-shaped retropubic TVT at mid-urethra versus flat TOT, the tape under the mid-urethra through both obturator foramina.

> **Key points to remember:**
> - Integral theory; loose tape at mid-urethra.
> - Type 1 macroporous monofilament polypropylene.
> - Delorme 2001; TOT anchors in obturator muscles; fewer bladder injuries than retropubic.
> - TOMUS: cure equivalent at 1 year (80.8% retropubic vs 77.7% transobturator).
> - TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
> - Cystoscopy after each TVT trocar pass; bladder perforation 4.5%.
> - Prefer retropubic TVT in ISD or fixed urethra.

SAQ 3 Burch colposuspension

P1

Type: SAQ (10 marks)
Source: 2022-Summer Paper 1, Q3b (related: 2017-Winter Paper 3, Q2)

Definition

- Retropubic suspension (**Burch, 1961**) of the paraurethral and para-bladder-neck vaginal wall to **Cooper's (iliopectineal) ligament**; Tanagho (1976) modified it.

Principle

- SUI (stress urinary incontinence) from **urethral hypermobility**: the urethra drops out of the retropubic position. Elevation restores **pressure transmission** and a backstop; scarring holds it.
- It treats hypermobility, not ISD (intrinsic sphincter deficiency).

Indications

- SUI or stress-predominant MUI (mixed urinary incontinence) with hypermobility, especially with **concomitant abdominal surgery** (sacrocolpopexy), **mesh refusal**, or poor vaginal access.
- **Contraindications:** immobile urethra or vagina, ISD alone, pure urgency incontinence.

Work-up

- Cough stress test, PVR (post-void residual), urinalysis; urodynamics only if SUI is not demonstrated.

Steps (open)

1. Low lithotomy, Foley; Pfannenstiel incision; enter the **space of Retzius**.
2. Vaginal finger elevates the vagina; sweep bladder medially. **Avoid midline dissection** over urethra.
3. Two to four 0 or 1-0 delayed-absorbable or non-absorbable sutures each side through vaginal wall (not lumen): distal at mid-urethra, proximal at bladder neck, placed laterally.
4. Pass through Cooper's ligament; tie with vaginal elevation, without tension; leave about **two fingerbreadths** between bladder neck and pubis.
5. **Cystoscopy:** ureteric efflux, no suture in bladder. - Laparoscopic and robotic versions match open short-term results.

Outcomes

- Cure 85–90% at 1 year, about 70% at 5 years (Cochrane).
- **SISTEr (2007):** autologous fascial sling more effective but more voiding dysfunction and UTI.
- **Ward and Hilton:** TVT (tension-free vaginal tape) equivalent to colposuspension; meta-analyses favour midurethral slings.
- **CARE (2006):** prophylactic Burch at sacrocolpopexy reduced postoperative SUI.

Complications

- Voiding dysfunction 2–27%, de novo storage symptoms 8–27%, **de novo prolapse** 2.5–27%; bladder or ureteric injury; bleeding.
- **Also know:** MMK (Marshall–Marchetti–Krantz; osteitis pubis 0.7–2.5%), paravaginal repair, VOS (vagino-obturator shelf); failure predictors: BMI (body mass index) >25, age >60, prior surgery.

Diagram to draw: sagittal pelvis with sutures from paravaginal fascia to Cooper's ligament.

> **Key points to remember:**
> - Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
> - Cure 85–90% at 1 year, ~70% at 5 years.
> - Niche: concomitant abdominal surgery, mesh avoidance.

SAQ 4 Management of perforated mid urethral sling

P1

Type: SAQ (10 marks)
Source: 2023-Winter Paper 1, Q3a

Terminology (IUGA/ICS, International Urogynecological Association / International Continence Society)

- **Perforation:** mesh in a hollow organ (bladder, urethra). **Exposure:** mesh seen through unhealed vaginal epithelium. **Extrusion:** passage out of a tissue. "Erosion" is no longer used.

- Two settings: **intraoperative trocar injury** and **late mesh perforation**.

A. Intraoperative trocar injury

- Bladder injury: 4.5% retropubic versus **0.6% transobturator**; registries 2.7–3.8%.
- **Detection:** cystoscopy with a **70° lens and full bladder distension** after every trocar pass.

Injury	Management
Bladder	Withdraw trocar, **repass more laterally**, confirm on cystoscopy, complete the sling; catheter overnight to 72 h. Benign; no later mesh perforation.
Urethra	**Do not place mesh** (AUA/SUFU, American Urological Association / Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction). Repair, catheter drainage; **staged** anti-incontinence surgery after healing.

B. Late perforation of bladder or urethra

- **Presentation:** recurrent UTI (urinary tract infection), haematuria, urgency, pain, stones on mesh, voiding dysfunction, fistula. Often months later.
- **Evaluation:** examination, **cystourethroscopy**, translabial ultrasound (best imaging for mesh); exclude fistula.

Management

- **Bladder:** endoscopic excision (scissors, resection or holmium laser) flush with the wall; or transvesical laparoscopic or robotic excision; open cystotomy if large.
- **Urethra:** transvaginal excision of mesh in the field and layered urethral repair; **Martius flap** if complex or fistula.
- **Continence after excision:** SUI recurs in many (44–100% after synthetic sling revision). Options: concomitant **autologous pubovaginal sling** (Shah 2013: good results) or staged. A damaged urethra predicts poor continence.
- Avoid urethral dilation after mesh slings; it can cause perforation.

Prevention

- Empty bladder; trocar hugging the pubis; bottom-up retropubic route; transobturator route when suitable.

Also know: Amid type I macroporous polypropylene integrates best; chronic pain or infection may need excision from both sides of the bone.

Diagram to draw: retropubic and transobturator trocar paths with bladder at risk.

> **Key points to remember:**
> - Bladder trocar injury: repass laterally, continue.
> - Urethral injury: abandon mesh; stage the repair.
> - Late perforation: excise mesh, repair, consider autologous sling.

SAQ 5 Female urethral diverticula

P1

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 1, Q3e

Definition

- **UD (urethral diverticulum):** a urine-filled periurethral sac within the periurethral fascia, connected to the urethra by an **ostium**.
- Third to seventh decades; prevalence up to 1–6%.

Aetiology

- Mostly **acquired**. **Routh theory** (most accepted): infection of **periurethral glands** leads to duct obstruction, suburethral abscess, rupture into the lumen and an epithelialised sac.
- Hence the ostium is **posterolateral at mid or distal urethra** in ≥90%.
- **Also know:** childbirth trauma theory (but 20–30% are nulliparous).

Clinical features

- "Three Ds" (dysuria, dyspareunia, post-void dribbling) in only about 5%.
- Commonest: frequency, urgency, pain, **recurrent UTI (urinary tract infection)**.
- Tender anterior vaginal wall mass; about one-third incontinent.
- **Complications:** stones (4–10%); 10% show metaplasia, dysplasia or carcinoma (**adenocarcinoma** commonest).

Evaluation

- Urine culture; **cystourethroscopy** (ostium seen in under half).
- **Imaging sensitivity: MRI (magnetic resonance imaging) 97%**, double-balloon urethrography 95%, ultrasound 82%, VCUG (voiding cystourethrogram) 74%.
- MRI shows saddlebag or circumferential extension.
- **Videourodynamics:** true SUI (stress urinary incontinence) versus dribbling from the sac.
- **Differential diagnosis:** Skene cyst (distal, distorts meatus, no communication), Gartner cyst, leiomyoma, caruncle, bulking agent.

Classification

- **L/N/S/C3 (Leach):** location, number, size, configuration, communication, continence.
- **Also know:** Leng–McGuire (absent fascia = pseudodiverticulum).

Management

- **Asymptomatic or unfit:** suppressive antibiotics, post-void stripping; counsel about cancer.
- **Symptomatic: transvaginal diverticulectomy.**
- **Principles:** 1. Sterile urine; topical oestrogen if atrophic. 2. **Inverted-U** anterior vaginal wall flap. 3. **Preserve periurethral fascia** as a separate layer. 4. Excise the **entire sac** and close the ostium. 5. **Watertight** urethral closure over a catheter (as small as 12 Fr). 6. **Multilayered, non-overlapping** closure; obliterate dead space. 7. **Martius flap** if poor tissue.
- **Bothersome SUI:** concomitant **autologous pubovaginal sling**; **never synthetic mesh**.
- **Also know:** Spence–Duckett marsupialisation for distal lesions only.
- Pericatheter VCUG at 14–21 days.

Complications

- Recurrence 10–22%; de novo SUI 7–16%; urethrovaginal fistula 1–8%; stricture; recurrent UTI.

Diagram to draw: sagittal section of a mid-urethral sac and its ostium.

Key points to remember:

- Periurethral gland infection (Routh); posterolateral ostium.
- MRI most sensitive (97%).
- Preserve fascia; non-overlapping layers; Martius flap.
- No synthetic mesh at diverticulectomy.

SAQ 6 Endoscopic procedure of SUI

P1

Type: SAQ (10 marks)
Source: Practice Set 13 Paper 1, Q3f

Read as endoscopic injection of urethral bulking agents.

Introduction

- **SUI (stress urinary incontinence)** results from urethral hypermobility and **ISD (intrinsic sphincter deficiency)**.
- **Bulking agents**, injected submucosally under cystoscopic vision, are the least invasive surgical option.

Mechanism

- Bulk at the proximal urethra improves **mucosal coaptation** and resistance.
- It does not correct hypermobility, but most SUI includes some ISD.

Indications

- ISD with a **fixed, immobile urethra** (low VLPP [Valsalva leak point pressure] <60 cmH_2O, historically).
- Women wanting a **low-risk** procedure; elderly or unfit; future pregnancy.
- Persistent leakage after a sling; neurogenic SUI (safer for bladder compliance than a sling).
- **Avoid in:** active UTI (urinary tract infection), untreated detrusor overactivity.

Agents

Agent	Note
Polyacrylamide hydrogel (Bulkamid)	Non-particulate
Polydimethylsiloxane (Macroplastique)	Silicone particles
Calcium hydroxylapatite; carbon-coated zirconium beads	Particulate
Bovine collagen (skin test); PTFE (polytetrafluoroethylene); autologous fat	Withdrawn or abandoned: allergy, migration, granuloma, embolism

Technique

1. Sterile urine; local anaesthesia; cystoscope (0–30° lens) in the urethra.
2. **Transurethral:** needle into the submucosa about 1–1.5 cm distal to the bladder neck; or **periurethral:** needle alongside the urethra, watched cystoscopically.
3. Inject at 2–4 sites until the mucosal cushions **coapt in the midline**.
4. Avoid an indwelling catheter (it moulds the bulk); use CISC (clean intermittent self-catheterisation) for retention.

Outcomes

- Early improvement is common, but cure is lower than with slings and **declines with time**; repeat injections are often needed.
- **Cochrane review (Kirchin 2017):** limited evidence of benefit.
- **AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction):** an option, with counselling on durability.
- **Men:** EAU (European Association of Urology): **do not offer** for post-prostatectomy incontinence.

Complications

- Transient retention, UTI, dysuria, haematuria, urgency.
- Sterile abscess, migration, erosion; may mimic a urethral diverticulum on imaging.

Diagram to draw: cystoscopic view of coapted mucosal cushions after injection.

> **Key points to remember:**
> - Bulking improves coaptation; does not correct hypermobility.
> - Best for fixed urethra with ISD, or unfit patients.
> - Less durable than slings; repeat injections common.
> - Not recommended for post-prostatectomy incontinence.

SAQ 7 Endometriosis in urology

P1

Type: SAQ (10 marks)
Source: Practice Set 22 SAQ Sheet, Q3d

Definition and epidemiology

- Endometriosis is endometrial glands and stroma outside the uterus. Urinary tract involvement occurs in about 1–5% of women with endometriosis, mostly with deep disease.
- Sites: **bladder** (commonest, about 80%) > ureter (about 15%) > kidney, urethra (rare).

Pathogenesis

- Retrograde menstruation (Sampson), coelomic metaplasia, lymphovascular spread, and **iatrogenic implantation** at pelvic surgery (e.g. caesarean section).
- Bladder lesions start on the **serosa** (vesicouterine pouch) and invade the detrusor.
- Ureteric disease: **extrinsic** (periureteric fibrosis, commoner) or intrinsic (muscularis or mucosa); usually distal, left more often.

Clinical features

- **Bladder:** cyclical haematuria or frequency, urgency and dysuria with menses; suprapubic pain. Differential: BPS (bladder pain syndrome), vesicouterine fistula (menouria).
- **Ureter:** often **silent**; hydronephrosis and loss of renal function may go unnoticed.

Evaluation

- Transvaginal and transabdominal ultrasound gauge size and depth; **renal ultrasound in every woman with deep endometriosis**.

- **MRI (magnetic resonance imaging):** extent; high T1 signal from blood products.
- **Cystoscopy:** limited value (disease is extravesical); may show a bluish submucosal lesion. Biopsy excludes carcinoma.
- CTU (computed tomography urography) and MAG3 (mercaptoacetyltriglycine) renogram for obstruction and split function.
- Staging: rASRM (revised American Society for Reproductive Medicine) I–IV; #Enzian records bladder and ureter.

Management (with gynaecologist)

- **Medical:** progestins (dienogest), combined pill, GnRH (gonadotrophin-releasing hormone) analogues. Symptomatic relief; not for obstructive ureteric disease.
- **Bladder: partial cystectomy** (full-thickness excision with margin), laparoscopic or robotic; stent ureters if near. TUR (transurethral resection) is incomplete, with recurrence and perforation risk.
- **Ureter:**
- Extrinsic, mild: ureterolysis.
- Intrinsic or stricture: segmental resection with ureteroureterostomy, or **reimplantation with psoas hitch or Boari flap**.
- Non-functioning kidney: nephrectomy.
- Treat all pelvic disease at the same operation to lower recurrence.
- **Also know:** risk factor for VVF (vesicovaginal fistula) and ureteric injury at hysterectomy; rare malignant change (clear-cell, endometrioid).

Diagram to draw: endometriotic nodule in the vesicouterine pouch invading the detrusor.

> **Key points to remember:**
> - Bladder commonest; cyclical haematuria.
> - Ureteric disease silent: screen kidneys.
> - Partial cystectomy, not TUR.

SAQ 8 Urethral caruncle.

P2

Type: SAQ (10 marks)
Source: 2016-Summer Paper 2, Q3f

Definition

- A benign, red, fleshy, **inflammatory polypoid lesion of the posterior lip of the distal female urethra**. Not a true tumour, but a benign mimic.
- Common; seen mainly in **postmenopausal women**.

Aetiology and pathogenesis

- **Oestrogen deficiency (GSM, genitourinary syndrome of menopause)**: atrophy and laxity of distal urethral mucosa.
- Related to **mucosal prolapse**: the everted posterior mucosa is chronically irritated, causing haemorrhage, necrosis and inflammatory growth.

Pathology

- Classical types: **papillomatous, angiomatous and granulomatous**; inflamed vascular stroma.

Clinical features

- Often asymptomatic, found on examination.
- **Spotting on underwear**, bleeding, pain, dysuria; may thrombose into a dark, tender mass.
- Examination: small, red, soft mass at the **6 o'clock** meatal margin; the rest of the meatus looks normal.

Differential diagnosis

Lesion	Distinguishing feature
Urethral mucosal prolapse	**Circumferential "doughnut"** around whole meatus; prepubertal girls and postmenopausal women
Urethral carcinoma	Firm, ulcerated, indurated, fixed; bleeding mass; biopsy
Urethral diverticulum	Mid or proximal anterior vaginal wall mass; dysuria, dyspareunia, dribbling; MRI (magnetic resonance imaging)
Skene gland cyst or abscess	Lateral to meatus; distorts it; does not communicate
Others	Melanoma, lymphoma, TB (tuberculosis), condyloma, intestinal metaplasia; postmenopausal bleeding from the uterus
- Caruncle is listed as a possible risk factor for female urethral cancer, so atypical lesions must be biopsied.	

Evaluation

- Clinical diagnosis; exclude a gynaecological bleeding source.
- **Excision biopsy** if atypical, indurated, ulcerated, enlarging or refractory; cystourethroscopy if in doubt.

Treatment

- **Conservative (first line)**: **topical vaginal oestrogen** (shrinks most), anti-inflammatory creams, sitz baths. Ask about breast or endometrial cancer before oestrogen.
- **Surgical excision** for large, symptomatic, thrombosed, refractory or suspicious lesions: 1. Grasp the tip and apply traction to expose the base. 2. Excise at the base. 3. Reapproximate mucosa with absorbable sutures if the defect is large. 4. Foley catheter for a few days.
- Complications: bleeding, recurrence, meatal stenosis.

Diagram to draw: meatus with a 6 o'clock caruncle compared with circumferential prolapse.

> **Key points to remember:**
> - Postmenopausal posterior-lip lesion; oestrogen deficiency.
> - Prolapse is circumferential; caruncle is focal.
> - Topical oestrogen first; excise if atypical or refractory.

SAQ 9 Female sexual dysfunction

P3

Type: SAQ (10 marks)
Source: Practice Set 1 Paper 3, Q3f

Definition and prevalence

- FSD (female sexual dysfunction): persistent desire, arousal, orgasm or pain problems causing **personal distress**.
- About **40%** of women in their lifetime; **HSDD (hypoactive sexual desire disorder) commonest**. Almost half of women with UI (urinary incontinence) have FSD.

Physiology and response models

- Arousal: pudendal afferents, pelvic parasympathetic efferents → **NO (nitric oxide)–cGMP (cyclic guanosine monophosphate)** → engorgement and lubrication. Oestrogen maintains tissues; testosterone drives desire.
- **Masters and Johnson** (linear): excitement, plateau, orgasm, resolution; Kaplan added desire. **Basson** (circular): desire is often responsive.

Classification

DSM-5 (Diagnostic and Statistical Manual, 2013)	ICSM (International Consultation on Sexual Medicine)/ISSWSH (International Society for the Study of Women's Sexual Health)
Female sexual interest/arousal disorder (≥3 of 6 criteria)	HSDD and arousal disorder kept separate (cognitive vs genital arousal)
Female orgasmic disorder	Orgasm disorders
Genito-pelvic pain/penetration disorder	Genital-pelvic pain dysfunction
–	PGAD (persistent genital arousal disorder)
- ≥6 months with distress; lifelong or acquired; generalised or situational.	

Causes

- Hormonal (menopause, GSM [genitourinary syndrome of menopause]), drugs (SSRIs [selective serotonin reuptake inhibitors], antihypertensives), diabetes, depression, relationship factors.
- Urological: incontinence, prolapse, sling dyspareunia, cystectomy nerve or clitoral injury.

Evaluation

- Clinician must ask. **PLISSIT** (permission, limited information, specific suggestions, intensive therapy); BSSC-W (Brief Sexual Symptom Checklist for Women) screen; **FSFI (Female Sexual Function Index, 19 items, six domains)**.
- Examination: atrophy, lichen sclerosus, cotton-swab test, pelvic floor tone. Laboratory tests rarely help.

Management

- Treat causes; education; sex therapy (sensate focus, mindfulness).

- **HSDD (premenopausal, FDA [US Food and Drug Administration] approved): flibanserin** 100 mg at bedtime (5-HT1A agonist/5-HT2A antagonist; alcohol caution); **bremelanotide** 1.75 mg subcutaneously on demand (melanocortin agonist). Postmenopausal: transdermal testosterone (off-label).
- **Arousal:** testosterone; PDE5 (phosphodiesterase type 5) inhibitors only with neurovascular disease.
- **Orgasm:** education, directed masturbation.
- **Pain/vestibulodynia:** pelvic floor physiotherapy, dilators, biofeedback, topical lidocaine, vaginal oestrogen; vestibulectomy if refractory.
- **Also know:** ospemifene, intravaginal prasterone (menopausal dyspareunia).

Diagram to draw: Basson circular sexual response model.

> **Key points to remember:**
> - About 40% prevalence; HSDD commonest; distress required.
> - DSM-5 merges desire and arousal, dyspareunia and vaginismus.
> - Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

SAQ 10 Duloxetin

P3

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 3, Q3e

Read as: duloxetine, mainly its role in stress urinary incontinence.

Introduction

- **Duloxetine** is an SNRI (serotonin–noradrenaline reuptake inhibitor), also an antidepressant.
- Licensed in Europe (not USA) for **moderate to severe SUI (stress urinary incontinence) in women**; off-label in men.

Mechanism

- During storage, bladder afferents drive glutamate excitation of pudendal motor neurons in **Onuf's nucleus (S2–S4)**, which keep the striated urethral sphincter closed.
- Serotonin (5-HT2) and noradrenaline (α1) amplify this glutamate drive. Duloxetine blocks their reuptake, so **sphincter tone and the sneeze-induced continence reflex increase**.
- In voiding, glutamate drive switches off, so retention does not occur.
- Central effects also reduce bladder activity.

Dose

- **40 mg twice daily**; starting at 20 mg twice daily for 2 weeks reduces nausea. Taper on stopping.

Evidence

- Phase III RCTs (randomised controlled trials) in women: about 50% median fall in incontinence episodes versus about 30–40% with placebo; better quality of life; **cure rare**.
- Benefit is temporary and lasts only while taken.
- Male PPI (post-prostatectomy incontinence): small, short-term benefit.

Guideline position

- **EAU (European Association of Urology):** may be offered to women for **temporary** improvement, not cure; warn about side effects. Men: offer for SUI (weak), and **inform of off-label use and adverse effects** (strong).
- **NICE (National Institute for Health and Care Excellence):** not first line; not routinely second line; only if surgery is unwanted or unsuitable.
- Adding PFMT (pelvic floor muscle training): evidence conflicting.

Adverse effects

- **Nausea** (commonest, early), dry mouth, dizziness, insomnia, constipation; **high discontinuation**.
- Raised blood pressure, hyponatraemia (elderly), suicidal ideation, discontinuation syndrome.

Contraindications and interactions

- MAOIs (monoamine oxidase inhibitors), including **methylene blue** (serotonin toxicity); CYP1A2 inhibitors (fluvoxamine, ciprofloxacin); hepatic impairment; severe renal impairment; uncontrolled hypertension or narrow-angle glaucoma.

Diagram to draw: Onuf's nucleus with 5-HT and noradrenaline terminals driving the pudendal nerve.

> **Key points to remember:**
> - SNRI acting on Onuf's nucleus to increase sphincter tone.
> - 40 mg twice daily; titrate from 20 mg to reduce nausea.
> - Temporary benefit only; approved in Europe, not USA.

SAQ 11 Acute urinary retention in females.

P4

Type: SAQ (10 marks)
Source: 2017-Summer Paper 4, Q3f

Introduction

- AUR (acute urinary retention) is sudden inability to void. It is uncommon in women and is often non-obstructive, so search for the cause.

Causes

Group	Examples
Obstructive	Large cystocele or prolapse; tight sling after incontinence surgery; urethral diverticulum, stricture; pelvic mass, **retroverted gravid uterus**, haematocolpos; cancer
Functional	**Fowler syndrome**; dysfunctional voiding; primary bladder neck obstruction (rare)
Neurogenic	MS (multiple sclerosis; may present first with retention), cauda equina, transverse myelitis, diabetes
Inflammatory	Cystitis, vulvovaginitis, **genital herpes**, Bartholin abscess
Drugs	Anticholinergics, opioids, tricyclics, calcium-channel blockers

Group	Examples
Postoperative/postpartum; psychogenic	Epidural, pain; diagnosis of exclusion

Fowler syndrome

- Young women (20–35) with painless retention and bladder capacity over 1 litre.
- **Urethral sphincter EMG (electromyography):** complex repetitive discharges, so the sphincter fails to relax. Urethral closure pressure >100 cm H_2O.

Evaluation

- **History:** drugs, surgery, delivery, neurological and bowel symptoms, genital lesions.
- **Examination:** pelvic (prolapse, mass); neurological (perineal sensation, anal tone).
- **Tests:** drained volume; urine culture; creatinine; pregnancy test; ultrasound.
- MRI (magnetic resonance imaging) spine if neurological signs; cystoscopy.
- **Later:** video-urodynamics (female obstruction, e.g. Qmax ≤12 mL/s with Pdet ≥20 cm H_2O); sphincter EMG.

Management

- **Immediate:** urethral catheter; suprapubic if urethral access fails.
- **Treat the cause:** stop drugs; treat infection or herpes; pessary or prolapse repair; reposition the uterus; sling division; diverticulectomy.
- **TWOC (trial without catheter)** after correction.
- **Persistent retention:**
- **CISC (clean intermittent self-catheterisation)** rather than an indwelling catheter.
- **SNM (sacral neuromodulation):** about 70% success in Fowler syndrome.
- Bethanechol not recommended. Bladder neck incision only with caution (incontinence).

Diagram to draw: flowchart: AUR → catheterise → find cause → treat → TWOC → CISC or SNM.

> **Key points to remember:**
>
> - Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
> - Fowler: abnormal sphincter EMG; SNM about 70%.
> - CISC for persistent retention.

Rapid Revision Sheet · Section 6

Female pelvic floor anatomy and pelvic organ prolapse: evaluation and surgical repair (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–S5.
- DeLancey: I suspension (uterosacral–cardinal; failure causes uterine prolapse), II attachment (ATFP), III fusion (urethral hypermobility); level I is the keystone.
- Always restore apical support.
- Enhorning 1961, DeLancey 1994, Petros & Ulmsten 1990, McGuire ISD 1978.
- Integral theory: mid-urethral sling; hammock: backing, not repositioning.
- Burch: Cooper's ligament; TOT: obturator foramen; PVS: bladder neck.
- Bulge is most specific; splinting means high grade.
- Central cystocele rugae lost, paravaginal preserved; POP-Q stage II = −1 to +1 cm (±1 cm of hymen).
- Sacrocolpopexy most durable; colpocleisis for frail elderly.
- Cystoscopy after repair; screen for occult SUI.
- Pessary first; sacrospinous hysteropexy 1.5–3 cm medial to the ischial spine.
- Manchester amputates the cervix: poor choice for future pregnancy.

Vesicovaginal fistula: management, timing of repair, eponymous techniques and recurrent VVF (LAQ)

- Image upper tracts; exclude ureteric fistula and cancer (biopsy before redo).
- Catheter trial only if small, fresh (<3 weeks), non-irradiated; works only if started within 3 weeks.
- Vaginal and abdominal routes both close about 90%.
- Early repair for clean iatrogenic VVF; closure comparable to delayed.
- Delay ~3 months for obstetric, ≥12 months for radiation fistula.
- First repair has the highest success.
- Interposition flap (Martius, peritoneum, omentum) for complex and recurrent VVF.
- Diversion after multiple failures or severe radiation.
- Latzko: colpocleisis, bladder not opened, vagina shortened.
- O'Conor: bivalve bladder to the fistula, interpose omentum.
- Martius for distal, omentum or peritoneum for apical VVF.

Urological complications of obstetric-gynaecological surgery: uretero- and urethrovaginal fistula (LAQ)

- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
- Ureteric injury is mostly distal and missed.
- UVF: distal ureteric injury at hysterectomy; continuous leak within 4 weeks with normal voiding.
- Orange tampon = UVF on double dye test; exclude coexisting VVF.
- Stent early (within 2 weeks); remove at 4–6 weeks.
- Reimplant with psoas hitch or Boari flap if stenting fails; success ≥90%; follow 30 months.
- Urethrovaginal fistula: diverticulectomy is the commonest Western cause; obstetric in developing countries.

- Look for a coexisting VVF (about 20%).
- Layered closure plus Martius flap; autologous sling only.
- Closure 90–95% (closure is easy); continence only 50–80% (continence is hard).

Female SUI: continence mechanism, classification, evaluation (UDS) and management, including male SUI surgery (LAQ)

- Levator ani = puborectalis, pubococcygeus, iliococcygeus; S3–5.
- DeLancey: I suspension, II attachment (ATFP), III fusion.
- Continence zone is the mid-urethra; submucosal plexus gives up to 30% of closure.
- Blaivas types 0, I, IIA, IIB, III; type III = ISD.
- Q-tip >30° = hypermobility; VLPP / ALPP <60 cmH_2O = ISD; MUCP ≤20 cmH_2O.
- Burch corrects position; slings and AUS correct function.
- VALUE (ValUE) trial: UDS not needed in uncomplicated (index) SUI.
- Use UDS when it changes treatment: mixed, failed surgery, voiding dysfunction.
- PFMT first; MUS first-line surgery.
- SISTEr (fascial sling > Burch); TOMUS (retropubic = transobturator).
- PVS for ISD; urethral injury → no mesh.
- Men: AUS gold standard; sling for mild, non-irradiated.
- AUS PRB 61–70 cmH_2O; deactivate 4–6 weeks.
- Radiation or severe male SUI: AUS first; MASTER: sling non-inferior.

Recto-urethral (recto-prostatic) fistula: etiopathogenesis, diagnosis and management (LAQ)

- Most acquired RUFs follow prostate cancer treatment.
- Martini: stage = size; grade = tissue (surgery, radiation, salvage).
- VCUG or RUG plus endoscopic biopsy to exclude cancer.
- Divert first; perineal (gracilis) or York-Mason for surgical fistulae; irradiated ones need imported tissue or diversion.

Leak point pressures and ALPP (SAQ)

- VLPP/ALPP: Pves at leak with Valsalva or strain, no detrusor contraction; start at 150 mL.
- ALPP low = weak sphincter; DLPP high = upper-tract danger.
- VLPP <60 ISD; 60–90 equivocal; ≥90 (>90) hypermobility.
- VLPP < CLPP; catheter size and volume alter it.
- DLPP >40: upper-tract danger; 85% upper-tract change (McGuire 1981).
- Lower DLPP is safer; no validated safe cut-off.

Mid-urethral slings: TOT and TVT (SAQ)

- Integral theory; loose tape at mid-urethra.
- Type 1 macroporous monofilament polypropylene.
- Delorme 2001; TOT anchors in obturator muscles; fewer bladder injuries than retropubic.
- TOMUS: cure equivalent at 1 year (80.8% retropubic vs 77.7% transobturator).
- TOT: less bladder injury (0.6% vs 4.5%), more groin pain (6.4% vs 1.3%); worse in ISD.
- Cystoscopy after each TVT trocar pass; bladder perforation 4.5%.
- Prefer retropubic TVT in ISD or fixed urethra.

Burch colposuspension (SAQ)

- Burch 1961: vaginal wall to Cooper's ligament; for hypermobility.
- Cure 85–90% at 1 year, ~70% at 5 years.
- Niche: concomitant abdominal surgery, mesh avoidance.

Management of perforated mid urethral sling (SAQ)

- Bladder trocar injury: repass laterally, continue.
- Urethral injury: abandon mesh; stage the repair.
- Late perforation: excise mesh, repair, consider autologous sling.

Female urethral diverticula (SAQ)

- Periurethral gland infection (Routh); posterolateral ostium.
- MRI most sensitive (97%).
- Preserve fascia; non-overlapping layers; Martius flap.
- No synthetic mesh at diverticulectomy.

Endoscopic procedure of SUI (SAQ)

- Bulking improves coaptation; does not correct hypermobility.
- Best for fixed urethra with ISD, or unfit patients.
- Less durable than slings; repeat injections common.
- Not recommended for post-prostatectomy incontinence.

Endometriosis in urology (SAQ)

- Bladder commonest; cyclical haematuria.
- Ureteric disease silent: screen kidneys.
- Partial cystectomy, not TUR. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Urethral caruncle. (SAQ)

- Postmenopausal posterior-lip lesion; oestrogen deficiency.
- Prolapse is circumferential; caruncle is focal.
- Topical oestrogen first; excise if atypical or refractory.

Female sexual dysfunction (SAQ)

- About 40% prevalence; HSDD commonest; distress required.
- DSM-5 merges desire and arousal, dyspareunia and vaginismus.
- Flibanserin and bremelanotide: only FDA-approved HSDD drugs (premenopausal).

Duloxetin (SAQ)

- SNRI acting on Onuf's nucleus to increase sphincter tone.
- 40 mg twice daily; titrate from 20 mg to reduce nausea.
- Temporary benefit only; approved in Europe, not USA. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Acute urinary retention in females. (SAQ)

- Look for prolapse, sling obstruction, pelvic mass, neurology and drugs.
- Fowler: abnormal sphincter EMG; SNM about 70%.
- CISC for persistent retention. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Section 7 · Voiding Dysfunction

Long answer questions (LAQ)

LAQ 1 Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management

P1 · P2 · P3 **merged from 4**

Asked as: P1 LAQ "Describe pathophysiology of OAB. Briefly outline management &describe nonmedical non-pharmacological management" · P2 SAQ "Overactive bladder and its drug therapy" · P3 SAQ "OAB Pharmacotherapy: Darifenacin vs Solifenacin vs Tolterodine, and Mirabegron" · P1 SAQ "Compare and contrast various detrusor relaxants in management of OAB"

Definition

- **OAB (overactive bladder), ICS (International Continence Society): urgency**, usually with frequency and nocturia, with or without UUI (urgency urinary incontinence), in the absence of UTI (urinary tract infection) or other obvious pathology.
- **OAB dry** versus **OAB wet** (with UUI).
- OAB is a **symptom** diagnosis. DO (detrusor overactivity) is a **urodynamic** finding; only 44–69% of patients with urgency show DO.

Pathophysiology

All mechanisms converge on **more afferent signal reaching awareness**: the bladder turns the signal up, or the CNS (central nervous system) fails to turn it down.

Mechanism	Explanation
Afferent (sensory)	Aδ fibres normally sense filling. Unmyelinated **C fibres** are "silent", but become spontaneously active and hypersensitive in pathological states
Urothelial	Urothelium acts as a transducer: mechanical, inflammatory or chemical stimuli release ATP (adenosine triphosphate), acetylcholine and other mediators onto suburothelial afferents
Myogenic	Patchy denervation causes receptor upregulation and spontaneous contraction; exaggerated **micromotions** cause urgency, and coordinated spread across the wall gives DO
Neurogenic / central	Forebrain and PAG–PMC (periaqueductal grey–pontine micturition centre) normally inhibit voiding; loss of this inhibition lets "afferent noise" be felt as urgency
Other	Bladder–bowel crosstalk (constipation), ischaemia–reperfusion, chronic inflammation, altered urinary microbiome

- **Risk factors and associations:** ageing, female sex, BOO (bladder outlet obstruction: an animal model, but only an indirect cause in humans; DO persists in up to 60% after TURP [transurethral resection of the prostate]), anxiety and depression, metabolic syndrome, poor toileting habits.

Evaluation

- **Minimum:** history, examination, **urinalysis**.

- Bladder diary (3–7 days; P2: 3 days), questionnaires, PVR (post-void residual) when indicated.
- Urodynamics and cystoscopy only for complicated or refractory cases, or if urodynamics will change an invasive decision.

Outline of management (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction]; not a compulsory sequence)

`Education + behavioural therapy (first line) ↓ (4–12 weeks) Antimuscarinic or β3-agonist (mirabegron, vibegron); combinations ↓ PTNS / onabotulinumtoxinA / SNM (third line) ↓ Augmentation cystoplasty or urinary diversion`

Line	Treatment
Behavioural	Education; reduce fluids by about 25%; cut caffeine; weight loss; **bladder training** (timed voids, interval increased by 15–30 minutes); **PFMT (pelvic floor muscle training)** with urge suppression: 50–80% fewer leaks
Drugs	**Antimuscarinics** (oxybutynin, solifenacin, extended-release trospium, darifenacin, fesoterodine, tolterodine); **β3-agonists** (mirabegron, vibegron); combinations
Third line	**PTNS (percutaneous tibial nerve stimulation)** weekly × 12; **onabotulinumtoxinA 100 U** intradetrusor; **SNM (sacral neuromodulation)** at S3 (success about 70–80%; revision >30% at 5 years)
Last resort	Augmentation cystoplasty or urinary diversion

Non-pharmacological management

1. Education and lifestyle

- Explain normal bladder function; set realistic goals.
- Reduce fluid intake by about 25% if excessive; cut caffeine; treat constipation.
- **Weight loss:** about 8 kg loss in obese women reduced UUI episodes.

2. Bladder training (BT) (breaks the cycle of urgency, frequent voiding and smaller capacity)

1. Keep a bladder diary.
2. Start a fixed interval (often 1 hour while awake).
3. Void by the clock, not by urge.
4. Increase by 15–30 minutes a week as control improves.
5. Target interval 3–4 hours. - Works by **uncoupling the void from the urge**.

3. PFMT and urge suppression

- Kegel exercises: quick 1–2 s flicks and 5–10 s holds, twice daily.
- A pelvic floor contraction **reflexly inhibits the detrusor**. Urge suppression: stop, stay still, squeeze quickly several times, relax, let the urge pass, then walk calmly to the toilet.
- PFMT and BT reduce incontinence episodes by **50–80%**.
- **Also know:** biofeedback helps patients who cannot find the muscle; electrical stimulation at 5–20 Hz inhibits the detrusor; timed and prompted voiding for cognitively impaired patients.

4. PTNS

- 34-gauge needle 3–4 cm above the medial malleolus; 20 Hz, 200 µs; confirm great-toe flexion or sole tingling. **30 min weekly for 12 weeks**, then maintenance.

- The posterior tibial nerve (L4–S3) shares sacral roots with the bladder. **SUmiT trial:** 55% vs 21% (sham) responders.

5. SNM

- Mechanism: stimulates **afferents** that modulate brainstem and forebrain control.
- **Test phase:** PNE (percutaneous nerve evaluation, 5–7 days) or **staged tined lead** at **S3** (anal "bellows" and great-toe flexion). ≥50% improvement on diary leads to implanting the pulse generator in the buttock.
- Success about 70–80%. **InSite trial:** 61% vs 42% (standard medical therapy). Revision >30% at 5 years; MRI (magnetic resonance imaging)-conditional rechargeable devices now available.

6. Intradetrusor onabotulinumtoxinA

- Cleaves SNAP-25 (synaptosomal-associated protein 25), blocking acetylcholine release; also reduces afferent signalling.
- **100 U in 10 mL at 20 sites** cystoscopically; effect lasts about 6–9 months.
- About 25% become completely dry. Risks: UTI about 20%, retention needing CISC (clean intermittent self-catheterisation). **ROSETTA:** greater UUI reduction than SNM at 6 months, but more UTI.

7. Last resort: augmentation cystoplasty or urinary diversion.

Drug therapy: place and mechanisms

- Drugs are second line, after behavioural therapy, as first-line pharmacotherapy (antimuscarinics and β3 agonists alongside each other; AUA/SUFU). Two main classes: **antimuscarinics** and **β3-adrenoceptor agonists**.
- **Antimuscarinics:** acetylcholine acts on **M3** receptors (raising intracellular calcium) to contract the detrusor; the commoner M2 enhances it. Blockade reduces involuntary contractions and afferent urgency signalling.
- Muscarinic side effects by subtype: **M1** (brain) cognition; **M2** heart rate; **M3** (glands, gut) dry mouth, constipation.
- Selectivity is by **receptor subtype** (M3) or **tissue** (bladder over salivary gland); tissue selectivity matters more.
- **β3 agonists:** β3 is the commonest (most highly expressed) adrenoceptor in human detrusor. Activation raises **cAMP (cyclic adenosine monophosphate)** and relaxes the detrusor during storage, sparing the voiding contraction.

Agents

Drug	Distinguishing feature	Usual dose
Oxybutynin	Non-selective; direct relaxant; metabolite causes most dry mouth; patch, gel	5 mg two to three times daily; ER (extended-release)
Tolterodine	Bladder (over salivary) selective; QT prolongation at high dose	2 mg twice daily (IR, immediate-release); ER 4 mg daily
Fesoterodine	Prodrug of tolterodine's active metabolite	4–8 mg daily
Solifenacin	10-fold M3 > M2; long half-life	5–10 mg daily
Darifenacin	About 30-fold M3-selective	7.5–15 mg daily

Drug	Distinguishing feature	Usual dose
Trospium	Quaternary; does not cross the blood–brain barrier (little brain entry)	20 mg twice daily
Mirabegron (β3)	Inhibits CYP2D6 (a liver enzyme): metoprolol interaction; small blood-pressure rise	25–50 mg daily
Vibegron (β3)	No CYP2D6 interaction	75 mg daily

Darifenacin vs solifenacin vs tolterodine

Feature	Darifenacin	Solifenacin	Tolterodine
Selectivity	**Most M3-selective (about 30-fold)**; not tissue-selective	M3 about 10-fold over M2; bladder-selective	Non-subtype-selective; **bladder-selective**
Dose	7.5–15 mg daily (ER)	5–10 mg daily	2 mg twice daily (IR); 4 mg daily (ER)
Half-life	Moderate (ER once daily)	**Long** (about 2–3 days): once daily	Short; active metabolite 5-HMT (5-hydroxymethyl tolterodine)
Metabolism	CYP3A4 (a liver enzyme) and CYP2D6	CYP3A4	CYP2D6 (and 3A4)
Dry mouth	Dose-related	Dose-related	Lower with ER
Constipation	**Highest** (gut M3)	Moderate–high	Lower
CNS	Spares M1: **preferred in elderly**	Better persistence	High dose: QT prolongation
Dose limit	Lower dose with CYP3A4 inhibitors or liver disease	5 mg in severe renal or moderate liver impairment	Halve with renal or liver impairment

- **Efficacy is similar**; response is individual, so switch agent or dose.
- **STAR trial:** flexible solifenacin beat tolterodine ER for urgency and incontinence.

Mirabegron (β3 agonist)

- First **β3-adrenoceptor agonist** for OAB; FDA (Food and Drug Administration) approval 2012.
- **Mechanism:** activation of adenylyl cyclase raises **cAMP**, relaxing the detrusor during **storage**. Reduces non-voiding contractions and afferent firing. The voiding contraction (muscarinic M3) is spared, so PVR barely rises.
- **Dose:** oral, extended release, once daily: start **25 mg**, increase to **50 mg** after 4–8 weeks. **25 mg maximum** in severe renal impairment (eGFR [estimated glomerular filtration rate] 15–29) or moderate hepatic impairment; avoid in end-stage renal disease and severe hepatic impairment.
- **Moderate CYP2D6 inhibitor:** raises levels of metoprolol, desipramine, flecainide; monitor digoxin.
- **Efficacy:** phase III trials (SCORPIO, ARIES, CAPRICORN): fewer incontinence episodes and micturitions than placebo. **TAURUS** (12 months): efficacy similar to tolterodine, with less dry mouth.
- **Adverse effects:** hypertension, tachycardia, headache, nasopharyngitis, UTI. **Dry mouth and constipation near placebo rates**, far below antimuscarinics.

- **Contraindicated:** severe uncontrolled hypertension (≥180/110 mmHg). Caution with arrhythmia, and with BOO when combined with antimuscarinics (retention).
- Other uses: add-on to α-blocker in men with BPH (benign prostatic hyperplasia); neurogenic detrusor overactivity.
- **Also know:** vibegron 75 mg (no CYP2D6 interaction).

Antimuscarinics versus β3 agonists

Feature	Antimuscarinics	β3 agonists
Efficacy	Similar	Similar (TAURUS: mirabegron ≈ tolterodine)
Dry mouth, constipation, blurred vision	Common; main reason for stopping	Near placebo
Cognition	Central M1 blockade; **dementia risk 10–65% higher with use >3 months**	No anticholinergic burden
Retention	Up to 3% in men	Low
Contraindications	Urinary or gastric retention, uncontrolled narrow-angle glaucoma	Uncontrolled (severe) hypertension (≥180/110 mmHg)

Choice of drug and special groups

- **Elderly and cognitive impairment:** antimuscarinic use >3 months is linked to 10–65% more new-onset dementia (SUFU white paper). Prefer a **β3 agonist** (no anticholinergic burden), or ER trospium, darifenacin or fesoterodine. Beers criteria: avoid antimuscarinics in dementia.
- ER forms cause less dry mouth. Poor persistence is the main problem; switch dose or drug before third-line therapy.
- **Combination:** solifenacin plus mirabegron beats each alone (**SYNERGY, SYNERGY II**), without extra cardiovascular effects; in non-responders it equals solifenacin 10 mg with less dry mouth (**BESIDE**).
- **Men:** α-blocker first; add an antimuscarinic (if PVR is low) or β3 agonist; retention up to 3%, so check PVR.
- **Also know:** vaginal oestrogen in atrophy; desmopressin for nocturnal polyuria; imipramine (children, mixed incontinence); intravesical oxybutynin; refractory OAB: onabotulinumtoxinA 100 U and neuromodulation.

Diagrams to draw

- Bladder showing urothelial afferents, C fibres and micromotions, with brainstem and forebrain (cortical) inhibitory pathways.
- Detrusor cell: acetylcholine → M3 → contraction; noradrenaline → β3 → cAMP → relaxation.
- Muscarinic subtypes M1 (brain), M2 (heart), M3 (bladder, gland, gut) with each drug's selectivity.

Key points to remember:

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- Behavioural therapy first: BT and PFMT with urge suppression give 50–80% fewer leaks; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP for storage-phase relaxation; voiding spared.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- STAR: solifenacin beat tolterodine ER.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: mirabegron (β3), trospium or darifenacin preferred over other antimuscarinics.
- Solifenacin + mirabegron combination works (SYNERGY).
- Refractory OAB: PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

LAQ 2 Neuromodulation for voiding dysfunction, including sacral neuromodulation

P2 · P3 · P4 **merged from 3**

Asked as: P4 LAQ "Neuromodulation for voiding dysfunction, including sacral neuromodulation" · P2 SAQ "Neuromodulation" · P3 SAQ "Sacral neuromodulation"

Taken as neuromodulation for lower urinary tract dysfunction.

Definition and history

- **Neuromodulation:** electrical stimulation of a peripheral nerve that alters the activity of central (reflex) pathways controlling the bladder. It acts on the nervous system (**afferents**), not directly on muscle.
- Direct muscle stimulation failed (painful, local). Stimulating **afferents** needs less current and affects the whole organ.
- **SNM (sacral neuromodulation)** (Tanagho and Schmidt, about 1985): continuous low-amplitude **S3** stimulation through a tined lead and IPG (implantable pulse generator). PTNS (percutaneous tibial nerve stimulation): FDA (Food and Drug Administration) approval 2000.
- **Rule:** exclude anatomical causes first (obstruction, stones, cancer, fistula). Stimulation cannot correct anatomy or restore muscle.

Mechanism

- Sacral roots are mixed nerves. Current excites **afferents to the spinal cord and forebrain** (areas of awareness and attention), altering PMC (pontine micturition centre) output. It acts on **afferent** nerves, not motor nerves.
- **OAB (overactive bladder):** heightened central control (afferent input) damps PMC firing, so there are fewer involuntary contractions.
- **Non-obstructive retention:** restores awareness of filling (sensation) and pelvic floor tone, so the PMC switches on and the **EUS (external urethral sphincter) relaxes**. Hence one therapy treats **both** opposite disorders.
- **Tibial:** the nerve carries **S2–S3** fibres; inhibition of OAB-type (C-fibre) reflexes is **opioid-mediated** (naloxone blocks it); long-term cortical plasticity.

Modalities

Modality	Target and method	Main use
SNM	Tined lead in **S3** foramen + implanted generator	Refractory OAB, NOUR (non-obstructive urinary retention), faecal incontinence
PTNS	34-gauge needle above medial malleolus	Refractory OAB
TTNS (transcutaneous tibial)	Surface patch, home use	OAB (less effective)
Implantable tibial	Small subcutaneous device over tibial nerve	OAB, avoids clinic visits
Pudendal	Tined lead in Alcock's canal	IDO (idiopathic detrusor overactivity), NDO (neurogenic DO), SNM failure (off-label)

Modality	Target and method	Main use
DGNS (dorsal genital nerve stimulation)	Dorsal penile/clitoral nerve	Experimental; NDO
IVES (intravesical electrical stimulation)	Intravesical cathode	Underactive bladder in children
SARS (sacral anterior root stimulator)	Anterior S2–S4 roots + dorsal rhizotomy	Complete SCI (spinal cord injury)

A. Sacral neuromodulation: indications

- **Third line** for OAB after behavioural and drug therapy (AUA [American Urological Association] and EAU [European Association of Urology] agree).

FDA approved	Not approved
Refractory UUI (urgency urinary incontinence) 1997	Bladder pain syndrome
Urgency–frequency 1999	Neurogenic bladder (small studies positive; low-level evidence)
Non-obstructive urinary retention (e.g. Fowler's syndrome) 1999	Pelvic pain
Faecal incontinence 2011	

- **Fowler's syndrome**: young women in retention with abnormal sphincter EMG (electromyography) activity; responds well to SNM.
- Retention needs some detrusor function; success falls with age.

SNM: technique

Test phase (with a bladder diary)

Test phase	PNE (percutaneous nerve evaluation)	Staged tined lead
Lead	Temporary, office, local anaesthesia (dislodges)	Permanent tined lead, theatre, fluoroscopy
Duration	5–7 days	Up to 4 weeks
Use	OAB	More reliable; higher conversion; preferred for retention (NOUR)

- **Landmark:** prone; needle into S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

Root	Pelvic response	Foot response
S2	—	Calf and foot (external) rotation
S3 (target)	**Bellows**, perineal sensation	**Great-toe flexion**
S4	Bellows/perineal only	None

- **Implant if ≥50% improvement** in a diary parameter (retention: ≥50% fewer catheterisations); IPG in the buttock.
- **Also know:** 10–16 Hz at sensory threshold; bilateral stimulation no better.

SNM: outcomes

- **InSite:** success 61% vs 42% with standard medical therapy at 6 months; about 70–80% success at 5 years (another source: about half keep benefit at ≥5 years).
- **ROSETTA:** onabotulinumtoxinA 200 U reduced UUI slightly more (−3.9 vs −3.3 episodes/day) with higher satisfaction, but caused more UTI (urinary tract infection) (24% vs 10%) and CIC (clean intermittent catheterisation, self-catheterisation); SNM removal 9% at 2 years.
- NOUR: most stop catheterising.

SNM: complications and contraindications

- Pain at the IPG site, lead migration, loss of effect, infection; **revision over 30% at 5 years**; costly (costliest option); battery replacement.
- Contraindications: failed test, inability to use the programmer, sacral anomaly.

B. Tibial nerve stimulation

- **PTNS technique:** 34-gauge needle **3–4 cm (three fingerbreadths) above the medial malleolus**, behind the tibia; ground pad on the same leg; 20 Hz, 0–10 mA. Confirm by **toe flexion/fanning** or sole tingling. **30 min weekly × 12 weeks**, then maintenance.
- **SUmiT** (n = 220, validated sham): response **55% vs 21%**. Extension: 77% relapse-free at 3 years with tapering maintenance.
- **OrBIT:** PTNS comparable to extended-release tolterodine, with fewer side effects.
- **TTNS:** home self-treatment; skin impedance limits amplitude, so less effective.
- **Implantable devices** remove clinic visits: **BlueWind** (miniature, external control; benefit sustained at 3 years); **eCoin** (coin-sized, battery-powered, office insertion under local anaesthesia).

C. Other targets

- **Pudendal nerve stimulation:** tined lead via a posterior approach to Alcock's canal. Better than SNM in a blinded crossover; **salvage after failed SNM**. Off-label.
- **DGNS:** purely sensory; acute DO suppression in SCI; **conditional (on-demand)** stimulation possible. Electrode fixation is difficult.
- **IVES (Katona):** intravesical cathode in saline activates Aδ afferents, producing detrusor contraction. Needs some afferent and muscle function; daily sessions; used in children; the sham-controlled trial was negative.
- **Brindley SARS:** stimulates anterior S2–S4 roots for voiding, defecation and erection. **Dorsal rhizotomy** abolishes reflex incontinence and DSD (detrusor–sphincter dyssynergia). The sphincter relaxes faster than the detrusor between bursts, giving **post-stimulus voiding**. Needs intact efferents.
- **Also know:** intravaginal stimulation (SUI, stress urinary incontinence); sacral dermatome stimulation; closed-loop SNM.

Comparison of third-line OAB options

	PTNS	OnabotulinumtoxinA 100 U	SNM
Invasiveness	Lowest	Moderate	Implant
Drawback	12 weekly visits	UTI, retention/CIC	Revision, cost
Retention	No role	Worsens	**Treats**

Recent advances

- Rechargeable stimulators (FDA 2019; ≥15 years); non-rechargeable devices lasting ≥10 years (2021).
- **Full-body (whole-body) MRI (magnetic resonance imaging)-conditional** systems at 1.5 T and 3 T.
- **Implantable tibial nerve stimulators** (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives; PTNS and pudendal stimulation as alternatives to SNM.
- Standardised tined-lead placement improves outcomes and lowers revision.
- Bilateral stimulation no better.

Diagram to draw

- Lead in the S3 foramen with IPG in the buttock, and PTNS needle above the medial malleolus.

> **Key points to remember:**
> - Afferent modulation explains SNM benefit in both OAB and retention.
> - S3: bellows plus great-toe flexion; implant if the test improves ≥50%.
> - SNM revision >30% at 5 years.
> - SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
> - ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.
> - Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
> - Full-body MRI-conditional devices at 1.5 T and 3 T.
> - Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
> - Standardised tined-lead placement improves outcomes and lowers revision.

LAQ 3 Pathophysiology, classification and management of LUTS.

P2

Type: LAQ (25 marks)
Source: 2024-Winter Paper 2, Q1 (medical and surgical management of BPH also answered in 2015-Summer Paper 2, Q1)

Definition

- **LUTS (lower urinary tract symptoms):** symptoms of the storage, voiding and post-micturition phases; not disease-specific; up to half of older men.
- **BPH (benign prostatic hyperplasia)** is histology; BPE is enlargement; BPO is prostatic obstruction; **BOO (bladder outlet obstruction)** has any cause. Not every man with LUTS has BPO.

Classification

1. By phase (ICS, International Continence Society)

Storage	Voiding	Post-micturition
Frequency, urgency, **nocturia** (waking ≥1 time to void), urgency incontinence	Weak or split stream, hesitancy, intermittency, straining, terminal dribble	Feeling of incomplete emptying, post-micturition dribble

- Storage symptoms form **OAB (overactive bladder)**: urgency ± urgency incontinence, usually with frequency and nocturia; most bothersome.

2. Functional (Wein)

	Because of the bladder	Because of the outlet
Failure to store	DO (detrusor overactivity), low compliance, hypersensitivity	Sphincter weakness (SUI, stress urinary incontinence)
Failure to empty	DUA (detrusor underactivity)	BPO, stricture, bladder neck contracture, sphincter dyssynergia

3. Severity: IPSS (International Prostate Symptom Score), 7 questions scored 0–5 plus a quality-of-life question: mild 0–7, moderate 8–19, severe 20–35.

4. By cause: prostate, bladder, urethra, neurological, polyuria, infection, stone, tumour.

Pathophysiology

- **Prostate:** BPH is an increase in cell number of stroma and glands in the **transition zone** and periurethral region. **Type 2 5α-reductase** in stromal fibroblasts makes **DHT (dihydrotestosterone)**; androgens are permissive, not mitogenic. Ageing, inflammation and metabolic syndrome contribute.
- **Static component:** gland bulk; the **capsule** transmits expansion to the urethra (TUIP, transurethral incision of the prostate, relieves obstruction without removing tissue).
- **Dynamic component: α1A-adrenergic** smooth-muscle tone; the basis of α-blockers.
- Size does not correlate with symptoms or obstruction.
- **Bladder response:** hypertrophy → compensation → decompensation. Collagen causes trabeculation and low compliance; DO causes storage symptoms; late detrusor failure causes retention. One-third keep storage symptoms after surgery.

- **Also know:** nocturnal polyuria (night output >33% of 24 h over age 65); BPO complications: AUR (acute urinary retention), UTI (urinary tract infection), stones, haematuria, renal failure.

Evaluation (EAU, European Association of Urology, men ≥40)

- **Basic:** history (comorbidity, drugs, sexual function), IPSS, DRE (digital rectal examination) and neurological check, **urinalysis (mandatory)**, PVR (post-void residual), PSA (prostate-specific antigen) if life expectancy >10 years.
- **Frequency–volume chart, 3 days:** for storage symptoms or nocturia.
- **Before treatment:** uroflowmetry (Qmax, maximum flow rate, <10 mL/s suggests BOO); ultrasound for volume and median lobe.
- **Selected:** creatinine and renal ultrasound if large PVR; cystoscopy for haematuria, stricture or before PUL (prostatic urethral lift).
- **Pressure–flow study** (separates BOO from DUA) before surgery if voided volume <150 mL, PVR >300 mL, age <50 or >80, or failed surgery.

Management: EAU algorithm

```
Bothersome LUTS? No -> watchful waiting, lifestyle advice Absolute surgical indication? Yes -> surgery Otherwise: Nocturnal polyuria -> lifestyle +/- desmopressin Storage predominant -> antimuscarinic or beta3-agonist Prostate >40 mL -> 5ARI (5-alpha-reductase inhibitor) +/- alpha1-blocker Others -> alpha1-blocker or PDE5 inhibitor Residual storage -> add antimuscarinic or beta3-agonist Failure or complication -> surgery
```

- **Watchful waiting:** education, fluid 1.5–2 L/day (none 2 h before bed), less caffeine and alcohol, double voiding, bladder training; yearly review.

Drug	Example	Mechanism and notes
α1-blocker	Tamsulosin 0.4 mg, silodosin 8 mg, alfuzosin 10 mg	Relax prostatic smooth muscle; acts in days; dizziness, ejaculatory dysfunction, IFIS (intraoperative floppy iris syndrome)
5ARI	Finasteride 5 mg, dutasteride 0.5 mg	Lower DHT; volume falls about 20% over 6–12 months; **PSA halves**; reduces AUR and surgery
Antimuscarinic	Solifenacin, tolterodine	Block detrusor M3 receptors; for storage symptoms if PVR <200 mL
β3-agonist	Mirabegron 50 mg, vibegron	Relax detrusor during storage
PDE5I (phosphodiesterase-5 inhibitor)	Tadalafil 5 mg daily	Smooth-muscle relaxation; useful with ED (erectile dysfunction)

- **MTOPS:** progression cut by doxazosin 39%, finasteride 34%, combination 66%; **CombAT** confirmed. Combine when volume >30–40 mL and PSA >1.5 ng/mL.

Surgery

- **Indications:** refractory retention, recurrent UTI, bladder stones, recurrent haematuria, BPO-related renal insufficiency; failed medical therapy.

Prostate volume	Options
<30 mL	TUIP, TURP (transurethral resection of the prostate)
30–80 mL	**TURP (reference standard)**, bipolar TURP, PVP (photoselective vaporisation of the prostate), enucleation, PUL, Rezum (water vapour), Aquablation
>80 mL	**HoLEP (holmium laser enucleation)** or thulium enucleation, open or robotic simple prostatectomy

- Ejaculation preserved: PUL, Rezum. Anticoagulated: PVP, HoLEP. PAE (prostatic artery embolisation) in selected men.
- **TURP complications:** bleeding, **TUR syndrome** (hyponatraemia with glycine; avoided by bipolar saline), retrograde ejaculation, stricture, incontinence.
- **Also know:** iTind; female LUTS follow OAB and SUI guidelines.

Diagram to draw: BPH → BPE → BPO → BOO → bladder changes → LUTS.

> **Key points to remember:**
> - ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
> - Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
> - 5ARI for >40 mL; MTOPS combination cut progression 66%.
> - TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

LAQ 4 Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male

P2

Type: LAQ (25 marks)
Source: Practice Set 1 Paper 2, Q2

Assumed meaning: cervical spinal cord injury (traumatic, operated, or from the surgery itself), giving a suprasacral lesion above T6.

Introduction

- A cervical cord lesion lies **between the PMC (pontine micturition centre) and the sacral cord**. It causes tetraplegia and the classic suprasacral pattern.
- Complete transection is rare; incomplete lesions are commonest. Manage by **urodynamics, not neurology**.

Pathophysiology

- The PMC normally coordinates detrusor contraction with sphincter relaxation. Cutting the descending pathway leaves a **spinal reflex**.
- **Why NDO (neurogenic detrusor overactivity) appears:** unmyelinated C-fibre afferents become sensitised (NGF (nerve growth factor)), sprout and drive a new sacral reflex. Bulbospinal inhibition is lost.

Phase	Detrusor	Outlet	Clinical
Spinal shock (usually 6–12 weeks; up to 1–2 years)	Areflexic, smooth bladder	Bladder neck closed; reduced closure pressure	**Retention**; overflow only
Recovery (suprasacral pattern)	**NDO**, no sensation below lesion	**Striated DSD (detrusor–sphincter dyssynergia)**; lesion above T7–T8 adds **smooth sphincter dyssynergia**	Reflex leakage, incomplete emptying, high pressure

- **Blaivas DSD types:** type 1, sphincter relaxes at peak pressure (incomplete lesions); type 2, sporadic contractions; type 3, crescendo–decrescendo obstruction (types 2–3 with complete lesions).
- **Consequences:** high storage and voiding pressure, VUR (vesicoureteric reflux) in 17–25%, hydronephrosis, stones (about 7% in 10 years), UTI (urinary tract infection). Untreated, about half of men with DSD deteriorate (worse if complete, continuous DSD).
- Upper tract risk also rises with low compliance (<12.5 mL/cm H_2O) and NDO for over a third of filling.
- **VUR:** treat the bladder first (lower the pressure); reflux surgery in a thick bladder is difficult.
- **Also know:** cervical myelopathy may instead give an underactive detrusor.

Autonomic dysreflexia

- Occurs with lesions **above T6** (cervical 60%), after spinal shock. Bladder or bowel distension triggers unopposed spinal sympathetic outflow, as medullary inhibition is lost.
- **Features:** pounding headache, severe hypertension (systolic rise ≥20 mmHg defines it at urodynamics), bradycardia, sweating and flushing above the lesion. It can cause stroke or

seizures.

- **Treatment:** sit up, **drain the bladder** (check catheter), then **nitroglycerin 2% paste** above the lesion; intravenous nitroprusside or labetalol if persistent. Prophylaxis with terazosin or prazosin. Monitor blood pressure during cystoscopy and urodynamics; use spinal or general anaesthesia.

Evaluation

- History, examination, **hand function** (decides self-catheterisation), transfers, carers.
- Urinalysis, creatinine, **renal ultrasound**. Men with good hand function can self-catheterise; others need carers or a suprapubic catheter.
- **VUDS (videourodynamics) after spinal shock**, with blood pressure monitoring: compliance, NDO, DSD, VUR, **DLPP (detrusor leak point pressure) ≥40 cm H_2O** = upper tract risk.
- AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2021: suprasacral lesions are "unknown risk" until tested; DSD or poor compliance = high risk.

Management

Goals: protect kidneys, low-pressure storage, low-pressure emptying, no infection, continence, social acceptability. Start with the simplest reversible option.

- **Spinal shock: CIC (clean intermittent catheterisation)** preferred; indwelling catheter early is acceptable. Defer irreversible procedures.

Aim	Options
Emptying	**CIC: gold standard**, volumes <500 mL; carer CIC in tetraplegics. **Suprapubic catheter** better than long-term urethral catheter. Credé/Valsalva not recommended
Storage (lower pressure)	**Antimuscarinics** ± mirabegron (β3 agonist); **intradetrusor onabotulinumtoxinA 200 U** if refractory (repeat when the effect wears off)
Refractory small bladder	**Augmentation ileocystoplasty** ± continent catheterisable channel (Mitrofanoff)
Men unable to do CIC	**External sphincterotomy** or intrasphincteric botulinum toxin → reflex voiding into condom catheter. High failure; DLPP ≥40 cm H_2O afterwards = failure. Urethral stents abandoned
Neuromodulation	**SARS (sacral anterior root stimulator, Brindley) + posterior sacral rhizotomy**: abolishes NDO and dysreflexia, allows voiding on demand
Last resort	Ileovesicostomy; bladder neck closure + suprapubic catheter; ileal conduit

- **Also know:** sacral neuromodulation is used for neurogenic bladder without SCI (spinal cord injury); hydrophilic catheters delay first UTI.

Complications and follow-up

- **UTI:** treat only symptomatic bacteriuria (5–14 days); no routine prophylaxis.
- Stones, VUR (treat the bladder first), bladder cancer (no screening cystoscopy, AUA/SUFU 2021).
- Lifelong: yearly review and renal function, upper tract imaging every 1–2 years, repeat urodynamics if anything changes.

`Cervical SCI → spinal shock (retention): CIC → VUDS at 3 months: NDO + DSD → antimuscarinic + CIC → Botox → augmentation → cannot CIC: sphincterotomy + condom / SARS / ileovesicostomy`

Diagram to draw: neural axis: PMC, lesion at cervical cord, sympathetic T10–L2, sacral S2–S4, with the resulting NDO + DSD.

> **Key points to remember:**
> - Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
> - Autonomic dysreflexia above T6: drain bladder first, nitrates next.
> - CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
> - DLPP ≥40 cm H_2O threatens the kidneys.

Short answer questions (SAQ)

SAQ 1 Botulinum toxin in urology

P1 · P2 · P3 · P4 **merged from 4**

Asked as: P1 SAQ "Botulinum toxin in urology: mechanism, indications and doses" · P2 SAQ "Botulinum neurotoxin and its role in urology" · P3 SAQ "Botulinum toxin in urology" · P4 SAQ "Botulinum toxin in urology"

"Botulimum" / "Botulinim" in the questions is read as botulinum toxin.

Introduction

- Neurotoxin of *Clostridium botulinum*, serotypes A–G. Urology uses **onabotulinumtoxinA (BoNT-A)**, the only licensed formulation; type B is short-acting.
- 150 kDa: a heavy chain (100 kDa; binding and entry) and a zinc-dependent light chain (50 kDa).

Mechanism of action

1. The heavy chain binds the cholinergic nerve terminal and the toxin is internalised.
2. The light chain (a zinc protease) enters the cytosol.
3. It **cleaves SNAP-25** (synaptosomal-associated protein 25), a SNARE (vesicle-docking) protein. Vesicles cannot fuse, so **acetylcholine release stops**: chemical denervation. - **Afferent action:** less ATP (adenosine triphosphate), substance P and CGRP (calcitonin gene-related peptide) release; fewer TRPV1 (capsaicin) and P2X3 (purinergic) sensory receptors on C fibres, so urgency and pain fall. - The effect lasts about 6–9 months and wanes as nerve terminals sprout, so injections are repeated; repeat injections stay effective and response is maintained.

Technique

- Flexible or rigid cystoscopy under local anaesthesia; diluted toxin injected into the detrusor at **about 20 sites (100 U, in 10 mL) or 30 sites (200 U, in 30 mL)**, usually sparing the trigone (reflux concern).
- Sterile urine, antibiotic cover; check PVR (post-void residual) at 2 weeks.

Urological uses: indications and doses

Indication	Dose and evidence
Idiopathic OAB (overactive bladder), third line	**100 U** (EMBARK): incontinence down about 59%, about 25% dry. **ABC** trial: equal to antimuscarinic, more complete dryness. **ROSETTA**: comparable to SNM (sacral neuromodulation), more UTI (urinary tract infection) (P2, P3: 200 U used, equal to or ≥ SNM; another source: slightly better than SNM)
NDO (neurogenic detrusor overactivity): SCI (spinal cord injury), MS (multiple sclerosis)	**200 U** (DIGNITY trials): less incontinence, larger capacity, lower detrusor pressure; works with or without antimuscarinics
Paediatric NDO (spina bifida)	About **10 U/kg (maximum 360 U)**, about 30 sites, in published (Campbell) series; FDA (Food and Drug Administration) label (2021, age ≥5) 6 U/kg, maximum 200 U; continence 65–87%, most avoid augmentation
DSD (detrusor–sphincter dyssynergia), underactive detrusor	Intrasphincteric injection 50–100 U, mainly in men unable to do CISC (clean intermittent self-catheterisation)
IC/BPS (interstitial cystitis/bladder pain syndrome)	100 U intradetrusor, often trigonal; AUA (American Urological Association) option, off-label
BPH (benign prostatic hyperplasia)	100–300 U intraprostatic: an early small trial was positive, but large placebo-controlled RCTs (randomised controlled trials) showed **no benefit over placebo (saline/sham)**; not recommended

- **Also know:** pelvic floor spasm; paediatric dysfunctional voiding (sphincter); chronic pelvic pain; chronic orchialgia (RCT negative); ED (erectile dysfunction) trials. Units of different toxin products are not interchangeable.

Complications

- **Retention needing CISC** (the main risk): 0–12% in OAB trials, but about 20% of women and up to 35% of men in practice; commoner in NDO. Each 10 mL rise in baseline PVR raises the risk about 27%. **The patient must accept CISC beforehand.**
- **UTI** about 20%; haematuria; rarely generalised weakness.

Contraindications

- Active UTI, myasthenia gravis, aminoglycosides, pregnancy, unwillingness to self-catheterise.

Recent advances

- Paediatric NDO: FDA approved onabotulinumtoxinA for children aged ≥5 years (2021); used from about 2 years of age.
- Other formulations: abobotulinumtoxinA and incobotulinumtoxinA under study for detrusor overactivity; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

Diagram to draw

- Cholinergic synapse (nerve terminal) showing SNAP-25 cleavage by the light chain.

Key points to remember:

- Cleaves SNAP-25; acts on efferent and afferent nerves.
- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U), approved for age ≥5 years (2021); used from about 2 years of age.
- Main risks: retention needing CISC, UTI.
- Other formulations (abobotulinumtoxinA, incobotulinumtoxinA) under study; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

SAQ 2 Urodynamic studies, including video urodynamics

P2 · P4 **merged from 3**

Asked as: P2 SAQ "Uro Dynamic Studies." · P2 SAQ "Vesicourodynamics" · P4 SAQ "Video Urodynamics"

"Vesicourodynamics" is taken as videourodynamics (VUDS): urodynamics with simultaneous fluoroscopic imaging of the bladder and urethra.

Definition and principles

- UDS (urodynamic studies) test bladder and urethral function in storage and voiding.
- ICS (International Continence Society): **decide the question first** and **reproduce the symptom**; a normal study does not exclude disease.
- Before invasive UDS: symptom score, 3-day **frequency–volume chart**, free uroflow and PVR (post-void residual); treat UTI (urinary tract infection) first.

Components of conventional UDS

Test	Method	Main values
Uroflowmetry	Voided volume ≥150 mL	Bell curve normal; Qmax <10 mL/s (men) suggests obstruction or weak detrusor; plateau = stricture
PVR	Bladder scan or catheter	Emptying
Filling cystometry	Bladder and rectal lines, fill with saline 30–100 mL/min (adults); **Pdet = Pves − Pabd**; zero at atmosphere, transducer at symphysis	Sensation, capacity, **DO (detrusor overactivity)**, **compliance** = ΔV/ΔPdet (<20 mL/cmH_2O impaired)
Leak point pressures	ALPP (abdominal LPP): cough or Valsalva at 150 mL	Low ALPP = weak sphincter (ISD, intrinsic sphincter deficiency)
	DLPP (detrusor LPP)	**>40 cmH_2O** risks upper tracts (McGuire)
Pressure–flow study	Void with lines in	High pressure–low flow = obstruction; low–low = underactivity
EMG (electromyography)	Perineal surface electrodes	DESD (detrusor–external sphincter dyssynergia)

Test	Method	Main values
UPP (urethral pressure profile)	Catheter withdrawn	Limited value
Videourodynamics	Above with contrast and fluoroscopy	VUR (vesicoureteric reflux), bladder neck, site of obstruction

Indices (men)

- **BOOI (bladder outlet obstruction index)** = PdetQmax − 2Qmax: **>40 obstructed**, 20–40 equivocal, <20 unobstructed.
- **BCI (bladder contractility index)** = PdetQmax + 5Qmax: **>150 strong**, 100–150 normal, **<100 weak**.
- Women: no validated nomogram (Nitti video criteria).

Indications for UDS

- Male LUTS (lower urinary tract symptoms) before surgery when unclear (young men, equivocal flow, failed TURP [transurethral resection of the prostate]); complex, mixed or recurrent incontinence.
- **Neurogenic bladder** (spinal injury, MS [multiple sclerosis], spina bifida), radiation cystitis, before augmentation.
- Children: neurogenic bladder, PUV (posterior urethral valves); fill at **5–10% of expected capacity per minute**; expected capacity = (age + 2) × 30 mL.

Video urodynamics (VUDS): definition and set-up

- **VUDS:** multichannel urodynamics (pressure–flow, with or without EMG) recorded at the same time as fluoroscopy, using **contrast as the filling fluid**.
- Vesical and rectal catheters (Pdet [detrusor pressure] = Pves − Pabd), flowmeter, EMG; C-arm (most flexible) or a tilting table.
- Position for the question: **SUI (stress urinary incontinence) views standing**; voiding in the usual position (women sitting, men standing).
- **Screen only at key moments** (provocation, pressure rise, voiding) to limit radiation.

What fluoroscopy adds

Situation	VUDS finding
NLUTD (neurogenic lower urinary tract dysfunction)	**Gold standard** (EAU); DESD and **DISD** (detrusor–internal sphincter dyssynergia, seen only on VUDS); trabeculation, diverticula
VUR	Pressure, volume and grade at onset; reflux "pops off" pressure, so compliance looks falsely safe
Bladder neck	Only way to see and assess it; **PBNO** (primary bladder neck obstruction) in young men and women
Female obstruction	**Nitti criteria:** radiographic narrowing between bladder neck and meatus during voiding, with a sustained detrusor contraction
Dysfunctional voiding	"Spinning-top" urethra at the external sphincter
SUI	Leakage with ALPP; open bladder neck at rest suggests ISD
Male obstruction	Localises prostatic obstruction, bladder neck contracture or stricture

- **PBNO types (young men):** I high pressure, low flow; II normal pressure, low flow with bladder-neck narrowing; III delayed bladder-neck opening.

Indications for VUDS

- Known or suspected NLUTD; unexplained retention in women; young men with LUTS; complex incontinence; prior radical pelvic surgery or radiation; diversion; renal transplant.
- **Children:** PUV, spinal dysraphism; contrast-enhanced ultrasound VUDS avoids radiation.

Urodynamic risk factors (upper-tract danger)

- Impaired compliance; DESD; DISD; high-pressure DO throughout filling; DLPP >40 cmH_2O; poor emptying with high storage pressure.

Limitations of VUDS

- Radiation, cost, space, availability, contrast allergy.
- EAU (European Association of Urology): optimum test in NLUTD. AUA (American Urological Association) guideline: lowest radiation dose that answers the question.

Also know

- Ambulatory UDS; bethanechol test; antibiotic prophylaxis for risk groups.

Diagrams to draw

- Normal trace: Pves, Pabd, Pdet, flow, EMG.
- VUDS trace beside a voiding cystourethrogram image.

> **Key points to remember:**
> - Pdet = Pves − Pabd.
> - Ask a question first; reproduce the symptom.
> - BOOI >40 obstructed; BCI <100 weak.
> - DLPP >40 cmH_2O: upper-tract risk.
> - VUDS = urodynamics + fluoroscopy with contrast.
> - VUDS is the only test for bladder neck dysfunction, PBNO and DISD.
> - Reflux pop-off makes compliance look falsely good.

SAQ 3 Uroflowmetry

P2 · P3 **merged from 2**

Asked as: P2 SAQ "Uroflowmetry." · P3 SAQ "Uroflowmetry"

Definition

- A non-invasive urodynamic test that records the **volume of urine voided per unit time** (mL/s). It reflects detrusor contraction and outlet resistance together.

Method

- Patient comes with a **comfortably full bladder** (normal desire) and voids in private, in the usual (preferred) position.
- **Flowmeters:** weight transducer (load cell), rotating disc, capacitance dipstick.
- **Voided volume ≥150 mL** (Qmax is unreliable below 125–150 mL). Repeat 2–3 times; ask whether the void was typical and compare with the frequency–volume chart.

- Measure **PVR** (post-void residual) by ultrasound afterwards.

Parameters

Parameter	Meaning
Qmax (maximum flow rate)	Most useful (key) value; normal men >15 mL/s, women >20 mL/s; read from the smoothed curve, ignoring spikes
Qave (average flow rate)	Volume ÷ flow time; less useful and less accurate for obstruction
Voided volume (VV)	Must be ≥150 mL
Flow time vs voiding time	Voiding time includes interruptions
Time to Qmax	Normally early, in the first third of the void
Voiding efficiency	VV ÷ (VV + PVR) × 100

Curve patterns (ICCS, International Children's Continence Society)

Pattern	Suggests
Bell-shaped	Normal
Tower (high, short)	Overactive bladder
Plateau (flattened, box, prolonged)	Fixed obstruction: **urethral stricture**, PUV (posterior urethral valves)
Prolonged, low, skewed	BPO (benign prostatic obstruction) or detrusor underactivity
Staccato (never reaches zero)	Sphincter overactivity, dysfunctional voiding
Intermittent / interrupted (reaches zero), straining	Detrusor underactivity with abdominal straining, obstruction, or DSD (detrusor–sphincter dyssynergia)

Interpretation

- **Qmax <10 mL/s:** obstruction likely (high specificity). **Qmax <15 mL/s:** more sensitive, less specific.
- **Nomograms / volume correction: Siroky** (men), **Liverpool** (men and women, centiles). Corrected Qmax = Qmax ÷ √VV.
- **Artefacts:** "wag" (moving stream), knocking the flowmeter, straining spikes. Read Qmax from the smoothed curve, not single spikes.
- Children: valid if voided volume ≥50% of expected capacity; accept Qmax only if the peak lasts ≥2 s.

Limitations

- **Cannot separate BOO (bladder outlet obstruction) from DUA (detrusor underactivity)**; only a pressure–flow study can.
- Varies with volume, position, anxiety and between days.

Uses

- BPH (benign prostatic hyperplasia): evaluation before active treatment and follow-up after medical or surgical treatment.
- Follow-up after **urethroplasty** or optical urethrotomy (stricture recurrence).

- Children: dysfunctional voiding, BBD (bladder and bowel dysfunction), after posterior urethral valve ablation; with patch EMG (electromyography).
- Women: voiding dysfunction, including before stress incontinence surgery.
- Pressure–flow study before BPH surgery if Qmax >10 mL/s (P2: EAU [European Association of Urology]; P3: AUA [American Urological Association], "may be done").

Diagram to draw

- Flow curves: normal bell, tower, plateau (stricture), prolonged low (BPO), interrupted/intermittent.

> **Key points to remember:**
> - Voided volume ≥150 mL; Qmax is the key value.
> - Plateau = fixed obstruction (stricture); staccato = sphincter; interrupted = weak detrusor or straining.
> - Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.
> - Cannot distinguish BOO from DUA.

SAQ 4 Detrusor underactivity and its management

P2 · P4 **merged from 2**

Asked as: P2 SAQ "Management of detrusor underactivity" · P4 SAQ "Detrussor underactivity."

Definition

- **DUA (detrusor underactivity), ICS (International Continence Society) 2002:** "a contraction of reduced strength and/or duration, resulting in prolonged bladder emptying and/or failure to achieve complete bladder emptying within a normal time span."
- **Acontractile detrusor:** no contraction.
- **UAB (underactive bladder):** the symptom complex of slow stream, hesitancy and straining, with or without incomplete emptying.
- DUA is common in elderly men and women having urodynamics: found in 40–48% of men over 70.

Pathophysiology and causes

- Failure may be **myogenic** (myocyte loss, fibrosis), **efferent** (poor nerve activation) or **afferent** (reduced sensation cannot sustain the voiding reflex).

Type	Examples
Neurogenic	Cauda equina, disc prolapse, spinal injury, diabetic neuropathy, multiple system atrophy, early stroke, radical pelvic surgery
Myogenic	Long-standing BOO (bladder outlet obstruction), diabetes, ischaemia, ageing
Drugs	Anticholinergics, opioids
Functional	Fowler syndrome, dysfunctional voiding

Evaluation

- Bladder diary, flow rate, **PVR (post-void residual)**, sacral neurological examination, spinal MRI if indicated (deficit), drug review.

- **Pressure-flow study is the only accepted (diagnostic) test:** low detrusor pressure with low Qmax.
- **BCI (bladder contractility index) = Pdet@Qmax (detrusor pressure at maximum flow) + 5 × Qmax:** <100 weak, 100–150 normal, >150 strong. Also PIP (projected isovolumetric pressure), Watts factor, stop test.
- Ambulatory urodynamics if no contraction is seen on the standard study (the "bashful bladder").

Management

- **Goals:** protect kidneys, prevent UTI (urinary tract infection) and stones, improve symptoms. No drug reliably improves contraction.

Aim / option	Details
Conservative drainage	Timed and double voiding; treat constipation; stop causative drugs
CISC (clean intermittent self-catheterisation)	**Mainstay** for high PVR; fewer infections than an indwelling catheter. Suprapubic catheter if CISC impossible
Lower outlet resistance	α-blockers (modest benefit); pelvic-floor biofeedback
Outlet surgery (TURP, transurethral resection of the prostate)	In selected obstructed men; little help in low-PVR men; may free catheter-dependent fit men
SNM (sacral neuromodulation)	For non-obstructive retention, especially Fowler syndrome; needs some detrusor function; intact sensation predicts response
Sacral anterior root stimulator (Brindley)	Complete spinal cord injury, with posterior rhizotomy
Parasympathomimetics (bethanechol, distigmine)	**Not recommended / not routine**: weak evidence; systemic side effects (bronchospasm, rarely cardiac arrest)
Intrasphincteric botulinum toxin	Experimental
Diversion	Mitrofanoff when urethral CISC fails; rarely ileal conduit

- Avoid Credé or Valsalva voiding (reflux risk).
- SNM and parasympathomimetics need some intact detrusor and sensory function; a fibrosed detrusor responds to nothing.

Investigational

- Intravesical electrotherapy, intravesical prostaglandins, TAC-302, latissimus dorsi detrusor myoplasty, muscle-derived stem cells.

Diagram to draw

- Pressure-flow plot on the Schäfer nomogram with BCI lines at 100 and 150.

> **Key points to remember:**
> - BCI = Pdet@Qmax + 5Qmax; <100 weak.
> - CISC is the mainstay; no effective drug; parasympathomimetics not recommended.
> - SNM helps non-obstructive retention (Fowler syndrome); intact sensation predicts response.

SAQ 5 Detrusor–external sphincter dyssynergia (DESD / DSD)

P1

Asked as:

- External Sphincter Dyssynergia — *Summer 2016 Paper I Q3a (SAQ)*
- Detrusor sphincter dyssynergia — *Practice Set 10 Paper I Q3f (SAQ)* (also answered in 2016-Summer Paper 1, Q3a)

Type: SAQ (10 marks)

Definition

- **DESD / DSD (detrusor–external sphincter dyssynergia):** the striated sphincter contracts involuntarily (involuntary contraction) during detrusor contraction, causing functional obstruction, high pressure and poor emptying.
- **It requires a lesion between the pons and the sacral cord:** suprasacral spinal cord injury (SCI), MS (multiple sclerosis), transverse myelitis, myelomeningocele. Brain lesions (stroke, Parkinson's) do not cause it (**never with brain lesions**).
- Lesions (at or) above T6 may add bladder-neck (**smooth sphincter**) dyssynergia and autonomic dysreflexia.

Blaivas classification

Type	EMG (electromyography) pattern	Lesion
1	Activity rises with detrusor pressure, then relaxes suddenly at peak	Incomplete
2	Sporadic contractions throughout voiding	Complete
3	Crescendo–decrescendo contraction; obstruction throughout	Complete
- Alternative: intermittent versus continuous (continuous means a complete lesion). Type has no relation to lesion level (type is unrelated to level).		

Diagnosis

- **Pressure–flow study with EMG:** EMG rises during the detrusor contraction, with high pressure and interrupted flow. Needle EMG is more reliable than surface.
- **VUDS (videourodynamics):** narrowed (narrow) membranous urethra with dilated posterior urethra; shows VUR (vesicoureteral reflux).
- **Exclude:** guarding reflex (normal); **pseudodyssynergia** (straining or suppressing a contraction); dysfunctional voiding (learned, no neurological lesion; Hinman syndrome in children).

Significance

- Worse when complete, continuous and **in men**: about half of untreated men get VUR, hydronephrosis, stones or urosepsis. Other risk factors: high filling pressure, indwelling catheter.
- **Management logic:** DSD is both a storage and an emptying failure. Lower bladder pressure, then empty it.

- DLPP (detrusor leak point pressure) ≥40 cmH_2O endangers the kidneys.

Management

- **First line:** CIC (clean intermittent catheterisation) plus antimuscarinic or β3-agonist; then intradetrusor botulinum toxin.
- **CIC impossible (men):** sphincterotomy or sphincteric botulinum toxin, with condom drainage. The goal is DLPP <40; DLPP ≥40 afterwards means failure; repeat procedures are common.
- Others: suprapubic catheter, Brindley stimulator with deafferentation, diversion. Urethral stents are abandoned; oral drugs disappoint. Credé and Valsalva voiding contraindicated.
- **Also know:** α-blockers for bladder-neck dyssynergia; lifelong upper-tract surveillance.

Diagram to draw: urodynamic trace with rising detrusor pressure (Pdet), EMG bursts and interrupted flow.

> **Key points to remember:**
> - Lesion between pons and sacral cord; never brain lesions.
> - Blaivas 1 incomplete; 2 and 3 complete.
> - Worst in men; DLPP ≥40 cmH_2O hostile.
> - CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

SAQ 6 EMG

P1

Type: SAQ (10 marks)
Source: Practice Set 9 Paper 1, Q3b

EMG is taken as urodynamic electromyography of the striated urethral/anal sphincter and pelvic floor.

Definition and principle

- EMG (electromyography) records the electrical potentials of depolarising striated muscle membranes, here the **external urethral sphincter and pelvic floor**, during filling and voiding.
- It shows **coordination** between detrusor and sphincter, not sphincter strength. The bladder neck (smooth muscle) is invisible to EMG.
- External anal sphincter activity mirrors the urethral sphincter, except in some neurological diseases.

Electrodes

Surface (patch)	Concentric needle
Perianal patches at 3 and 9 o'clock (ICS [International Continence Society] standard: two skin electrodes plus reference)	Needle into the sphincter (girls beside the meatus; boys transperineal)
Comfortable; poorer signal; movement artefact	Best signal; painful; dislodges
Coordination only	Also motor unit analysis: acute versus chronic denervation

Normal pattern

- Filling: activity **rises gradually** (recruitment, guarding reflex).
- Voiding: EMG **falls silent before** detrusor pressure rises and stays silent until flow ends.
- **Lag time** (uroflow with patch EMG in children): relaxation to flow 2–6 s is normal; <2 s suggests DO (detrusor overactivity); >6 s suggests primary bladder neck dysfunction.

Abnormal patterns

Finding	Meaning
DSD (detrusor–sphincter dyssynergia)	Sphincter contracts during detrusor contraction; needs a **suprasacral** cord lesion (SCI [spinal cord injury], MS [multiple sclerosis]). Blaivas types 1–3
Dysfunctional voiding	Same tracing without neurological lesion (learned)
Pseudodyssynergia	EMG rise from straining or trying to suppress a contraction
Denervation potentials	Sacral cord or cauda equina lesion; MSA (multiple system atrophy)
Complex repetitive discharges, decelerating bursts	**Fowler syndrome** (retention in young women); predicts SNM (sacral neuromodulation) response

Clinical uses

- Neurogenic bladder (spina bifida, SCI); voiding dysfunction in children; female retention; before sphincter surgery.
- **Limitation:** surface EMG misdiagnoses some dysfunctional voiders; **VUDS (videourodynamics)** is more accurate.

Diagram to draw: tracing of Pdet, flow and EMG: EMG silent during normal voiding versus active in DSD.

> **Key points to remember:**
> - EMG judges coordination, not strength.
> - True DSD needs a suprasacral lesion.
> - Lag time 2–6 s normal.
> - Fowler syndrome: complex repetitive discharges on needle EMG.

SAQ 7 Bladder dysfunction non genitourinary systemic disease

P2

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 2, Q3a

Taken as lower urinary tract dysfunction caused by systemic (non-genitourinary) diseases.

Principle

- The **site of the lesion**, not the disease name, predicts the bladder:
- Brain above the pons: **DO (detrusor overactivity), no DSD (detrusor–sphincter dyssynergia)**.
- Spinal cord: DO **with DSD**.
- Peripheral nerves or roots: **areflexia, reduced sensation, high PVR (post-void residual)**.

- **Urodynamics** before treatment, because symptoms do not predict findings.

Diabetes mellitus (most important)

- **Diabetic cystopathy:** sensory then motor autonomic neuropathy. Impaired sensation leads to large capacity, poor contractility, low flow and later high PVR and UTI (urinary tract infection).
- **DO is in fact the commonest urodynamic finding** (early disease or coexisting cerebrovascular disease); hypocontractility comes late.
- Mechanisms: glycation end products, sorbitol pathway, oxidative stress, polyuria.
- Differentiate from BOO (bladder outlet obstruction) by pressure–flow study. Treatment: **timed voiding**, CIC (clean intermittent catheterisation), cautious antimuscarinics.

Other systemic diseases

Group	Disease	Usual finding
Neurological	Stroke	Acute retention, then DO; no true DSD
	Parkinson disease	DO; pseudodyssynergia; TURP (transurethral resection of prostate) possible after excluding MSA (multiple system atrophy)
	Multiple sclerosis	DO about 38%, DO with DSD 29%, areflexia 26%; changes over time
Infective	HIV/AIDS	DO, DSD; areflexia with CMV (cytomegalovirus)
	Herpes zoster/simplex, Guillain–Barré	Retention, areflexia; reversible
	Spinal tuberculosis, Lyme, HTLV-1	DO with DSD or areflexia
Autoimmune and connective tissue	SLE (systemic lupus erythematosus)	DO; lupus cystitis
	Systemic sclerosis	Areflexia, low compliance
Infiltrative	Amyloidosis (transthyretin)	Low sensation, underactivity, SUI (stress urinary incontinence)
Nutritional	B12 deficiency, alcohol (thiamine)	Mixed; reversible with replacement
Endocrine	Hyperthyroidism	Poor emptying; resolves with treatment

Also know: myasthenia gravis; schizophrenia; gastroparesis.

Management principles

- Treat the disease; reversible therapy.
- Storage: antimuscarinics (trospium, or mirabegron in elderly brains), botulinum toxin, neuromodulation.
- Emptying: CIC; avoid Credé with DSD.

Diagram to draw: neuraxis with lesion levels and the bladder pattern at each.

Key points to remember:

- Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
- Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
- Urodynamics before treating; reversible treatment first.

SAQ 8 Urinary bladder compliance

P3

Type: SAQ (10 marks)
Source: 2017-Summer Paper 3, Q3e

Definition

- **Compliance** is the change in volume for a change in detrusor pressure: **C = ΔV / ΔPdet**, in **mL/cmH_2O**.
- It reflects **low-pressure storage** during filling.

Physiological basis

- **Viscoelastic wall**: smooth muscle, collagen (about 50% of the wall) and elastin (about 2%). Coiled collagen straightens as the bladder fills.
- More collagen or less elastin makes the wall stiff.

Measurement (filling cystometry)

- ICS (International Continence Society) points: Pdet at **start of filling** and at **cystometric capacity** (or before a leak-causing contraction), **excluding detrusor contractions**.
- **No agreed normal value**; compliance depends on capacity. Below **20 mL/cmH_2O** (some use 10–20) suggests poor accommodation. In children, normal is often taken as **>15 mL/cmH_2O**.

Pitfalls

Pitfall	Effect	Fix
Fast filling	False low compliance	Slow or stop filling; pressure falls
Sustained low-amplitude detrusor contraction	Mimics low compliance	Stop filling: pressure returns to baseline
Pop-off: VUR (vesicoureteric reflux), diverticulum, leaking outlet	Falsely good compliance	Video-urodynamics; occlude outlet
- A small bladder can give a low number without danger (**compliance paradox**). **Absolute storage pressure matters more than the ratio.**		

Causes of low compliance

- **Neurogenic**: spina bifida, spinal cord injury, lower motor neuron and decentralisation injury (radical pelvic surgery).
- **Outlet obstruction**: PUV (posterior urethral valves), long-standing BPO (benign prostatic obstruction).
- **Structural**: radiation cystitis, tuberculosis, schistosomiasis, multiple bladder operations.

Consequences and DLPP

- **DLPP (detrusor leak point pressure)**: lowest Pdet at which leakage occurs without contraction or strain.
- **McGuire (1981)**: DLPP **>40 cmH_2O** in myelodysplastic children led to hydronephrosis or VUR in 85%. Not a validated safe limit (20 cmH_2O is more sensitive); aim as low as possible.
- A strong outlet keeps the patient dry but transmits high pressure to the kidneys.

Management

- **Clean intermittent catheterisation + antimuscarinics** (first line).
- **Intradetrusor botulinum toxin**.
- **Augmentation cystoplasty** if refractory; vesicostomy in infants.
- Repeat urodynamics to confirm safety.

Diagram to draw: cystometrogram: normal flat versus steep rising filling curve.

> **Key points to remember:**
> - C = ΔV/ΔPdet; <20 mL/cmH_2O suggests poor compliance.
> - McGuire: DLPP >40 cmH_2O, 85% upper-tract damage.
> - Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

Rapid Revision Sheet · Section 7

Overactive bladder: pathophysiology, pharmacotherapy and non-pharmacological management (LAQ)

- OAB is defined by urgency; DO is a urodynamic finding.
- Final pathway: increased afferent signal (urothelial, myogenic) or failed central inhibition.
- Behavioural therapy first: BT and PFMT with urge suppression give 50–80% fewer leaks; antimuscarinic or β3-agonist next.
- M3 mediates contraction; β3 raises cAMP for storage-phase relaxation; voiding spared.
- Equal efficacy; β3 avoids dry mouth and dementia risk (10–65% with antimuscarinics >3 months).
- Darifenacin most M3-selective; tolterodine bladder-selective; solifenacin longest half-life.
- STAR: solifenacin beat tolterodine ER.
- Mirabegron 25 mg → 50 mg; avoid if blood pressure ≥180/110; CYP2D6 interaction (metoprolol).
- Elderly: mirabegron (β3), trospium or darifenacin preferred over other antimuscarinics.
- Solifenacin + mirabegron combination works (SYNERGY).
- Refractory OAB: PTNS (SUmiT 55% vs 21%), SNM (InSite 61% vs 42%), Botox 100 U.

Neuromodulation for voiding dysfunction, including sacral neuromodulation (LAQ)

- Afferent modulation explains SNM benefit in both OAB and retention.
- S3: bellows plus great-toe flexion; implant if the test improves ≥50%.
- SNM revision >30% at 5 years.
- SUmiT: PTNS 55% vs sham 21%; 12 weekly sessions.
- ROSETTA: Botox 200 U slightly better for UUI, more UTI and CIC.
- Rechargeable and long-life non-rechargeable stimulators, lasting at least 10 years.
- Full-body MRI-conditional devices at 1.5 T and 3 T.
- Implantable tibial nerve stimulators (BlueWind, eCoin) and transcutaneous tibial stimulation as less invasive alternatives.
- Standardised tined-lead placement improves outcomes and lowers revision.

Pathophysiology, classification and management of LUTS. (LAQ)

- ICS: storage, voiding, post-micturition; IPSS 0–7, 8–19, 20–35.
- Static (bulk, capsule) plus dynamic (α1A) obstruction; size ≠ symptoms.
- 5ARI for >40 mL; MTOPS combination cut progression 66%.
- TURP 30–80 mL; HoLEP or simple prostatectomy for >80 mL.

Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male (LAQ)

- Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
- Autonomic dysreflexia above T6: drain bladder first, nitrates next.
- CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
- DLPP ≥40 cm H_2O threatens the kidneys.

Botulinum toxin in urology (SAQ)

- Cleaves SNAP-25; acts on efferent and afferent nerves.

- 100 U idiopathic OAB; 200 U NDO; spare trigone.
- Children: 10 U/kg (max 360 U) in series; FDA label 6 U/kg (max 200 U), approved for age ≥5 years (2021); used from about 2 years of age.
- Main risks: retention needing CISC, UTI.
- Other formulations (abobotulinumtoxinA, incobotulinumtoxinA) under study; doses are not interchangeable between products.
- Injection-free delivery under research: intravesical liposome-encapsulated toxin and other instillation vehicles.

Urodynamic studies, including video urodynamics (SAQ)

- Pdet = Pves − Pabd.
- Ask a question first; reproduce the symptom.
- BOOI >40 obstructed; BCI <100 weak.
- DLPP >40 cmH_2O: upper-tract risk.
- VUDS = urodynamics + fluoroscopy with contrast.
- VUDS is the only test for bladder neck dysfunction, PBNO and DISD.
- Reflux pop-off makes compliance look falsely good.

Uroflowmetry (SAQ)

- Voided volume ≥150 mL; Qmax is the key value.
- Plateau = fixed obstruction (stricture); staccato = sphincter; interrupted = weak detrusor or straining.
- Qmax <10 mL/s suggests BOO but cannot exclude underactivity; Siroky and Liverpool nomograms.
- Cannot distinguish BOO from DUA.

Detrusor underactivity and its management (SAQ)

- BCI = Pdet@Qmax + 5Qmax; <100 weak.
- CISC is the mainstay; no effective drug; parasympathomimetics not recommended.
- SNM helps non-obstructive retention (Fowler syndrome); intact sensation predicts response.

Detrusor–external sphincter dyssynergia (DESD / DSD) (SAQ)

- Lesion between pons and sacral cord; never brain lesions.
- Blaivas 1 incomplete; 2 and 3 complete.
- Worst in men; DLPP ≥40 cmH_2O hostile.
- CIC plus detrusor relaxation first; sphincterotomy if CIC impossible.

EMG (SAQ)

- EMG judges coordination, not strength.
- True DSD needs a suprasacral lesion.
- Lag time 2–6 s normal.
- Fowler syndrome: complex repetitive discharges on needle EMG.

Bladder dysfunction non genitourinary systemic disease (SAQ)

- Lesion level predicts pattern: brain DO; cord DO + DSD; peripheral areflexia.
- Diabetic cystopathy: low sensation, large capacity, high PVR; DO common early.
- Urodynamics before treating; reversible treatment first.

Urinary bladder compliance (SAQ)

- C = ΔV/ΔPdet; <20 mL/cmH_2O suggests poor compliance.
- McGuire: DLPP >40 cmH_2O, 85% upper-tract damage.
- Pop-offs (VUR, diverticulum) hide true pressure; use video-urodynamics.

Section 8 · Nocturia & Male Incontinence

Short answer questions (SAQ)

SAQ 1 Post-prostatectomy incontinence

P2

Asked as:

- Post prostatectomy incontinence. — *Summer 2021 Paper II Q3a (SAQ)*
- Prostatectomy incontinence — *Practice Set 11 Paper II Q3f (SAQ)*

Type: SAQ (10 marks)

(Practice Set 11 Paper II Q3f: also answered in 2021-Summer Paper 2, Q3a)

Definition and incidence

- PPI (post-prostatectomy incontinence) is mainly **SUI (stress urinary incontinence)** after RP (radical prostatectomy), the commonest cause of male SUI. Rates vary with definition ("no pad" vs "safety pad") and reporting; expert no-pad continence is in the mid-90s % at 1 year; continence improves for up to 2 years.

Pathophysiology

- RP removes the internal sphincter (bladder neck, prostatic urethra), so continence depends (entirely) on the external **rhabdosphincter**.
- Causes: apical injury to rhabdosphincter or pudendal branches (sphincter or nerve injury at the apex), short membranous urethra, afferent denervation; also detrusor overactivity, low compliance, **VUAS (vesicourethral anastomotic stenosis)**.
- Nerve sparing and posterior reconstruction (Rocco stitch) give **earlier, not better final** continence.

Evaluation

- History (type, severity, radiotherapy, cancer status), bladder diary, **24-hour pad weight**: mild <200 g, moderate 200–400 g, severe >400 g (another source: mild <150 g, moderate <400 g, severe more; thresholds vary).
- **Standing cough test**, grade (grades) 0 to 4 (0–4: leak reported but not seen; delayed drops; early drops; delayed stream; early persistent stream).
- Urinalysis, creatinine, PVR (post-void residual), PSA (prostate-specific antigen). **Cystoscopy** excludes VUAS and judges (shows) residual sphincter coaptation; urodynamics when needed (if mixed symptoms).

Management

1. **Wait 6–12 months** (improves up to 2 years). Supervised **PFMT (pelvic floor muscle training)** with biofeedback, ideally preoperative (started before surgery); pads, condom catheter or (penile) clamp.
2. **Duloxetine** (off-label). **Bulking agents are not advised.**
3. **Surgery** (AUA/SUFU [American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction] 2019, EAU [European Association of Urology]):

Situation	Choice
Mild–moderate, no radiotherapy	**Male sling**: AdVance / AdVance XP (repositions the bulbar urethra; needs a coapting sphincter) or adjustable systems (ATOMS [adjustable transobturator male system])
Severe, or any radiotherapy	**AUS (artificial urinary sphincter), AMS 800**: gold standard; cuff pressure 61–70 cm H_2O
ProACT balloons	Expert centres; avoid after radiotherapy

- **MASTER trial:** male sling non-inferior to AUS at 12 months.
- AUS: social continence (0–1 pad) 55–77%; revision 10–29% at 3 years; lasts (lifespan) 8–10 years. Erosion risks: radiotherapy (risk higher after radiotherapy), prior erosion, urethroplasty (use a **transcorporal cuff** for fragile urethra).
- **Also know:** climacturia; treat VUAS first.

Diagram to draw: AMS 800 with cuff, scrotal pump and pressure-regulating balloon.

> **Key points to remember:**
> - Wait 6–12 months with PFMT before surgery.
> - Sling for mild–moderate; AUS for severe or irradiated.
> - MASTER: sling non-inferior to AUS at 12 months.

SAQ 2 Desmopressin (DDAVP)

P3

Asked as:

- Desmopressin. — *Winter 2023 Paper III Q3b (SAQ)*
- DDAVP — *Practice Set 3 Paper III Q3c (SAQ)*

Type: SAQ (10 marks)
(also answered in 2023-Winter Paper 3, Q3b)

Pharmacology

- DDAVP (**1-deamino-8-D-arginine vasopressin**, desmopressin), a synthetic analogue of AVP (arginine vasopressin/antidiuretic hormone).
- Deamination and D-arginine substitution prolong action (give longer action) and make it a **selective V2 agonist (selective V2 agonism) without pressor (V1) effect**.
- **Mechanism:** V2 receptor on collecting-duct cells raises cAMP (cyclic adenosine monophosphate), inserting **aquaporin-2**; water is reabsorbed, so urine is less and concentrated (V2 receptor → cAMP → aquaporin-2 insertion → water reabsorption → less, concentrated urine).
- Endothelial V2 action releases von Willebrand factor and factor VIII (haemostatic use).

Formulations

Form	Dose
Tablet	0.2–0.4 mg at bedtime, about 1 h before bed (enuresis)
Oral lyophilisate (melt)	120–240 µg, about 1 h before bed
Low-dose nasal spray (nocturia)	0.83 or 1.66 µg, 30 min before bed

Form	Dose
Sublingual ODT (orally disintegrating tablet)	27.7 µg women, 55.3 µg men
IV / subcutaneous	0.3 µg/kg (bleeding)

Uses (in urology)

1. **Monosymptomatic nocturnal enuresis:** first line with the alarm. **Rule of thirds**: one-third full, one-third partial, one-third no response. Acts (works) at once but relapses on stopping. Restrict fluids in the evening (avoid evening fluid); add oxybutynin if partial.
2. **Nocturia from nocturnal polyuria** (NPi [nocturnal polyuria index] >20–33%, age-dependent). EAU (European Association of Urology): offer below (under) 65; low dose above (over) 65 with ≥2 voids. Add to α-blocker for refractory nocturia in BPH (benign prostatic hyperplasia).
3. **Neurogenic nocturia** (multiple sclerosis, MSA [multiple system atrophy]).
4. **Bleeding:** uraemic platelet dysfunction, von Willebrand disease (including acquired, in Wilms tumour). Tachyphylaxis after about 3 (repeated) doses.
5. **Diagnosis:** central versus nephrogenic DI (diabetes insipidus).

Adverse effects

- **Hyponatraemia** (most important): headache, nausea, confusion, seizures.
- Risk factors: **age ≥65**, baseline sodium <135 mmol/L, CrCl (creatinine clearance) <60 mL/min, higher dose; women are more sensitive.
- Most events occur in the **first 2–3 weeks**.

Monitoring and contraindications

- Sodium at baseline, then within 1 week and at 1 month (EAU adds day 3), and after dose change.
- **Avoid:** hyponatraemia, polydipsia, heart failure, SIADH (syndrome of inappropriate antidiuretic hormone), significant renal impairment, **loop diuretics**.

Diagram to draw: V2 receptor–cAMP–aquaporin-2 pathway in the collecting duct.

> **Key points to remember:**
> - Selective V2 agonist; no pressor effect.
> - Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
> - Hyponatraemia: age ≥65, low sodium; check sodium early.

Rapid Revision Sheet · Section 8

Post-prostatectomy incontinence (SAQ)

- Wait 6–12 months with PFMT before surgery.
- Sling for mild–moderate; AUS for severe or irradiated.
- MASTER: sling non-inferior to AUS at 12 months. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Desmopressin (DDAVP) (SAQ)

- Selective V2 agonist; no pressor effect.
- Enuresis: 0.2–0.4 mg tablet or 120–240 µg melt; rule of thirds.
- Hyponatraemia: age ≥65, low sodium; check sodium early. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Section 9 · Andrology & Sexual Medicine

Long answer questions (LAQ)

LAQ 1 Spermatogenesis, azoospermia and non-obstructive azoospermia

P1 · P2 · P3 **merged from 3**

Asked as: P1 LAQ "Spermatogenesis, azoospermia, non-obstructive azoospermia and maturation arrest" · P2 SAQ "Azoospermia and non-obstructive azoospermia" · P3 SAQ "Spermatogenesis"

Maturation arrest is taken as spermatogenic maturation arrest on testicular histology.

Spermatogenesis: definition

- Production of haploid spermatozoa from diploid spermatogonia in the seminiferous tubules: about **64 days** (42–76); about 123 million sperm a day.

Hormonal control: HPG (hypothalamic–pituitary–gonadal) axis

- The hypothalamus releases **GnRH (gonadotropin-releasing hormone)** in pulses; pulsatility is essential (constant agonist exposure shuts the axis down). Kisspeptin triggers puberty.
- **LH (luteinising hormone)** acts on Leydig cells to make testosterone (about 5 mg/day). **FSH (follicle-stimulating hormone)** acts on Sertoli cells.
- Feedback: testosterone (and oestradiol) on LH; **inhibin B** from Sertoli cells on FSH. So FSH and inhibin B report the tubules, and LH reports the Leydig cells.
- **Intratesticular testosterone** is about 50-fold higher than serum (held by **ABP, androgen-binding protein**) and is essential for postmeiotic maturation. Exogenous testosterone suppresses LH and so stops spermatogenesis.
- The testis is kept 2–4 °C below core temperature by the pampiniform countercurrent exchange.

Sertoli cells and the blood–testis barrier

- Nurse cells: nourish germ cells; secrete ABP, inhibin and fluid; phagocytose residual bodies.
- **Sertoli–Sertoli tight junctions** divide the tubule into a **basal** compartment (spermatogonia) and an **adluminal** compartment (spermatocytes, spermatids). Post-meiotic cells are "foreign" to the immune system; the barrier and local tolerance protect them.

Phases of spermatogenesis

Phase	Events
1. Proliferation (spermatocytogenesis)	**Dark type A (Ad)** reserve stem cells self-renew (GDNF [glial cell line–derived neurotrophic factor] niche); **pale type A (Ap)** divide every 16 days into type B spermatogonia, then primary spermatocytes. Daughter cells stay joined by cytoplasmic bridges
2. Meiosis	Primary spermatocytes cross into the adluminal compartment. Meiosis I (reduction division; synapsis and crossing-over) gives secondary spermatocytes; meiosis II gives 4 haploid round spermatids

Phase	Events
3. Spermiogenesis	No division: acrosome from Golgi, flagellum from centriole, nucleus condenses to about 10%, mitochondria form the midpiece, cytoplasm shed
4. Spermiation	Sertoli cell releases sperm into the lumen

- **Cycle, stage, wave:** six stages in humans; cycles are arranged as a helical wave, so production is continuous.
- Defective meiotic recombination causes maturation arrest and sperm aneuploidy.

Epididymal maturation and transport

- Testicular sperm are immotile. During 2–12 days of epididymal transit they gain **progressive motility** and fertilising ability; they are stored in the cauda.
- **Capacitation** and the acrosome reaction occur in the female tract.

Clinical relevance of spermatogenesis

- A semen sample reflects events about 3 months earlier; reassess 3 months after treatment.
- Disruptors: heat (varicocele, cryptorchidism), exogenous testosterone, chemotherapy, radiation.

Azoospermia: definition

- No sperm in the **centrifuged** ejaculate on **at least two** semen analyses; 10–15% of infertile men.
- **NOA (non-obstructive azoospermia):** azoospermia due to **failure of spermatogenesis** (testicular failure); about 60% of azoospermic men.

Classification

Type	Causes	Clues
Pre-testicular (hypogonadotropic)	Kallmann syndrome, pituitary tumour, hyperprolactinaemia, **exogenous testosterone or anabolic steroids**	Low FSH, LH, testosterone
Testicular (NOA, spermatogenic failure)	Klinefelter, Y microdeletion, cryptorchidism, mumps, chemotherapy or radiation, idiopathic	Small soft testes, **FSH >7.6 IU/L**, long axis ≤4.6 cm
Post-testicular (OA, obstructive azoospermia)	CBAVD (congenital bilateral absence of the vas deferens), epididymal obstruction, vasectomy, hernia repair, EDO (ejaculatory duct obstruction)	Normal testes, **FSH ≤7.6**, dilated epididymis
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume; sperm in post-ejaculate urine

Causes of NOA

- **Genetic: Klinefelter (47,XXY)**, the commonest; **Y microdeletion** (AZF [azoospermia factor] a, b, c); other karyotype anomalies.
- **Acquired:** cryptorchidism, mumps orchitis, torsion, **chemotherapy or radiation**, trauma, varicocele.
- **Idiopathic:** commonest overall.

- *Note:* hypogonadotropic hypogonadism is pre-testicular and treatable medically (hCG [human chorionic gonadotropin] ± FSH).

Evaluation

- History (mumps, surgery, chemotherapy, drugs, testosterone use); examination (testis size, **vasa**, epididymis, varicocele).
- **Hormones:** FSH (reflects germ-cell mass), LH, testosterone, oestradiol, prolactin.
- NOA: small soft testes, palpable vasa, normal ejaculate volume. **FSH >7.6 IU/L and testis long axis ≤4.6 cm** point to NOA; FSH ≤7.6 with normal testes (>4.6 cm) suggests obstruction.

```
Azoospermia (2 centrifuged samples) ├─ Vasa absent → CBAVD → CFTR testing, both partners; renal scan/US ├─ Volume <1 mL → post-ejaculate urine: sperm = retrograde; none → TRUS for EDO └─ Normal volume: FSH ≤7.6 IU/L + testis long axis >4.6 cm → obstructive FSH >7.6 + testis ≤4.6 cm (small) → spermatogenic failure
```

- **Genetics (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine): karyotype** and **Y microdeletion** in all azoospermic men with NOA, with counselling. **AZFa or AZFb complete deletion: no sperm, do not operate**; AZFc: sperm often found. **CFTR (cystic fibrosis transmembrane conductance regulator)** testing in CBAVD.
- **TRUS (transrectal ultrasound):** seminal vesicle >1.5 cm wide or a midline cyst suggests EDO.
- **Testis biopsy:** combined with sperm extraction and freezing.

Histology patterns in NOA

Pattern	Histology / meaning	Micro-TESE retrieval
Hypospermatogenesis	All stages present, fewer cells	Best
Maturation arrest	Halts at one germ-cell stage	About 40%
SCO (Sertoli cell–only)	No germ cells	About 30%

Maturation arrest

Definition

- A histological pattern of **spermatogenic failure** in which germ cells develop normally up to one stage and **the sequence halts** at a premeiotic, meiotic or postmeiotic step. No mature sperm, or very few, are made.

Classification

- **Early:** arrest at spermatogonia or **primary spermatocyte** (meiotic failure).
- **Late:** arrest at the **spermatid** stage.
- **Complete** (all tubules) versus **incomplete** (foci progress to sperm): incomplete gives better retrieval.

Causes

- **Genetic:** complete **AZFb deletion** causes arrest; meiotic recombination defects in about half of men with maturation arrest; karyotype anomalies.
- **Acquired:** heat (varicocele, cryptorchidism), chemotherapy, radiation, drugs, orchitis; often idiopathic.

Clinical features and diagnosis

- Azoospermia or severe oligospermia, normal semen volume, palpable vasa.
- **Testis size and FSH are often normal**, so it can mimic obstruction (FSH ≤7.6 IU/L, long axis >4.6 cm).
- **Diagnosis needs testicular biopsy** (Bouin's fixative, never formalin), usually done as sperm extraction.
- Karyotype and Y microdeletion testing in all azoospermic men (AUA/ASRM).

Management

- Optimise: treat varicocele, stop exogenous testosterone; off-label hormonal stimulation (clomiphene, hCG) unproven.
- **Micro-TESE** with ICSI: retrieval **about 40%** (versus 30% in SCO).
- **Complete AZFb deletion: retrieval highly unlikely; do not operate.**
- No sperm: donor sperm or adoption. Round spermatid injection is experimental.
- Genetic counselling: sons inherit Y deletions.

Management of azoospermia

1. Pre-testicular (hypogonadotropic)

- Stop exogenous testosterone. **hCG 1500–5000 IU 2–3 times weekly ± FSH**; pulsatile GnRH in hypothalamic failure; dopamine agonist for prolactinoma.

2. Obstructive

- Microsurgical **vasovasostomy** or **vasoepididymostomy** (clear watery fluid or sperm parts in vasal fluid: vasovasostomy; thick toothpaste-like fluid without sperm: vasoepididymostomy).
- **TURED (transurethral resection of the ejaculatory ducts)** for EDO.
- CBAVD or unreconstructable: sperm retrieval by **MESA (microsurgical epididymal sperm aspiration)**, PESA (percutaneous epididymal sperm aspiration) or TESA + **ICSI (intracytoplasmic sperm injection)**; freeze sperm at reconstruction.

3. Non-obstructive

- **Optimisation (before retrieval):** stop exogenous testosterone; repair a clinical varicocele (evidence weak; no definitive evidence in NOA); off-label hormonal therapy (clomiphene, anastrozole, hCG) for low testosterone.
- **Micro-TESE (microdissection testicular sperm extraction) + ICSI:**
- **Principle:** spermatogenesis in NOA is **focal**. Under ×20–25 magnification, **larger, dilated, opaque tubules** are chosen; the least tissue is removed and the blood supply is preserved.
- Sperm found in **about 50%**; better than conventional TESE (Klinefelter 68%, Sertoli cell–only 30%).

Predictor	Retrieval (%)
Post-orchidopexy	74
AZFc deletion	70
Klinefelter	68
Post-chemotherapy	53
AZFa or AZFb complete deletion	**0: do not operate**

- FSH and testis size **do not** reliably predict retrieval.
- **ICSI:** fresh or frozen testicular sperm. Live birth per cycle is modest; miscarriage is higher (sperm DNA damage). Male offspring of AZF deletion carriers inherit the deletion.

4. Ejaculatory

- Retrograde: sympathomimetics, or sperm from alkalinised urine. Anejaculation: vibratory or electroejaculation.

5. Failure (no sperm): donor sperm or adoption; genetic counselling.

Also know

- Spermatogonial stem cell renewal by GDNF; in-vitro spermatogenesis and stem-cell transplantation are experimental.

Diagrams to draw

- Cross-section of a seminiferous tubule showing basal and adluminal compartments with germ cells maturing towards the lumen.
- Hypothalamic–pituitary–testis axis with the three levels of failure.
- Opened testis under the microscope: plump tubules versus thin Sertoli cell–only tubules.
- Maturation arrest: seminiferous tubule showing germ cells up to spermatocytes with no spermatids or sperm.

> **Key points to remember:**
>
> - Spermatogenesis about 64 days; Ap divide every 16 days; six stages.
> - Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
> - FSH → Sertoli, LH → Leydig; tight junctions between Sertoli cells form the blood–testis barrier.
> - Confirm azoospermia on two centrifuged samples.
> - FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
> - Karyotype and Y microdeletion in all.
> - AZFa/b complete deletion: do not operate (no retrieval).
> - NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.
> - Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
> - FSH and testis size may be normal in maturation arrest: biopsy diagnoses.
> - Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

LAQ 2 Azoospermia: testicular biopsy and surgical sperm retrieval techniques

P1 · P2 · P3 · P4 **merged from 4**

Asked as: P4 LAQ "Define Azoospermia and describe types of testicular biopsies and surgical sperm retrival." · P2 SAQ "Sperm retrieval techniques (sperm extraction in obstructive azoospermia; testicular aspiration and assisted reproduction)" (asked as "Sperm extraction in Obstructive azoospermia", "Testicular aspiration of sperms and assisted reproduction" and "Sperm Retrival techniques") · P1 SAQ "Testicular biopsy and TESA (testicular sperm aspiration)" (asked as "TESA" and "Testicular biopsy") · P3 SAQ "Sperm retrieval techniques"

Definitions

- **Azoospermia:** no sperm in the **centrifuged pellet** of the ejaculate on **at least two** semen analyses. Found in about 1% of men and 10–15% of infertile men.
- Not **cryptozoospermia** (rare sperm in pellet) or **aspermia** (no ejaculate).
- **OA (obstructive azoospermia):** no sperm in the centrifuged ejaculate despite **normal spermatogenesis**, from blockage of the ducts.
- **TESA (testicular sperm aspiration):** percutaneous needle aspiration of seminiferous tubules under local anaesthesia to retrieve sperm for **ICSI (intracytoplasmic sperm injection)**, which needs only one sperm per oocyte.

Classification

Type	Causes	Clues
Pre-testicular	Kallmann syndrome, pituitary disease, **exogenous testosterone**	Low FSH (follicle-stimulating hormone), LH (luteinising hormone)
Testicular: **NOA (non-obstructive azoospermia)**	Klinefelter, Y microdeletion, cryptorchidism, chemotherapy, mumps	Small testes, high FSH
Post-testicular: **OA**	CBAVD (congenital bilateral absence of the vas deferens), vasectomy, epididymal or ejaculatory duct obstruction, hernia-repair injury	Normal testes and FSH
Ejaculatory	Retrograde ejaculation, anejaculation	Low volume; sperm in post-ejaculate urine

Evaluation (brief)

- Absent vasa (CBAVD): CFTR (cystic fibrosis transmembrane conductance regulator) testing of both partners. Volume <1 mL: post-ejaculate urine, then TRUS (transrectal ultrasound).
- **FSH ≤7.6 IU/L and testis long axis >4.6 cm: obstruction likely;** FSH >7.6 and axis ≤4.6 cm: spermatogenic failure.
- **Karyotype and Y microdeletion** in NOA. **AZF (azoospermia factor) a or b complete deletion: no sperm, do not operate.**

Testicular biopsy

Indications

- Azoospermia with **normal testicular size, palpable vasa and normal FSH**: separates OA from NOA (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine).
- Since ICSI needs one sperm, biopsy is now **diagnostic and therapeutic**: plan it as TESE (testicular sperm extraction) and **cryopreserve** sperm.
- **Not needed:** CBAVD with normal FSH (spermatogenesis always present); high antisperm antibodies after iatrogenic vasal injury. Do not operate with complete AZFa or AZFb deletion.

Types

Type	Method	Use	Limits
Open biopsy	Small incision in the albuginea, pea-sized tissue	**Gold standard** for histology and retrieval	Open procedure
Percutaneous core biopsy	14 G biopsy gun under cord block	Fresh retrieval in OA; core of tissue	Blind; may injure epididymis or testicular artery; avoid in scarred scrotum
FNA (fine-needle aspiration), i.e. TESA	23 G needle, suction, several passes	OA retrieval; backup to epididymal methods	Few tubules, poor architecture; low yield in NOA
Microdissection biopsy (micro-TESE)	Operating microscope, plump tubules	NOA	Long operation

- **Also know:** multi-site FNA "mapping" in NOA.

Open biopsy technique

1. Local anaesthesia with **spermatic cord block** (or general/spinal); choose the larger testis.
2. 2 cm transverse scrotal incision; open tunica vaginalis.
3. Stay sutures; 3–4 mm albugineal incision avoiding surface vessels (loupes).
4. Excise extruding tubules with iris scissors, **no forceps**.
5. Fix in **Bouin's, Zenker's or glutaraldehyde, never formalin** (distorts histology).
6. **Touch prep** at once, then wet squash prep for sperm; if negative, contralateral biopsy.
7. Close albuginea and tunica vaginalis separately; no antibiotics.

Histology patterns

Pattern	Meaning
Normal spermatogenesis	Suggests obstruction
Hypospermatogenesis	All stages, fewer cells
Maturation arrest	Stops at a germ-cell stage
SCO (Sertoli cell–only)	No germ cells
Tubular sclerosis/hyalinisation	End-stage (e.g. Klinefelter)

Surgical sperm retrieval: indications

- **OA with normal spermatogenesis:** CBAVD, unreconstructable obstruction, failed or declined vasectomy reversal, back-up at reconstruction (vasoepididymostomy; frozen for IVF [in vitro fertilisation]), or failed epididymal aspiration.
- **NOA (testicular failure):** only testicular retrieval works; TESA gives poor yield; **micro-TESE (microdissection testicular sperm extraction)** preferred.

- **Anejaculation** (spinal cord injury): vibratory stimulation or electroejaculation first; retrieval after these fail.
- All retrieved sperm need **ICSI**, because sperm from obstructed systems have poor motility; cryopreserve spare sperm.

Techniques of sperm retrieval

Technique	Method / site	Best for	Merits, limits and notes
PESA (percutaneous epididymal sperm aspiration)	Fine needle into caput epididymis, local anaesthesia	OA	Simple; motile but few sperm; may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule opened under microscope at 16–25×; micropipette by **capillary action**, never suction	**OA (best)**	Many (most) motile sperm, multiple frozen vials; needs microsurgical skill
TESA	Testis, 23 G needle with suction, many passes	OA	Least invasive; immature, immotile sperm, small yield; poor in NOA
Percutaneous core biopsy	14 G biopsy gun	OA	Fresh retrieval; core of tissue; blind; vessel injury
Conventional TESE	Open biopsy, one or more sites	OA, NOA	When epididymis is absent or empty; gold standard for tissue diagnosis; blind sampling
Micro-TESE (Schlegel 1999)	Testis bivalved (opened in an avascular plane); **larger, opaque tubules** picked under the microscope	**NOA (best)**	Most sperm, least tissue; sperm in about half; long operation
Vasal aspiration	Vasotomy, testicular end flushed	At reconstruction	

MESA technique

1. Testis delivered; epididymis inspected at 16–25×.
2. Dilated tubule opened with a microknife; fluid checked at once.
3. No motile sperm: close and **move proximally** (start distally), up to the efferent ductules (proximal sperm are more motile; the most motile are often in the efferent ductules).
4. Collect by **capillary action** into micropipettes, never suction; 4–12 pipettes.
5. **Cryopreserve** in several aliquots; close tubule with 10-0 suture.

TESA technique

1. Spermatic cord block with local anaesthetic.
2. Testis fixed; **23 G needle** (or angiocatheter) on a high-suction syringe.
3. Multiple passes in different directions draw tubules into the needle (often 10–20 passes needed; P1 table: 3–4 passes).
4. Immediate wet examination for sperm; repeat until sperm are found. **Cryopreserve** spare sperm.

TESA: advantages and limitations

- **Advantages:** office procedure, no microsurgery, cheap, repeatable.

- **Limitations:** few tubules with poor architecture (not diagnostic histology); small numbers of immature, immotile sperm; often too few to freeze; blind.

Micro-TESE logic

- Spermatogenesis in NOA is **focal**. Plump, larger tubules hold sperm; thin white ones are Sertoli cell–only.
- Embryologist examines tissue in theatre; map positive sites; if negative, explore the other testis.
- It removes the least tissue for the most sperm, and is better than conventional TESE.

Choice: epididymal sperm first (MESA) in OA; testicular (TESA/TESE) if epididymal sperm are absent or retrieval fails; micro-TESE in NOA.

Results

- OA: retrieval almost always succeeds; good pregnancy rates.
- NOA: micro-TESE finds sperm in **about half**; fine-needle aspiration yields far fewer sperm than micro-TESE. Retrieval depends on the cause:

Condition	Micro-TESE retrieval
Post-orchidopexy	74%
AZFc deletion	70%
Klinefelter	68%
Post-chemotherapy	53%
Maturation arrest	40%
SCO	30%
AZFa/AZFb deletion	**0%: do not operate**

- **Before retrieval in NOA:** stop testosterone; repair clinical varicocele; off-label clomiphene, anastrozole or hCG (human chorionic gonadotropin).

Assisted reproduction and use of retrieved sperm

- **IUI (intrauterine insemination):** washed sperm into the uterus; mild male factor.
- **IVF:** oocytes and sperm combined in the laboratory.
- **ICSI:** one sperm injected into each oocyte; essential for surgically retrieved (testicular) sperm (poor motility).
- Frozen retrieved sperm equal fresh, except possibly in spinal cord injury.
- NOA pregnancies: more miscarriage (sperm aneuploidy, DNA damage); genetic counselling. No sperm: donor sperm or adoption.
- ICSI bypasses natural selection: risk of transmitting genetic defects. Offer karyotype, Y-microdeletion and CFTR testing as indicated before ICSI.
- Female burden: ovarian stimulation, multiple pregnancy.

Complications

- **Haematoma** (commonest), pain, infection (rare), testicular vascular injury (TESA, core biopsy).
- **Inadvertent epididymal biopsy** (most serious: causes obstruction); epididymal obstruction (PESA).

- Testicular atrophy or transient fall in testosterone (TESE).

Also know

- Testicular sperm may beat ejaculated sperm when DNA fragmentation is high.
- Post-mortem retrieval needs prior consent.
- Microfluidic sperm sorting.

Diagrams to draw

- Testis and epididymis showing PESA, MESA and TESA sites; testis with 23 G needle on a suction syringe and multiple passes; flow to ICSI.
- Open biopsy: small albugineal window with extruding tubules snipped off.
- Micro-TESE: bivalved testis with plump (dilated) spermatogenic tubules beside thin Sertoli cell–only tubules.

> **Key points to remember:**
>
> - Azoospermia: no sperm in centrifuged pellet on two samples.
> - Biopsy indication: azoospermia, normal testes, vasa and FSH.
> - Open biopsy is the gold standard; fix in Bouin's, never formalin; no forceps; touch prep.
> - OA: normal FSH and testes; MESA is best (capillary action, never suction; move proximally): many motile sperm, multiple vials.
> - TESA: 23 G needle, suction, local anaesthesia; for OA; TESA/TESE when epididymal sperm absent.
> - NOA: micro-TESE best (sperm in about half); TESA yield poor.
> - AZFa/AZFb deletion: 0% retrieval, do not operate.
> - All surgically retrieved sperm need ICSI; cryopreserve extras.
> - Haematoma commonest; epididymal biopsy most serious.

LAQ 3 Surgical management of ED and penile prostheses (types, biofilms)

P2 · P3 · P4 **merged from 3**

Asked as: P2 LAQ "Surgical management of erectile dysfunction and penile prosthesis" (asked as "Discuss surgical management of erectile dysfunction", "Penile Prosthesis" and "Describe various penile prosthesis in ED") · P3 SAQ "Types of penile prosthesis. What are biofilms in penile prosthesis?" · P4 SAQ "Penile implants."

Introduction

- Surgery for ED (erectile dysfunction) means **penile prosthesis** (main operation) and, rarely, **penile vascular surgery**.
- Prosthesis: most effective, but last-line, treatment of ED; **satisfaction 80–90%**. First devices: Scott inflatable (1972), Small–Carrion rod (1978).
- Lue's goal-directed approach: surgery is for men who fail, cannot use or decline oral PDE5Is (phosphodiesterase type 5 inhibitors), intracavernosal injection, intraurethral alprostadil or a VED (vacuum erection device).
- Detailed vascular testing is needed only before vascular surgery.

Indications and contraindications for a prosthesis

- **Indications:** ED not responding to, contraindicated for, or unacceptable with (refusal of) oral PDE5Is, injections or vacuum devices; common after prostatectomy, pelvic radiotherapy, diabetes.
- Corporal (cavernosal) fibrosis: after ischaemic priapism (**>36 h**), infection or implant removal; early implant preserves length.
- **Peyronie disease with ED**; penile or pelvic trauma.
- **Contraindications:** active skin, urinary (UTI [urinary tract infection]) or systemic infection; uncontrolled diabetes; unrealistic expectations; reversible ED.
- In insensate men (spinal cord injury) use an inflatable device, because rods erode.

A. Penile vascular surgery

	Arterial revascularisation	**Venous surgery**
Defect	Arteriogenic (poor inflow)	Veno-occlusive (venous leak)
Work-up	Duplex; selective pudendal angiography (also checks the inferior epigastric donor)	DICC (dynamic infusion cavernosometry and cavernosography) locates the leak
Technique	Inferior epigastric artery to dorsal artery (revascularisation) or to deep dorsal vein (arterialisation)	Ligation or embolisation of deep dorsal, superficial dorsal or crural veins; crural plication
Selection	Age <55, non-smoker, non-diabetic, no venous leak, focal internal pudendal stenosis; best in young men after pelvic or perineal trauma	Not defined
Status	Strictly selected men	Investigational
Complications	Glans hyperaemia (commonest), shunt thrombosis, inguinal hernia	Glans numbness, shortening, curvature, skin necrosis

B. Penile prosthesis: types

Type	Devices / examples	Features	Best for
Malleable (semirigid) rods	Tactra (nitinol core), Genesis (braided silver), Rigi 10 (stainless steel)	Two bendable rods; simple, cheap; almost never fail (mechanical failure exceedingly rare; P3: no mechanical failure); always firm; erode more in insensate men	Poor manual dexterity; cost
Mechanical rods	Spectra (discontinued)	Articulating segments	–
2-piece inflatable	Ambicor (discontinued)	Cylinders + scrotal pump; no reservoir; poor flaccidity	–
3-piece inflatable	AMS 700 CX/LGX/CXR; Coloplast Titan, Titan Touch, Titan Narrow; Rigicon Infla 10	Two cylinders, scrotal pump, abdominal reservoir; **best rigidity and true flaccidity**; most popular	Most men

Three-piece inflatable details

- **Expansion:** girth only (CX, Titan) or girth and length (LGX, up to 25%; Infla 10 AX). **Narrow** cylinders (CXR, Titan Narrow) for fibrotic corpora.
- **Material:** AMS and Rigicon use silicone–fabric–silicone layers; Titan uses polyurethane (Bioflex).
- **Coatings:** InhibiZone (rifampicin + minocycline) on AMS; hydrophilic coating on Titan, dipped in antibiotic.
- **Reservoir:** space of Retzius, or **ectopic submuscular** after pelvic surgery or cystectomy; lockout valve prevents autoinflation.

Consent

- The erection may be shorter (pseudocapsule scar); the aim is a firm penis for intercourse; sensation and ejaculation may reduce; infection or mechanical failure may need reoperation.

Preoperative preparation

- Treat phimosis and skin, fungal and urine infection; sterile urine; stop anticoagulants and aspirin.
- Blood glucose **<200 mg/dL on the day of surgery** (more important than HbA1c [glycated haemoglobin]).
- Shave in theatre; chlorhexidine–alcohol skin preparation (about 40% fewer wound infections than povidone-iodine); MRSA (methicillin-resistant *Staphylococcus aureus*) decolonisation if a carrier.
- IV vancomycin or a cephalosporin plus gentamicin, 1 hour before incision, for up to 24 hours.

Operative steps (3-piece, penoscrotal approach)

`Catheter → upper scrotal incision → expose corpora lateral to urethra → stay sutures → corporotomy → dilate proximally to ischial tuberosity, distally under glans (instruments dorsolateral; 7→13 mm) → goalpost test for crossover → measure → cylinders equal to or 1 cm shorter → test inflation → close`

`corporotomies → pump in dependent subdartos (dartos) pouch → reservoir in space of Retzius → connect tubing → test inflation → irrigate → close in layers`

- Dilate with Hegar or Brooks dilators, instruments pointing dorsolaterally; check for crossover or perforation; cylinders sized to corporal length (equal to or 1 cm shorter than measured length).
- **Approaches:** penoscrotal (commonest; secures the pump; urethra at risk); infrapubic (reservoir placed under direct vision; dorsal nerve at risk); subcoronal or ventral penile for rods.
- Malleable rods are placed slightly shorter: 0.5–1 cm shorter than the measured length; girth chosen with the two-dilator test.
- **Surgical principles:** antibiotics, glucose control, no-touch technique.
- **Intraoperative problems:** crossover (redilate laterally); crural perforation (suture sling); distal urethral perforation or urethral injury (abandon that side for 5–6 months); fibrotic corpora (cavernotomes, narrow cylinders); curvature (modelling for Peyronie's disease); floppy glans (glans fixation).
- **Postoperative:** mummy wrap with glans visible; cylinders partly inflated; drain out by 24 hours; activation at 4–6 weeks.

Biofilm

- **Definition:** a structured microbial community in a self-made **polysaccharide matrix**, attached to the implant.
- **Stages:** 1. Conditioning film of host proteins coats the device within minutes. 2. Reversible, then irreversible bacterial adhesion. 3. Microcolonies. 4. Maturation into a 3D structure with water channels and quorum sensing. 5. Dispersal of free bacteria, causing clinical infection.
- **Organisms:** coagulase-negative staphylococci commonest; also *S. aureus*, Gram-negatives, *Candida*.
- **Why antibiotics fail:** the matrix blocks penetration, bacteria are slow-growing (persisters), and they survive up to **1000–1500 times** normal killing levels.
- **Relevance:** organisms from the first operation lie dormant; revising an uninfected device gives about 10% infection, cut to about 3% by **antiseptic washout**.

Complications

- **Mechanical failure** (commonest; inflatables): repair in about **15% at 5 years, 30% at 10 years**.
- **Infection (1–3%)**: biofilm on the device.
- **Prevention:** glucose <200 mg/dL; treat skin and urine; decolonise MRSA carriers; intravenous vancomycin or cephalosporin plus gentamicin (up to 24 h); chlorhexidine–alcohol preparation; **antibiotic-coated devices halve infection** (InhibiZone rifampin–minocycline; hydrophilic Titan dipped in antibiotic); **no-touch technique** (under 0.5%).
- **Infected implant:** pus, fixed pump or exposed parts. Remove **all** components.
- **Mulcahy salvage:** remove all parts; wash with dilute povidone-iodine (3 minutes) then saline or antibiotic; change gloves, drapes and instruments; insert a new device at once. Success **84%**; fails with early onset, aggressive organisms, cellulitis. Delayed reimplantation loses length.
- **Intraoperative:** crural perforation, crossover, urethral injury.

- **Others:** erosion or extrusion (catheterisation, rods in insensate men); reservoir erosion into bladder or bowel; reservoir herniation; pump migration; autoinflation; penile shortening; **glans necrosis** (diabetes, smoking, radiation, previous implant); glans hypermobility; cylinder aneurysm.

Special situations (also know)

- Ischaemic priapism >36 hours: early implantation preserves length.
- Transplant recipients: no extra infection.
- After phalloplasty: single cylinder, high erosion.
- Peyronie's disease: modelling.

Outcomes

- Patient and partner satisfaction **80–90%**, the highest of any ED treatment; the main complaint is shorter length.

Recent advances

- Minimal-dilation technique; flat ectopic reservoirs; penile transplantation. Regenerative therapies (Li-ESWT [low-intensity shock wave therapy], stem cells) remain experimental.

Diagrams to draw

- 3-piece inflatable prosthesis: two cylinders, scrotal pump, reservoir in the space of Retzius.
- Biofilm life cycle.

> **Key points to remember:**
> - Prosthesis is the last line but gives 80–90% satisfaction.
> - 3-piece inflatable gives the best rigidity and flaccidity.
> - Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
> - Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
> - Biofilm: microbes in a matrix; antibiotics fail.
> - Infection 1–3%; infected implant: remove all parts; Mulcahy salvage 84%.
> - Repairs 15% at 5 years.
> - Arterial revascularisation only for young men with post-traumatic focal stenosis.

LAQ 4 Priapism: diagnosis and management

P2 · P3 **merged from 2**

Asked as: P3 LAQ "Priapism: Diagnosis and Management" (also asked as P3 SAQ "Priapism") · P2 SAQ "Management of priapism"

Definition

- Full or partial erection lasting **≥4 hours** beyond sexual stimulation, or unrelated to it (AUA/SMSNA [American Urological Association/Sexual Medicine Society of North America] 2021–2022).

Classification (types)

	Ischaemic (low flow)	**Stuttering (recurrent ischaemic)**	**Non-ischaemic (high flow)**
Share	**95%**	Subset of ischaemic	<5%
Lesion	Veno-occlusion	Repeated (recurrent) ischaemic episodes	Cavernosal artery–sinusoid (arterio-sinusoidal) fistula after trauma
Pain, rigidity	Painful, fully rigid	Painful, recurrent, often nocturnal	Painless, partial
Typical cause	ICI (intracavernosal injection), drugs, SCD (sickle cell disease)	SCD	Perineal or needle trauma
Urgency	**Emergency**	Emergency per episode + prophylaxis	Not an emergency

Aetiology

- **Ischaemic:** ICI (papaverine, phentolamine, PGE1 [prostaglandin E1]), including recreational use; α-blockers, trazodone, antipsychotics, cocaine; heparin or warfarin withdrawal; **SCD** (29–42% of affected men in their lifetime; 63% of paediatric ischaemic priapism); leukaemia (1–5%, high white cell count); penile metastases; scorpion and spider venom.
- **Non-ischaemic:** straddle or coital injury, pelvic fracture, needle injury. Onset is often delayed, when the clot over the arterial tear lyses.

Pathophysiology

- **Ischaemic:** failed outflow acts like a compartment syndrome. Hypoxia, hypercarbia and acidosis cause sinusoidal thrombosis, smooth-muscle necrosis and TGF-β (transforming growth factor beta)-driven fibrosis, so ED (erectile dysfunction) follows. Irreversible damage starts within hours.
- **Stuttering (molecular):** low basal NO (nitric oxide) from eNOS (endothelial NO synthase) dysfunction downregulates PDE5 (phosphodiesterase 5). When an erection starts, cGMP (cyclic guanosine monophosphate) is not broken down, so relaxation is unopposed. Low Rho-kinase and high adenosine add to this. This explains why daily PDE5 inhibitors prevent episodes (they reset PDE5 regulation).
- **Non-ischaemic:** unregulated, well-oxygenated arterial inflow, so fibrosis is unusual.

Diagnosis

- **History:** duration, pain, drugs, trauma, SCD or blood disease, prior episodes, baseline erectile function.
- **Examination:** rigid tender corpora with a soft glans and spongiosum (ischaemic); partial, non-tender (high flow); perineal bruising; vital signs before phenylephrine.
- **Corporal blood gas** (cornerstone; decides the type); high flow resembles arterial blood:

	Colour	pO2 (mmHg)	pCO2 (mmHg)	pH
Ischaemic	Dark red	**<30**	**>60**	**<7.25**
Flaccid (mixed venous)	Medium red	40	50	7.35
Non-ischaemic (arterial)	Bright red	>90	<40	7.40

- **PDUS (penile duplex ultrasound, colour Doppler)** when the gas is equivocal: ischaemic = absent cavernosal flow or PSV (peak systolic velocity) <50 cm/s with no diastolic flow; high flow = PSV >50 cm/s, may show the fistula. Angiography only for embolisation.
- Blood count, reticulocytes, haemoglobin electrophoresis, coagulation, toxicology: in parallel, never delaying treatment.

Management of ischaemic priapism (AUA/SMSNA 2021)

``` Counsel on ED risk → dorsal penile block <36 h: aspiration (19 G, penoscrotal junction, 3 or 9 o'clock) ± saline irrigation + phenylephrine 100–500 µg/mL, 1 mL every 3–5 min, up to 1 h (monitor blood pressure, pulse: reflex bradycardia) → fails → distal shunt (Winter, T-shunt, Al-Ghorab) ± tunnelling → proximal shunt (Quackels) or penoscrotal decompression

> 36 h or refractory → early (immediate) penile prosthesis (or analgesia and counselling) ```

- Aspiration alone succeeds in under 1 in 4. Cold (10 °C) saline irrigation was better than body temperature in an RCT (randomised controlled trial); avoid chilled saline in SCD.
- Ice, oral agents, hydration and IV (intravenous) fluids are adjuncts only; in SCD, intracavernosal treatment still comes first, with concurrent haematology care.

| Shunt | Method |
|---|---|
| **Winter** (distal) | Tru-Cut (biopsy) needle through glans; least effective |
| **Ebbehoj / T-shunt** (distal) | Scalpel through glans; T-shunt rotates blade 90° |
| **Al-Ghorab** (distal, open) | Dorsal glans incision; excise corporal (tunica) tips |
| **Snake tunnelling** | 7/8 Hegar dilator passed proximally through distal shunt |
| **Quackels / Sacher** (proximal) | Cavernosum to spongiosum at bulb; stagger ≥1 cm |
| **Grayhack / Barry** (proximal) | Cavernosum to saphenous / dorsal vein |
| **Penoscrotal decompression** | Proximal corporotomy, clot suction (evacuation); allows prosthesis |

- **ED by duration** (Bennett and Mulhall, SCD series): ≤12 h 0%; 12–24 h 22%; 24–36 h 56%; ≥36 h 100%.
- **Early prosthesis** (Zacharakis 2014): less shortening and fewer revisions than delayed insertion: shortening 3% vs 40%, revision 9% vs 27%, satisfaction 96% vs 60%.

## Stuttering priapism

- An episode lasting ≥4 hours is treated as ischaemic; home self-injection of phenylephrine.
```

- **Prophylaxis:** daily PDE5 inhibitor (Burnett 2014 RCT: primary endpoint missed, but promising), pseudoephedrine, etilefrine; hormonal agents (GnRH [gonadotropin-releasing hormone] agonists, antiandrogens, finasteride) at the cost of libido. **Also know:** digoxin; baclofen.

Non-ischaemic priapism (AUA/SMSNA 2022)

- Observation, perineal compression and ice; about 60% resolve. Reassess at 4 weeks.
- Persistent: PDUS, then **selective cavernosal artery embolisation** (resolution ≥75%, recurrence 30–40%). For recurrence, repeat embolisation with non-resorbable PVA (polyvinyl alcohol), not surgery. Surgical ligation is a last resort (ED about 50%).

Complications

- ED, corporal fibrosis, penile shortening; shunt-related oedema, haematoma, urethral injury or fistula, skin necrosis, infection.

Diagram to draw: distal (glans) and proximal shunt sites on a penile outline.

> **Key points to remember:**
> - Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
> - Aspiration plus phenylephrine first; distal shunt ± tunnelling before proximal.
> - ED risk 0% ≤12 h, 100% ≥36 h; early prosthesis beyond 36 h or if refractory.
> - High flow: observe (about 60% resolve), then embolise; repeat embolisation rather than surgery.

LAQ 5 Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED

P1

Asked as:

- Write in detail about physiology of erection. How will you evaluate a diabetic with ED? — *Practice Set 5 Paper I Q2 (LAQ)*
- Neurotransmitters in erectile function — *Practice Set 12 Paper I Q3e (SAQ)*

Type: LAQ (25 marks); also asked as SAQ (10 marks)

(erection physiology also in Practice Set 5 Paper 1, Q2)

Introduction

- Erection is **cavernous smooth-muscle relaxation**: low cytosolic Ca^{2+} dilates helicine arteries and sinusoids; expanding sinusoids compress emissary veins (veno-occlusion).
- Pathways: parasympathetic **S2–S4** cavernous nerves (erection), sympathetic **T11–L2** (detumescence), pudendal (sensation, ischiocavernosus rigidity).

Functional anatomy

- **Corpora cavernosa:** sinusoids lined by endothelium, separated by **trabecular smooth muscle**. Flaccid tone depends on smooth-muscle contraction.
- **Tunica albuginea:** inner circular and outer longitudinal layers. **Emissary veins** pass obliquely between them and are compressed when the corpora expand: the **veno-occlusive mechanism**.
- **Arteries:** internal pudendal → common penile → **cavernous (helicine branches)**, dorsal and bulbourethral arteries. Accessory pudendal arteries in about one-third.
- Spongiosum and glans have a thin or absent tunica, so no veno-occlusion; they stay low-pressure.

Haemodynamic phases

Phase	Events	Intracavernous pressure
Flaccid	Smooth muscle contracted; low inflow	Low
Latent (filling)	Arterial dilation; **inflow peaks**	Barely rises
Tumescence	Sinusoids fill; venous outflow compressed	Rising
Full erection	Maximal veno-occlusion	About 100 mmHg
Rigid	**Ischiocavernosus** contraction (pudendal nerve)	Several hundred mmHg
Detumescence	Initial, slow, fast phases	Falls to baseline

Molecular mechanism and peripheral neurotransmitters

- **Relaxation (erection):** NO (nitric oxide) from **nNOS (neuronal nitric oxide synthase)** in cavernous nerves initiates erection; endothelial NOS (eNOS) sustains it.
- NO → soluble guanylyl cyclase → **cGMP (cyclic guanosine monophosphate)** → protein kinase G (PKG) → K^+ channels open, Ca^{2+} channels close, Ca^{2+} sequestered → **low cytosolic Ca^{2+}** → relaxation.

- Second route: PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide) → **cAMP (cyclic adenosine monophosphate)** → protein kinase A.
- **PDE5 (phosphodiesterase type 5)** degrades (breaks down) cGMP and ends relaxation (ends / terminates erection); PDE5 inhibitors block it.

Peripheral pro-erectile transmitters

Transmitter	Source / mechanism
NO (nitric oxide)	Principal. **nNOS (neuronal NOS)** initiates; **eNOS (endothelial)** sustains. NO → soluble guanylyl cyclase → **cGMP** → PKG → K^+ channels open, Ca^{2+} channels close → relaxation
ACh (acetylcholine)	Indirect: endothelial NO release, inhibits adrenergic nerves; atropine does not block erection
PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide), CGRP (calcitonin gene-related peptide)	**cAMP** → PKA (protein kinase A); VIP role minimal in man
CO, H2S, EDHF (endothelium-derived hyperpolarising factor)	Gaseous/endothelial relaxants via cGMP or K^+ channels

- O2 is a NOS substrate: low flaccid Po2 (~35 mmHg) limits NO; erect Po2 ~90 mmHg.
- **Also know:** carbon monoxide and hydrogen sulfide also raise cGMP; acetylcholine acts indirectly through endothelial NO (atropine does not block erection).

Anti-erectile transmitters (flaccidity and detumescence)

- **Contraction (flaccidity): noradrenaline (α1)** (principal), endothelin-1, angiotensin II → raised Ca^{2+} → calmodulin → myosin light-chain kinase → contraction.
- **Noradrenaline (α1)**: principal; α2 prejunctional reduces noradrenaline and NO release.
- **Endothelin-1** (ET-A), angiotensin II (AT1), PGF2α, thromboxane A2: Gq → IP3 → Ca^{2+} rise → Ca^{2+}–calmodulin → MLC (myosin light-chain) kinase → contraction.
- **Rho-kinase** inhibits myosin (MLC) phosphatase (calcium sensitisation) and maintains (sustaining) tone.
- **Detumescence:** NO release stops, PDE5 degrades cGMP and sympathetic discharge contracts the smooth muscle, so emissary veins reopen.
- Gap junctions (connexin-43) synchronise the smooth muscle.

Neural control

Pathway	Origin	Role
Parasympathetic	**S2–S4** → pelvic plexus → cavernous nerves	Erection
Sympathetic	**T11–L2** → hypogastric nerves	Detumescence, emission
Somatic	Pudendal nerve (Onuf nucleus)	Penile sensation; ischiocavernosus rigidity

- **Three types: psychogenic** (brain, chiefly by reducing sympathetic tone), **reflexogenic** (genital touch, sacral arc; preserved in upper cord injury), **nocturnal** (REM [rapid eye movement] sleep).
- **Central:** MPOA (medial preoptic area) and **PVN (paraventricular nucleus)**. Dopamine and oxytocin promote; serotonin, prolactin, GABA (γ-aminobutyric acid) and opioids inhibit.

Central neurotransmitters (PVN [paraventricular nucleus], MPOA [medial preoptic area])

Promote	Inhibit
Dopamine (D2 in PVN; apomorphine)	**Serotonin** (5-HT1A, 5-HT2; but 5-HT2C pro-erectile)
Oxytocin	GABA (γ-aminobutyric acid)
Central NO	Opioids (high dose), cannabinoids
Central noradrenaline; melanocortins (MC4R)	**Prolactin**

- Dopamine and oxytocin converge on NO in the PVN. Noradrenaline is pro-sexual centrally but anti-erectile peripherally.

Clinical relevance

- PDE5 inhibitors, intracavernosal alprostadil (cAMP), papaverine, phenylephrine for priapism (α1), SSRIs (selective serotonin reuptake inhibitors) cause dysfunction, hyperprolactinaemia lowers libido.

Diagram to draw: NO–cGMP pathway in the cavernous smooth-muscle cell.

Evaluation of a diabetic with ED (erectile dysfunction)

Why diabetics: ED is commoner, earlier and more severe in DM (diabetes mellitus). Mechanisms: endothelial NO loss, autonomic neuropathy, arteriopathy, smooth-muscle loss and fibrosis (veno-occlusive failure), AGEs (advanced glycation end products) and low testosterone. ED is a **sentinel of cardiovascular disease**.

1. History

- Onset, duration (≥3 months), severity, morning erections, libido, ejaculation; partner.
- **IIEF-5 (International Index of Erectile Function, SHIM):** 5–7 severe, 8–11 moderate, 12–16 mild-to-moderate, 17–21 mild, 22–25 no ED.
- Diabetes: type, duration, **glycaemic control**, complications (retinopathy, nephropathy, neuropathy, foot disease), autonomic symptoms (postural giddiness, gastroparesis, bladder dysfunction).
- Cardiovascular: angina, exercise tolerance, smoking, hypertension, dyslipidaemia, nitrates.
- Drugs: antihypertensives, antidepressants. Depression screen.

2. Examination

- BMI, waist, blood pressure, peripheral pulses.
- Genitalia: Peyronie plaques, testicular size, gynaecomastia.
- Neurological: genital sensation, **bulbocavernosus reflex**, foot sensation. Penile neuropathy can exist without foot neuropathy.

3. Laboratory

- **HbA1c (glycated haemoglobin)** and fasting glucose, **lipid profile**, creatinine/eGFR (estimated glomerular filtration rate), urine albumin.
- **Morning total testosterone**; if low, repeat with free testosterone, LH (luteinising hormone), prolactin. TSH (thyroid-stimulating hormone) if indicated.

4. Cardiovascular risk stratification (Princeton Consensus)

- **High risk** (unstable angina, recent infarction, uncontrolled hypertension, some arrhythmias): defer sexual activity and treatment; cardiology referral. **Indeterminate:**

cardiac assessment (e.g. stress test), then restratify. **Low:** start treatment.

5. Specialised tests (selected cases)

- **CIS (combined intracavernous injection and stimulation):** alprostadil 10–20 µg; a rigid erection excludes major vascular disease.
- **Penile duplex Doppler:** PSV (peak systolic velocity) **<25 cm/s = arterial insufficiency** (>35 normal); EDV (end-diastolic velocity) **>5 cm/s** and RI (resistive index) **<0.75 = veno-occlusive dysfunction**.
- **NPTR (nocturnal penile tumescence and rigidity), RigiScan:** separates psychogenic from organic ED.
- **Neurophysiological:** bulbocavernosus reflex latency, dorsal nerve conduction velocity, biothesiometry.
- DICC (dynamic infusion cavernosometry and cavernosography) and angiography: rarely, as diffuse diabetic vascular disease is not suited to revascularisation.

```
Diabetic with ED → history + IIEF-5 + examination → HbA1c, lipids,
testosterone, renal function → Princeton cardiac risk → treat risk factors →
PDE5 inhibitor trial → failure → duplex/CIS → ICI or prosthesis
```

> **Key points to remember:**
> - nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
> - Noradrenaline α1 is the main anti-erectile transmitter.
> - Dopamine/oxytocin promote; serotonin/prolactin inhibit.
> - Rigid phase: ischiocavernosus, suprasystolic pressure.
> - Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
> - Diabetic ED: check HbA1c, testosterone and cardiac risk.

LAQ 6 Discuss the role of urologist in reproductive endocrinology and male infertility practice.

P2

Type: LAQ (25 marks)
Source: 2022-Summer Paper 2, Q1

Introduction

- **Infertility:** no conception after 12 months of regular unprotected intercourse; about **15% of couples** (WHO, World Health Organization). The male contributes about half, yet many men are never examined.
- The urologist (andrologist) has five roles: **diagnose** the male factor, **treat endocrine causes**, **correct surgical causes**, **retrieve sperm for ART (assisted reproductive technology)**, and protect **general health and future fertility**.
- Commonest diagnoses: idiopathic, **varicocele** (commonest correctable), obstruction.

1. Diagnostic role

- **History:** duration, coital timing (semen is best after 1–2 days' abstinence), partner's age (fecundity falls after 35), cryptorchidism, mesh herniorrhaphy, RPLND (retroperitoneal lymph node dissection), diabetes, spinal injury, smoking, cannabis, opioids, **exogenous testosterone or anabolic steroids**.
- **Examination:** androgenisation; **testis size** (reflects spermatogenic mass); epididymis; **both vasa present?**; varicocele (standing, Valsalva).
- **Semen analysis**, at least two. WHO 6th edition (2021) 5th centiles: volume 1.4 mL, concentration 16 million/mL, total count 39 million, progressive motility 30%, normal forms 4%. These describe fertile men, not fertility cut-offs.
- **Genetics:** karyotype and **Y-chromosome microdeletion** (AZF, azoospermia factor) in non-obstructive azoospermia or severe oligospermia; **CFTR (cystic fibrosis transmembrane conductance regulator)** testing of both partners in CBAVD (congenital bilateral absence of the vas deferens).
- **Imaging:** scrotal US (ultrasound); **TRUS (transrectal US)** if volume <1 mL (ejaculatory duct obstruction, EDO), after post-ejaculate urine excludes retrograde ejaculation; renal US in vasal agenesis.

2. Endocrine evaluation (the urologist's reproductive endocrinology)

- Spermatogenesis needs a **high intratesticular testosterone (T)**. Morning total T + SHBG (sex hormone-binding globulin) + albumin gives bioavailable T; low is below about **280–300 ng/dL**; repeat borderline values (T is pulsatile and circadian).

Test	What it tells
LH (luteinising hormone)	Localises low T: high → testicular (Leydig) failure; low/normal → pituitary or hypothalamus
FSH (follicle-stimulating hormone) / inhibin B	Germ-cell mass. Azoospermia with **FSH >7.6 IU/L and testis long axis ≤4.6 cm → spermatogenic failure**; FSH ≤7.6 and >4.6 cm → obstruction
Oestradiol (E2)	**T (ng/dL) : E2 (pg/mL) <10** → aromatase excess (obesity)

Test	What it tells
Prolactin	Only with low T, low libido/ED (erectile dysfunction), headache or field defect
TSH (thyroid-stimulating hormone), cortisol, 17-hydroxyprogesterone	Only if signs; CAH (congenital adrenal hyperplasia)

3. Medical (endocrine) treatment

Condition	Mechanism	Treatment
Hypogonadotropic hypogonadism (Kallmann, pituitary)	Testis normal, signal missing: most treatable cause	**hCG (human chorionic gonadotropin) 1500–5000 IU 2–3 times weekly**, add hMG (human menopausal gonadotropin) 75 IU or recombinant FSH; pulsatile GnRH (gonadotropin-releasing hormone) pump for hypothalamic failure. Sperm may take 1–2 years
Anabolic steroids / exogenous T	Negative feedback suppresses LH → intratesticular T falls → azoospermia	Stop; hCG ± clomiphene to speed recovery. **Never give T to a man wanting fertility** (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine)
Hyperprolactinaemia	Prolactin inhibits GnRH	Repeat level; pituitary MRI (magnetic resonance imaging) if high; **cabergoline** (fewer side effects than bromocriptine)
Partial hypogonadism, obesity	Aromatisation of T to E2 suppresses LH	Weight loss; **clomiphene 25 mg daily** (off-label), **anastrozole 1 mg daily** or letrozole if T:E2 <10
CAH	Adrenal androgens suppress gonadotropins; testicular adrenal rest tumours	Glucocorticoids
Primary testicular failure (Klinefelter 47,XXY)	Germ cells absent or few	No drug restores sperm; exogenous T worsens it; **micro-TESE** (microdissection testicular sperm extraction)

4. Surgical role

Problem	Operation
Clinical varicocele + abnormal semen	**Microsurgical subinguinal varicocelectomy** or embolisation: improves semen, DNA fragmentation and pregnancy. Not for US-only varicocele
Vasal or epididymal obstruction	**Vasovasostomy / vasoepididymostomy** (microsurgical intussusception)
EDO	**TURED** (transurethral resection of ejaculatory ducts)
Non-obstructive azoospermia	**Micro-TESE**, best technique; no operation for **AZFa/AZFb** deletions (no sperm found)
Unreconstructable obstruction	PESA/MESA (percutaneous/microsurgical epididymal sperm aspiration), TESA (testicular sperm aspiration)
Retrograde ejaculation; anejaculation	Sympathomimetics, urine sperm retrieval; **penile vibratory stimulation**, electroejaculation

5. Partner of the ART team, counselling and men's health

- Choose the least invasive route: natural conception, IUI (intrauterine insemination), IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection). **Cryopreserve sperm** at retrieval.
- **Fertility preservation:** sperm banking before chemotherapy, radiotherapy or RPLND; **onco-TESE** if azoospermic.
- **Genetic counselling:** AZFc deletion passes to sons; CFTR risk to offspring; paternal age ≥40 years raises offspring risk.
- **Infertility is a marker of general health:** higher risk of **testis cancer** (scan every azoospermic man), later comorbidity and mortality.

Recent advances: sperm DNA fragmentation testing; microfluidic sperm selection; genomic panels; prepubertal testicular tissue cryopreservation (experimental).

Also know: antisperm antibodies; globozoospermia and ciliary dyskinesia (ICSI); androgen receptor defects (high T and LH).

Diagram to draw: hypothalamic–pituitary–testis axis with feedback loops (T and E2 on LH; inhibin B on FSH).

> **Key points to remember:**
> - FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
> - Exogenous testosterone is a contraceptive: never give it for infertility.
> - Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
> - Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

Short answer questions (SAQ)

SAQ 1 Male reproductive immunology and immunobead test

P1 · P3 **merged from 2**

Asked as: P1 SAQ "Male reproductive immunology" · P3 SAQ "Immunobead test for semen"

Introduction

- Haploid germ cells appear at puberty, after self-tolerance is established, so their antigens are "foreign". The testis is therefore **immune-privileged**.

Immune privilege

Anatomical barrier: blood–testis barrier (BTB)

- **Tight junctions between Sertoli cells**, formed at puberty, divide a **basal compartment** (spermatogonia, plasma-like fluid) from an **adluminal compartment** (later spermatocytes, spermatids). If the barrier is breached, sperm antigens meet the immune system.

Functional barrier

- A small antigen leak at the **rete testis, efferent ductules and epididymis** induces suppressor T cells and tolerance; a large load provokes immunity.
- Sertoli and local mediators: galectin-1, TGF-β (transforming growth factor-β), interferon-γ, soluble Fc receptor, **androgens**.

- Insult before puberty does not induce antibodies; the barrier also limits chemotherapy (isolated testicular relapse).

Antisperm antibodies (ASA)

- **Causes:** obstruction (**vasectomy** commonest); torsion, trauma, biopsy; infection (orchitis, prostatitis, STI [sexually transmitted infection]); heat (varicocele, cryptorchidism); genetic (human leukocyte antigen HLA-B28).
- **Effects:** agglutination, poor motility, failure to cross (poor penetration of) cervical mucus, failure to bind the zona and impaired fertilisation; head-bound antibody matters most.
- **Indications to test:** sperm agglutination or isolated asthenozoospermia; unexplained infertility; after vasectomy reversal.

Tests for ASA: immunobead test (IBT) and MAR

- The IBT detects ASA bound to sperm (direct) or free in body fluids (indirect).
- **Principle:** polyacrylamide beads coated with rabbit anti-human **IgG, IgA or IgM**.

Test	Method	Use
Direct IBT	Patient's motile sperm washed free of seminal plasma; mixed with beads coated with anti-human IgG/IgA; motile sperm with attached beads counted	Sperm-bound antibody: % bound, class and site (head/tail)
Indirect IBT	Antibody-free donor sperm incubated with heat-inactivated patient serum, seminal plasma or female serum/cervical mucus; then direct IBT done	Free antibody; patient sperm too few or immotile; female partner
MAR (mixed antiglobulin reaction)	Fresh unwashed semen + immunoglobulin (Ig) G (or IgA)-coated latex beads + antiglobulin (anti-IgG); beads stick to antibody-coated motile sperm; mixed sperm–bead clumps counted	Quick, cheap screen; IgG/IgA only

- Report: **% of motile sperm with beads**, immunoglobulin class, and **site** (head, midpiece, tail). Tail-tip binding is ignored.
- **WHO (World Health Organization) 2010 (5th edition) threshold: ≥50% of motile sperm bound is clinically significant.**
- **Head binding** and **IgA** matter most (IgA is produced locally and blocks mucus penetration; IgM is too large to enter semen).
- Both direct tests need motile sperm.
- **Also know:** ELISA (enzyme-linked immunosorbent assay) and flow cytometry; serum tests correlate poorly with sperm-bound antibody.

Management of ASA

Treatment	Comment
Condoms	Lack good evidence
Corticosteroids	Two controlled trials conflict; risk of avascular necrosis; not recommended
Sperm washing + intrauterine insemination (IUI)	Bypasses cervical mucus; for mild binding; modest benefit, lacks good evidence (washing removes unbound, not bound, antibody)

Treatment	Comment
IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)	Preferred, usual treatment; fertilisation largely unaffected, bypassing antibody effects

Infection and inflammation

- **Leukocytospermia:** >1 × 10^6 peroxidase-positive leukocytes/mL. Reactive oxygen species and cytokines damage sperm.
- **Orchitis:** mumps (post-pubertal, atrophy); bacterial (*Chlamydia*/gonococcus if young, *E. coli* if older); tuberculous; **autoimmune**, primary (antibodies with azoospermia) or secondary to vasculitis.
- **Also know:** sperm DNA fragmentation testing; prostatitis harms sperm via oxidative stress and ASA; seminal plasma suppresses female tract immunity.

Diagrams to draw

- Seminiferous tubule with Sertoli tight junctions and compartments.
- Motile sperm with beads attached at head and tail.

> **Key points to remember:**
> - BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
> - ASA: vasectomy commonest; ≥50% bound significant; head-bound IgA matters most.
> - Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
> - Treatment of choice: IVF/ICSI.
> - Leukocytospermia >1 million/mL.

SAQ 2 Premature ejaculation

P1 · P3 **merged from 2**

Asked as: P1 SAQ "Premature Ejaculation" · P3 SAQ "Premature ejaculation."

Definition

- **ISSM (International Society for Sexual Medicine) unified definition (2014):** all three needed: 1. Ejaculation always or nearly always within **about 1 minute** of vaginal penetration (lifelong), or a clinically significant, bothersome fall in latency, often to **≤3 minutes** (acquired). 2. Inability to delay ejaculation on all or nearly all penetrations. 3. Negative personal consequences: distress, frustration, avoidance of intimacy.
- **AUA/SMSNA (American Urological Association/Sexual Medicine Society of North America) 2020:** lifelong PE (premature ejaculation) within about **2 minutes**.
- **Types:** lifelong, acquired, variable and subjective (normal latency).

Physiology and aetiology

- Ejaculation = **emission** (sympathetic T10–L2) then **expulsion** (somatic S2–S4, pudendal). **Serotonin brakes and dopamine accelerates**.
- **Lifelong PE:** neurobiological and genetic; 5-HT2C hyposensitivity and/or 5-HT1A hypersensitivity.

- **Acquired:** performance anxiety, relationship problems, **ED (erectile dysfunction)**, chronic prostatitis, hyperthyroidism, drug withdrawal.

Evaluation

- Diagnosis is by **history**: latency, control, bother; lifelong or acquired.
- **IELT (intravaginal ejaculatory latency time):** self-estimate in clinic; stopwatch in trials. Population median 5.4 minutes.
- **PEDT (Premature Ejaculation Diagnostic Tool):** 5 items; score ≥11 suggests PE, 9–10 probable PE. Also PEP (Premature Ejaculation Profile) and IPE (Index of Premature Ejaculation).
- IIEF-5 (International Index of Erectile Function) for ED; examination; TSH (thyroid-stimulating hormone) only if suggested.

Office algorithm (diagram to draw):

1. Confirm PE.
2. Secondary to ED? → treat the cause.
3. Otherwise → classify lifelong or acquired.

Management

- **Psychosexual education** for all; behavioural therapy (stop–start, squeeze).
- Acquired PE: treat the cause first (ED, prostatitis, thyroid).

Drug	Dose and timing
Dapoxetine (short-acting SSRI [selective serotonin reuptake inhibitor]; only licensed oral drug)	30–60 mg, 1–3 h before intercourse
Paroxetine (strongest delay) 10–40 mg, sertraline, fluoxetine, citalopram	Daily, off-label; full effect 2–3 weeks; do not stop abruptly
Clomipramine	12.5–50 mg daily or 3–4 h before
Topical lidocaine–prilocaine spray (licensed in Europe 2014)	5–30 min before (P1: at least 5 min before); condom avoids partner numbness
PDE5Is (phosphodiesterase type 5 inhibitors)	With ED, alone or with SSRI
Tramadol	Low evidence; addiction risk

- SSRIs raise synaptic serotonin, which delays ejaculation. Avoid in bipolar disorder.
- **Also know:** high dropout from drug therapy.

> **Key points to remember:**
> - ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
> - Lifelong is neurobiological; acquired is usually secondary.
> - Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

SAQ 3 Oligospermia and oligoasthenozoospermia

P2 · P3 **merged from 2**

Asked as: P3 SAQ "Oligospermia and oligoasthenozoospermia" (asked as "Oligoanaesthospermia" and "Oligospermia") · P2 SAQ "Oligospermia."

Assumed meaning (oligoanaesthospermia): oligoasthenozoospermia (low sperm count with poor motility).

Definition (WHO [World Health Organization] 2021, 6th edition, 5th centiles)

- **Oligozoospermia:** sperm concentration below the lower reference limit: **WHO 2021 (6th edition) 16 million/mL** (5th centile; total count 39 million per ejaculate). Earlier thresholds were 20 million/mL (1999, the legacy cut-off) and 15 million/mL (2010).
- **Severity:** mild to moderate 5–16 million/mL; **severe oligozoospermia <5 million/mL**; cryptozoospermia: sperm only in a centrifuged pellet.

Parameter (WHO 2021 5th centiles)	Lower reference value
Concentration	16 million/mL (total 39 million)
Volume	1.4 mL
Progressive motility	30%
Total motility	42%
Normal forms (strict)	4%
Vitality	54%

- These centiles come from fertile men; they are a reference, not pass/fail fertility cut-offs. Define oligospermia in context: read a single value with testis size, FSH (follicle-stimulating hormone) and repeated values.

Causes

- **Varicocele** (commonest correctable; heat and oxidative stress) and **idiopathic** (largest group).
- **Pre-testicular (hormonal):** hypogonadotropic hypogonadism, **exogenous testosterone or anabolic steroids**, obesity (raised oestradiol), hyperprolactinaemia, opioids.
- **Testicular: varicocele**, cryptorchidism, mumps orchitis, torsion, chemotherapy or radiotherapy, gonadotoxins; genetic (Klinefelter, Klinefelter mosaic, **AZFc [azoospermia factor c] deletion**).
- **Post-testicular (motility):** infection or **leucocytospermia** (over 1 million/mL), antisperm antibodies, partial ejaculatory duct obstruction (EDO; low volume), PCD (primary ciliary dyskinesia).
- **Lifestyle:** smoking, alcohol, cannabis, heat, drugs, gonadotoxins.

Evaluation

- **At least two semen analyses, 2–3 weeks apart**, same abstinence (WHO 2–7 days of abstinence; parameters peak at 1–2 days).
- History and examination: testis size/volume, varicocele, vasa, secondary sex characters.
- Morning **testosterone, LH (luteinising hormone), FSH**; oestradiol and prolactin if testosterone is low.
- **Under 5 million/mL** (especially with raised FSH or small testes): **karyotype and Y-microdeletion** testing (AUA/ASRM [American Urological Association/American Society for Reproductive Medicine] 2020 guideline).
- Selected: vitality test (eosin or hypo-osmotic swelling), antisperm antibodies, semen culture, scrotal Doppler, TRUS (transrectal ultrasonography) if volume is low (<1.0 mL), DNA fragmentation.

Management

- **General:** stop smoking, weight loss, avoid heat; **stop exogenous testosterone** and gonadotoxins; treat infection.
- **Varicocelectomy** (microsurgical subinguinal) for a **palpable** varicocele with abnormal semen; it improves semen parameters and pregnancy rates. Never treat a subclinical one.
- **Medical:**
- Hypogonadotropic: hCG (human chorionic gonadotropin) ± FSH.
- Idiopathic with low testosterone: off-label clomiphene 25 mg daily, or anastrozole 1 mg daily if the testosterone:oestradiol ratio is low (<10).
- Antioxidants: weak evidence.
- **ART (assisted reproductive technology):** mild: IUI (intrauterine insemination); severe or failed IUI: **IVF/ICSI (in vitro fertilisation/intracytoplasmic sperm injection)**. Bank sperm in AZFc carriers because counts may fall.
- Spermatogenesis takes about 64 days, so judge response (reassess) after 3–6 months.

Diagram to draw: hypothalamic–pituitary–testis axis with sites of treatment.

> **Key points to remember:**
> - WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
> - Karyotype and Y-microdeletion if under 5 million/mL.
> - Repair palpable varicoceles only; never give testosterone.

SAQ 4 Management of Peyronie's disease

P1

Type: SAQ (10 marks)
Source: 2017-Winter Paper 1, Q3a

Definition

- **PD (Peyronie's disease)** (AUA (American Urological Association) 2015): acquired fibrosis of the tunica albuginea, with pain, deformity, ED (erectile dysfunction) and/or distress. Prevalence 3–20%; up to half worsen untreated.

Phases

- **Acute (active, first 12–18 months):** pain, worsening deformity; aim to stop progression. **Stable:** pain gone, deformity unchanged for ≥3–6 months; aim to straighten.

Evaluation

- History: pain, deformity, Dupuytren disease, ability to penetrate. **Key question:** "If your penis were straight, would your current rigidity allow intercourse?" (predicts postoperative function).
- Plaque, SPL (stretched penile length); **office intracavernosal injection ± duplex Doppler ultrasound** before any invasive treatment (curvature by goniometer, calcification, vascular status).
- **Kelami grading:** ≤30° mild, 31–60° moderate, >60° severe; note hourglass or hinge. PDQ (Peyronie's Disease Questionnaire).

Non-surgical treatment

Option	Evidence/position
NSAIDs (non-steroidal anti-inflammatory drugs)	For pain (only oral agent AUA supports)
Oral vitamin E, tamoxifen, potaba, pentoxifylline	No reliable curvature benefit; not recommended
CCH (collagenase *Clostridium histolyticum*)	Stable disease, curve 30–90°, intact erections. **IMPRESS I/II:** 34% improvement vs 18% placebo. Up to 4 cycles (2 injections of 0.58 mg) with modelling. Risk: corporal fracture
Verapamil, interferon α-2b (intralesional)	May offer; stabilise and modestly improve
Traction therapy	Gains length, reduces curvature; useful in acute phase
ESWT (extracorporeal shock wave therapy)	Pain only; no effect on curvature

Surgery

- **Indications:** stable disease (≥12 months, stable ≥6 months), deformity preventing intercourse, failed conservative therapy.

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Erection adequate (± PDE5I (phosphodiesterase-5 inhibitor))? ├─ No → penile prosthesis (inflatable) + manual modelling; │ residual >30° → tunical incision; graft if defect >2 cm └─ Yes → curve <60–70°, no hourglass/hinge → plication (Nesbit, 16-dot, tunica albuginea plication) curve >60–70°, hourglass, hinge, short penis → plaque incision/partial excision + graft (pericardium, small intestinal submucosa)
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- **Plication** shortens the convex side: simple, low ED risk, but loses length.
- **Grafting** restores length and girth but carries higher ED risk (5–54%).

Diagram to draw: Nesbit wedge excised opposite the plaque.

> **Key points to remember:**
> - Ask the "straight penis" question; duplex before surgery.
> - CCH: IMPRESS, stable 30–90° curve, intact erections.
> - Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

SAQ 5 Haematospermia

P1

Asked as:

- Hematospermia — *Winter 2025 Paper I Q3c (SAQ)*
- Hematospermia — *Practice Set 9 Paper I Q3d (SAQ)* (also answered in 2025-Winter Paper 1, Q3c)

Type: SAQ (10 marks)

Definition

- Blood in the ejaculate. It alarms patients but is usually **benign and self-limiting**, most often from non-specific prostatic or seminal vesicle inflammation.

- **Primary (idiopathic):** no cause found, often young men. **Secondary:** a cause is identified.
- **Pseudo-haematospermia:** blood from the partner (partner's blood; condom test) or haematuria.

Causes

Group	Examples
Iatrogenic (commonest identifiable)	Prostate biopsy (about 10–50%, lasts about a month), TURP (transurethral resection of the prostate), radiotherapy, BPH (benign prostatic hyperplasia) procedures
Inflammation and infection	Prostatitis, vesiculitis, urethritis, STI (sexually transmitted infection), **genitourinary tuberculosis**, schistosomiasis
Obstruction and cysts	Ejaculatory duct obstruction or calculi; seminal vesicle cyst (Zinner syndrome) or utricle cyst
Tumours	**Prostate cancer**; rarely urethral, bladder or seminal vesicle
Vascular	Prostatic varices, BPH vessels, AV (arteriovenous) malformation
Systemic	Hypertension, bleeding disorders, anticoagulants, liver disease; also abstinence, trauma

Evaluation

- **History:** episodes, recent biopsy, LUTS (lower urinary tract symptoms), haematuria, TB (tuberculosis) contact, anticoagulants.
- **Examination:** blood pressure, genitalia, **DRE (digital rectal examination)**.
- **Basic tests:** urinalysis and culture, semen culture, STI screen, **PSA (prostate-specific antigen)** in men over 40, coagulation if indicated.

Who needs full work-up (full work-up if): age **≥40**, persistent beyond several weeks or recurrent, haematuria, abnormal DRE or PSA, risk factors for TB or cancer, or a distressed patient.

- **TRUS (transrectal ultrasound):** cysts, calculi, dilated seminal vesicles (>1.5 cm) or ejaculatory ducts (>2.3 mm) suggest obstruction.
- **MRI (magnetic resonance imaging) pelvis:** best for seminal vesicle haemorrhage, cysts and prostate cancer.
- **Cystoscopy and urine cytology**; early-morning urine for acid-fast bacilli.

Management

- **Young, single episode, normal tests: reassure**.
- **Treat the cause:** antibiotics or antitubercular drugs; control BP (blood pressure); correct coagulopathy.
- **Persistent prostatic bleeding:** 5-ARI (5-alpha-reductase inhibitor), e.g. finasteride.
- **Refractory or obstructive:** TURED (transurethral resection of ejaculatory ducts), cyst unroofing or aspiration, **seminal vesiculoscopy** (4–6 Fr small scope through the ejaculatory duct or utricle, irrigation, laser for stones).

Diagram to draw: algorithm: <40 and single episode → reassure; ≥40 or persistent → PSA, TRUS/MRI, cystoscopy.

Key points to remember:

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
- Exclude prostate cancer and genitourinary tuberculosis.
- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

SAQ 6 Avanafil

P1

Type: SAQ (10 marks)
Source: Practice Set 21 SAQ Sheet, Q3e

Introduction

- Second-generation oral **PDE5I (phosphodiesterase type 5 inhibitor)**; **FDA (US Food and Drug Administration) approved 2012**; Europe 2013. Structure differs most from the other PDE5Is.

Mechanism

- Sexual stimulation releases NO (nitric oxide), which raises cGMP (cyclic guanosine monophosphate) and relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; avanafil blocks this.
- **Enhances, does not initiate, erection**: stimulation is needed.
- **More selective for PDE5** over PDE6 (retina) and PDE1 than sildenafil, so **no visual disturbance** in the comparison table.

Pharmacokinetics (Campbell Table 72.7)

Feature	Avanafil	Sildenafil	Tadalafil
Time to peak	**0.3–0.5 h (fastest)**	0.8 h	2 h
Half-life	3–5 h	3–5 h	17.5 h
Fatty food	Reduces absorption	Reduces	No effect
Doses	**50, 100, 200 mg**	25–100 mg	5–20 mg
Blue vision	No	Yes	Rare

- Taken about 15–30 minutes before sex; start 100 mg; maximum once daily. Metabolised by CYP3A4 (cytochrome P450 3A4).

Efficacy

- Effective in general ED (erectile dysfunction), **diabetes** and **after nerve-sparing radical prostatectomy** (REVIVE trial programme): improves IIEF (International Index of Erectile Function) erectile scores and successful-intercourse rates, some within 15 minutes. Efficacy equivalent across PDE5Is; no head-to-head superiority.

Adverse effects

- Headache, flushing, nasal congestion, dyspepsia; back pain rare.

Contraindications and interactions

- **Nitrates in any form**: absolute; no nitrate within 12 hours of a dose.
- Avoid strong CYP3A4 inhibitors (ketoconazole, ritonavir); maximum 50 mg with moderate inhibitors.
- **α-blockers:** start 50 mg once stable (additive hypotension).
- Caution: aortic stenosis, recent myocardial infarction or stroke, hypotension, retinitis pigmentosa. Not recommended in severe renal impairment, dialysis or severe hepatic impairment.
- **Also know:** proposed prophylaxis for recurrent ischaemic priapism; can cause priapism rarely.

Advantages

- **Fastest onset**, fewer visual and muscle effects; suits on-demand spontaneous use.

Diagram to draw: NO–cGMP pathway in cavernosal smooth muscle with the PDE5 block.

> **Key points to remember:**
> - Peak 0.3–0.5 h; doses 50, 100, 200 mg.
> - Highly PDE5-selective: no blue vision.
> - Nitrates absolutely contraindicated.

SAQ 7 CRE & BEP

P2

Type: SAQ (10 marks)
Source: Practice Set 14 Paper 2, Q3f

"CRE" is read as carbapenem-resistant Enterobacterales, and BEP as bleomycin, etoposide and cisplatin chemotherapy for germ cell tumours; the two are answered separately.

Part A: CRE (carbapenem-resistant Enterobacterales)

- **Definition:** Enterobacterales (Klebsiella pneumoniae, Escherichia coli, Enterobacter) resistant to at least one carbapenem, or producing a carbapenemase.

Mechanism	Example
Class A carbapenemase	KPC (Klebsiella pneumoniae carbapenemase)
Class B metallo-β-lactamase	**NDM (New Delhi metallo-β-lactamase)**: common in India
Class D	OXA-48
Non-enzymatic	ESBL (extended-spectrum β-lactamase) or AmpC plus porin loss

- **Urological risk:** catheters, stents, prolonged stay, prior carbapenem or fluoroquinolone use, transrectal biopsy, ICU (intensive care unit) care.
- **Treatment** (by culture and carbapenemase type; ID (infectious disease) input):
- KPC, OXA-48: ceftazidime–avibactam; meropenem–vaborbactam or imipenem–relebactam for KPC.
- **NDM:** ceftazidime–avibactam plus aztreonam, or cefiderocol.
- Cystitis: single-dose aminoglycoside, fosfomycin; polymyxins as last resort. Avoid tigecycline for UTI (urinary tract infection) (low urine levels).

- **Prevention:** antibiotic stewardship, contact precautions, rectal screening of high-risk patients, early catheter removal, transperineal rather than transrectal biopsy.

Part B: BEP (bleomycin, etoposide, cisplatin)

- Standard first-line chemotherapy for metastatic **GCT (germ cell tumour)**; replaced PVB (less neuromuscular toxicity).
- **Usual 21-day cycle:** cisplatin 20 mg/m² days 1–5, etoposide 100 mg/m² days 1–5, bleomycin 30 units days 1, 8, 15.

IGCCCG (International Germ Cell Cancer Collaborative Group) risk	Regimen
Good	BEP ×3 or EP ×4
Intermediate/poor	BEP ×4 (VIP ×4 if lungs poor)
Stage I NSGCT (non-seminomatous GCT), high risk	BEP ×1 adjuvant

- **Toxicity:** bleomycin lung fibrosis (keep FiO_2 low at later RPLND (retroperitoneal lymph node dissection)) and Raynaud phenomenon; cisplatin nephrotoxicity, ototoxicity, neuropathy; myelosuppression; infertility (bank sperm); late cardiovascular disease and second cancers.

Diagram to draw: BEP 21-day cycle timeline.

> **Key points to remember:**
> - CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
> - BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
> - Bleomycin lung: low FiO_2 and restricted fluids at surgery.

SAQ 8 Genetic causes of male infertility. Enumerate and classify.

P3

Type: SAQ (10 marks)
Source: 2017-Summer Paper 3, Q3d

Introduction

- Genetic defects underlie many cases of **azoospermia** and **severe oligozoospermia**. They decide **sperm retrieval odds** and **transmission to offspring** through ICSI (intracytoplasmic sperm injection).

Classification

Group	Condition	Key features
1. Chromosomal, numerical	**Klinefelter syndrome (47,XXY)**	**Commonest genetic cause**; 1 in 500–1000; small testes, high FSH (follicle-stimulating hormone), low testosterone, usually azoospermia. Non-mosaic in 80–90%. Testicular sperm in about 50% (micro-TESE, microdissection testicular sperm extraction)
	47,XYY; **46,XX male** (SRY gene translocated to X)	XX male: azoospermia, no sperm retrievable

Group	Condition	Key features
2. Chromosomal, structural	**Robertsonian and reciprocal translocations**, inversions, ring Y	Oligozoospermia; recurrent miscarriage
3. Y chromosome microdeletions (Yq, AZF (azoospermia factor))	**AZFa** (USP9Y, DDX3Y), **AZFb**, **AZFc** (DAZ genes)	Complete AZFa or AZFb deletion: **no sperm at retrieval**. AZFc: sperm often in ejaculate or testis; sons inherit the deletion
4. Single-gene: excurrent ducts	**CFTR (cystic fibrosis transmembrane conductance regulator) mutations**: CBAVD (congenital bilateral absence of vas deferens); idiopathic ejaculatory duct obstruction	Low-volume azoospermia, normal spermatogenesis; **test both partners** (ΔF508)
5. Single-gene: hypothalamic–pituitary	**Kallmann syndrome** (KAL1, anosmia); GnRH (gonadotropin-releasing hormone) receptor mutations	Low LH (luteinising hormone) and FSH; treat with hCG ± FSH
6. Androgen action	**Androgen receptor** mutation (long CAG repeat); 5α-reductase type 2 deficiency	High testosterone and LH, poor sperm
7. Sperm structure and motility	**Globozoospermia** (SPATA16, DPY19L2: no acrosome); macrocephalic sperm (AURKC); **PCD (primary ciliary dyskinesia)**/Kartagener (DNAH5, DNAI1: immotile sperm)	ICSI; macrocephalic sperm: ICSI not advised (aneuploidy)

Genetic testing indications

- **Karyotype and Y microdeletion:** non-obstructive azoospermia and severe oligozoospermia (**<5 million/mL**, ASRM/AUA (American Society for Reproductive Medicine/American Urological Association) 2021; EAU (European Association of Urology) uses <10 million/mL for karyotype).
- **CFTR:** CBAVD, idiopathic obstructive azoospermia; screen the partner; counsel before ICSI.
- **Also know:** Noonan syndrome, myotonic dystrophy.

Diagram to draw: Y chromosome long arm showing AZFa, b, c.

> **Key points to remember:**
> - Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
> - AZFa or AZFb complete deletion: do not attempt retrieval.
> - CBAVD = genital form of cystic fibrosis; test both partners.

SAQ 9 Ejaculatory duct obstruction

P3

Type: SAQ (10 marks)
Source: Practice Set 2 Paper 3, Q3b

Assumed meaning: ejaculatory duct obstruction (EDO).

Anatomy

- Intraprostatic continuations of the seminal vesicles, joined by the vasal ampullae; they lack muscle and open at the **verumontanum**. Seminal vesicles give about 60% of the ejaculate and its fructose.

Aetiology

- **Congenital (commonest):** midline (Müllerian/utricular) cyst, Wolffian duct cyst, atresia. Idiopathic EDO carries CFTR (cystic fibrosis transmembrane conductance regulator) mutations in about half.
- **Acquired:** infection, instrumentation, duct stones, compression.
- **Zinner syndrome:** seminal vesicle cyst, ipsilateral renal agenesis, EDO.
- Complete bilateral EDO causes under 5% of azoospermia. Partial EDO gives oligoasthenospermia.

Clinical features

- Infertility, painful ejaculation, haematospermia, perineal pain.

Diagnosis

Semen (complete bilateral EDO)

- Volume **<1.0 mL**, **acidic pH**, low or absent **fructose**, azoospermia.
- Palpable vas, normal FSH (follicle-stimulating hormone) and testes.
- Exclude retrograde ejaculation and CBAVD (congenital bilateral absence of the vas deferens).

Investigations

Test	Method / finding
TRUS (transrectal ultrasound)	Seminal vesicle AP diameter **>1.5 cm**, duct >2.3 mm, midline cyst. Low specificity
Seminal vesicle aspiration	**≥3 sperm per high-power field** confirms obstruction
Vesiculography	Contrast into vesicles; invasive
Chromotubation	Indigo carmine into vesicles; dye at the veru on cystoscopy

Management

TURED (transurethral resection of the ejaculatory ducts)

- For azoospermic or severely oligoasthenospermic men, after the couple declines IVF (in vitro fertilisation).
- **Steps:** TRUS-guided aspiration (cryopreserve sperm) → instil indigo carmine with contrast → 24-Fr resectoscope, finger in rectum → resect the veru until dye effluxes. Spare bladder neck, sphincter and rectum. Catheter overnight.
- **Results:** volume rises in most; sperm appear in about half. Cold-knife incision alone re-obstructs.

Complications

- **Urine reflux** into the ducts, **epididymitis**, retrograde ejaculation, rectal injury, re-obstruction.

Alternatives

- **Seminal vesiculoscopy** (4–6 Fr via utricle): avoids reflux; restenosis unknown. Sperm retrieval with ICSI (intracytoplasmic sperm injection).

Diagram to draw: sagittal prostate with midline cyst compressing the ducts at the veru.

> **Key points to remember:**
> - Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
> - TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
> - TURED: sperm return in about half; reflux and epididymitis are its price.

SAQ 10 Tadalafil

P3

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 3, Q3b

Pharmacology

- Selective **PDE5 (phosphodiesterase type 5) inhibitor**; FDA (Food and Drug Administration) approved 2003.
- **Mechanism:** sexual stimulation releases NO (nitric oxide), raising cGMP (cyclic guanosine monophosphate), which relaxes cavernosal smooth muscle. PDE5 breaks down cGMP; tadalafil blocks this. It **augments, not initiates,** erection.
- Structurally distinct from sildenafil; cross-reacts with **PDE11**, little with PDE6 (retina).

Property	Tadalafil (vs sildenafil)
Time to peak	2 h (0.8 h)
Half-life	**17.5 h** (3–5 h)
Fatty food	No effect (reduces absorption)
Window	Widest (up to 36 h)

Uses and doses

1. **ED (erectile dysfunction):** 10–20 mg on demand, or 2.5–5 mg once daily. Intercourse success about 70%; 40–50% in diabetes and after prostatectomy.
2. **BPH (benign prostatic hyperplasia) with LUTS (lower urinary tract symptoms), with or without ED: 5 mg daily**, the only PDE5 inhibitor licensed. IPSS (International Prostate Symptom Score) falls about 6 points (placebo 3.6) **without Qmax (peak flow rate) change**; mechanism is NO–cGMP relaxation, perfusion and reduced afferent activity, not de-obstruction. Equal to tamsulosin for IPSS. EAU (European Association of Urology) 2022 and AUA (American Urological Association) 2021: option for moderate–severe LUTS.
3. **Also know:** fixed finasteride–tadalafil (FDA 2021); penile rehabilitation (benefit unproven); prevention of stuttering priapism (PDE5I paradox); pulmonary arterial hypertension.

Adverse effects

- Headache, dyspepsia, **back pain and myalgia** (characteristic), flushing, nasal congestion. Visual effects rare.
- NAION (non-arteritic anterior ischaemic optic neuropathy): no proven link; stop if sudden visual loss.

Contraindications and interactions

- **Nitrates in any form: absolute.** After tadalafil, no nitrates for **48 h** (24 h for sildenafil).
- Caution: α-blockers (start lowest dose of the added drug), aortic stenosis, hypotension. CYP3A4 inhibitors (ketoconazole, ritonavir) raise levels.
- Not studied: MI (myocardial infarction) or stroke within 6 months, unstable angina, BP <90/50 or >170/100 mmHg, retinitis pigmentosa.

Diagram to draw: NO–cGMP–PDE5 pathway in cavernosal smooth muscle.

> **Key points to remember:**
> - Half-life 17.5 h; food-independent; back pain.
> - 5 mg daily for LUTS: IPSS improves, Qmax does not.
> - No nitrates for 48 h after a dose.

SAQ 11 Retrograde ejaculation

P3

Type: SAQ (10 marks)
Source: Practice Set 3 Paper 3, Q3f

Definition

- RE (retrograde ejaculation): semen passes backwards into the bladder at orgasm because the **bladder neck fails to close**; orgasm is preserved, ejaculate absent or low volume.

Physiology

Phase	Event	Control
Emission	Vas, seminal vesicles, prostate deliver semen to posterior urethra	Sympathetic T10–L2
Ejection	**Bladder neck closes**, external sphincter relaxes, bulbospongiosus contracts	Somatic S2–S4 + sympathetic reflex
Orgasm	Cerebral perception	Cortical

Causes

- **Anatomical: TURP (transurethral resection of prostate)** 42–100%, TUIP (incision) 5–45%; most other resections and enucleations; previous bladder neck reconstruction. Ejaculation-sparing techniques, PUL (prostatic urethral lift) and Aquablation preserve ejaculation.
- **Neurogenic: diabetic autonomic neuropathy** (commonest medical), RPLND (retroperitoneal lymph node dissection; less with nerve sparing), colorectal surgery, sympathectomy, aortic surgery, spinal cord injury, multiple sclerosis.
- **Drugs:** α-blockers (silodosin, tamsulosin), antipsychotics, antidepressants, thiazides.
- **Congenital:** Müllerian duct cyst, Wolffian duct anomaly, prune belly syndrome.

Evaluation

- Low-volume or absent ejaculate with orgasm; infertility.
- **Post-ejaculatory urinalysis:** void, ejaculate, void again; centrifuge. **>5–10 sperm per HPF (high-power field)** confirms RE; significant if urinary sperm approach or exceed

antegrade sperm.
- Sperm and fructose in urine separate RE from **failure of emission**; TRUS (transrectal ultrasound) if EDO (ejaculatory duct obstruction) suspected.

Treatment

- Reassure if fertility not desired; stop offending drug.
- **Sympathomimetics** (close bladder neck): **pseudoephedrine** 60–120 mg (120 mg 6-hourly the day before and twice on the day); midodrine, ephedrine, phenylpropanolamine. Side effects: dry mouth, hypertension.
- **Imipramine** 25 mg twice daily; antihistamines (brompheniramine) add to sympathomimetics.
- **Bladder neck reconstruction:** poor results; collagen injection experimental.
- **Sperm retrieval:** alkalinise urine (sodium bicarbonate), void or catheterise after orgasm, wash sperm in buffer → IUI (intrauterine insemination), IVF (in vitro fertilisation) or ICSI (intracytoplasmic sperm injection).
- **Prevention:** nerve-sparing RPLND, ejaculation-preserving procedures for BPH (benign prostatic hyperplasia).

Diagram to draw: bladder neck open at ejaculation with semen entering the bladder.

> **Key points to remember:**
> - TURP and diabetic neuropathy are the commonest causes.
> - Post-ejaculatory urine: sperm and fructose confirm RE.
> - Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

SAQ 12 CBAVD

P3

Type: SAQ (10 marks)
Source: Practice Set 4 Paper 3, Q3f

Definition

- **CBAVD (congenital bilateral absence of the vas deferens):** both vasa absent, causing **obstructive azoospermia** with normal spermatogenesis.
- Often a **genital form of CF (cystic fibrosis)**: nearly all men with CF lack both vasa.

Genetics

- **CFTR (cystic fibrosis transmembrane conductance regulator)**: chloride channel gene; more than 1600 mutations.
- Severe mutation on both alleles: CF. **Mild mutation on one or both alleles: CBAVD.** Commonest severe mutation **ΔF508**; the **5T variant** (intron 8) is a common mild allele.
- A minority have other mesonephric duct gene defects instead, with more **renal agenesis**.

Clinical features

- **Vasa not palpable** (Meacham: a presumed vas slipping away three times is absent).
- Caput epididymis present and distended; body and tail often absent.
- Normal testis size, FSH (follicle-stimulating hormone) and testosterone.

- Seminal vesicles absent or hypoplastic: **semen volume <1.0 mL, acidic, fructose-negative**, azoospermic.
- Sinus or chest symptoms may reveal mild CF.

Evaluation

- Semen analysis; FSH; **TRUS (transrectal ultrasound)** for seminal vesicles.
- **Renal ultrasound** (renal agenesis in about 1 in 10).
- **CFTR testing of the man and his female partner** (carrier rate about 1 in 20 in northern Europeans). Panels cover 25–40 mutations, so a negative panel leaves residual risk.
- Testis biopsy unnecessary when FSH and testis size are normal.

Management

- **Genetic counselling**: if the partner carries a mutation, offspring may have CF; PGT (preimplantation genetic testing) or donor sperm can be offered.
- Reconstruction is impossible, so **surgical sperm retrieval + ICSI (intracytoplasmic sperm injection)**:

Technique	Points
MESA (microsurgical epididymal sperm aspiration)	Best: many motile sperm; cryopreserve several vials
PESA (percutaneous epididymal aspiration)	Local anaesthetic; fewer sperm
TESA/TESE (testicular aspiration/extraction)	If epididymal retrieval fails
- ICSI results are good; one study reported more miscarriage and stillbirth.	
- **Also know:** CUAVD (unilateral absence) is a Wolffian duct defect with ipsilateral renal agenesis.	

Diagram to draw: absent vas and seminal vesicle with intact testis and caput epididymis.

> **Key points to remember:**
> - CBAVD = mild CFTR mutation; test both partners.
> - Low-volume, acidic, fructose-negative azoospermia; normal FSH.
> - Renal ultrasound; MESA + ICSI.

SAQ 13 Sertoli cell only syndrome

P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3a (non-obstructive azoospermia also answered in 2025-Winter Paper 2, Q3a)

Definition

- SCOS (Sertoli cell only syndrome), **germ cell aplasia** (Del Castillo syndrome): seminiferous tubules lined only by Sertoli cells, with **complete loss of spermatogonia**. Leydig cells usually normal.
- The worst histological pattern of NOA (non-obstructive azoospermia).

Types

Type	Feature
Pure (complete, congenital)	All tubules lack germ cells
Mixed (focal, acquired)	Some tubules keep foci of spermatogenesis, the basis of sperm retrieval

Aetiology

- **Developmental:** primordial germ cells fail to migrate into the testis.
- **Genetic: complete AZFa (azoospermia factor a) deletion** gives pure SCOS; **AZFc** may give SCOS with foci; Klinefelter (47,XXY).
- **Acquired:** cryptorchidism, chemotherapy or radiation, mumps orchitis, torsion; idiopathic.

Clinical features

- Normal virilisation; **azoospermia** with normal ejaculate volume; small soft testes.

Evaluation

- Semen: azoospermia on two centrifuged samples.
- **FSH (follicle-stimulating hormone) high**, **inhibin B low** (Sertoli marker); LH (luteinising hormone) normal or high; testosterone normal or low.
- **FSH >7.6 IU/L and testis long axis ≤4.6 cm** suggest NOA.
- **Karyotype and Y microdeletion** before surgery.
- **Histology** confirms, usually at sperm retrieval.

Pattern	Meaning	Micro-TESE retrieval
Hypospermatogenesis	All stages, fewer	Best
Maturation arrest	Stops at one stage	About 40%
SCOS	No germ cells	**About 30%**

Management

- **Counsel** on genetic findings.
- **Complete AZFa or AZFb deletion:** retrieval essentially nil; do not operate; donor sperm or adoption.
- **Micro-TESE (microdissection testicular sperm extraction):** at ×20–25, **larger, opaque, yellow tubules** (foci of spermatogenesis) are selected; SCOS tubules are thin, white and stringy. Live births are reported even in SCOS.
- Sperm used for **ICSI (intracytoplasmic sperm injection)**.
- Optimise first: stop exogenous testosterone; hormonal stimulation evidence weak.
- **Also know:** spermatogonial stem cell transplantation and in-vitro spermatogenesis are experimental.

Diagram to draw: tubule cross-section: normal layered epithelium versus Sertoli cells only.

> **Key points to remember:**
> - Complete loss of spermatogonia; high FSH, low inhibin B.
> - Complete AZFa deletion: pure SCOS, no retrieval.
> - Micro-TESE retrieval about 30%: worst histology, not hopeless.

SAQ 14 Oxidative stress and male infertility

P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3b

Definition

- **Oxidative stress (OS):** imbalance where ROS (reactive oxygen species) exceed the antioxidant capacity of semen. The term **MOSI (male oxidative stress infertility)** has been proposed.
- Low ROS levels are **physiological**: needed for capacitation, hyperactivation, acrosome reaction and sperm–oocyte fusion.

Sources of ROS

Source	Examples
Leucocytes (main)	Infection, prostatitis, epididymitis, leucocytospermia
Immature sperm	Excess residual cytoplasm
Varicocele	Heat and hypoxia
Lifestyle	**Smoking**, alcohol, obesity, heat
Others	Ageing, diabetes, pollutants, radiation

Why sperm are vulnerable

- Membrane rich in **polyunsaturated fatty acids** → lipid peroxidation.
- Little cytoplasm, so few antioxidant enzymes; limited DNA repair.
- **Seminal plasma antioxidants:** superoxide dismutase, catalase, glutathione peroxidase; taurine, vitamins C and E, zinc.

Effects

- Lipid peroxidation → loss of **motility** and membrane fluidity.
- **Sperm DNA fragmentation** → poor embryo quality, miscarriage, lower ART (assisted reproductive technology) success.
- Mitochondrial damage and apoptosis.

Evaluation

- Semen analysis: peroxidase-positive WBC (white blood cells) **>1 million/mL** = leucocytospermia.
- **Direct ROS:** chemiluminescence (luminol).
- **Indirect:** total antioxidant capacity, malondialdehyde (lipid peroxidation), ORP (oxidation-reduction potential).
- **DNA fragmentation tests:** TUNEL (terminal deoxynucleotidyl transferase dUTP nick-end labelling), SCSA (sperm chromatin structure assay), SCD (sperm chromatin dispersion), Comet.
- Use selectively (unexplained infertility, recurrent miscarriage, failed ART); not routine.

Management

- **Lifestyle:** stop smoking, reduce weight and alcohol, avoid heat.

- **Treat cause:** antibiotics for infection; **varicocelectomy** for clinical varicocele (reduces DNA fragmentation).
- **Antioxidants** (vitamins C and E, CoQ10 [coenzyme Q10], zinc, selenium, carnitine): Cochrane review, low-quality evidence; **MOXI trial (2020)** showed no gain in semen parameters, DNA fragmentation or live birth. Avoid excess: **reductive stress** also harms.
- High DNA fragmentation with failed ICSI (intracytoplasmic sperm injection): **testicular sperm** has less fragmentation.

Diagram to draw: balance of ROS sources against seminal antioxidants, with effects on sperm.

> **Key points to remember:**
> - Leucocytes and immature sperm are main ROS sources.
> - Lipid peroxidation and DNA fragmentation are the key injuries.
> - Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

SAQ 15 Dhat syndrome

P3

Type: SAQ (10 marks)
Source: Practice Set 8 Paper 3, Q3c

Definition

- **Dhat syndrome:** a **culture-bound syndrome** of South Asia. The man believes he is losing **semen ("dhat")** in urine or nocturnal emissions, and attributes weakness, fatigue and sexual problems to it.
- Term coined by **Wig** (India); listed in ICD-10 (International Classification of Diseases, 10th revision) and in the DSM-5 (Diagnostic and Statistical Manual, 5th edition) glossary of cultural concepts of distress.
- Related syndromes: **Shen-kui** (China), **Koro** (genital retraction).

Cultural basis

- Traditional belief that semen is a **precious vital fluid** made from many drops of blood; its loss is felt to drain body and mind.

Epidemiology

- Young men, often unmarried or newly married, rural, lower education; common in urology and sexual medicine clinics.

Clinical features

- **Somatic:** fatigue, weakness, aches, poor appetite.
- **Psychological:** anxiety, low mood, guilt (about masturbation), poor concentration, sleep loss.
- **Sexual:** fear of ED (erectile dysfunction) or PE (premature ejaculation), worry about penis size and fertility.
- Whitish urine is usually phosphaturia, oxaluria or normal secretions.
- **Comorbidity: depression** and **anxiety** in many; somatisation.

Evaluation

- Non-judgmental sexual history and belief system.
- Examination of genitalia; urine microscopy (phosphates, oxalates, sperm, pus cells).
- Exclude organic disease: **UTI (urinary tract infection), prostatitis, STI (sexually transmitted infection), diabetes**, true spermatorrhoea.
- Screen for depression and anxiety.

Management

1. Counselling (main)

- **Sex education:** semen is continuously produced; nocturnal emission and masturbation are normal and harmless.
- Reassure without dismissing.

2. Psychological therapy: CBT (cognitive behavioural therapy) for misbeliefs and anxiety.

3. Drugs

- **SSRIs (selective serotonin reuptake inhibitors)** for depression, anxiety or associated PE; short anxiolytic course if needed.
- Treat genuine ED or PE.

4. Avoid harm

- Avoid unnecessary "tonics", antibiotics and repeated tests; they reinforce the belief.
- Refer to psychiatry when depression is significant.

Prognosis

- Good with counselling; drop-out is common.

Diagram to draw: flowchart: complaint of semen loss → exclude organic cause → counsel → treat depression or anxiety.

> **Key points to remember:**
> - Culture-bound semen-loss anxiety of South Asia.
> - Depression and anxiety common; exclude UTI, prostatitis, STI.
> - Counselling and sex education first; SSRIs for comorbidity.

SAQ 16 Male menopause (andropause, late-onset hypogonadism)

P4

Asked as:

- Male menopause. — *Summer 2017 Paper IV Q3e (SAQ)*
- Andropause — *Winter 2017 Paper IV Q3e (SAQ)*
- Male menopause — *Summer 2019 Paper IV Q3f (SAQ)*

Type: SAQ (10 marks)

Read as: late-onset hypogonadism (testosterone deficiency of ageing men, "andropause"). Same topic as 2017-Summer Paper 4, Q3e and 2017-Winter Paper 4, Q3e; full answer given here.

Definition

- **Symptoms plus consistently low serum testosterone** in an ageing man.
- "Menopause" (or "Andropause") is a misnomer: the fall is gradual and partial. Preferred terms: PADAM (partial androgen deficiency of the ageing male), LOH (late-onset hypogonadism).

Pathophysiology

- Testosterone falls slowly from middle age.
- Testosterone in the 70s is about one-third lower than in young men. (Another source: mean testosterone in the seventh decade is about **one-third lower** than in young men.)
- **Testis:** fewer Leydig cells, lower sperm output; FSH (follicle-stimulating hormone) rises about threefold.
- **Centre:** blunted feedback, irregular GnRH (gonadotrophin-releasing hormone) pulses.
- **SHBG (sex hormone-binding globulin) rises with age**, so free testosterone (bioavailable [free + albumin-bound] testosterone) falls more.
- Worse with obesity, diabetes and **opioids**.

Clinical features

- **Sexual (most specific):** low libido, ED (erectile dysfunction), fewer morning erections.
- Fatigue, low muscle mass, osteoporosis, anaemia, low mood.
- **ADAM (Androgen Deficiency in the Aging Male)** questionnaire: sensitive, not specific.

Diagnosis

- **Morning (7–11 am) total testosterone, twice (repeated).** AUA (American Urological Association): **<300 ng/dL** (≈10.4 nmol/L). EAU (European Association of Urology): about **12 nmol/L**; calculated free testosterone if borderline. Calculated bioavailable testosterone when SHBG is abnormal.
- **LH (luteinising hormone):** high in primary, low or normal in secondary hypogonadism. Check **prolactin** if LH is low (Low LH → **prolactin**); pituitary MRI (magnetic resonance imaging) if testosterone <150 ng/dL or prolactin high.
- Baseline: PSA (prostate-specific antigen), DRE (digital rectal examination), haematocrit.

Treatment

- Lifestyle first: weight loss, exercise, treat comorbidities.
- **TRT (testosterone replacement therapy)** for symptoms + confirmed low level: gels, injections (enanthate every 2–3 weeks; undecanoate every 10–14 weeks), oral undecanoate. Stop if no benefit at 3–6 months (another source: at about 3 months).
- **Contraindications:**
- Prostate or breast cancer.
- **Wish for fertility** (TRT stops sperm production [suppresses spermatogenesis]; use hCG [human chorionic gonadotrophin] or clomiphene).
- Haematocrit >50% (raised haematocrit).
- Untreated severe sleep apnoea; recent MI (myocardial infarction) or stroke.

Monitoring

- At 3–6 months, then yearly: testosterone, **haematocrit** (stop/hold if >54%), PSA.

Safety evidence

- **TRAVERSE (2023):** 5,246 men aged 45–80 at high cardiovascular risk. TRT was **non-inferior for major cardiac events**; more atrial fibrillation and pulmonary embolism.
- **Testosterone Trials**: modest sexual benefit, **no cognitive benefit**.

Diagram to draw: hypothalamic–pituitary–testis axis with ageing defects.

> **Key points to remember:**
> - Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
> - SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
> - TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

SAQ 17 Klinefelter's syndrome.

P4

Type: SAQ (10 marks)
Source: 2018-Summer Paper 4, Q3b

Definition and genetics

- **47,XXY**: commonest sex chromosome aneuploidy and **commonest genetic cause of male infertility**; 1 in 500–1000 male births.
- Non-mosaic in **80–90%**; the rest mosaic **46,XY/47,XXY** (milder, more sperm) or 48,XXXY variants. Extra X from non-disjunction (maternal age).
- Underdiagnosed: most never diagnosed; increasingly detected on NIPT (non-invasive prenatal testing).

Pathophysiology

- Seminiferous tubules **hyalinise** progressively from puberty; germ cells are lost. Leydig cells fail, so testosterone falls. Puberty starts on time, then stalls.

Clinical features

Age	Features
Infant	Usually normal; micropenis, cryptorchidism, hypospadias
Adolescent/adult	**Small firm testes (<3.5 cm, 1–4 mL)**, gynaecomastia, tall stature with long legs (only ~30%), sparse hair, female fat distribution
Neurodevelopment	Language, learning and attention difficulties

Hormones and tests

- **High FSH (follicle-stimulating hormone) and LH (luteinising hormone), low or low-normal testosterone, high oestradiol** (raised oestradiol:testosterone ratio). Inhibin B low.
- Semen: **azoospermia** in most (~8% have ejaculated sperm).
- **Karyotype** confirms (AUA/ASRM, American Urological Association/American Society for Reproductive Medicine, guideline: karyotype all men with non-obstructive azoospermia or severe oligozoospermia).

Management

- **Fertility: micro-TESE** (microdissection testicular sperm extraction) + ICSI (intracytoplasmic sperm injection). Sperm found in **~40–50%** (68% in one series); offspring are usually chromosomally normal.
- Spermatogenesis declines with age, so retrieve early (late adolescence or young adulthood). Some give clomiphene, anastrozole or hCG (human chorionic gonadotropin) beforehand.
- **TRT (testosterone replacement therapy):** for hypogonadism (bone, muscle, mood, libido). It **suppresses spermatogenesis**, so offer sperm retrieval or banking **before** starting.
- Gynaecomastia: reduction mammoplasty if persistent. Speech, educational and psychological support.

Associated risks

- Osteoporosis, metabolic syndrome, diabetes, VTE (venous thromboembolism), autoimmune disease.
- **Mediastinal (extragonadal) germ cell tumours**; male breast cancer (~1% lifetime).

Diagram to draw: hypothalamic–pituitary–testis axis showing high FSH and LH, low testosterone.

> **Key points to remember:**
> - 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
> - Small firm testes, high FSH and LH, azoospermia.
> - Micro-TESE ~50% success; retrieve sperm before testosterone.
> - Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

SAQ 18 Semen analysis nomenclature.

P4

Type: SAQ (10 marks)
Source: 2018-Summer Paper 4, Q3e

Collection (WHO, World Health Organization, 2021, 6th edition)

- Masturbation into a clean wide-mouthed non-toxic container (special condom if needed); complete sample.
- Abstinence **2–7 days** (evidence: 1–2 days best). Keep at 20–37 °C; analyse within **1 hour**, after liquefaction (~30 min).
- At least **two samples, 2–3 weeks apart**, because of variability.

Lower reference limits (5th centile, fertile men)

Parameter	Lower limit
Volume	**1.4 mL**
Concentration	**16 million/mL**
Total count	**39 million/ejaculate**
Total motility	**42%**
Progressive motility	**30%**

Parameter	Lower limit
Normal forms (strict)	**4%**
Vitality	**54%**
Peroxidase-positive leucocytes	<1 million/mL

- These are reference values, not fertility cut-offs. The 6th edition grades motility as progressive, non-progressive or immotile.

Nomenclature

Term	Meaning
Normozoospermia	All parameters at or above reference
Oligozoospermia	Total count (or concentration) below reference
Asthenozoospermia	Progressive motility below reference
Teratozoospermia	Normal forms below reference
OAT (oligoasthenoteratozoospermia)	All three abnormal
Azoospermia	No sperm, **including centrifuged pellet**
Cryptozoospermia	Sperm absent in fresh smear, present only after centrifugation
Aspermia	No ejaculate (dry orgasm)
Hypospermia	Volume below reference
Necrozoospermia	Low vitality (many dead sperm)
Leucocytospermia	Peroxidase-positive white cells above 1 million/mL
Haemospermia	Blood in ejaculate
Globozoospermia	Round-headed sperm without acrosome

Notes

- **Low volume or aspermia:** post-ejaculatory urine for retrograde ejaculation; TRUS (transrectal ultrasound) for EDO (ejaculatory duct obstruction).
- **Vitality:** eosin–nigrosin, or **HOS (hypo-osmotic swelling)** test, which keeps sperm usable for ICSI (intracytoplasmic sperm injection).
- Only azoospermia is definitive; values overlap between fertile and infertile men.
- **Also know:** Kruger strict morphology; DNA fragmentation; antisperm antibodies (MAR, mixed antiglobulin reaction); semen pH no longer recommended.

Diagram to draw: normal sperm (head, midpiece, tail) beside a WHO reference box.

> **Key points to remember:**
> - WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
> - Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
> - Two samples 2–3 weeks apart; abstinence 2–7 days.

Rapid Revision Sheet · Section 9

Spermatogenesis, azoospermia and non-obstructive azoospermia (LAQ)

- Spermatogenesis about 64 days; Ap divide every 16 days; six stages.
- Intratesticular testosterone 50 × serum; exogenous testosterone causes azoospermia.
- FSH → Sertoli, LH → Leydig; tight junctions between Sertoli cells form the blood–testis barrier.
- Confirm azoospermia on two centrifuged samples.
- FSH ≤7.6 IU/L and testis >4.6 cm → obstructive; FSH >7.6 and testis ≤4.6 cm suggest NOA.
- Karyotype and Y microdeletion in all.
- AZFa/b complete deletion: do not operate (no retrieval).
- NOA: micro-TESE + ICSI; sperm in about half; SCO 30%.
- Maturation arrest: arrest at one stage; early (spermatocyte) or late (spermatid).
- FSH and testis size may be normal in maturation arrest: biopsy diagnoses.
- Micro-TESE retrieval in maturation arrest about 40%; none with complete AZFb deletion.

Azoospermia: testicular biopsy and surgical sperm retrieval techniques (LAQ)

- Azoospermia: no sperm in centrifuged pellet on two samples.
- Biopsy indication: azoospermia, normal testes, vasa and FSH.
- Open biopsy is the gold standard; fix in Bouin's, never formalin; no forceps; touch prep.
- OA: normal FSH and testes; ICSI essential.
- OA: MESA (capillary action); MESA is best: many motile sperm, multiple vials.
- Move proximally; capillary action, never suction.
- TESA: 23 G needle, suction, local anaesthesia; for OA with normal spermatogenesis.
- TESA/TESE when epididymal sperm absent.
- NOA: micro-TESE best; finds sperm in about half of NOA; TESA yield poor.
- AZFa/AZFb deletion: 0% retrieval (no sperm), do not operate.
- All surgically retrieved (testicular) sperm need ICSI; cryopreserve extras.
- Haematoma commonest; epididymal biopsy most serious.

Surgical management of ED and penile prostheses (types, biofilms) (LAQ)

- Prosthesis is the last line but gives 80–90% satisfaction.
- 3-piece inflatable gives the best rigidity and flaccidity.
- Malleable rods: nitinol (Tactra), silver (Genesis); almost never fail.
- Glucose <200 mg/dL, chlorhexidine–alcohol, coated device, no-touch technique.
- Biofilm: microbes in a matrix; antibiotics fail.
- Infection 1–3%; coated devices and no-touch technique; infected implant: remove all parts; Mulcahy salvage 84%.
- Satisfaction 80–90%; repairs 15% at 5 years.
- Arterial revascularisation only for young men with post-traumatic focal stenosis.

Priapism: diagnosis and management (LAQ)

- Blood gas decides the type: ischaemic pO2 <30, pCO2 >60, pH <7.25.
- Aspiration plus phenylephrine first; distal shunt ± tunnelling before proximal.
- ED risk 0% ≤12 h, 100% ≥36 h; consider early prosthesis beyond 36 h.

- Beyond 36 h or refractory: early penile prosthesis (or analgesia and counselling).
- High flow: observe (about 60% resolve), then embolise; repeat embolisation rather than surgery.

Physiology of erection, neurotransmitters, and evaluation of the diabetic with ED (LAQ)

- nNOS initiates, eNOS maintains; PDE5 ends (terminates) erection.
- Noradrenaline α1 is the main anti-erectile transmitter.
- Dopamine/oxytocin promote; serotonin/prolactin inhibit.
- Rigid phase: ischiocavernosus, suprasystolic pressure.
- Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
- Diabetic ED: check HbA1c, testosterone and cardiac risk.

Discuss the role of urologist in reproductive endocrinology and male infertility practice. (LAQ)

- FSH >7.6 IU/L + testis ≤4.6 cm = spermatogenic failure.
- Exogenous testosterone is a contraceptive: never give it for infertility.
- Hypogonadotropic hypogonadism: hCG ± FSH; most treatable cause.
- Varicocelectomy only for palpable varicocele with abnormal semen; micro-TESE for non-obstructive azoospermia.

Male reproductive immunology and immunobead test (SAQ)

- BTB = Sertoli tight junctions; plus active tolerance (galectin-1, TGF-β, androgens).
- ASA: vasectomy commonest; ≥50% motile sperm bound is significant; head-bound IgA matters most.
- Direct IBT = sperm-bound antibody; indirect = antibody in serum, seminal plasma or mucus.
- Treatment of choice: IVF/ICSI.
- Leukocytospermia >1 million/mL.

Premature ejaculation (SAQ)

- ISSM: about 1 min (lifelong), ≤3 min (acquired) plus poor control plus distress.
- Lifelong is neurobiological; acquired is usually secondary.
- Dapoxetine 30–60 mg on demand; paroxetine gives the greatest delay.

Oligospermia and oligoasthenozoospermia (SAQ)

- WHO 2021: concentration 16 million/mL, progressive motility 30%; severe <5 million/mL.
- Karyotype and Y-microdeletion if under 5 million/mL.
- Repair palpable varicoceles only; never give testosterone.

Management of Peyronie's disease (SAQ)

- Ask the "straight penis" question; duplex before surgery.
- CCH: IMPRESS, stable 30–90° curve, intact erections.
- Surgery: stable ≥12 months; plication <60–70°, graft complex curves, prosthesis if ED.

Haematospermia (SAQ)

- Usually benign; prostate biopsy is the commonest identifiable cause.
- Full work-up if ≥40, persistent or recurrent, or associated haematuria (or with haematuria).
- Exclude prostate cancer and genitourinary tuberculosis.

- TRUS: seminal vesicle >1.5 cm or ejaculatory duct >2.3 mm suggests obstruction.

Avanafil (SAQ)

- Peak 0.3–0.5 h; doses 50, 100, 200 mg.
- Highly PDE5-selective: no blue vision.
- Nitrates absolutely contraindicated. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

CRE & BEP (SAQ)

- CRE: NDM common in India; ceftazidime–avibactam + aztreonam for metallo-β-lactamase.
- BEP ×3 good risk, ×4 intermediate/poor; EP ×4 if bleomycin avoided.
- Bleomycin lung: low FiO_2 and restricted fluids at surgery. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Genetic causes of male infertility. Enumerate and classify. (SAQ)

- Klinefelter is the commonest genetic cause; micro-TESE finds sperm in about half.
- AZFa or AZFb complete deletion: do not attempt retrieval.
- CBAVD = genital form of cystic fibrosis; test both partners.

Ejacultory duct obstruction (SAQ)

- Low volume (<1 mL), acidic, fructose-negative azoospermia with palpable vasa.
- TRUS: vesicle >1.5 cm; aspiration ≥3 sperm per high-power field confirms.
- TURED: sperm return in about half; reflux and epididymitis are its price.

Tadalafil (SAQ)

- Half-life 17.5 h; food-independent; back pain.
- 5 mg daily for LUTS: IPSS improves, Qmax does not.
- No nitrates for 48 h after a dose.

Retrograde ejaculation (SAQ)

- TURP and diabetic neuropathy are the commonest causes.
- Post-ejaculatory urine: sperm and fructose confirm RE.
- Pseudoephedrine first; else urinary sperm retrieval for assisted reproduction.

CBAVD (SAQ)

- CBAVD = mild CFTR mutation; test both partners.
- Low-volume, acidic, fructose-negative azoospermia; normal FSH.
- Renal ultrasound; MESA + ICSI.

Sertoli cell only syndrome (SAQ)

- Complete loss of spermatogonia; high FSH, low inhibin B.
- Complete AZFa deletion: pure SCOS, no retrieval.
- Micro-TESE retrieval about 30%: worst histology, not hopeless.

Oxidative stress and male infertility (SAQ)

- Leucocytes and immature sperm are main ROS sources.
- Lipid peroxidation and DNA fragmentation are the key injuries.
- Leucocytospermia >1 million/mL; antioxidant benefit unproven (MOXI).

Dhat syndrome (SAQ)

- Culture-bound semen-loss anxiety of South Asia.
- Depression and anxiety common; exclude UTI, prostatitis, STI.
- Counselling and sex education first; SSRIs for comorbidity. *All the key points from this section in one place. Read this in the last 30 minutes before the exam.*

Male menopause (andropause, late-onset hypogonadism) (SAQ)

- Symptoms + low morning testosterone twice (AUA <300 ng/dL; EAU 12 nmol/L).
- SHBG rises with age; LH separates primary from secondary; prolactin if LH low.
- TRT harms fertility; watch haematocrit; TRAVERSE: cardiac non-inferiority.

Klinefelter's syndrome. (SAQ)

- 47,XXY; non-mosaic 80–90%; 1 in 500–1000.
- Small firm testes, high FSH and LH, azoospermia.
- Micro-TESE ~50% success; retrieve sperm before testosterone.
- Risks: mediastinal germ cell tumour, breast cancer, osteoporosis.

Semen analysis nomenclature. (SAQ)

- WHO 2021: 1.4 mL, 16 million/mL, 39 million, 42%/30%, 4%, 54%.
- Cryptozoospermia = sperm only in pellet; aspermia = no ejaculate.
- Two samples 2–3 weeks apart; abstinence 2–7 days.